MUHS final-year BPTh (IV BPTh), 2012 syllabus

Neuro Physiotherapy

Complete exam notes in syllabus order: Chapters 1 to 4, assessment, adult and paediatric conditions, the treatment programme and exam skills.



Built from	20 MUHS question papers, Summer 2016 to Summer 2026 (code 51412)
Arrangement	The order of the MUHS syllabus, so the notes follow the order your college teaches
Contents	45 pages with past questions, worked long cases, answer skeletons and self-tests
Content checked	20 October 2026
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/neuro/

Educational notes for exam preparation. Not medical advice. Written in original words from current textbooks and dated guidelines; each page lists its sources.

How to use these notes

1. **Follow your classes.** The parts are in the order of the MUHS syllabus. Each part opens with its pages and their page numbers.
2. **Read a topic once**, slowly, including the "Why?" boxes.
3. **Close it and recall.** Write the keywords and draw the diagrams on paper from memory, then compare.
4. **Test yourself** with the questions at the end of each topic. Cover the answer before you look.
5. **Come back** on day 1, day 7 and day 21. Testing yourself again beats re-reading.

> **What the real paper looks like (20 papers, 2016–2026)**
>
> 80 marks in 3 hours. **No MCQs**, even though the syllabus printed a scheme with them.
>
> - Short answers: any 5 of 6 at **3 marks**, and any 5 of 6 at **7 marks**.
> - Long answers: two questions, each any 1 of 2 at **15 marks**. Almost every long case is a patient with a neurological condition; most ask for ICF, a functional diagnosis or impairments, goals and management.
>
> Practical: long case 45, short case 20, spots 10, journal 5. See the exam skills pages at the end.

How each topic is laid out

Each topic opens with its syllabus heading, priority, the sessions it was asked in (LAQ = 15-mark long answer, 7m and 3m = short answers; S = Summer, W = Winter, for example S24 = Summer 2024), the usual mark split, and the guidelines it was checked against. Page references such as (p. 123) point to related pages in this book. Abbreviations are listed at the end.

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1 Applying the ICF model in neurological conditions: domains, relationships, goals and red flags

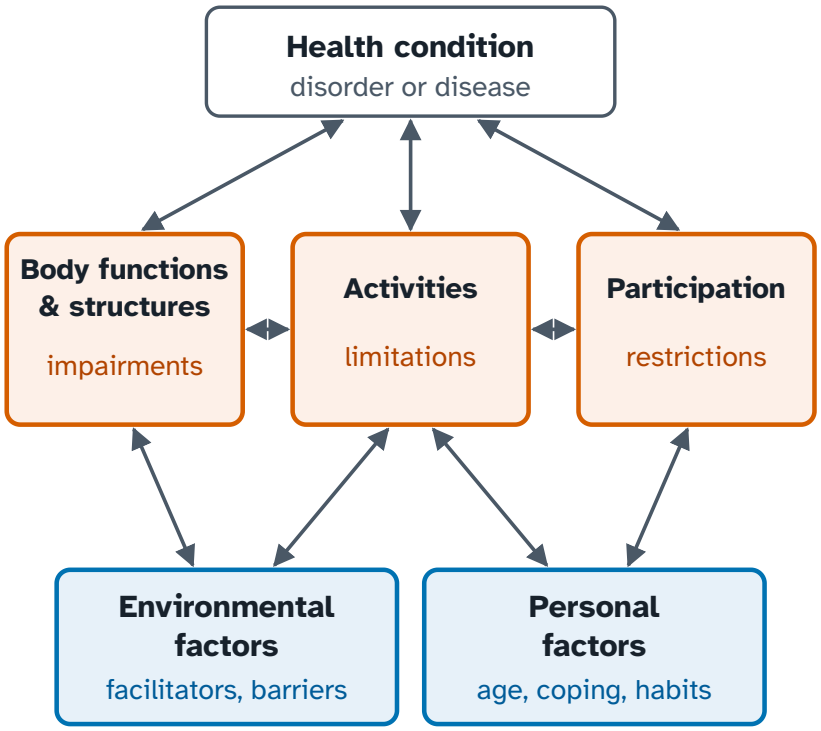
> The domains of ICF were a 7-mark question in S19, and nearly every long case in 2016–2026 asked for ICF, a functional diagnosis or impairments with clinical reasoning. Learn the model once, with its domains, relationships and goal setting, and you can apply it to any patient on any page of these notes.

Syllabus	Ch 1 Features of the ICF model (bio, psycho and social) to plan goals: a. bidirectional relationships among the domains; b. environmental and personal factors as facilitators and barriers; c. capacity and performance; d. patient-specific goals and expected outcome with clinical reasoning; e. documentation of disability and functioning; red flags
Priority	**High yield.** In 1 of 20 papers (2016–2026) as a 7-mark question (domains of ICF), and inside almost every long case.
Asked as	7m S19
Usual split	7 (define ICF 1, the components with their chapters 4, qualifiers or an example 2); in long cases ICF carries 5–7 marks
Checked against	WHO, Towards a Common Language for Functioning, Disability and Health: ICF (beginner's guide, 2002) for components, chapters, qualifiers, capacity and performance, and principles. Full list under Sources (p. 10). Content checked 20 October 2026.

1 The ICF model

- **ICF** (International Classification of Functioning, Disability and Health; WHO, 2001) describes health and disability as an interaction between a health condition and contextual factors.
- It is **biopsychosocial**: it joins the medical model (disability caused by disease, fixed by treatment) and the social model (disability created by society and the environment).
- **Functioning** is the positive side (body functions, activities, participation); **disability** is the negative side (impairments, activity limitations, participation restrictions).
- **Principles:** universality (it applies to everyone, not only disabled people), parity (no split by cause into "physical" and "mental"), neutrality (neutral wording), and environmental influence.

The ICF model



Draw this first in every ICF answer. The arrows go both ways: every box can change every other box.

2 Components and domains of ICF (7 marks in S19)

Component (code letter)	Domains (chapters)	Neuro example
Body functions (b): physiological and mental functions	Mental; sensory functions and pain; voice and speech; cardiovascular, haematological, immunological and respiratory; digestive, metabolic and endocrine; genitourinary and reproductive; neuromusculoskeletal and movement-related; skin	Muscle power (b730), muscle tone (b735), control of voluntary movement (b760), gait pattern (b770)
Body structures (s): anatomical parts	Nervous system; eye and ear; voice and speech; cardiovascular, immunological and respiratory; digestive, metabolic and endocrine; genitourinary and reproductive; structures related to movement; skin	Brain, spinal cord, peripheral nerve
Activities and participation (d): execution of tasks and involvement in life situations	Learning and applying knowledge; general tasks and demands; communication; mobility; self-care; domestic life; interpersonal interactions and relationships; major life areas (education, work); community, social and civic life	Transferring (d420), walking (d450), washing (d510), paid work (d850)
Environmental factors (e): the physical, social and attitudinal world	Products and technology; natural environment and human-made changes; support and relationships; attitudes; services, systems and policies	Mobility aids (e120), immediate family (e310)
Personal factors	Not classified: age, sex, education, coping style, habits, motivation	A motivated young teacher; a smoker

Qualifiers

- A code becomes a classification when a **qualifier** is added after the point: 0 no problem, 1 mild, 2 moderate, 3 severe, 4 complete problem (8 not specified, 9 not applicable).
- For example, b730.3 is a severe impairment of muscle power.
- Environmental factors use a minus scale for barriers and a plus for facilitators: e120.2 is a moderate barrier, e120+2 a moderate facilitator.

3 Bidirectional relationships and their clinical use

- **Impairment to activity:** spasticity and weak dorsiflexors limit walking.
- **Activity back to impairment:** not walking causes weakness, contracture and low fitness (secondary impairments).
- **Participation to body functions:** losing a job or friends lowers mood and activity, which worsens fatigue and fitness.
- **Environment to activity:** a ramp or walking aid improves mobility; stairs and an Indian-style toilet make it harder.
- **Personal factors:** motivation and coping change how much practice happens and how much is recovered.

Clinical use: treat at more than one level at once (for example, strengthen, practise walking, and change the home), and predict secondary problems before they happen.

4 Environmental and personal factors: facilitators and barriers

Factor	As a facilitator	As a barrier
Products and technology	Walking aid, orthosis, wheelchair, grab rails, western commode	No aids; Indian-style toilet; poorly fitted orthosis
Built and natural environment	Ground-floor home, lift, ramps	Stairs (a fifth-floor flat without a lift), uneven roads, heat
Support and relationships	Joint family, caring spouse, supportive employer	Living alone; carer at work all day
Attitudes	Family encourages independence	Overprotection; stigma about disability
Services, systems and policies	Nearby rehabilitation centre, insurance or government schemes, disability certificate	Distance and cost of therapy; loss of income
Personal factors	Young, motivated, educated	Older, depressed, smoker, low health literacy

5 Capacity and performance

- **Capacity:** the highest level a person can achieve in a standard environment, without help or devices (what the person *can* do in the clinic).
- **Performance:** what the person actually does in their usual environment, with whatever help and devices they really use (what the person *does* at home).
- **The gap:** if capacity is greater than performance, something in the environment is a barrier; if performance is better than capacity, the environment is helping.
- **Example:** a man after stroke walks 50 m with a stick in the clinic (capacity) but never leaves his first-floor flat because of the stairs (performance). Treatment: stair training and a rail, not only more gait training.

6 Patient-specific goals with clinical reasoning

- Set goals **with** the patient and family, from what matters to them (work, toilet independence, walking to the temple or market).
- Make them **SMART:** specific, measurable, achievable, relevant and time-bound.
- **Short-term goals** (2–6 weeks): impairments and basic activities. **Long-term goals** (months): activities and participation.
- Give the reason for each goal from the ICF (the clinical reasoning) and the expected outcome from the prognosis.

Weak goal	SMART goal
Improve balance	Berg score from 38 to 46 in 6 weeks, so he can stand to cook safely

Weak goal	SMART goal
Improve walking	Walks 200 m outdoors with a stick in under 4 minutes within 8 weeks, to reach the bus stop
Return to work	Back at the shop for half days with a chair at the counter by 3 months

7 Documentation and red flags

Documentation of disability and functioning

- Record each ICF level with measures (see scales (p. 21)), the functional diagnosis, the goals and the dates.
- **ICF codes with qualifiers** can summarise problems; **ICF Core Sets** (short lists of the most relevant categories) exist for conditions such as stroke, spinal cord injury, multiple sclerosis, brain injury and cerebral palsy.
- Written records support the disability certificate and access to schemes.

Red flags: stop, and refer urgently

Sign	Think of
Sudden face droop, arm weakness, speech or vision loss, severe sudden headache	Stroke or TIA: call emergency services at once
Headache, vomiting, drowsiness, falling GCS, unequal pupils	Raised intracranial pressure, shunt failure
Saddle numbness, new bladder or bowel loss, weakness in both legs	Cauda equina or spinal cord compression
Rapidly spreading weakness, weak cough, breathlessness lying flat, difficulty swallowing	GBS, myasthenic crisis, MND with breathing failure
Pounding headache and high blood pressure in SCI at T6 or above	Autonomic dysreflexia
Calf swelling and pain, sudden breathlessness or chest pain	Deep vein thrombosis, pulmonary embolism
Fever with neck stiffness and confusion	Meningitis or encephalitis
Night pain, weight loss, history of cancer	Tumour or metastasis

8 Writing ICF in a long case

1. **Spot the clues** in the question: age, job, family, home, the timing, and each symptom.
2. **Draw the model** and fill two or three points into each box.
3. **Make a table** with three columns: component, findings from the case, and clinical reasoning (why it happens or why it matters).
4. **Write a one-line functional diagnosis** that links the health condition, the main impairments, the activity limitations and the participation restriction.
5. Go on to goals that come straight from the table.

Every condition page in these notes has a worked example in section 10 (for example stroke (p. 50), cerebral palsy (p. 275), GBS (p. 144)).

Recall check. Name the five components of the ICF and two chapters under each of the first four.

9 Exam corner

Past questions (MUHS, 2016–2026)

S19 Q2(d), 7 marks

Explain domains of ICF.

Long cases, every paper

ICF with clinical reasoning, functional diagnosis, or structural and functional impairments (see the condition pages).

Keywords to underline

biopsychosocial functioning and disability body functions and structures activities and participation environmental factors personal factors qualifiers capacity and performance facilitators and barriers bidirectional SMART goals

Answer skeleton: domains of ICF (7 marks)

1. **[Definition, 1]** WHO 2001, biopsychosocial, functioning and disability.
2. **[Model, 1]** The diagram with two-way arrows.
3. **[Components and chapters, 4]** Body functions, body structures, activities and participation, environmental factors (with their chapters), personal factors.
4. **[Qualifiers and an example, 1]** 0–4 scale; a neuro example such as b735.2.

Draw these

- The ICF model (section 1)

Viva questions

1. Why is the ICF called biopsychosocial?
2. What is the difference between an activity limitation and a participation restriction?
3. What is the difference between capacity and performance?
4. Why are personal factors not classified?
5. Give two environmental facilitators and two barriers for a wheelchair user in India.
6. What does b730.3 mean?

Linked practical spots

- In the practical long case, present your findings in ICF order and end with the functional diagnosis (see the practical exam (p. 358)).

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Differentiate capacity and performance.**

Answer

Capacity is the highest level a person can do in a standard environment without help or devices; performance is what the person actually does in the usual environment with the help and devices actually used. If capacity is higher, the environment is a barrier; if performance is higher, the environment is a facilitator.

3 m **Give examples of environmental facilitators and barriers in a patient with paraplegia.**

Answer

Facilitators: a light wheelchair, ground-floor home with ramps, western toilet with rails, supportive family and employer, disability schemes. Barriers: stairs, Indian-style toilet, uneven roads, crowded transport, cost of equipment, negative attitudes.

7 m **Explain domains of ICF. (S19)**

Answer

ICF (WHO, 2001) is a biopsychosocial classification of functioning and disability; draw the model. **Body functions** (mental; sensory and pain; voice and speech; cardiovascular, haematological, immunological and respiratory; digestive, metabolic and endocrine; genitourinary and reproductive; neuromusculoskeletal and movement-related; skin). **Body structures** (the matching eight chapters, from the nervous system to skin). **Activities and participation** (learning and applying knowledge; general tasks; communication; mobility; self-care; domestic life; interpersonal relationships; major life areas; community, social and civic life). **Environmental factors** (products and technology; natural environment; support and relationships; attitudes; services, systems and policies). **Personal factors** (not classified). Qualifiers 0–4 show severity; environmental factors use + for facilitators. Example: in stroke, b735.2 moderate tone impairment, d450.3 severe walking limitation.

LAQ **Practice: using a 65-year-old woman with Parkinson's disease (Hoehn and Yahr III) who has stopped going to the temple after two falls, explain the ICF with its bidirectional relationships and write two SMART goals. (7 + 5 + 3)**

Answer

ICF (7): health condition PD; body functions: bradykinesia, rigidity, postural instability, fear of falling; activities: turning, walking outdoors, steps; participation: temple visits, social life; environment: crowded temple steps (barrier), supportive daughter (facilitator); personal: older, anxious. **Relationships (5):** postural instability causes falls; falls cause fear and less walking; less walking causes weakness and lower fitness; isolation lowers mood and motivation; a handrail and a companion restore participation. **Goals (3):** "TUG under 13.5 s in 8 weeks"; "walks to the temple with her daughter once a week by 3 months".

MCQ **In ICF, walking is coded under:**

A. Body functions

B. Body structures

C. Activities and participation

D. Environmental factors

Answer

C. d450.

MCQ **Which ICF component is not classified?**

A. Environmental factors

B. Personal factors

C. Body structures

D. Participation

Answer

B.

MCQ **e120+2 means:**

A. A moderate barrier

B. A moderate facilitator

C. A severe impairment

D. A complete limitation

Answer

B. The plus sign marks a facilitator.

Last recall check. Draw the ICF model, list the chapters of each component, and write one SMART goal for a stroke patient.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. International Classification of Functioning, Disability and Health (ICF). Geneva: WHO; 2001.
3. World Health Organization. Towards a Common Language for Functioning, Disability and Health: ICF (beginner's guide). Geneva: WHO; 2002.

2 Motor control: the theories behind neurological physiotherapy

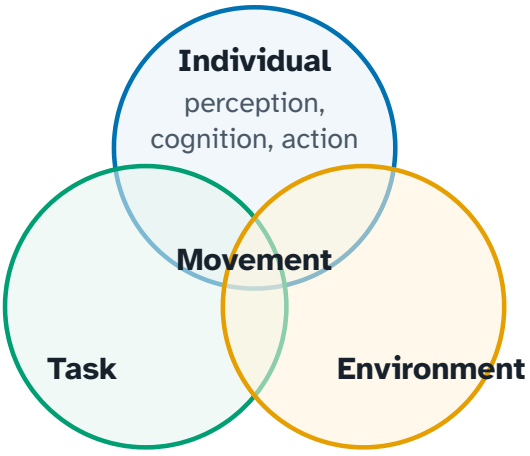
> "Explain any two theories of motor control" was a 7-mark question in three papers (W19, S21, S23). Each theory answers one question: how does the nervous system organise movement? The older answers (reflexes, a hierarchy) produced the neurophysiological approaches; the newer ones (systems, dynamic systems, ecological) produced task-oriented training. Learn each theory as idea, limitation and clinical use.

Syllabus	Ch 2 Theoretical basis of motor control and learning to understand various neurophysiotherapeutic approaches
Priority	**Medium yield.** In 3 of 20 papers (2016–2026): 3 seven-mark questions.
Asked as	7m W19 S21 S23
Usual split	7 (two theories, about 3.5 marks each: idea, limitation, clinical implication)
Checked against	Shumway-Cook and Woollacott, Motor Control: Translating Research into Clinical Practice; Bernstein's problem of degrees of freedom; Gibson's ecological approach. Full list under Sources (p. 14). Content checked 20 October 2026.

1 What motor control means

- **Motor control** is the ability to regulate or direct the mechanisms essential to movement.
- Movement emerges from three things acting together: the **individual** (perception, cognition, action systems), the **task** and the **environment**.
- A theory of motor control matters because it decides what the therapist treats: reflexes, tone, or the person practising a task in a real setting.

Movement emerges from the individual, the task and the environment



The systems view: change any of the three, and the movement changes. Treatment therefore works on the person, the task and the setting.

2 The theories

Theory	Key idea	Limitation and clinical use
Reflex theory (Sherrington)	Reflexes are the building blocks of movement; complex movement is a chain of reflexes triggered by sensory input	**Limitation:** cannot explain movement without sensory input, fast movements, or new movements. **Use:** assess reflexes; use sensory stimuli to bring out or inhibit movement (Rood)

Theory	Key idea	Limitation and clinical use
Hierarchical theory (Hughlings Jackson, Magnus)	Control is top-down: the cortex controls the brainstem, which controls the spinal cord. A lesion of higher centres releases primitive reflexes; development follows a fixed sequence	**Limitation:** lower levels can also control higher ones (spinal reflexes can dominate), and not every adult movement goes through a hierarchy. **Use:** the neurophysiological approaches (Bobath, Brunnstrom, Rood) that inhibit abnormal reflexes and follow developmental sequences
Motor programming theory	Movement can be produced by a stored central program or central pattern generator, even without sensory input (for example the spinal pattern generator for stepping)	**Limitation:** a program cannot alone explain how movement adapts to body and environment. **Use:** retrain whole movement patterns, not single muscles; treadmill stepping uses spinal pattern generators
Systems theory (Bernstein)	The body is a mechanical system with many **degrees of freedom**; movement results from many systems working together, controlled through synergies, with forces from the body and environment	**Limitation:** less detail on how the nervous system and environment interact. **Use:** assess and treat impairments in all systems (strength, range, sensation, perception) as well as the nervous system; train functional synergies
Dynamic systems theory	Movement **self-organises** from interacting parts; preferred patterns are stable "attractor states", and a change in one variable (a control parameter such as speed) can shift behaviour to a new pattern (walk to run)	**Limitation:** can underplay the role of the nervous system. **Use:** find the control parameter that shifts an abnormal pattern (speed, support, task demand); practise at transition points
Ecological theory (Gibson)	Perception and action are linked: people perceive the environment in terms of what it allows them to do (**affordances**), and act to achieve goals	**Limitation:** less on the nervous system's organisation. **Use:** practise real tasks in varied real environments; teach the person to read the environment (kerbs, crowds)

Current practice draws on several theories at once: the **systems approach** sees movement as arising from the person, the task and the environment, which is the basis of **task-oriented training**.

Recall check. Which theory explains why a patient who cannot lift his foot in lying can still step on a treadmill?

3 From theory to approach

Theory	Approach that grew from it	What the therapist does
Reflex and hierarchical	Neurofacilitation: Rood, Bobath, Brunnstrom, PNF	Changes tone and reflexes with handling and sensory input; follows developmental sequences
Systems, dynamic systems, ecological; motor learning	Task-oriented training; Motor Relearning Programme; CIMT	Teaches the person to practise real tasks, adapts the task and environment, treats impairments that limit the task

Details of each approach are on the neurotherapeutic approaches page (p. 209).

4 Exam corner

Past questions (MUHS, 2016–2026)

W19 Q2(c), S21 Q2(b), S23 Q2(d)
Explain theories of motor control (any two).

Pages for this topic: plasticity and motor learning (p. 15), neurotherapeutic approaches (p. 209).

Keywords to underline

reflex chaining hierarchy central pattern generator degrees of freedom self-organisation attractor state affordance individual, task, environment task-oriented training

Answer skeleton: any two theories (7 marks)

1. **[Introduction, 1]** Define motor control; individual, task and environment.
2. **[Theory 1, 3]** Proponent, key idea, limitation, clinical implication.
3. **[Theory 2, 3]** The same four points. Pick a contrasting pair: hierarchical and systems.

Draw these

- The individual, task and environment circles (section 1)
- The hierarchy: cortex over brainstem over spinal cord, with arrows

Viva questions

1. Why can reflex theory not explain all movement?
2. What is the degrees-of-freedom problem?
3. What is a central pattern generator?
4. What is an affordance?
5. Which approaches came from the hierarchical theory?

Linked practical spots

- None directly; theory supports the viva on approaches.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **What is the dynamic systems theory?**

Answer

Movement self-organises from many interacting parts; stable preferred patterns are attractor states; changing a control parameter (such as speed) can shift behaviour to a new pattern. Clinically, find and change the parameter that moves a patient out of an abnormal pattern.

3 m **Give two limitations of the reflex theory.**

Answer

It cannot explain movement without sensory input (deafferented animals still move), fast movement sequences that are too quick for reflex chains, or the creation of new movements.

7 m **Explain any two theories of motor control.**

Answer

Hierarchical theory: top-down control from cortex to brainstem to spinal cord; higher lesions release primitive reflexes; limitation: lower levels also influence higher ones; use: Bobath, Brunnstrom, Rood. **Systems theory (Bernstein):** the body as a mechanical system with many degrees of freedom, controlled through synergies, with movement arising from many systems and external forces; use: assess and treat all systems and train functional tasks. Close with: current practice combines theories in task-oriented training.

LAQ **Explain how your choice of motor control theory changes the treatment of a patient with hemiplegia. (Practice question)**

Answer
Under a hierarchical view, treatment inhibits abnormal reflexes and tone and follows developmental sequences (Bobath, Brunnstrom). Under a systems or task-oriented view, treatment assesses all impairments (strength, range, sensation, balance), practises real tasks (sit to stand, reaching, walking) with many repetitions, varies the task and environment, and uses feedback. Current evidence favours task-oriented practice, borrowing techniques from older approaches when they help.

MCQ **The "degrees of freedom" problem was described by:**

A. Sherrington

B. Bernstein

C. Hughlings Jackson

D. Gibson

Answer
B.

MCQ **"Affordance" is a key idea in:**

A. Reflex theory

B. Hierarchical theory

C. Ecological theory

D. Motor programming theory

Answer
C. Gibson's ecological theory links perception and action.

MCQ **Task-oriented training is based mainly on:**

A. Reflex theory

B. Systems and ecological theories

C. Hierarchical theory only

D. Rood's approach

Answer
B.

Last recall check. Write six theories with one clinical use each.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Shumway-Cook A, Woollacott MH. *Motor Control: Translating Research into Clinical Practice*. Wolters Kluwer (chapter on theories of motor control).
3. Bernstein N. *The Co-ordination and Regulation of Movements*. Pergamon; 1967.
4. Gibson JJ. *The Ecological Approach to Visual Perception*. Houghton Mifflin; 1979.

3 Neuroplasticity, recovery, motor learning and skill acquisition

> This chapter came up in 9 papers, almost all as 3-mark questions: define neuroplasticity, brain changes and synaptogenesis, the stages of motor learning (three times), factors that help motor learning, and intrinsic vs extrinsic feedback, plus a 7-mark question on feedback. One idea links them: the brain changes with what the person practises, so therapy must give enough meaningful, repeated, challenging practice with the right feedback.

Syllabus	Ch 3 Plasticity of the intact and injured brain; spontaneous recovery; environment and behaviour; adaptation of motor performance; muscle adaptation; strength training and conditioning; skill acquisition (information, instruction, demonstration, feedback, practice)
Priority	**High yield.** In 9 of 20 papers (2016–2026): 1 seven-mark and 8 three-mark questions.
Asked as	7m S25 · 3m S16 · W17 · S18 · S21 · S22 · W24 · W25 · S26
Usual split	3 (a definition with three points, or three stages, or three factors); 7 (role and types of feedback with examples)
Checked against	Kleim and Jones, principles of experience-dependent plasticity (2008); Levin et al., recovery vs compensation (2009); Ada et al., strength training after stroke (2006); Fitts and Posner stages; Schmidt and Lee on feedback and practice. Full list under Sources (p. 20). Content checked 20 October 2026.

1 Neuroplasticity

Asked for 3 marks in S16 (define), S18 (brain changes) and S26 (neuroplasticity and synaptogenesis).

- **Neuroplasticity** is the ability of the nervous system to change its structure and function in response to experience, learning, training or injury.
- It happens in the **intact brain** (learning a skill) and after a **lesion** (recovery and reorganisation). It can be **adaptive** (helps function) or **maladaptive** (for example spasticity, pain, or a habit of not using the weak arm).

Brain changes in neuroplasticity (S18)

Level	Change
Synapse	Stronger or weaker transmission at existing synapses (long-term potentiation and depression); unmasking of silent synapses
Neuron	**Synaptogenesis** (new synapses); new dendritic branches and spines; **axonal sprouting** from intact neurons into areas that lost their input
White matter	Changes in myelination with practice
Cortical maps	The area representing a trained part enlarges; areas next to a lesion take over lost functions; the other hemisphere may be recruited

Synaptogenesis (S26): the formation of new synapses between neurons. It is fastest in early development, continues with learning throughout life, and after injury intact neurons sprout and form new synapses (reactive synaptogenesis), which therapy can shape.

Principles of experience-dependent plasticity (Kleim and Jones, 2008)

Principle	Meaning for therapy
Use it or lose it	Functions not used decline
Use it and improve it	Training a function strengthens it

Principle	Meaning for therapy
Specificity	The skill trained is the skill that changes
Repetition matters	Lasting change needs many repetitions
Intensity matters	Training must be challenging enough
Time matters	Different changes happen at different times after injury
Salience matters	Practice must be meaningful to the person
Age matters	Plasticity is greater in the young brain
Transference	Training one skill can help related skills
Interference	Learning one pattern (a compensation) can block learning another

Recall check. Which plasticity principles explain why CIMT works?

2 Recovery after a brain lesion

- **Spontaneous (early) recovery:** swelling settles, blood flow returns to the penumbra around the lesion, and distant areas that were "switched off" by the lesion (diaschisis) start working again. Most of this happens in the first weeks to months.
- **Experience-dependent recovery:** reorganisation driven by practice, continuing for months to years.
- **Recovery vs compensation** (Levin et al., 2009): *recovery* means doing the movement in the same way as before the injury; *compensation* means doing the task in a new way (for example leaning the trunk to reach). Compensation gets tasks done but can block true recovery.
- **Learned non-use:** after early failed attempts, the person stops using the weak limb; the brain area shrinks. CIMT reverses this.
- **Environment and behaviour:** an active, stimulating environment with chances to practise (an "enriched" environment) supports recovery; long periods of inactivity on a ward slow it.

Muscle adaptation after neurological injury

- Loss of motor units and weakness; muscle fibres become smaller and fibre types change.
- Muscles kept in a shortened position lose sarcomeres and become stiffer, so contractures form.
- Changes in tendons and connective tissue add non-neural stiffness to the reflex part of tone.
- So treat muscle length and strength as well as the nervous system.

3 Strength training and physical conditioning

- Strength training after stroke increases strength and improves activity, and does **not** increase spasticity (Ada et al., 2006).
- Use progressive resistance, functional strengthening (sit-to-stand, step-ups) and power training, with enough load to fatigue the muscle in a set.
- Aerobic conditioning improves fitness, walking and health in most neurological conditions; doses are on each condition page: stroke (p. 50), SCI (p. 77), MS (p. 133), Parkinson's disease (p. 164), CP (p. 275).
- Watch blood pressure, fatigue, heat (MS), autonomic dysreflexia (SCI) and overwork (GBS, DMD).

Classic answer vs current evidence: strengthening spastic muscles		
Classic Avoid resisted exercise in spasticity because effort increases tone and abnormal patterns.	**Current** Strength training improves strength and activity without increasing spasticity; weakness, not spasticity, limits most function.	**Write in the exam** "Progressive strength and functional training; it does not increase spasticity."

4 Motor learning

- **Motor learning** is a set of processes, linked to practice or experience, that leads to relatively permanent changes in the capability for skilled movement.
- **Performance vs learning:** performance is what you see during practice; learning is shown by **retention** (doing it later) and **transfer** (doing it in a new setting).

Stages of motor learning (3 marks in S21, W24 and W25)

Stage (Fitts and Posner)	What the learner is like	What the therapist does
Cognitive	Works out what to do; many errors; slow, variable, needs full attention	Clear instructions, demonstration, frequent feedback, a quiet setting, some guidance
Associative	Refines how to do it; fewer errors; more consistent	Less feedback, more practice, start to vary the task
Autonomous	Automatic; little attention needed; can do two tasks at once	Varied, real-world practice; dual tasks; little feedback

Other models: **Gentile** (getting the idea of the movement, then fixation or diversification); **Bernstein** (freezing the degrees of freedom, releasing them, then exploiting them).

Factors that help motor learning (3 marks in W17)

- **Practice:** enough repetitions; distributed rather than massed; variable and random order for retention and transfer; whole-task practice where possible.
- **Feedback:** right type and amount, reduced as learning progresses.
- **Attention and motivation:** meaningful goals; an external focus (on the effect of the movement, for example "reach the cup") rather than an internal focus on the body part.
- **Task-specific and context-specific** practice in a setting like real life.
- **Demonstration and mental practice**; good sleep and rest.

5 Skill acquisition: instruction, demonstration, feedback and practice

Feedback (3 marks in S22; 7 marks in S25)

Type	Meaning	Examples
Intrinsic	From the person's own senses during and after movement	Feeling the foot hit the ground; seeing the cup move; balance from the inner ear
Extrinsic (augmented)	Added from outside to supplement intrinsic feedback	Therapist's words, a mirror, video, EMG biofeedback, force platforms, virtual reality
Knowledge of results (KR)	Extrinsic feedback about the outcome	"You walked 10 metres in 12 seconds"
Knowledge of performance (KP)	Extrinsic feedback about the movement pattern	"Your knee locked back in stance"

- **Timing:** concurrent (during) or terminal (after); immediate or delayed.
- **Schedule:** frequent feedback helps the early stage, but too much creates dependence; fade it, give summary feedback after several trials, or give it only when errors exceed a set band.
- **Role in neuro-rehabilitation (S25):** gives information to correct errors, motivates, reinforces correct movement, and replaces lost intrinsic feedback (for example visual feedback for poor proprioception, EMG biofeedback for weak muscles).

Instruction, demonstration and practice

- **Instructions:** short, one or two key points, with an external focus.
- **Demonstration:** show the whole task first; observational learning activates motor areas.
- **Practice:** blocked practice helps early performance; random and variable practice gives better retention and transfer (contextual interference). Part practice suits complex tasks with separable parts; whole practice suits continuous tasks like walking.
- **Mental practice** adds to physical practice.

Recall check. Give an example of knowledge of results and knowledge of performance for sit-to-stand practice.

6 Exam corner

Past questions (MUHS, 2016–2026)

S25 Q2(b)
Explain the role and types of feedback in neuro-rehabilitation.

3-mark questions
Define neuroplasticity (S16); three factors that facilitate motor learning (W17); brain changes in neuroplasticity (S18); stages of motor learning (S21); extrinsic and intrinsic feedback (S22); motor learning and its stages (W24); three stages of motor learning (W25); neuroplasticity and synaptogenesis (S26).

Pages for this topic: motor control (p. 11), neurotherapeutic approaches (p. 209), stroke (p. 50).

Keywords to underline

neuroplasticity synaptogenesis axonal sprouting cortical reorganisation use it or lose it repetition and intensity learned non-use cognitive, associative, autonomous KR and KP retention and transfer

Answer skeleton: feedback in neuro-rehabilitation (7 marks)

1. **[Define, 1]** Information about movement used to correct and learn.
2. **[Types, 3]** Intrinsic vs extrinsic; KR vs KP; concurrent vs terminal; immediate vs delayed; summary, faded, bandwidth.
3. **[Role, 2]** Information, motivation, reinforcement, replacing lost sensation.
4. **[Examples, 1]** Mirror, EMG biofeedback, video, verbal cues; fade as learning progresses.

Draw these

- A learning curve with performance rising and the three stages marked

Viva questions

1. Define neuroplasticity.
2. What is diaschisis?

3. What is the difference between recovery and compensation?
4. Name the stages of motor learning.
5. Why should feedback be reduced as learning progresses?
6. What is contextual interference?

Linked practical spots

- EMG biofeedback unit; mirror therapy box.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Define neuroplasticity.**

Answer
The ability of the nervous system to change its structure and function in response to experience, learning, training or injury: through changes in synaptic strength, new synapses, sprouting and cortical reorganisation.

3 m **Write the three stages of motor learning.**

Answer
Cognitive (understanding the task; many errors, needs attention and frequent feedback); associative (refining the movement; fewer errors); autonomous (automatic, consistent, can dual-task) (Fitts and Posner).

7 m **Explain the role and types of feedback in neuro-rehabilitation.**

Answer
Feedback is information about movement. Intrinsic (own senses) and extrinsic (therapist, mirror, video, biofeedback). Extrinsic is KR (outcome) or KP (pattern); concurrent or terminal; immediate or delayed; summary, faded or bandwidth schedules. Role: corrects errors, motivates, reinforces, substitutes for lost sensation. Give more early and fade it to avoid dependence. Examples: EMG biofeedback for a weak dorsiflexor; mirror for posture; KR on walking speed.

LAQ **Using motor learning principles, plan training of sit-to-stand for a patient 6 weeks after stroke. (Practice question)**

Answer
Analyse the task and missing components; give a short instruction with an external focus ("bring your nose over your toes and push to stand") and a demonstration; practise the whole task with many repetitions per session (blocked at first, then varied: different chair heights, speeds, with and without arms); give KP early (trunk lean, weight on the weak leg), then fade to summary KR (number of stands in 30 seconds); add a home practice schedule; progress to dual tasks and real settings.

MCQ **"Your heel touched first this time" is an example of:**

A. Intrinsic feedback
B. Knowledge of results
C. Knowledge of performance
D. Concurrent visual feedback

Answer
C. It describes the movement pattern.

MCQ **Which practice schedule gives the best retention and transfer?**

A. Blocked, constant practice

B. Random, variable practice

C. Watching only

D. Practice with feedback after every trial

Answer

B. Contextual interference improves learning, though early performance looks worse.

MCQ **The principle "the skill trained is the skill that changes" is:**

A. Transference

B. Specificity

C. Salience

D. Interference

Answer

B. Kleim and Jones (2008).

Last recall check. List Kleim and Jones' ten principles, the three stages of motor learning and the types of feedback.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Kleim JA, Jones TA. Principles of experience-dependent neural plasticity: implications for rehabilitation after brain damage. *J Speech Lang Hear Res* 2008;51:S225–39.
3. Levin MF, Kleim JA, Wolf SL. What do motor "recovery" and "compensation" mean in patients following stroke? *Neurorehabil Neural Repair* 2009;23:313–9.
4. Ada L, Dorsch S, Canning CG. Strengthening interventions increase strength and improve activity after stroke: a systematic review. *Aust J Physiother* 2006;52:241–8.
5. Fitts PM, Posner MI. *Human Performance*. Brooks/Cole; 1967. Schmidt RA, Lee TD. *Motor Learning and Performance*. Human Kinetics.
6. Shumway-Cook A, Woollacott MH. *Motor Control: Translating Research into Clinical Practice* (motor learning and recovery of function).

4 Quality of life scales, independence measures and other outcome measures

> Scales came up in 6 papers as 3-mark questions: the Glasgow Coma Scale four times, the Modified Ashworth Scale twice, and the ASIA scale and Hoehn and Yahr stages once each, and every long-case answer is stronger with a named measure. Pick a measure for each ICF level you are treating, score it at the start and again later, and write the numbers in your goals.

Syllabus	Ch 4 Quality of life scales and independence measures; General item 4: assessment tools named in the syllabus (GCS, MMSE, Ashworth, DGI, BBS, Functional Reach, TUG, Barthel, SF-36, STREAM, Brunnstrom, Fugl-Meyer, ASIA, UPDRS, EDSS)
Priority	**High yield.** In 6 of 20 papers (2016–2026): 8 three-mark questions (GCS 4, Modified Ashworth 2, ASIA 1, Hoehn and Yahr 1).
Asked as	3m W16 · S19 ×3 · S20 · W20 · W21 · S26
Usual split	3 (the scale with its grades or items, plus one line on use or interpretation)
Checked against	Barthel index interpretation (Shah et al., 1989); Berg Balance Scale, Timed Up and Go (Shumway-Cook et al., 2000) and Dynamic Gait Index (Shumway-Cook et al., 1997) cut-offs; STREAM structure (Daley et al., 1999); Glasgow Coma Scale, Modified Ashworth Scale, ISNCSCI and Hoehn and Yahr as checked on their condition pages. Full list under Sources (p. 27). Content checked 20 October 2026.

1 Why and how to use outcome measures

- **Purposes:** describe the problem and its severity, predict outcome, set measurable goals, show change after treatment, and communicate with the team.
- **Match the ICF level:** body functions (tone, strength, motor control, cognition), activities (balance, walking, self-care) and participation and quality of life.
- **Use them properly:** standard instructions, equipment and scoring; same tester and conditions on retest; record the date and score.
- Put the numbers into goals: "Berg from 38 to 46 in 6 weeks".

What makes a good measure

Property	Meaning
Reliability	Gives the same result when repeated (test–retest) and between testers (inter-rater)
Validity	Measures what it claims to measure
Responsiveness	Detects real change over time
Minimal clinically important difference	The smallest change that matters to the patient
Floor and ceiling effects	Too many people score at the lowest or highest possible score, so change cannot show
Feasibility	Time, equipment, training, cost; language and literacy (use validated Indian versions where they exist)

2 Independence measures

Barthel index

- **Ten items:** feeding, bathing, grooming, dressing, bowels, bladder, toilet use, transfers (bed to chair), mobility on level surfaces, stairs.

- Items are scored in steps of 5 (0, 5, 10 or 15) by how much help is needed; total **0–100**. It weights mobility and continence heavily.
- **Interpretation (Shah et al., 1989):** 0–20 total dependence; 21–60 severe; 61–90 moderate; 91–99 slight dependence; 100 independent in these tasks (not free of disability).
- Quick, valid and widely used, but has a ceiling effect in milder patients.

Functional Independence Measure (FIM)

- **18 items:** 13 motor (self-care, sphincter control, transfers, locomotion) and 5 cognitive (communication, social cognition).
- Each item 1 (total assistance) to 7 (complete independence); total **18–126**.
- More detailed than Barthel and includes cognition; needs training to score.

Modified Rankin Scale (stroke outcome)

Grade	Meaning
0	No symptoms
1	No significant disability despite symptoms; usual activities
2	Slight disability: cannot do all previous activities, but looks after own affairs
3	Moderate disability: needs some help, walks without assistance
4	Moderately severe: cannot walk or attend to bodily needs without help
5	Severe: bedridden, incontinent, needs constant care
6	Dead

3 Quality of life scales

- **Health-related quality of life** is the person's own view of how health affects physical, mental and social well-being; it is asked, not observed.
- **SF-36 (generic):** 36 questions in 8 domains: physical functioning, role limits from physical health, bodily pain, general health, vitality, social functioning, role limits from emotional problems, and mental health. Each domain scores 0–100 (higher is better), with physical and mental summary scores.
- **EQ-5D (generic):** 5 dimensions (mobility, self-care, usual activities, pain or discomfort, anxiety or depression) plus a 0–100 visual scale of overall health.
- **Disease-specific:** Stroke Impact Scale; PDQ-39 for Parkinson's disease; MS quality of life scales. They pick up changes that matter in that condition.

4 Quick reference: scales named in the syllabus

Scale	Measures and scoring	Interpretation or use
Glasgow Coma Scale	Consciousness: eyes 1–4, verbal 1–5, motor 1–6; total 3–15	Severe 3–8, moderate 9–12, mild 13–15 (section 5)
Mini-Mental State Examination	Cognition: orientation, registration, attention, recall, language and copying; 30 points	Low scores (commonly 23 or less) suggest impairment; adjust for age and education; the Hindi Mental State Examination suits rural, non-literate older Indians
Modified Ashworth Scale	Resistance to passive movement: 0, 1, 1+, 2, 3, 4	Grades spasticity (section 5)
Berg Balance Scale	14 balance tasks, each 0–4; total 0–56	Below about 45 suggests a fall risk; 41–56 independent, 21–40 walks with help, 0–20 wheelchair level

Scale	Measures and scoring	Interpretation or use
Functional Reach Test	Forward reach along a ruler at shoulder height without moving the feet	Less than about 15 cm (6 inches) suggests a high fall risk
Timed Up and Go	Time to stand from a chair, walk 3 m, turn, return and sit	13.5 s or more suggests a fall risk in community-dwelling older adults
Dynamic Gait Index	8 walking tasks (speed changes, head turns, obstacles, stairs), each 0–3; total 0–24	19 or less suggests a fall risk
Barthel index	10 ADL items; 0–100	Section 2
SF-36	8 domains of health-related quality of life; 0–100 each	Section 3
STREAM	30 items: 10 arm and 10 leg movements (0–2 each) and 10 basic mobility tasks (0–3 each); 70 points	Motor recovery and mobility after stroke
Brunnstrom stages	Six stages of motor recovery after stroke, from flaccidity through synergies to isolated movement	See approaches (p. 209) and stroke (p. 50)
Fugl-Meyer Assessment	Motor function (arm 66, leg 34), plus sensation, balance, range and pain sections	The research standard for motor recovery after stroke
ASIA Impairment Scale	Grades A–E from the ISNCSCI examination	Completeness of spinal cord injury (section 5)
MDS-UPDRS	4 parts: non-motor and motor daily-living experiences, motor examination (part III), motor complications; items 0–4	Severity and progression of Parkinson's disease
EDSS	0 to 10 in half steps	Disability in MS, mainly by walking; 6.0 needs a stick for 100 m (see MS (p. 133))

5 The four scales asked for 3 marks

Glasgow Coma Scale (W16, S19, S20, S26)

Response	Scores
Eye opening (E, 1–4)	4 spontaneous; 3 to sound; 2 to pressure; 1 none
Verbal response (V, 1–5)	5 oriented; 4 confused; 3 words; 2 sounds; 1 none
Motor response (M, 1–6)	6 obeys commands; 5 localises; 4 normal flexion; 3 abnormal flexion; 2 extension; 1 none

Total 3–15: severe 3–8, moderate 9–12, mild 13–15. Report the parts ("E2 V2 M4 = 8"); the motor score predicts outcome best. Used in head injury, stroke, meningitis and any reduced consciousness. Full detail: brain injury (p. 66).

Modified Ashworth Scale (S19, W21)

Grade	Meaning
0	No increase in tone
1	Slight increase: a catch and release, or minimal resistance at the end of the range
1+	Slight increase: a catch followed by minimal resistance through less than half of the remaining range
2	More marked increase through most of the range, but the part moves easily
3	Considerable increase; passive movement is difficult
4	The part is rigid in flexion or extension

Patient relaxed; move the part through its full range in about one second; grade the first movement. Full detail: tone (p. 216).

ASIA Impairment Scale (S19)

- **A, complete:** no sensory or motor function in S4–S5.
- **B, sensory incomplete:** sensation but no motor function below the level, including S4–S5.
- **C, motor incomplete:** motor function below the level, but less than half of the key muscles below the neurological level reach grade 3.
- **D, motor incomplete:** half or more of those key muscles reach grade 3 or more.
- **E, normal:** normal sensation and power in a person who had deficits.

Full detail with sacral sparing: spinal cord (p. 77).

Hoehn and Yahr stages (W20)

- **1:** one side only. **2:** both sides, balance normal. **3:** both sides, some postural instability, physically independent. **4:** severe disability, still walks or stands without help. **5:** wheelchair-bound or bedridden unless helped.
- The modified scale adds 1.5 (one side plus trunk) and 2.5 (both sides, recovers on the pull test). Full detail: Parkinson's disease (p. 164).

6 Choosing measures by condition

Condition	Body functions	Activity and participation
Stroke	Fugl-Meyer, STREAM, Brunnstrom stage, Modified Ashworth, MMSE or MoCA	Berg, TUG, 10-metre walk, Barthel or FIM, modified Rankin, Stroke Impact Scale
Spinal cord injury	ISNCSCI with the ASIA Impairment Scale (AIS) grade, Modified Ashworth	Spinal Cord Independence Measure, walking index, wheelchair skills
Parkinson's disease	MDS-UPDRS part III, Hoehn and Yahr	Berg or Mini-BESTest, TUG, gait speed, PDQ-39
Multiple sclerosis	EDSS, fatigue scales	TUG, walking tests, MS quality of life scales
Brain injury	GCS, Rancho level	FIM, Glasgow Outcome Scale
Cerebral palsy	Modified Ashworth or Tardieu, range of motion	GMFCS level, Gross Motor Function Measure

Recall check. Pick one body-function measure and one activity measure for a stroke patient, and say what each tells you.

7 Exam corner

Past questions (MUHS, 2016–2026)

W16 Q1(c), 3 marks

Components of the Glasgow Coma Scale; two conditions where it is used.

S19 Q1(a), S20 Q1(b), 3 marks

Glasgow Coma Scale.

S26 Q1(d), 3 marks

Interpreting the Glasgow Coma Scale.

S19 Q1(e), W21 Q1(f), 3 marks

Modified Ashworth Scale.

S19 Q1(c), 3 marks

ASIA scale.

W20 Q1(a), 3 marks

Write the Hoehn and Yahr scale.

Keywords to underline

reliability validity responsiveness ICF level Barthel 0–100 FIM 18–126 SF-36 eight domains Berg 56 TUG 13.5 s GCS 3–15 MAS 1+ AIS A–E

Answer skeleton: any scale (3 marks)

1. What it measures and in whom (one line).
2. The grades or items with scoring (a small table).
3. Interpretation (cut-offs or severity) or one clinical use.

Draw these

- The GCS table and the Modified Ashworth grades from memory

Viva questions

1. What is the difference between reliability and validity?
2. Name the ten items of the Barthel index.
3. What does a ceiling effect mean? Give an example.
4. Which scale would you use to predict falls in an older adult?
5. What are the domains of the SF-36?
6. What is the difference between grades 1 and 1+ on the Modified Ashworth Scale?

Linked practical spots

- Score a scale on a mock patient: Berg, TUG (stopwatch, chair, 3 m line), Functional Reach (ruler on the wall), Barthel from a case description.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the components of the Glasgow Coma Scale and two conditions where it is used. (W16)**

Answer

Eye opening 1–4 (spontaneous, to sound, to pressure, none); verbal 1–5 (oriented, confused, words, sounds, none); motor 1–6 (obeys, localises, normal flexion, abnormal flexion, extension, none); total 3–15, severe 3–8. Used in head injury and stroke (also meningitis, overdose).

3 m **Write the Modified Ashworth Scale. (S19, W21)**

Answer
0 no increase; 1 catch and release or minimal resistance at the end of range; 1+ catch then minimal resistance through less than half the range; 2 more marked increase through most of the range, part moves easily; 3 considerable increase, passive movement difficult; 4 rigid.

7 m **Describe independence measures used in neurological rehabilitation.**

Answer
Purpose (measure dependence in daily activities, set goals, show change). **Barthel index:** 10 items, 0–100, Shah bands (0–20 total, 21–60 severe, 61–90 moderate, 91–99 slight dependence, 100 independent); quick but has a ceiling effect. **FIM:** 18 items (13 motor, 5 cognitive), each 1–7, total 18–126; includes cognition, needs training. **Modified Rankin Scale:** 0–6 global disability after stroke. Condition-specific: Spinal Cord Independence Measure. Mention reliability, validity, responsiveness and the use of scores in goals.

LAQ **Practice: a 60-year-old man is 3 months after a left MCA stroke, walks with a stick and needs help to bathe. Choose and justify outcome measures for each ICF level, and write two goals using them. (5 + 5 + 5)**

Answer
Body functions (5): Fugl-Meyer (motor recovery), Modified Ashworth (spasticity), MMSE or MoCA (cognition). **Activities and participation (5):** Berg and TUG (balance and fall risk), 10-metre walk (speed), Barthel or FIM (independence), Stroke Impact Scale (quality of life). **Goals (5):** "TUG from 22 s to under 15 s in 6 weeks"; "Barthel from 75 to 90 in 8 weeks, bathing with supervision only".

MCQ **The maximum score on the Berg Balance Scale is:**

A. 24
B. 56
C. 100
D. 126

Answer
B. 24 is the Dynamic Gait Index, 100 the Barthel index and 126 the FIM.

MCQ **On the GCS, abnormal (decorticate) flexion scores:**

A. M2
B. M3
C. M4
D. M5

Answer
B.

MCQ **A measure that gives the same score when repeated by two testers is:**

A. Valid

B. Responsive

C. Reliable

D. Specific

Answer

C. Inter-rater reliability.

Last recall check. Write GCS, MAS, AIS and Hoehn and Yahr from memory, the Barthel items, and the Berg, TUG and DGI cut-offs.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Shah S, Vanclay F, Cooper B. Improving the sensitivity of the Barthel Index for stroke rehabilitation. *J Clin Epidemiol* 1989;42:703–9.
3. Berg KO, et al. Measuring balance in the elderly: validation of an instrument. *Can J Public Health* 1992;83(Suppl 2):S7–11.
4. Shumway-Cook A, Brauer S, Woollacott M. Predicting the probability for falls in community-dwelling older adults using the Timed Up and Go Test. *Phys Ther* 2000;80:896–903.
5. Shumway-Cook A, et al. Predicting the probability for falls in community-dwelling older adults. *Phys Ther* 1997;77:812–9 (Dynamic Gait Index).
6. Duncan PW, et al. Functional reach: predictive validity in a sample of elderly male veterans. *J Gerontol* 1992;47:M93–8.
7. Daley K, Mayo N, Wood-Dauphinee S. Reliability of scores on the Stroke Rehabilitation Assessment of Movement (STREAM) measure. *Phys Ther* 1999;79(1).
8. Folstein MF, Folstein SE, McHugh PR. "Mini-mental state". *J Psychiatr Res* 1975;12:189–98; Ganguli M, et al. A Hindi version of the MMSE for a largely illiterate rural elderly population in India. *Int J Geriatr Psychiatry* 1995;10:367–77.
9. Ware JE, Sherbourne CD. The MOS 36-item short-form health survey (SF-36). *Med Care* 1992;30:473–83.

Part G

Assessment for every condition

Evaluation, investigations, movement dysfunction and outcome tools

G Neurological assessment and investigations 29

G Cognition, perception, apraxia and aphasia 38

G Movement dysfunction in daily tasks: bed mobility to reaching 44

G Neurological assessment and investigations: UMN vs LMN, reflexes, sensation, cranial nerves, EMG and nerve conduction

> Upper vs lower motor neuron lesions came up in 6 papers as a 3-mark question, and the same skills are needed to assess every long case. Decide first whether the problem is in the upper or the lower motor neuron, because that decides the treatment: tone and movement control for an upper motor neuron lesion, and strengthening, protection and sensory care for a lower motor neuron lesion.

Syllabus	General: evaluation, interpretation of investigations and clinical reasoning for the functional diagnosis; choosing assessment tools for arousal, cognition, sensation, tone, strength, locomotion and balance (applies to every condition)
Priority	**High yield.** In 6 of 20 papers (2016–2026): 6 three-mark questions on UMN and LMN lesions; the basis of every long-case assessment.
Asked as	3m S16 · W16 · S20 · W21 · W22 · S24
Usual split	3 (six differences in a table, or six features of a UMN lesion, each with one line)
Checked against	International Standards for Neurological Classification of Spinal Cord Injury (key sensory points); the MRC scale for muscle strength; standard clinical and electrodiagnostic texts for reflex levels and EMG and nerve conduction patterns. Full list under Sources (p. 37). Content checked 20 October 2026.

1 The order of a neurological assessment

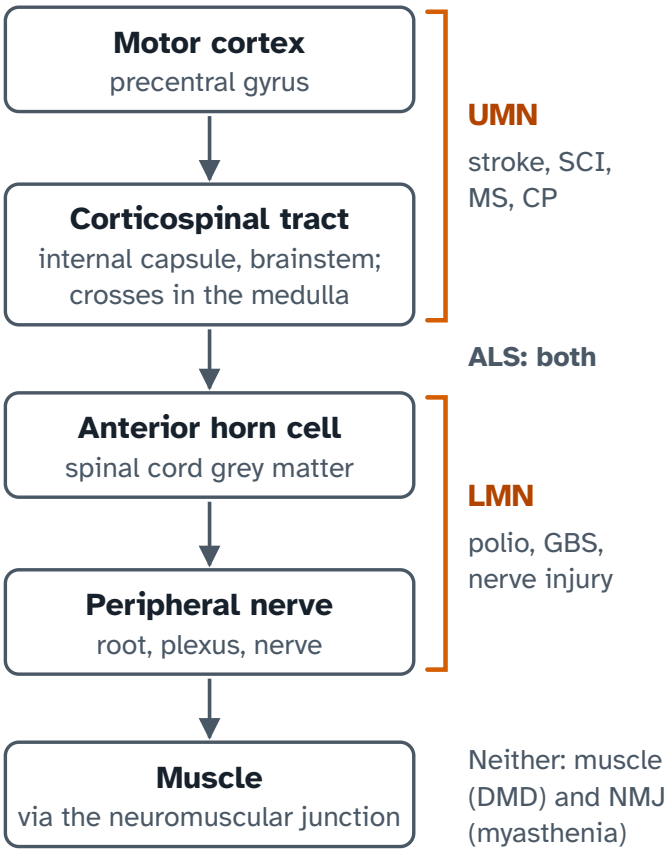
1. **History:** chief complaints; onset (sudden or gradual), duration and progression; past, family, personal and social history; drugs; red flags.
2. **General examination:** vital signs, level of consciousness (GCS), built, posture, attitude of the limbs, deformities, skin and pressure areas, aids in use.
3. **Higher functions:** orientation, memory, speech, cognition (MMSE or MoCA), perception (see cognition and perception (p. 38)).
4. **Cranial nerves** (section 6).
5. **Motor system:** bulk (girth), tone, power, reflexes, coordination, involuntary movements (sections 3 and 4).
6. **Sensory system:** superficial, deep and combined sensations; dermatomes (section 5).
7. **Balance and gait** (see balance and coordination (p. 223)).
8. **Function:** bed mobility, lying to sitting, sit to stand, walking, reaching and manipulation; ADL (Barthel index).
9. **Outcome measures** for the condition (see scales (p. 21)) and the **investigations** (section 7).
10. **Sum up:** problem list, ICF and functional diagnosis, goals and plan (see ICF (p. 4)).

2 Upper vs lower motor neuron lesions

Asked for 3 marks in S16, S20 and W21 (differences) and in W16, W22 and S24 (signs or features of a UMN lesion).

- The **upper motor neuron (UMN)** runs from the motor cortex down the corticospinal tract to the anterior horn of the spinal cord (or to the cranial nerve nuclei).
- The **lower motor neuron (LMN)** runs from the anterior horn cell (or cranial nerve nucleus) through the root, plexus and peripheral nerve to the muscle.

Where UMN and LMN lesions lie



Draw this beside a UMN vs LMN answer. The anterior horn cell is where the two meet: the UMN ends there and the LMN starts there.

Feature	UMN lesion	LMN lesion
Site	Motor cortex, internal capsule, brainstem or spinal cord tracts	Anterior horn cell, root, plexus or peripheral nerve
Weakness	Groups of muscles, a whole limb or one side; in the arm the extensors and in the leg the flexors are weaker; fine movements lost	Individual muscles in the territory of the root or nerve
Tone	Increased: spasticity (clasp-knife)	Reduced: flaccid
Wasting	Slight and late (disuse)	Marked and early
Fasciculation	Absent	May be present, especially in anterior horn cell disease
Deep reflexes	Brisk, with clonus	Reduced or absent
Plantar response	Extensor (Babinski sign)	Flexor or absent
Abdominal reflexes	Lost	Present, unless their own segments are involved
EMG and nerve conduction	No denervation; normal nerve conduction	Denervation potentials; slowed or absent conduction in nerve lesions
Examples	Stroke, spinal cord injury, MS, CP	Polio, GBS, nerve injury, Bell's palsy

Memory hook

UMN goes up, LMN goes down

- **UMN:** tone up, reflexes up, toes up (Babinski).
- **LMN:** tone down, reflexes down, muscle bulk down.

Features of a UMN lesion (W16, W22, S24)

- **Negative features** (what is lost): weakness in a pyramidal pattern, loss of fine and skilled movements, slowness and early fatigue.
- **Positive features** (what is released): spasticity, brisk reflexes with spread and clonus, the Babinski sign and Hoffmann's sign, flexor or extensor spasms, associated reactions and mass synergy patterns.
- Loss of the abdominal and cremasteric reflexes; little wasting and no fasciculation.
- **Exception:** straight after an acute lesion (cerebral or spinal shock) the limb is flaccid with absent reflexes; spasticity appears over days to weeks.

Why it changes physiotherapy

- **UMN:** manage tone, train movement control and tasks, prevent contracture, and strengthen (strength training does not increase spasticity).
- **LMN:** strengthen each muscle according to its grade, protect weak or denervated muscles from overstretching with splints, care for numb skin, and watch for reinnervation.

Recall check. Give six differences between UMN and LMN lesions without looking at the table.

3 Motor examination

- **Bulk:** look for wasting and fasciculation; measure girth at fixed distances from a bony point on both sides.
- **Tone:** passive movement at slow and fast speeds; spasticity (velocity-dependent, clasp-knife), rigidity (lead-pipe or cogwheel), hypotonia; grade spasticity with the Modified Ashworth Scale (see tone (p. 216)).
- **Power:** MRC grades for LMN and muscle disease. In UMN lesions, isolated muscle testing is less useful; record voluntary control and synergy (Brunnstrom stage, Fugl-Meyer) instead.
- **Coordination:** finger–nose, heel–shin, rapid alternating movements, rebound (see ataxia (p. 155)).
- **Involuntary movements:** tremor (at rest in Parkinson's disease; on action or intention in cerebellar disease), chorea, athetosis, dystonia, ballismus, myoclonus, tics.

MRC scale for muscle power

Grade	Meaning
0	No contraction
1	Flicker or trace of contraction, no movement
2	Active movement with gravity eliminated
3	Active movement against gravity
4	Active movement against gravity and resistance
5	Normal power

4 Reflexes

Deep tendon reflexes

Reflex	Root	Nerve
Biceps	C5, C6	Musculocutaneous
Brachioradialis (supinator)	C5, C6	Radial
Triceps	C7 (C6–C8)	Radial
Knee (patellar)	L3, L4 (L2–L4)	Femoral
Ankle (Achilles)	S1 (S1, S2)	Tibial

Memory hook

1, 2, buckle my shoe

- **S1–2** buckle my shoe: ankle.
- **L3–4** kick the door: knee.
- **C5–6** pick up sticks: biceps and brachioradialis.
- **C7–8** lay them straight: triceps.

Grading: 0 absent; 1+ reduced (trace); 2+ normal; 3+ brisk (can be normal); 4+ very brisk with clonus. Compare the two sides, and use reinforcement (clenching the teeth, or pulling the hooked fingers apart: Jendrassik's manoeuvre) before calling a reflex absent.

Superficial reflexes

- **Abdominal** (about T8–T12): stroke each quarter towards the umbilicus; the umbilicus moves towards the stroke. Lost in UMN lesions.
- **Cremasteric** (L1–L2): stroke the inner thigh; the testis rises.
- **Plantar** (S1): stroke the outer sole from the heel towards the little toe and across; the toes normally curl down.
- **Anal** (S2–S4): touching the perianal skin contracts the anal sphincter; important in spinal cord injury.

Pathological signs

- **Babinski sign:** the big toe extends and the other toes fan out on plantar stimulation (normal in infants).
- **Hoffmann's sign:** flicking the nail of the middle finger makes the thumb and index finger flex.
- **Clonus:** a quick, sustained dorsiflexion of the ankle produces rhythmic beats; sustained clonus means a UMN lesion.
- Primitive reflexes reappearing in adults (grasp, sucking, palmomental) suggest frontal lobe damage.

5 Sensory examination

Test with the eyes closed, compare with the normal side, and move from the numb area towards normal skin.

Sensation	Pathway	How to test
Light touch	Dorsal columns and spinothalamic tracts	Cotton wisp; the patient says "yes" when touched
Pain	Spinothalamic (crosses in the cord)	Pin: sharp or dull
Temperature	Spinothalamic	Tubes of warm and cold water

Sensation	Pathway	How to test
Joint position sense	Dorsal columns (cross in the medulla)	Move a finger or toe up or down, holding its sides
Vibration	Dorsal columns	128 Hz tuning fork on bony points, distal first
Combined (cortical) sensations	Parietal cortex	Two-point discrimination, stereognosis, graphaesthesia, localisation, extinction (see cognition and perception (p. 38))

Patterns of sensory loss

- **Glove and stocking:** polyneuropathy (diabetes, alcohol).
- **One dermatome:** root lesion (disc prolapse).
- **One nerve territory:** peripheral nerve injury.
- **A level on the trunk:** spinal cord lesion.
- **One half of the body:** brain lesion (thalamus, internal capsule, cortex).
- **Pain and temperature lost, touch kept:** syringomyelia, anterior cord syndrome; **Brown-Séquard:** power and joint sense lost on one side, pain and temperature on the other (see spinal cord (p. 77)).
- **Saddle anaesthesia** with bladder or bowel change: cauda equina (an emergency).

Dermatomes: key sensory points (ISNCSCI)

- **Arm:** C5 outer side of the elbow crease; C6 thumb; C7 middle finger; C8 little finger; T1 inner side of the elbow crease; T2 apex of the axilla.
- **Trunk:** T4 nipple line; T6 xiphisternum; T10 umbilicus; T12 middle of the inguinal ligament.
- **Leg:** L2 middle of the front of the thigh; L3 inner knee (medial femoral condyle); L4 medial malleolus; L5 top of the foot at the third toe joint; S1 outer heel; S2 middle of the back of the knee; S3 ischial tuberosity; S4–5 around the anus.

6 Cranial nerves

Nerve	Quick test	Sign of damage
I Olfactory	Smell a familiar odour, each nostril	Loss of smell (head injury)
II Optic	Acuity, visual fields by confrontation, pupil reaction to light	Field loss (hemianopia), poor acuity
III, IV, VI Eye movements	Follow a finger in an H pattern; pupils	Double vision, squint; III: ptosis and a large pupil
V Trigeminal	Face sensation in 3 divisions; clench the jaw; corneal reflex	Numb face; jaw deviates to the weak side on opening
VII Facial	Raise eyebrows, close eyes tight, show teeth, puff cheeks	LMN: whole half of the face; UMN: lower face only (see facial nerve (p. 124))
VIII Vestibulocochlear	Whisper, Rinne and Weber tests; head impulse and positional tests	Deafness, vertigo, nystagmus (see vestibular (p. 116))
IX, X Glossopharyngeal, vagus	Say "aah"; gag reflex; swallow; voice	Uvula pulled away from the weak side; choking, nasal voice
XI Accessory	Shrug the shoulders; turn the head against resistance	Weak trapezius and sternocleidomastoid
XII Hypoglossal	Put out the tongue	Tongue deviates to the weak side; wasting and fasciculation in LMN lesions

7 Investigations and how to read them

Test	What it shows	Example
Nerve conduction: demyelinating pattern	Slow conduction velocity, long latencies, conduction block, late or absent F-waves	GBS (AIDP), hereditary neuropathy type 1
Nerve conduction: axonal pattern	Small responses (low amplitude) with near-normal velocity	Diabetic and alcoholic neuropathy, axonal GBS
Normal sensory responses with weakness	The lesion lies before the dorsal root ganglion	Root avulsion, radiculopathy, MND, polio
Needle EMG: neurogenic	Fibrillations and positive sharp waves at rest (from about 2–3 weeks); large, long, polyphasic units; fewer units recruited	Nerve injury, MND, polio
Needle EMG: myopathic	Small, short, polyphasic units recruited early, even with weak effort	DMD, polymyositis
Repetitive nerve stimulation	Falling responses (decrement) in myasthenia; rising after exercise in Lambert–Eaton syndrome	Myasthenia gravis
Evoked potentials	Slowed conduction in central pathways	MS (visual evoked potentials)
CT	Bleeding, fractures, swelling; quick first scan	Haemorrhagic stroke, head injury
MRI	Infarcts, demyelination, the spinal cord, tumours	MS plaques, cord injury, myelopathy
CSF	Protein, cells, organisms	GBS: high protein with normal cells; meningitis
Blood tests	Creatine kinase (CK); antibodies; sugar, B12	Very high CK in DMD; acetylcholine receptor antibodies in myasthenia

Read reports for the physiotherapy meaning: the level and side of the lesion, whether nerves are recovering, and what that means for the prognosis and the plan.

8 Exam corner

Past questions (MUHS, 2016–2026)

S16 Q1(d), 3 marks

Write a comparison between upper and lower motor neuron lesions.

W16 Q1(a), 3 marks

Enumerate any 3 signs of an upper motor neuron lesion.

S20 Q1(c), 3 marks

Any 3 differences between upper and lower motor neuron lesions (printed as "lower motor and lower motor").

W21 Q1(e), 3 marks

Enumerate three differences between UMN and LMN lesions.

W22 Q1(d), 3 marks

Features of UMN lesion.

S24 Q1(f), 3 marks

Write clinical features of UMN lesion.

Keywords to underline

corticospinal tract anterior horn cell spasticity clasp-knife hyperreflexia clonus Babinski sign flaccid wasting fasciculation areflexia denervation

Answer skeleton: UMN vs LMN (3 marks)

1. One line defining each (where the UMN and LMN run).
2. A table of at least six differences: weakness pattern, tone, wasting, fasciculation, deep reflexes, plantar response.
3. An example of each, and the shock-phase exception.

Draw these

- The motor pathway with the UMN and LMN parts marked (section 2)
- A reflex arc: receptor, sensory neuron, cord, motor neuron, muscle
- A dermatome map, front and back

Viva questions

1. Where does the upper motor neuron end?
2. Why is the plantar response extensor in a UMN lesion?
3. Why can an acute UMN lesion look flaccid?
4. Which reflex tests S1, and which tests C7?
5. Which sensations travel in the dorsal columns?
6. How does an EMG tell a nerve problem from a muscle problem?

Linked practical spots

- Reflex hammer, 128 Hz tuning fork, two-point discriminator, pin and cotton wisp, tape for girth.
- EMG and nerve conduction reports to interpret (see the practical exam (p. 358)).

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write three differences between UMN and LMN lesions. (S20, W21)**

Answer

Tone: increased (spasticity) in UMN, reduced (flaccid) in LMN. Reflexes: brisk with clonus in UMN, reduced or absent in LMN. Plantar: extensor (Babinski) in UMN, flexor or absent in LMN. Also: wasting slight in UMN, marked in LMN; fasciculation absent in UMN, may be present in LMN.

3 m **Write the clinical features of a UMN lesion. (W22, S24)**

Answer

Weakness in a pyramidal pattern with loss of fine movements; spasticity (clasp-knife); brisk reflexes and clonus; extensor plantar (Babinski) and Hoffmann's sign; lost abdominal and cremasteric reflexes; little wasting and no fasciculation; associated reactions and synergies. Early after onset there may be a flaccid shock phase.

7 m **Describe the examination of the sensory system in a neurological patient.**

Answer
Preparation (quiet room, eyes closed, compare sides, numb to normal). Superficial: light touch (cotton), pain (pin), temperature (tubes). Deep: joint position sense, vibration (128 Hz fork), deep pressure. Combined: two-point discrimination, stereognosis, graphaesthesia, localisation, extinction. Chart the findings on a dermatome map, name the pattern (glove and stocking, dermatomal, nerve territory, sensory level, hemisensory, dissociated) and give its meaning for physiotherapy: skin protection, balance training with vision, sensory re-education.

LAQ **Practice from the S16 ALS case: a 36-year-old dentist has wasting of the small hand muscles, fasciculations in the arm, hyperreflexia in both arms, hyporeflexia in both legs and bilateral Babinski signs. Sort the findings into UMN and LMN signs and write the assessment you would do. (5 + 10)**

Answer
UMN signs (5): hyperreflexia in the arms, Babinski signs. **LMN signs:** wasting, fasciculations, reduced leg reflexes. Both together point to ALS. **Assessment (10):** history (onset, progression, falls, cramps, swallowing, breathing at night, work); bulk, girth and fasciculations; tone; MRC grades of key muscles, grip and pinch; deep, superficial and pathological reflexes; sensation (expected normal); coordination and fine hand tasks; balance and gait; breathing (vital capacity, cough); function (ALSFRS-R, Barthel); EMG and nerve conduction reports.

MCQ **Which is a sign of an LMN lesion?**

A. Clonus
B. Babinski sign
C. Fasciculation
D. Clasp-knife spasticity

Answer
C.

MCQ **The knee jerk tests mainly:**

A. C5–C6
B. C7–C8
C. L3–L4
D. S1–S2

Answer
C.

MCQ **Small, short, polyphasic motor units recruited early on EMG suggest:**

A. Myopathy
B. Nerve injury
C. Motor neuron disease
D. A UMN lesion

Answer
A.

Last recall check. Write the UMN vs LMN table, the five deep reflexes with their roots, and the dermatome key points of the leg.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. American Spinal Injury Association. International Standards for Neurological Classification of Spinal Cord Injury (ISNCSCI): worksheet and key sensory points.
3. Medical Research Council. Aids to the examination of the peripheral nervous system (MRC muscle scale).
4. Standard clinical neurology and electrodiagnostic texts for reflex levels, sensory pathways, cranial nerve tests and EMG and nerve conduction patterns (root levels of reflexes differ slightly between texts; the common teaching values are given).

G Cognition, perception, apraxia and aphasia: assessment and physiotherapy

> These problems came up in 7 papers: the types of apraxia were a 7-mark question three times, and aphasia, neglect, figure-ground discrimination and combined cortical sensations were 3-mark questions. They matter because a patient who cannot plan a movement, understand an instruction or notice one side of the world will not benefit from exercises taught in the usual way: you must change how you teach.

Syllabus	General: assessment of arousal and cognition; cortical sensations; perception (applies to stroke, brain injury and other conditions)
Priority	**Medium yield.** In 7 of 20 papers (2016–2026): 3 seven-mark and 6 three-mark questions.
Asked as	7m S23 · S24 · W24 · 3m W18 ×2 · S19 · S22 · S23 · S25
Usual split	7 (define apraxia 1, types with lesion and example 4–5, assessment and treatment 1–2); 3 (types with one line each, or a test)
Checked against	Classical neuropsychology of apraxia, aphasia and neglect; standard tests (line bisection, cancellation, MMSE, MoCA). Full list under Sources (p. 43). Content checked 20 October 2026.

1 Apraxia

Asked for 7 marks in S23, S24 and W24, and for 3 marks in S22.

Apraxia is the inability to carry out learned, purposeful movements on command, even though strength, sensation, coordination, comprehension and willingness are adequate. It usually follows a left parietal or frontal lesion.

Type	What fails	Example
Ideomotor	Cannot do a movement on command or by imitation, though the person knows what to do and may do it automatically (left parietal)	Cannot show how to comb hair when asked, but combs spontaneously in the morning
Ideational (conceptual)	Loses the idea or sequence of a task; misuses objects (left parietal, diffuse damage)	Puts toothpaste on the comb; tries to drink tea before pouring it
Limb-kinetic	Clumsy, imprecise fine finger movements (premotor cortex)	Cannot button a shirt or pick up a coin, without weakness
Constructional	Cannot copy, draw or build two- or three-dimensional shapes (parietal, either side)	Cannot copy a cube or arrange blocks to a pattern
Dressing	Cannot dress correctly (often right parietal, with neglect)	Puts the shirt on inside out or one arm only
Buccofacial (oral)	Cannot do facial movements on command (left frontal; often with Broca's aphasia)	Cannot blow out a candle when asked, but does it when the candle is lit
Gait apraxia	Cannot start or organise walking though leg movements are normal in lying (frontal lobe, hydrocephalus)	Feet seem "glued to the floor"

Assessment

- Ask the person to pantomime tool use ("show me how you brush your teeth"), imitate meaningless and meaningful gestures, use real objects, and perform a multistep task (making tea).
- Note errors: wrong movement, body part used as the tool, wrong order, perseveration.

Physiotherapy

- Practise **real tasks with real objects in their natural setting** (drink from a real cup in the ward), rather than abstract movements on command.

- **Strategy training:** teach the person to talk themselves through each step, with pictures or written step cards.
- **Errorless learning and chaining:** guide the correct movement; teach the steps one at a time (forward or backward chaining).
- Physical guidance rather than long verbal instructions; reduce cues gradually.
- Gesture training for ideomotor apraxia.

2 Aphasia

Asked for 3 marks in S23 (Broca's and Wernicke's) and S25 (aphasia and its types).

Aphasia is a loss or impairment of language (speaking, understanding, reading or writing) from brain damage, usually of the left hemisphere. It differs from **dysarthria**, which is a motor speech disorder with normal language.

Type	Speech and comprehension	Lesion
Broca's (expressive, non-fluent)	Slow, effortful, short phrases with missing small words (telegraphic); comprehension fairly good; repetition poor; the person knows and is frustrated	Left inferior frontal gyrus; often with right arm weakness
Wernicke's (receptive, fluent)	Fluent but empty or meaningless speech with wrong and made-up words; poor comprehension and repetition; often unaware of the problem	Left posterior superior temporal gyrus
Conduction	Fluent, comprehension good, but repetition very poor	Arcuate fasciculus
Global	All language functions severely affected	Large left MCA territory lesion
Anomic	Fluent with good comprehension, but word-finding difficulty	Various, often mild
Transcortical motor or sensory	Like Broca's or Wernicke's, but repetition is preserved	Areas around the language centres

How physiotherapy changes

- Speak slowly in short, simple sentences; one instruction at a time; ask yes or no questions.
- Use gestures, demonstration, pictures and physical guidance; check understanding by watching the action, not by "yes".
- Reduce noise and distraction; give time to respond; do not finish sentences for the person or speak as if to a child.
- Work with the speech and language therapist; use the same communication aids.

3 Unilateral neglect

Asked for 3 marks in S19 (assessment of unilateral visual neglect).

- **Unilateral neglect** is a failure to report, respond or orient to stimuli on the side opposite a brain lesion, not caused by weakness or sensory loss. It is commonest and most severe after right parietal lesions (left neglect).
- **Signs:** eats food from one half of the plate; shaves or dresses one side; bumps into door frames on the left; ignores people on the left; the head and eyes turn to the right; may deny the weak arm belongs to them.
- **Neglect vs hemianopia:** in hemianopia the person knows about the field loss and turns the head to compensate; in neglect the person is unaware and does not compensate.

Assessment (S19)

Test	How	Sign of neglect
Line bisection	Mark the middle of several horizontal lines	Marks shifted towards the good side
Cancellation tests (star, letter, Albert's lines)	Cross out all targets on a sheet	Targets missed on the neglected side
Drawing and copying	Draw a clock, copy a flower or house	One half missing or crowded

Test	How	Sign of neglect
Reading and writing	Read a paragraph; write a sentence	Starts reading mid-line; writing drifts to one side
Functional observation	Eating, dressing, wheelchair use; Catherine Bergego Scale	One side ignored in daily tasks

Physiotherapy

- Visual scanning training: anchor the eyes on a coloured line at the edge of the neglected side, then scan across.
- Approach and talk from the neglected side once the person can tolerate it; place items progressively towards that side.
- Limb activation: move the neglected arm in the neglected space; trunk rotation towards the neglected side.
- Other options with some evidence: prism adaptation, mirror therapy; the evidence for neglect treatments remains limited.
- Safety: check the neglected limb's position (arm caught in wheels, foot off the footplate).

4 Perceptual disorders and cortical sensations

Figure-ground discrimination (3 marks in W18)

The inability to pick out an object from its background. The person cannot find a white shirt on a white bed, a brake on the wheelchair, or a step edge on a patterned floor (usually right parietal-occipital damage). **Assess** by asking the person to find an item in a cluttered drawer or picture. **Treat** by reducing clutter, adding colour contrast (bright tape on brakes and step edges), scanning practice, and gradually busier settings.

Combined (cortical) sensations (3 marks in W18)

Sensations that need intact primary sensation plus integration in the parietal cortex:

- **Stereognosis:** recognising an object (coin, key) by touch with the eyes closed.
- **Graphaesthesia:** recognising a number or letter drawn on the skin.
- **Two-point discrimination:** telling one touch from two close points.
- **Tactile localisation:** pointing to where you were touched.
- **Barognosis:** judging weight; recognising texture.
- **Bilateral simultaneous stimulation:** feeling touches on both sides at once (one side "disappears" in extinction).

Other perceptual problems

- **Body scheme disorders:** poor awareness of body parts and their relations (somatoagnosia), right-left confusion, finger agnosia.
- **Spatial relations:** position in space, depth perception (misjudging steps and kerbs), topographical disorientation (getting lost).
- **Agnosias:** visual object agnosia; prosopagnosia (faces).
- **Anosognosia:** unawareness or denial of the paralysis.
- **Pusher syndrome:** see the stroke page (p. 50).

5 Cognition

- **Domains:** arousal, attention, memory, executive function (planning, problem-solving, judgement), processing speed.
- **Screening tools:** Mini-Mental State Examination (MMSE, 30 points; low scores suggest impairment, adjusted for education); Montreal Cognitive Assessment (MoCA, 30 points, more sensitive to mild and executive problems).

- **How physiotherapy changes:** short, structured sessions at the same time each day; one-step instructions; repetition; errorless learning; memory aids (diary, checklists, phone reminders); a quiet room; involve family; check safety awareness before independent practice.

Recall check. How would you teach a transfer to a patient with ideational apraxia, and to one with Wernicke's aphasia?

6 Exam corner

Past questions (MUHS, 2016–2026)

S23 Q2(f), S24 Q2(d), W24 Q2(f)
Apraxia and its types (with examples in S23).

3-mark questions
Figure-ground discrimination and combined cortical sensations (W18); assessment of unilateral visual neglect (S19); apraxia and its types (S22); Broca's and Wernicke's aphasia (S23); aphasia and its types (S25).

Pages for this topic: stroke (p. 50), brain injury (p. 66), multiple sclerosis (p. 133), scales (p. 21).

Keywords to underline

apraxia ideomotor ideational constructional dressing apraxia Broca's Wernicke's fluent vs non-fluent neglect line bisection cancellation figure-ground stereognosis

Answer skeleton: apraxia and its types (7 marks)

1. **[Define, 1]** Inability to perform learned purposeful movement despite adequate strength, sensation and comprehension.
2. **[Types, 4]** Ideomotor, ideational, limb-kinetic, constructional, dressing, buccofacial, gait: lesion and an example each.
3. **[Assessment and physiotherapy, 2]** Pantomime, imitation, real objects, sequencing; real tasks in context, strategy training, errorless learning, chaining, guidance.

Draw these

- A line bisection result and a cancellation sheet with left-side omissions
- The left hemisphere with Broca's and Wernicke's areas and the arcuate fasciculus

Viva questions

1. Differentiate ideomotor and ideational apraxia.
2. How does Broca's aphasia differ from Wernicke's?
3. How do you tell neglect from hemianopia?
4. What is stereognosis, and how do you test it?
5. How do you give instructions to a patient with aphasia?

Linked practical spots

- Line bisection and cancellation sheets; objects for stereognosis; MMSE form.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Differentiate Broca's and Wernicke's aphasia.**

Answer
Broca's: non-fluent, effortful, telegraphic speech; comprehension fairly good; aware; left inferior frontal lesion. Wernicke's: fluent but meaningless speech with paraphasias; poor comprehension; often unaware; left posterior superior temporal lesion. Repetition is poor in both.

3 m **How do you assess unilateral visual neglect?**

Answer
Line bisection (marks shifted to the good side); cancellation tests (targets missed on one side); clock drawing and figure copying (half missing); reading (starts mid-line); observation of eating, dressing and wheelchair use (Catherine Bergego Scale).

7 m **Describe apraxia and its types with examples.**

Answer
Definition; ideomotor (cannot comb on command but does it automatically); ideational (wrong sequence or object use: toothpaste on the comb); limb-kinetic (clumsy buttoning); constructional (cannot copy a cube); dressing (shirt inside out); buccofacial (cannot blow out a candle on command); gait apraxia (feet glued to the floor). Assessment and physiotherapy as in the skeleton.

LAQ **A patient with a right parietal stroke eats only the right half of his meal and keeps hitting his left arm on door frames. Plan assessment and physiotherapy. (Practice case)**

Answer
Assessment: line bisection, cancellation, drawing, reading; observe daily tasks; separate neglect from hemianopia; check sensation and anosognosia. Physiotherapy: visual scanning with an anchor on the left; limb activation in left space; trunk rotation to the left; approach and place items from the left as tolerated; mirror therapy or prism adaptation if available; safety for the left arm and foot in the wheelchair; teach family; repeat tests to track change.

MCQ **A patient puts toothpaste on his comb. This is:**

A. Ideomotor apraxia

B. Ideational apraxia

C. Constructional apraxia

D. Limb-kinetic apraxia

Answer
B. The concept and sequence of object use are lost.

MCQ **Recognising a key placed in the hand with the eyes closed tests:**

A. Graphaesthesia

B. Stereognosis

C. Barognosis

D. Two-point discrimination

Answer
B.

MCQ **Fluent speech with poor comprehension and made-up words suggests:**

A. Broca's aphasia

B. Wernicke's aphasia

C. Dysarthria

D. Conduction aphasia

Answer

B.

Last recall check. List seven types of apraxia with an example, six aphasias, and five neglect tests.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Folstein MF, et al. "Mini-mental state". *J Psychiatr Res* 1975;12:189–98. Nasreddine ZS, et al. The Montreal Cognitive Assessment. *J Am Geriatr Soc* 2005;53:695–9.
3. Structure and classical content: Zoltan, *Vision, Perception, and Cognition*; O'Sullivan and Schmitz, *Physical Rehabilitation* (cognitive and perceptual dysfunction); Lundy-Ekman, *Neuroscience*.

G Movement dysfunction in daily tasks: bed mobility, lying to sitting, standing up, walking, balance, reaching and manipulation

> The syllabus asks you to know how movement fails in seven everyday tasks after damage to the nervous system. For each task learn three things: its essential components, the common problems and compensations (stroke is the usual example), and how to practise it; this is the backbone of every management answer.

Syllabus	General item 3: manifestation of movement dysfunction following disease or trauma of the central or peripheral nervous system: bed mobility, lying to sitting, standing up and sitting down, walking, balance, reaching, manipulation
Priority	**Syllabus.** Not asked on its own in 2016–2026; it supports every long-case management answer and the motor relearning questions.
Asked as	Not asked on its own in 2016–2026
Usual split	Within long cases: task analysis and training of the main problem task
Checked against	Task-oriented training principles (motor relearning programme, Carr and Shepherd); trial of trunk restraint during reaching after stroke (Michaelsen et al., 2006); outcome measures as on the scales page. Full list under Sources (p. 48). Content checked 20 October 2026.

1 How movement fails

- **Impairments:** weakness and slowness, loss of dexterity and coordination, altered tone, sensory loss, shortened soft tissues, poor balance.
- **Adaptations (compensations):** the person uses the stronger side, the arms, a wider base or a stiff trunk to get the task done. Some compensations are useful; others block recovery (learned non-use of the weak arm, all weight on the strong leg).
- **Training rule:** analyse the task, find the missing components, practise them and then the whole task many times in varied real settings, and measure the result (see motor relearning (p. 209) and motor learning (p. 15)).

2 Bed mobility (rolling and bridging)

- **Essential components:** turning the head and reaching the arm across; flexing the top hip and knee; rotating the trunk; bridging (lifting the pelvis) to move up the bed and onto a bedpan.
- **Common problems:** cannot bring the weak arm or leg across; pulls on the bed rail with the strong arm; cannot bridge because of weak hip extensors.
- **Training:** practise rolling to both sides with the weak limbs placed to help; bridging with the weak foot loaded; reduce reliance on the rail.

3 Lying to sitting and sitting to lying

- **Essential components:** roll to side lying, lower the legs over the edge, push up through the lower arm while side-flexing the neck and trunk upwards; the reverse to lie down.
- **Common problems:** pulls straight up with the strong arm; legs stay on the bed; poor sitting balance at the end.
- **Training:** practise each part (side lying, legs off, push up), then the whole task; vary bed height and side; practise lying down with control.

4 Standing up and sitting down

- **Essential components:** feet placed back (behind the knees); trunk moves forward by flexing at the hips with a straight spine; knees move forward over the feet; then hips and knees extend together to stand; sitting down reverses this with controlled lowering.
- **Common problems:** weight on the strong leg; weak foot left forward; not enough forward lean; pushes up with the arms; drops into the chair when sitting down.
- **Training:** foot placement with the weak foot back; practise with a higher seat, then lower; repetitions in sets; step-ups and leg strengthening; time it (for example the five times sit-to-stand test).

5 Walking

- **Essential components:** stance: hip extension, a controlled small knee bend at loading then extension, forward roll over the ankle, push-off; swing: knee bending, hip flexion and ankle dorsiflexion to clear the foot; a small side-to-side shift of the pelvis.
- **Common problems:** short stance on the weak leg, knee hyperextension or buckling, little push-off, stiff knee and foot drop in swing, circumduction, slow speed.
- **Training:** walking practice with intensity and variety, treadmill, stepping drills, strengthening, an AFO when needed; measure speed and distance (see gait training (p. 232)).

6 Balance

- **Essential components:** keeping the centre of mass over the base of support while still and while moving; anticipatory adjustments before a movement; reactive steps or strategies after a push; using vision, vestibular and body sensation.
- **Common problems:** wide base, weight kept on the strong side, stiffening, holding on with the arms, fear of falling.
- **Training:** reaching beyond arm's length in sitting and standing, stepping in all directions, practice on varied surfaces, perturbations, dual tasks; measure with Berg, TUG or Mini-BESTest (see balance (p. 223)).

7 Reaching

- **Essential components:** shoulder flexion with the scapula rotating upwards and moving forward, elbow extension, a wrist ready for grasp, and trunk movement only for targets beyond arm's length.
- **Common problems:** the shoulder hitches up instead of the scapula rotating; the elbow stays bent; the trunk leans forward to make up for a short reach.
- **Training:** reaching to targets at different heights and distances; limiting forward trunk lean during practice (trunk restraint improved reaching in a trial after stroke; Michaelsen et al., 2006); strengthening shoulder and elbow; task practice with real objects.

8 Manipulation (grasp, release and hand use)

- **Essential components:** wrist extension during grasp; opening the hand to match the object; thumb opposition; grip force scaled to the object; release; moving objects within the hand.
- **Common problems:** weak wrist extensors so the grip is weak; the hand cannot open (flexor tone or weak extensors); poor thumb opposition; too much or too little grip force; poor release.
- **Training:** graded object tasks (size, weight, texture), wrist-extension practice in function, constraint-induced movement therapy or intensive bimanual practice for suitable patients, splints only for a stated aim (see orthoses (p. 249)).

Recall check. For standing up, list the essential components, two common compensations after stroke and two training methods.

9 Exam corner

Where this is asked

Inside other questions

Motor relearning programme (S18, W18 on the approaches page); "management with emphasis on trunk control and ambulation" in long cases; hemiplegic gait and the spastic hand on the stroke page.

Keywords to underline

task analysis essential components compensation learned non-use feet back, trunk forward scapular upward rotation trunk restraint wrist extension in grasp whole-task practice

Answer skeleton: training any task

1. Essential components of the task.
2. What is missing and how the patient compensates.
3. Practice of missing components, then the whole task with repetition and variety; feedback; measure.

Draw these

- Stick figures of sit to stand: feet back, trunk forward, knees forward, extension

Viva questions

1. Why should the feet be placed back before standing up?
2. What is learned non-use?
3. How does a stroke patient compensate when reaching?
4. Why is wrist extension important for grip?
5. Name one measure for sit to stand and one for balance.

Linked practical spots

- Demonstrate task analysis and training of sit to stand or reaching on a model patient.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the essential components of sit to stand.**

Answer
Feet placed back behind the knees; trunk moves forward by hip flexion with a straight spine; knees move forward over the feet; hips and knees then extend together to upright standing.

3 m **List common compensations during reaching after stroke.**

Answer
Shoulder hitching (scapular elevation) instead of upward rotation; elbow kept flexed; forward trunk lean to extend the reach; using the other hand instead.

7 m **Describe the training of standing up and sitting down in a hemiplegic patient.**

Answer
Analyse the task (foot placement, forward lean, knee advancement, extension, controlled descent) and identify problems (weight on the strong side, weak foot forward, too little lean, arm pushing). Train: place the weak foot back, practise forward lean with reaching, start from a higher seat and lower it, repeated practice in sets, discourage arm use, practise sitting down slowly, strengthen hip and knee extensors with step-ups; vary seats (chair, bed, toilet, car); measure with a timed sit-to-stand test.

LAQ **Practice: a 62-year-old man 6 weeks after a left MCA stroke cannot get out of bed alone, stands up only by pulling on a rail, and reaches with a shoulder hitch. Analyse these three tasks and plan training. (15)**

Answer
Lying to sitting: pulls straight up; train rolling to side lying, legs over the edge, push up through the left arm, with the right leg helped and then not. **Sit to stand:** weight on the left, right foot forward, pulls on rail; train right foot back, forward lean, raised seat, repetitions, then lower seat, no rail. **Reaching:** shoulder hitch, trunk lean; train reaching to targets within arm's length with trunk kept back, scapular upward rotation and elbow extension practice, then functional reaching with objects. Measure: timed sit to stand, reach distance, independence in bed mobility.

MCQ **Before standing up, the feet should be:**

A. Forward of the knees

B. Behind the knees

C. Wide apart in front

D. Crossed

Answer
B.

MCQ **Learned non-use means:**

A. Loss of nerve supply

B. Not using the weak limb because the strong one is easier, which then blocks recovery

C. Muscle shortening

D. Neglect of one side

Answer
B.

MCQ **A weak grip in hemiplegia is often due to weakness of the:**

A. Wrist extensors

B. Biceps

C. Deltoid

D. Hip abductors

Answer
A. Wrist extension positions the finger flexors to grip.

Last recall check. For each of the seven tasks write the components, one common problem and one training method.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Carr JH, Shepherd RB. *Neurological rehabilitation: optimizing motor performance*. 2nd ed. Edinburgh: Churchill Livingstone; 2010.
3. Michaelsen SM, Dannenbaum R, Levin MF. Task-specific training with trunk restraint on arm recovery in stroke: randomized control trial. *Stroke* 2006;37:186–92.

Part 5A

Physiotherapy management: adult conditions

37 hours theory, 95 practical

5A(a)	Stroke	50
5A(b)	Acquired brain injury: trauma and space-occupying lesions	66
5A(c)	Spinal cord disorders: traumatic and non-traumatic	77
5A(d)	Peripheral nerve injuries and the brachial plexus	94
5A(d)	Peripheral neuropathies and nerve root lesions	107
5A(e)	Vestibular disorders	116
5A(f)	Facial (VII) nerve palsy and other cranial nerves	124
5A(g)	Multiple sclerosis	133
5A(g)	Guillain–Barré syndrome	144
5A(h)	Cerebellar disease and ataxia	155
5A(i)	Parkinson's disease and other extrapyramidal disorders	164
5A(j)	Anterior horn cell diseases: MND, SMA and poliomyelitis	177
5A(k)	Myopathies and myasthenia gravis	186
5A(l)	Disorders of the autonomic nervous system	193
5A(m)	Psychosomatic pain and paralysis	198

5A(a) Stroke: types, arterial syndromes, assessment and physiotherapy

> Stroke was in 18 of the 20 papers (2016–2026): a long case in 11 of them, and a 7-mark question on hemiplegic gait, the hemiplegic shoulder, the spastic hand, the acute stage, motor relearning or the circle of Willis. Every answer follows one path: where the lesion is, what it does to movement and daily life, and the task practice that drives recovery.

Syllabus	Ch 5A(a) Stroke: cerebral circulation, types of stroke and manifestations, assessment and management
Priority	**Tier 1.** In 18 of 20 papers (2016–2026): 11 long cases, 13 seven-mark and 5 three-mark questions.
Asked as	LAQ W17 · S18 · S19 · S20 · S21 · W21 · W22 · S23 · S24 · W24 · W25 · 7m S16 · W16 · S17 · S18 · W18 · S19 · S20 · S21 · S22 ×2 · S24 · S25 · S26 · 3m S18 · S20 · W24 · S26 ×2
Usual split	15 (ICF with clinical reasoning, short- and long-term goals, treatment: 5 + 5 + 5, 5 + 7 + 3 or 7 + 8); 7-mark gait, shoulder, hand or acute stage
Checked against	AHA/ASA guidelines for acute ischaemic stroke (2026) and for adult stroke rehabilitation and recovery (2026); NICE stroke rehabilitation guideline NG236 (2023); AHA/ASA physical activity statement for stroke survivors (2014); AVERT (2015), EXCITE (2006), EPOS (2010) and CLOTS 3 (2013) trials; ICMR National Stroke Registry. Full list under Sources (p. 64). Content checked 19 October 2026.

1 Definition

- **Stroke (WHO):** rapidly developing clinical signs of focal (or global) disturbance of brain function, lasting more than 24 hours or leading to death, with no apparent cause other than a vascular one.
- **Current tissue-based definition (AHA/ASA, 2013):** death of brain, spinal cord or retinal cells caused by ischaemia, shown on imaging or pathology, or by symptoms lasting 24 hours or more; bleeding into the brain or subarachnoid space is included.
- **Transient ischaemic attack (TIA):** a short episode of neurological dysfunction from focal ischaemia **without** infarction on imaging. A TIA is a warning: the risk of stroke is highest in the next few days.
- **Hemiplegia** is paralysis of one side of the body; **hemiparesis** is weakness of one side. Both are on the side **opposite** the brain lesion, because the motor pathway crosses in the medulla.

2 Classification and the cerebral circulation

By pathology

Type	Mechanism	Typical clue
Ischaemic: thrombotic	Clot on an atherosclerotic plaque in a large artery	Onset during sleep or rest; stepwise worsening; earlier TIAs
Ischaemic: embolic	Clot travels from the heart (atrial fibrillation, rheumatic valve disease) or the carotid	Sudden, maximal at onset, often during activity
Ischaemic: lacunar	Small deep penetrating artery blocked (hypertension, diabetes)	Pure motor or pure sensory picture; no cortical signs
Haemorrhagic: intracerebral	Small artery ruptures into brain tissue (hypertension)	Headache, vomiting, falling consciousness, raised BP
Haemorrhagic: subarachnoid	Aneurysm bursts into the space around the brain	"Worst headache of my life", neck stiffness

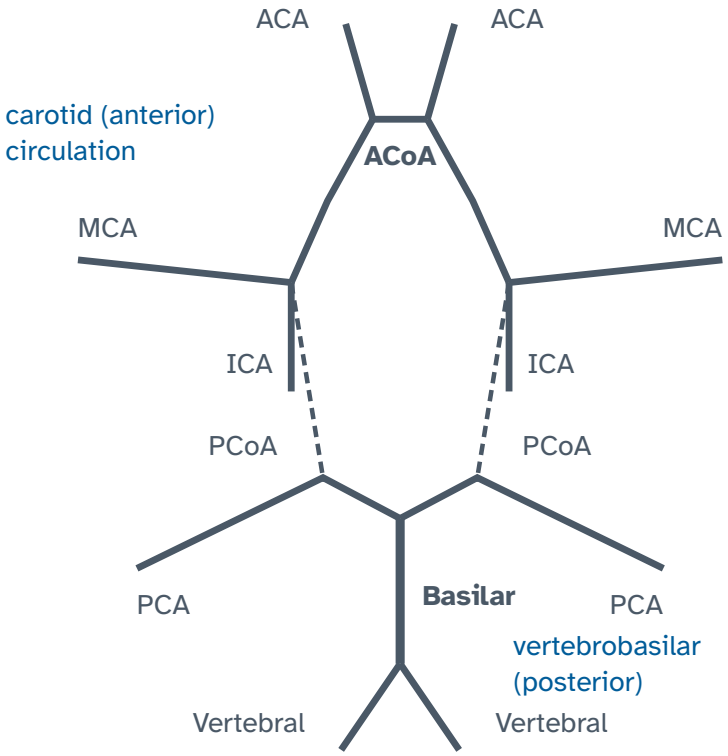
In Western series about 85% of strokes are ischaemic. Indian registries record a larger share of haemorrhages, about a fifth to a third of all strokes.

By time course and recovery phase

- Classic terms: TIA, stroke in evolution (still worsening), completed stroke.
- **Recovery phases** (Stroke Recovery and Rehabilitation Roundtable, 2017): hyperacute 0–24 hours; acute 1–7 days; early subacute 7 days to 3 months; late subacute 3–6 months; chronic after 6 months.

The circle of Willis (7 marks in S20)

The circle of Willis, seen from below



A ring of arteries at the base of the brain. The communicating arteries (dashed) let blood cross over if one feeding artery narrows slowly. ACA anterior cerebral, ACoA anterior communicating, ICA internal carotid, MCA middle cerebral, PCoA posterior communicating, PCA posterior cerebral.

? Text version

> Front: right and left ACA joined by the ACoA. Each ICA divides into the ACA (forward) and MCA (sideways). Each ICA joins the PCA on its side through the PCoA. Back: the two vertebral arteries join to form the basilar artery, which divides into the two PCAs.

For the 7-mark answer: draw the ring; list its vessels; say what it supplies (forebrain from the carotids, brainstem, cerebellum and occipital lobes from the vertebrobasilar system); and give its function: a safety connection between the right and left sides and between the front and back circulations. Most aneurysms causing subarachnoid haemorrhage arise at its junctions, especially the anterior communicating artery.

Arterial syndromes (types of stroke, 3 marks in S26)

Artery	Main features (opposite side unless stated)
Middle cerebral (the commonest)	Weakness and sensory loss of face and arm more than leg; homonymous hemianopia; **aphasia** if the dominant (usually left) side; **neglect** and visuospatial problems if the right side

Artery	Main features (opposite side unless stated)
Anterior cerebral (3 marks in S26)	Weakness and sensory loss of the **leg more than the arm**; urinary incontinence; apathy and slowed thinking (abulia); grasp reflex; difficulty starting gait
Posterior cerebral	Homonymous hemianopia; visual agnosia; memory loss; thalamic pain syndrome
Vertebrobasilar (brainstem, cerebellum)	**Crossed signs** (cranial nerve palsy on the same side, limb weakness on the other); vertigo, double vision, dysarthria, dysphagia, ataxia; "locked-in" syndrome with basilar occlusion
Lacunar (small deep arteries)	Pure motor, pure sensory, sensorimotor, ataxic hemiparesis, or dysarthria with a clumsy hand; no aphasia, neglect or field loss

Oxfordshire (Bamford) classification by clinical picture: total anterior circulation (weakness, field loss and a higher cortical deficit together), partial anterior circulation (two of the three, or a cortical deficit alone), lacunar, and posterior circulation strokes.

3 Causes and risk factors

- **Non-modifiable:** older age, male sex, family history, previous stroke or TIA.
- **Modifiable:** in the INTERSTROKE study (32 countries, 2016), ten factors accounted for about 90% of stroke risk: hypertension (the most important), smoking, diabetes, physical inactivity, poor diet, psychosocial stress and depression, abdominal obesity, alcohol, cardiac causes such as atrial fibrillation, and abnormal blood lipids.
- **Causes to think of in a young patient:** rheumatic heart disease, cerebral venous thrombosis (for example after childbirth), tuberculous meningitis with vasculitis, arterial dissection, drug misuse, clotting disorders.

Indian context: in the ICMR hospital-based stroke registry (30 hospitals, 2020–2022) about **1 in 7 patients was younger than 45**, and the average age was about 59. Uncontrolled hypertension, tobacco in any form, and diabetes are the targets in nearly every Indian case.

4 Pathophysiology to impairments

1. **Ischaemia:** blood flow stops, so the central core of brain tissue dies within minutes. Around it, the **penumbra** is starved but still alive for some hours; reopening the artery early saves it ("time is brain").
2. **Haemorrhage:** the blood clot compresses and irritates the brain and raises intracranial pressure.
3. Damage to the motor cortex or its pathway (the corticospinal tract through the internal capsule) produces an **upper motor neuron syndrome**:
    - **Negative features:** weakness, loss of dexterity and selective movement, fatigue.
    - **Positive features:** spasticity, brisk reflexes, clonus, Babinski sign, associated reactions, and movement locked into **synergy patterns**.
    - **Adaptive features:** stiffness and contracture of muscles kept short; **learned non-use** of the affected arm; general deconditioning.
4. Other areas damaged according to the artery: sensation, vision, speech, swallowing, perception (neglect, apraxia, pushing), cognition and mood.
5. **Recovery:** swelling settles and nearby or opposite brain areas take over (neuroplasticity). Plasticity is **use-dependent**: repeated, task-specific practice strengthens the new pathways, and disuse lets them fade. Most spontaneous recovery happens in the first 3 months.

? Why does the arm usually recover less than the leg after an MCA stroke?

> The MCA supplies the part of the motor cortex controlling the face and arm, while the leg area lies on the inner surface supplied by the ACA. The hand also depends heavily on the direct corticospinal pathway for fine, independent finger movement, whereas walking can use other pathways and both sides of the brain.

Recall check. Name three negative, three positive and two adaptive features of the upper motor neuron syndrome.

5 Clinical features

- **Sudden onset;** the public warning sign is **BE-FAST**: Balance, Eyes, Face drooping, Arm weakness, Speech difficulty, Time to call for help.
- **Motor:** hemiplegia or hemiparesis; tone is often low at first (flaccid stage), then spasticity develops in the antigravity muscles; synergy patterns; brisk reflexes and an extensor plantar response.
- **Sensory and visual:** loss of touch and position sense, homonymous hemianopia.
- **Communication and swallowing:** aphasia, dysarthria, dysphagia (aspiration risk).
- **Perception and cognition:** neglect, apraxia, pusher syndrome, memory and attention problems (see cognition and perception (p. 38)).
- **Others:** shoulder pain and subluxation, bladder incontinence, depression, emotionalism, fatigue, central post-stroke pain.

Synergy patterns (flexion synergy: 3 marks in S18)

Synergy	Components
Upper limb flexion	Scapular retraction and elevation; shoulder abduction and external rotation; **elbow flexion** (the strongest part); forearm supination; wrist and finger flexion
Upper limb extension	Scapular protraction; shoulder adduction and internal rotation (**pectoralis major**, the strongest part); elbow extension; forearm pronation; wrist and finger flexion
Lower limb flexion	Hip flexion (the strongest part), abduction and external rotation; knee flexion; ankle dorsiflexion and inversion; toe extension
Lower limb extension	Hip extension, **adduction** and internal rotation; knee extension; ankle **plantarflexion** and inversion; toe flexion

The typical spastic posture combines the stronger components: arm held in flexion with the shoulder adducted and internally rotated; leg in extension with the foot pointed and turned in.

Pusher syndrome (3 marks in S20 and W24)

The patient **pushes with the non-paretic arm and leg towards the paretic side** and resists any attempt to correct the posture, so they fall towards the weak side in sitting and standing. It is linked with a disturbed sense of upright body posture, often after right-hemisphere or thalamic lesions, and is graded with the Scale for Contraversive Pushing. Treatment uses **vision to show what is vertical** (a mirror, a door frame, the therapist's arm), reaching towards the strong side, and sitting or standing with the strong side against a wall or high table, which removes the pushing surface.

6 Investigations

Test	What it shows	Physio meaning
CT scan (first test)	Rules out bleeding before clot-busting treatment; infarct appears dark later	Know the type, side and size before mobilising
MRI (diffusion imaging)	Shows an infarct within minutes; small brainstem and lacunar lesions	Lesion site predicts the deficits to expect
CT or magnetic resonance angiography; carotid Doppler	Blocked large vessel; carotid narrowing	Candidate for thrombectomy or carotid surgery
ECG, Holter, echocardiogram	Atrial fibrillation; clot or valve disease in the heart	Monitor heart rate and rhythm during exercise

Test	What it shows	Physio meaning
Blood sugar, lipids, clotting	Risk factors; reason for bleeding	Low sugar can mimic stroke; anticoagulants mean bruising risk
NIHSS and GCS	Stroke severity (NIHSS 0–42); level of consciousness	Severity guides prognosis and how active the first sessions can be

7 Medical and surgical management

- **Stroke unit care** by a specialist team reduces death and dependency for every type of stroke.
- **Ischaemic stroke (AHA/ASA 2026):** clot-dissolving drug (tenecteplase or alteplase) within **4.5 hours** of onset, and in selected patients up to **9 hours** when imaging shows brain still worth saving; **mechanical thrombectomy** for a blocked large artery, within 6 hours and in selected patients up to **24 hours**; antiplatelet drugs (a short course of two antiplatelets after a minor stroke); later statins, blood-pressure control, anticoagulation for atrial fibrillation, and carotid surgery for a tight narrowing.
- **Haemorrhagic stroke:** lowering blood pressure, reversing anticoagulants, and surgery when needed (removing the clot, decompressive craniectomy, clipping or coiling an aneurysm).
- **Swallowing:** nothing by mouth until a swallow screen is passed.

> **What this means for physio**
>
> After thrombolysis or thrombectomy, follow the unit's protocol for observations and blood-pressure limits before mobilising. After a craniectomy, the side without bone must not take pressure; the patient wears a helmet when mobile.

8 Complications and prevention

Complication	What to look for	Physio role
Aspiration pneumonia	Cough on eating, wet voice, fever, crackles	Upright positioning, chest care, early mobility; follow the swallow plan
Deep vein thrombosis	Swollen, painful calf	Early mobility; **intermittent pneumatic compression** reduces DVT in immobile patients (CLOTS 3); stockings alone do not (CLOTS 1)
Shoulder subluxation and pain	Gap below the acromion; pain on movement	Support and positioning; never pull on the arm (section 12)
Spasticity and contracture	Tight elbow and finger flexors, calf, adductors	Positioning, daily stretching and active use; refer for botulinum toxin
Pressure injury	Redness over sacrum, heels, hips	Turning, positioning, early sitting
Falls	Poor balance, neglect, impulsivity, low BP	Balance training, safe transfers, a falls programme
Depression, fatigue	Low mood, poor engagement, tiredness	Graded activity, goals the patient chooses; refer
Complex regional pain (shoulder-hand syndrome)	Swollen, painful, discoloured hand with painful shoulder	Elevation, gentle active movement, desensitisation; refer early
Learned non-use	Arm ignored even when some movement returns	Task practice that forces use; CIMT when eligible

9 PT assessment

Subjective

- Date and type of stroke, side, hand dominance, treatment given (thrombolysis, surgery); risk factors and medicines.
- Function before the stroke; job, home (stairs, lift, **Indian or Western toilet**, floor-level sitting), family support, patient's own goals.

Objective

- **Consciousness and cognition:** GCS; MMSE or MoCA; ability to follow commands.
- **Communication and perception:** aphasia; neglect (line bisection, cancellation tests); pushing; apraxia.
- **Sensation:** touch, pain, position sense, stereognosis.
- **Tone:** Modified Ashworth Scale (0, 1, 1+, 2, 3, 4).
- **Motor control:** Brunnstrom stages (below); Fugl-Meyer Assessment (arm 66 points, leg 34); STREAM; grip and pinch.
- **Trunk and balance:** sitting and standing balance; Berg Balance Scale (out of 56; below about 45 suggests a fall risk); Functional Reach; Timed Up and Go.
- **Gait:** observational gait analysis by phase; 10-metre walk test (speed); 6-minute walk (endurance).
- **Shoulder:** subluxation (finger-breadths below the acromion), pain on movement (0–10), range.
- **Function:** Barthel Index (0–100) or FIM; bed mobility, transfers, stairs.

Brunnstrom stages of motor recovery

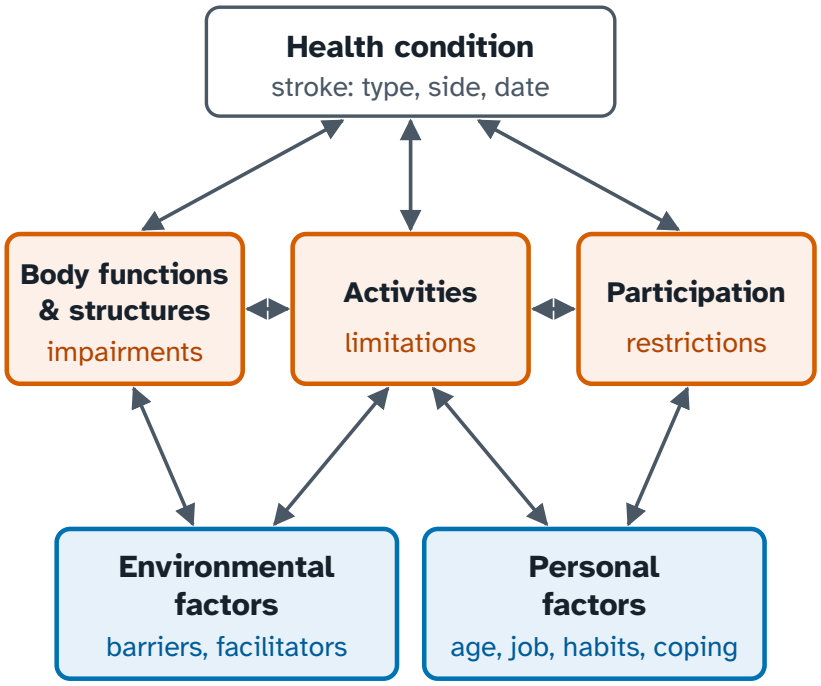
Stage	What you see
1	Flaccid; no voluntary movement
2	Spasticity begins; synergies appear as associated reactions or minimal voluntary movement
3	Voluntary movement only within synergies; spasticity at its peak
4	Some movements out of synergy (for example hand behind the back); spasticity declining
5	More complex movements independent of synergy; spasticity waning
6	Isolated joint movements with near-normal coordination; spasticity minimal

Not every patient passes through every stage, and recovery can stop at any stage.

10 PT diagnosis by ICF

Every stroke long case asks for ICF with clinical reasoning, or for impairments, before goals and treatment.

The ICF model (draw this first in every case answer)



Six boxes, all arrows two-way. Draw it, then fill each box from the case.

Worked example from a real paper (S23, 5 + 7 + 3 marks)

> "Write down ICF with clinical reasoning for a case of Lt. MCA infarct diagnosed 3 months back. Patient is 55 year old male and he is an auto driver by profession and a chronic diabetic. Plan short and long term goals with treatment strategies for the same. Patient walks independently with the help of a stick but has difficulty carrying out activities of daily living with his upper limb."

Spot the ICF clues before you write

Left MCA means a right hemiparesis, arm more than leg, and probably **aphasia** (left hemisphere). 3 months means the end of the early subacute phase: recovery is slowing, so practice must be intensive. auto driver means his livelihood needs a working right hand and driving clearance. chronic diabetic means foot care, sugar checks and slower healing. walks with a stick means the leg has recovered more than the arm.

ICF component	From this case	Clinical reasoning
Health condition	Left MCA ischaemic stroke, 3 months; type 2 diabetes	MCA territory: face and arm worse than leg
Body structure	Infarct in the left fronto-parietal cortex and internal capsule	Corticospinal tract damage gives the UMN picture
Body function impairments	• Right arm weakness with poor selective hand movement (Brunnstrom stage 3–4) • Spasticity of elbow, wrist and finger flexors (MAS 1+ to 2) • Reduced sensation and position sense in the right hand • Reduced standing balance; gait with reduced right stance and foot clearance • Possible speech difficulty (aphasia); risk of shoulder pain • Reduced exercise tolerance	• Hand area of the cortex lies in MCA territory • Spasticity and weakness block reach and grasp • Sensory loss leads to learned non-use • Weak hip and ankle control and spasticity of the calf change gait

ICF component	From this case	Clinical reasoning
Activity limitations	Eating, dressing, buttoning, bathing and toileting with the right hand; walking long distances and on rough ground; climbing into the auto	Poor reach, grasp and release; slow, effortful gait
Participation restrictions	Cannot drive the auto rickshaw, so income is lost; social and family roles reduced	Driving needs both hands, quick reactions and medical clearance
Environmental factors	**Facilitators:** family support, walking stick, hospital rehab. **Barriers:** loss of income; ask about stairs, Indian toilet, distance to the clinic.	Barriers become goals; facilitators are used in the plan
Personal factors	55-year-old man, sole or main earner; diabetic; ask about smoking, tobacco and motivation	Return to work is his strongest motivator

One-line functional diagnosis

A 55-year-old diabetic auto driver, 3 months after a left MCA infarct, with right arm weakness, flexor spasticity and sensory loss, and mildly impaired balance and gait. These limit self-care with the right hand and community walking, and stop him from driving his auto. Family support is a facilitator; loss of income is a barrier.

Practice cases from the papers: W17 watchman with a left MCA infarct and circumduction gait; S18 bank manager two weeks after a right MCA infarct; S20 housewife with an anterior-cerebral picture (leg more than arm); W25 civil engineer with a right MCA infarct who cannot stand. ICF code numbers are not needed.

Recall check. Draw the six ICF boxes from memory and fill them for the W25 civil engineer in the exam corner.

11 Problem list and goals

Goals for the S23 auto driver

Problem	Short-term (4 weeks)	Long-term (3 months)
Right hand use	Reaches and grasps a cup and brings it to the mouth	Eats, dresses and grooms using both hands
Spasticity	Full passive wrist and finger extension kept; MAS not worse	Active wrist and finger extension for release of objects
Balance and gait	Berg score up by about 5 points; walks 200 m with the stick	Walks in the community without a stick; climbs stairs safely
Fitness	Walks 10–15 minutes at a moderate effort	Aerobic exercise 20–60 minutes, 3–5 days a week
Work	Knows the driving rules and his risk factors	Assessed for return to driving, or an alternative job with the family

Write goals as SMART: specific, measurable, achievable, relevant and timed. Agree them with the patient and family.

12 PT management

Principles in every phase: **task-specific practice** (practise the real task), **repetition and intensity**, progression, feedback, and practice outside therapy time. NICE (2023) recommends offering at least **3 hours a day, 5 days a week** of combined therapy to those who can take part.

Phase 1: acute stage (7 marks in S17)

- **Timing:** start once medically stable. The AVERT trial (2015) found that very early, high-dose out-of-bed activity in the first 24 hours reduced the chance of a good outcome at 3 months, so begin with short, frequent sessions and build up.
- **Positioning** every 2 hours: lying on the affected side (shoulder brought forward, not lain on), lying on the unaffected side (affected arm forward on a pillow), supine (pillow under the affected shoulder and hip), and

supported sitting as soon as possible. Head up 30° when there is a swallowing or chest risk.

- **Chest care:** deep breathing, assisted cough, positioning; watch for aspiration.
- **Shoulder protection:** support the arm in sitting; teach staff and family never to lift the patient by the arm.
- **Range of motion** of all affected joints daily, gently, within pain-free range.
- **Early functional training:** rolling to both sides, lying to sitting over the edge of the bed, sitting balance, sit to stand, standing, and short walks with help.
- **DVT prevention:** active movements, early mobility, intermittent pneumatic compression when immobile.

Phase 2: subacute rehabilitation

- **Trunk and balance:** reaching in sitting beyond arm's length, weight shift in standing, stepping, perturbations; a formal falls-prevention programme for every patient (AHA/ASA 2026); tai chi and dual-task training are reasonable.
- **Sit to stand and transfers:** feet back, trunk forward, equal weight; practise from different heights.
- **Gait training:** overground walking practice, treadmill training with or without body-weight support, an **ankle-foot orthosis** for foot drop, and walking aids only as needed.
- **Upper limb:** task practice of reach, grasp, release and manipulation; **constraint-induced movement therapy** for those with some active wrist and finger extension (in the EXCITE trial the higher-functioning group had at least 20° active wrist and 10° finger extension; a mitt was worn on the strong hand for about 90% of waking hours, with 6 hours a day of task practice for 2 weeks); **mirror therapy** added to usual therapy (reasonable, AHA/ASA 2026); electrical stimulation of wrist and finger extensors; bilateral arm training.
- **Strength and fitness:** progressive resistance training and aerobic exercise (doses in phase 3).
- **Sensory and perceptual training:** sensory re-education of the hand; visual scanning for neglect; the pusher programme (section 5).
- **Spasticity:** positioning, daily stretching and active use; botulinum toxin injection into overactive arm or leg muscles, followed by stretching and task practice, is recommended for focal spasticity (AHA/ASA 2026, Class 1).

Phase 3: chronic stage and community

Exercise prescription for stroke survivors (AHA/ASA, 2014)

Type	Dose
Aerobic (walking, cycling, arm or arm-leg ergometer)	**40–70%** of heart rate reserve (RPE 11–14), **3–5 days** a week, **20–60 minutes** (or several 10-minute bouts), with 5–10 minutes of warm-up and cool-down
Strength	**1–3 sets of 10–15 repetitions** of 8–10 exercises for the major muscle groups, 2–3 days a week
Flexibility	Static stretches held 10–30 seconds, 2–3 days a week
Balance and coordination	Balance activities, tai chi, yoga, 2–3 days a week

- Home exercise programme; walking in the community; return to work and driving with assessment; secondary prevention (section 14).

Hemiplegic gait and its training (7 marks in S18, S19, S21, S22 and S25)

Phase	What goes wrong	Main reason
Stance	Short stance on the weak leg; knee snaps back (hyperextension) or stays bent; trunk leans; little hip extension at the end of stance; weak push-off	Weak hip extensors and quadriceps, spastic or tight calf, poor balance
Swing	Stiff knee; foot drop and inversion; **circumduction** or hip hiking to clear the foot	Weak hip flexors and dorsiflexors, spastic quadriceps and calf (extensor synergy)
Overall	Slow, short, asymmetric steps; arm held in flexion; high energy cost	Weakness, spasticity, poor selective control, fear of falling

- **Training:** treat the cause (strengthen hip extensors, quadriceps and dorsiflexors; stretch the calf); practise each part (weight shift onto the weak leg, step-ups, stepping forwards and backwards, knee control in small ranges); then the whole task: **repetitive walking practice** overground and on a treadmill, faster as able; AFO or electrical stimulation of the peroneal nerve for foot drop; walking outdoors, on slopes and with obstacles.
- **Through the phases of recovery (S25):** in the flaccid stage the patient stands and steps in the parallel bars with knee support; in the synergy stage walking is dominated by the extensor synergy (stiff knee, pointed foot, circumduction), so training breaks the synergy (knee bending with the hip extended, dorsiflexion with the knee straight); as selective control returns, the gait becomes faster and more symmetrical and training moves to the community.

The hemiplegic shoulder (7 marks in S16 and S26)

- **Problems:** inferior subluxation in the flaccid stage (the weight of the arm stretches the capsule); pain; spasticity of the internal rotators and adductors; shoulder-hand syndrome.
- **Prevent:** support the arm on a pillow, arm trough or lap tray in sitting; correct positioning in bed; handle the arm with the scapula moved forward; **never pull on the arm** during transfers.
- **Treat:** range of motion within the pain-free range with the scapula mobilised first; active and task practice as movement returns; electrical stimulation of supraspinatus and deltoid for subluxation; botulinum toxin for a spastic, painful shoulder (reasonable, AHA/ASA 2026); manage shoulder-hand syndrome early.
- **Do not use overhead pulley exercises**: the 2026 AHA/ASA guideline classes them as harmful because they increase shoulder pain.

Classic answer vs current guideline: overhead pulleys

Classic	Current (AHA/ASA 2026)	Write in the exam
Overhead pulleys and self-assisted overhead exercises to keep shoulder range.	Recommended against (Class 3: Harm) for patients with subluxation, weakness or shoulder pain.	"Gentle pain-free range with the scapula mobilised; avoid overhead pulleys, which cause shoulder pain."

The spastic hand (7 marks in W16 and S24; 3 techniques: 3 marks in S17)

- **Stretch and position:** slow, sustained stretch of wrist and finger flexors; weight-bearing through the open hand in sitting.
- **Reduce tone before practice:** slow rhythmic rotation of the trunk and shoulder, prolonged icing, reflex-inhibiting positions (see managing tone (p. 216)).
- **Active practice:** grasp and release of objects of different sizes; electrical stimulation of wrist and finger extensors; mirror therapy; CIMT if eligible.
- **Medical:** botulinum toxin to finger and wrist flexors followed by therapy.
- **Splints:** resting splints and taping do **not** prevent wrist and finger spasticity (AHA/ASA 2026, no benefit). Use a splint only for a specific reason, such as a developing contracture or hygiene, and check the skin.

The flaccid upper limb (7 marks in S22)

- Protect the shoulder (support, positioning, careful handling).
- Stimulate activity: tapping, quick stretch, brushing and vibration over the muscles to be facilitated (Rood); weight-bearing through the arm; placing and holding the arm in different positions.
- Electrical stimulation of shoulder and wrist muscles; mirror therapy; bilateral arm training.
- Active-assisted, then active task practice as the first movements return.

Motor Relearning Programme for the upper limb (7 marks in W18)

Carr and Shepherd's programme trains real tasks in four steps: (1) **analyse** the task and find the missing components; (2) **practise the missing components** (for example forward reach with elbow extension, wrist extension while grasping, opening the hand to release); (3) **practise the whole task** (reach, grasp, move and release a glass); (4)

transfer the training into daily life with practice outside therapy. Avoid unnecessary muscle activity and do not let the strong hand take over. More detail on the neurotherapeutic approaches (p. 209) page.

By impairment

Impairment	Technique and dose	Why
Weakness	Progressive resistance, 1–3 sets of 10–15, 2–3 days a week; functional strengthening (sit to stand, step-ups)	Strength links closely with function after stroke
Poor selective arm control	Task practice of reach, grasp and release, many repetitions a day; CIMT when eligible	Use-dependent plasticity; overcomes learned non-use
Spasticity	Positioning, sustained stretch, active use; botulinum toxin plus therapy	Prevents contracture and frees movement
Poor balance	Reaching, weight shift, stepping and perturbation training; dual-task practice	Reduces falls and improves walking
Slow, asymmetric gait	Repetitive overground and treadmill walking; AFO for foot drop	Walking practice is what improves walking
Neglect	Visual scanning to the neglected side; cues; approach from that side	Draws attention to the ignored space
Low fitness	Aerobic training 40–70% heart rate reserve, 20–60 minutes, 3–5 days a week	Improves endurance and lowers vascular risk

Recall check. Write the gait problems in stance and swing, the three main shoulder problems, and the CIMT entry criteria.

13 Precautions, contraindications and stop criteria

Take care

- Check the unit's blood-pressure limits, especially after thrombolysis or a haemorrhage; avoid exercise when resting BP is above about 200/110 mmHg.
- No oral food or drink until a swallow screen is passed; position upright.
- Protect the shoulder; never pull the affected arm; no overhead pulleys.
- Craniectomy: no pressure on the side without bone; helmet when mobile.
- Neglect and sensory loss: check the skin and position of the affected limbs; approach from the neglected side in training, not in dangerous tasks.
- Anticoagulants: gentle handling; watch for bruising.
- Diabetes: check sugar before exercise if on insulin; inspect the feet.

Stop and call for help if

- New or worsening weakness, speech or vision change, severe sudden headache, falling consciousness (a new stroke or bleeding)
- Chest pain, palpitations, breathlessness or dizziness
- A seizure
- Signs of DVT or pulmonary embolism

14 Education, lifestyle and home programme

Secondary prevention

- Take blood-pressure, sugar, cholesterol and antiplatelet or anticoagulant medicines every day; check BP regularly.
- Stop smoking and all tobacco (bidi, gutka, khaini); limit alcohol; less salt and fried food; more vegetables and fruit.
- Regular exercise as prescribed; reduce sitting time.
- Know the warning signs (BE-FAST) and call for help at once if they come back.

Caregiver training and home

- Safe transfers and positioning; how to support the shoulder; how to help without taking over.
- Home changes: a commode or raised toilet seat if there is an Indian toilet; grab bars; a firm chair with arms; remove loose rugs; a bed at a height that allows the feet to rest flat on the floor.
- A written home exercise programme with pictures: arm task practice, standing balance, walking time each day.
- Return to work and driving only after assessment; look for adapted or alternative work early.

15 Prognosis and discharge criteria

- **Most recovery happens in the first 3 months**, and it continues more slowly for months after; improvement in function is possible even in the chronic stage with training.
- **Early predictors of arm recovery:** in the EPOS study (2010), patients with some voluntary finger extension and shoulder abduction within 72 hours had a 98% chance of regaining some dexterity at 6 months, against 25% for those without.
- **Poorer outcome:** older age, severe stroke, loss of consciousness at onset, severe neglect, poor sitting balance, persistent incontinence, little social support.
- **Discharge:** safe transfers and mobility at the expected level; caregiver trained; home adapted; home programme understood; follow-up and community rehabilitation arranged.

16 Exam corner

Past questions (MUHS, 2016–2026)

S23 Q4(b), 5 + 7 + 3 marks

The worked example in section 10.

W25 Q3(a), 3 + 5 + 7 marks

45-year-old right-dominant civil engineer, right MCA fronto-parietal infarct 3 months ago; cannot stand; arm activities limited by spasticity; works from a home office on the fifth floor; social smoker and drinker. Structural and functional impairments; short- and long-term goals; techniques to improve upper limb control.

W24 Q4(b), 7 + 8 marks

45-year-old CEO with right-sided weakness for one month, difficulty walking and with daily activities. ICF and physiotherapy.

S24 Q4(a), 15 marks

55-year-old with a right MCA infarct 3 months ago and difficulty using the arm. ICF with reasoning; goals and management.

W21 Q4(a), 5 + 10 marks

53-year-old teacher, haemorrhagic stroke 2 days ago with left weakness, conscious, cannot stand or walk. ICF; short- and long-term physiotherapy.

S21 Q3(a), 5 + 10 marks

40-year-old diabetic, hypertensive woman with a left ACA infarct 2 months ago, grade 2 spasticity on the right, second floor without a lift. ICF; short- and long-term interventions.

S20 Q4(a), 5 + 10 marks

65-year-old housewife with heaviness of the left side, untreated hypertension, tone higher in the leg than the arm. ICF and management.

S19 Q3(a), 5 + 10 marks

33-year-old after craniotomy for a right thrombotic stroke 6 months ago, past DVT, left hand difficulty in daily activities, circumduction gait. ICF; goals and management.

S18 Q3(a), 2 + 5 + 8 marks

54-year-old bank manager, right MCA infarct (internal capsule) 2 weeks ago; spasticity, voluntary control 0 in the arm and 2 in the leg, facial palsy, stands with support; smoker, hypertensive. Lifestyle modification; goals; management of impairments.

W17 Q3(b), 7 + 8 marks

45-year-old watchman, left MCA infarct 6 months ago: paraesthesia, facial involvement, arm voluntary control grade 3, poor dynamic standing balance, circumduction gait. Impairments with reasoning; short-term goals and management.

7-mark questions

Hemiplegic shoulder (S16); spastic hand (W16, S24); acute stage of CVA (S17); spastic hemiplegic gait (S18, S22); motor relearning for the arm (W18); hemiplegic gait (S19, S21); circle of Willis (S20); flaccid upper limb (S22); hemiplegic gait through the phases of recovery (S25); shoulder in stroke (S26).

3-mark questions

Flexion synergy of the spastic arm (S18); pusher syndrome (S20, W24); anterior cerebral artery syndrome (S26); types of stroke (S26).

Related

Cerebellar stroke with Berg 38 (W22) is on the ataxia (p. 155) page; aphasia (S23, S25) is on the cognition and perception (p. 38) page.

Pages for this topic: neurotherapeutic approaches (p. 209), managing tone (p. 216), transfers and gait training (p. 232), orthoses (p. 249), scales (p. 21).

Keywords to underline

contralateral hemiparesis arm more than leg (MCA) leg more than arm (ACA) UMN syndrome synergy pattern Brunnstrom stage Modified Ashworth task-specific practice use-dependent plasticity learned non-use circumduction no overhead pulleys Berg Balance Scale secondary prevention

Answer skeleton: any stroke long case

1. Restate the case: age, job, side and artery, type of stroke, time since onset (phase), risk factors, present function.
2. **[ICF, 5]** ICF diagram; table with clinical reasoning (artery → deficits); one-line functional diagnosis.
3. **[Goals, 3–5]** Short-term (2–4 weeks) and long-term (3–6 months), SMART, linked to his job and home.
4. **[Management, 7–8]** Principles (task-specific, intensive); by phase or by problem: positioning and shoulder care, trunk and balance, sit to stand and transfers, gait training with AFO, arm training (MRP, CIMT, mirror), spasticity, fitness dose; education, secondary prevention and caregiver training; precautions.

Draw these

- The circle of Willis (section 2)
- The ICF model, filled from the case (section 10)
- A stick figure of the spastic posture: arm flexed and adducted, leg extended with the foot pointed

Viva questions

1. Why is the arm weaker than the leg after an MCA stroke?
2. What is the penumbra?
3. Describe Brunnstrom stage 3 and stage 4.
4. Why should you never pull a hemiplegic patient by the arm?
5. Who is eligible for constraint-induced movement therapy?
6. What is pusher syndrome, and how do you treat it?

Linked practical spots

- Modified Ashworth, Brunnstrom, Fugl-Meyer and Berg as scale spots; an ankle-foot orthosis as an orthosis spot.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Clinical features of anterior cerebral artery syndrome.**

Answer
Opposite-side weakness and sensory loss of the leg more than the arm; urinary incontinence; apathy and slowed thinking; grasp reflex; difficulty starting to walk.

3 m **Components of the flexion synergy of the spastic upper limb.**

Answer
Scapular retraction and elevation; shoulder abduction and external rotation; elbow flexion (strongest); forearm supination; wrist and finger flexion.

7 m **Describe hemiplegic gait and its physiotherapy management.**

Answer
Stance: short weight-bearing on the weak side, knee hyperextension or flexion, reduced hip extension and push-off. Swing: stiff knee, foot drop and inversion, circumduction or hip hiking. Overall slow, short, asymmetric steps. Causes: weakness, spasticity (extensor synergy), poor control, contracture, poor balance. Management: strengthen hip extensors, quadriceps, dorsiflexors; stretch the calf; part practice (weight shift, step-ups, knee control); whole-task repetitive walking overground and on a treadmill; AFO or peroneal stimulation; community walking.

MCQ **A patient 6 weeks after stroke has painful shoulder subluxation. Which is contraindicated?**

A. Arm trough in sitting

B. Electrical stimulation of supraspinatus and deltoid

C. Overhead pulley exercises

D. Pain-free range with scapular mobilisation

Answer
C. Overhead pulleys increase shoulder pain; the 2026 AHA/ASA guideline recommends against them.

MCQ **Which finding at 72 hours best predicts that the arm will regain some dexterity?**

A. Normal reflexes

B. Some voluntary finger extension and shoulder abduction

C. Low tone

D. Intact pain sensation

Answer
B. In EPOS, patients with both had a 98% chance of some dexterity at 6 months.

MCQ **A patient leans and pushes towards the weak side and resists correction. The best first strategy is:**

A. Pull him upright repeatedly

B. Use a mirror or door frame to show vertical and reach towards the strong side

C. Strap him to the chair

D. Wait until it resolves

Answer
B. Visual feedback of vertical and reaching to the strong side retrain the sense of upright; forcing correction increases pushing.

LAQ **A 45-year-old right-handed civil engineer had a right MCA fronto-parietal infarct 3 months ago. He cannot stand and has difficulty in all upper limb activities because of spasticity. He works from a home office on the 5th floor, and smokes and drinks socially. Write the structural and functional impairments, short- and long-term goals, and techniques to improve upper limb control. (3 + 5 + 7, W25)**

Answer
Impairments (3): structural: right fronto-parietal infarct. Functional: left hemiplegia, arm more than leg; upper limb spasticity; poor sitting and standing balance; likely left-sided neglect and sensory loss after a right parietal lesion; risk of shoulder pain and subluxation.

Goals (5): short-term (4–6 weeks): steady sitting while reaching; sit-to-stand and standing with support; full pain-free shoulder range; left hand used to hold paper while writing. Long-term (3–6 months): walk indoors with an AFO and a cane; use the left hand as a helper at the computer; resume home-office work; stop smoking.

Upper limb control (7): support the arm on a trough or table and position it daily; prolonged stretch, not overhead pulleys; weight-bearing through the open hand; many repetitions of task-specific reaching, grasping and release at the desk; mirror therapy; modified constraint-induced therapy once he has 20° of wrist and 10° of finger extension; electrical stimulation of wrist and finger extensors; botulinum toxin with stretching for focal spasticity, if the doctor advises it.

Last recall check. Draw the circle of Willis and the ICF model, list the gait problems by phase, and write the aerobic exercise dose.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Prabhakaran S, et al. 2026 Guideline for the early management of patients with acute ischemic stroke (AHA/ASA). *Stroke* 2026. doi:10.1161/STR.0000000000000513 (time windows checked through published summaries).

3. Richards LG, Ifejika NL, Stein J, et al. 2026 Guideline for adult stroke rehabilitation and recovery (AHA/ASA). *Stroke* 2026. doi:10.1161/STR.0000000000000536 (recommendations on pulleys, botulinum toxin, splints, balance and mirror therapy checked through published summaries).
4. National Institute for Health and Care Excellence. Stroke rehabilitation in adults (NG236), 2023.
5. Billinger SA, et al. Physical activity and exercise recommendations for stroke survivors (AHA/ASA scientific statement). *Stroke* 2014;45:2532–53.
6. AVERT Trial Collaboration. Efficacy and safety of very early mobilisation within 24 h of stroke onset. *Lancet* 2015;386:46–55.
7. Wolf SL, et al. Effect of constraint-induced movement therapy on upper extremity function 3 to 9 months after stroke: the EXCITE randomized clinical trial. *JAMA* 2006;296:2095–104.
8. Nijland RH, et al. Presence of finger extension and shoulder abduction within 72 hours after stroke predicts functional recovery: early prediction of functional outcome after stroke (EPOS). *Stroke* 2010;41:745–50.
9. CLOTS Trials Collaboration. Effectiveness of intermittent pneumatic compression in reduction of risk of deep vein thrombosis in patients who have had a stroke (CLOTS 3). *Lancet* 2013;382:516–24.
10. O'Donnell MJ, et al. Global and regional effects of potentially modifiable risk factors associated with acute stroke in 32 countries (INTERSTROKE). *Lancet* 2016;388:761–75.
11. Sacco RL, et al. An updated definition of stroke for the 21st century (AHA/ASA). *Stroke* 2013;44:2064–89.
12. Bernhardt J, et al. Agreed definitions and a shared vision for new standards in stroke recovery research: the Stroke Recovery and Rehabilitation Roundtable. *Int J Stroke* 2017;12:444–50.
13. ICMR National Stroke Registry Programme, hospital-based registry 2020–2022 (as reported in *Int J Stroke* and national press); Hyderabad stroke registry for the share of haemorrhages.
14. Structure and classical content: O'Sullivan, Schmitz and Fulk, *Physical Rehabilitation*; Carr and Shepherd, *Neurological Rehabilitation*; Brunnstrom, *Movement Therapy in Hemiplegia*; Cash's Textbook of Neurology for Physiotherapists.

5A(b) Traumatic brain injury and space-occupying lesions: GCS, Rancho levels, raised ICP and physiotherapy

> Brain injury was a long case in 5 papers, and the Glasgow Coma Scale was a 3-mark question four times. The cases run from a man unresponsive at GCS 5 to an agitated patient at Rancho level IV and a teacher with a frontal subdural haematoma at GCS 12. Every answer starts by placing the patient on these scales, because the level of consciousness and cognition decides what physiotherapy can do and how to keep the brain safe while doing it.

Syllabus	Ch 5A(b) Acquired brain injury: trauma and pathology (S.O.L., space-occupying lesions)
Priority	**Tier 1.** In 11 of 20 papers (2016–2026): 5 long cases, 2 seven-mark and 7 three-mark questions.
Asked as	LAQ S17 · S23 · S24 · W24 · S26 · 7m S18 · S20 · 3m W16 · S19 · S20 · W22 · S24 · S25 · S26
Usual split	15 (ICF with clinical reasoning 5, goals 5–7, management 3–5; S26 asked goals and treatment techniques)
Checked against	Glasgow Coma Scale (Teasdale and Jennett, 1974; structured approach, 2014); Brain Trauma Foundation guidelines for severe TBI (4th edition, 2016); Rancho Los Amigos Levels of Cognitive Functioning. Full list under Sources (p. 75). Content checked 20 October 2026.

1 Definition

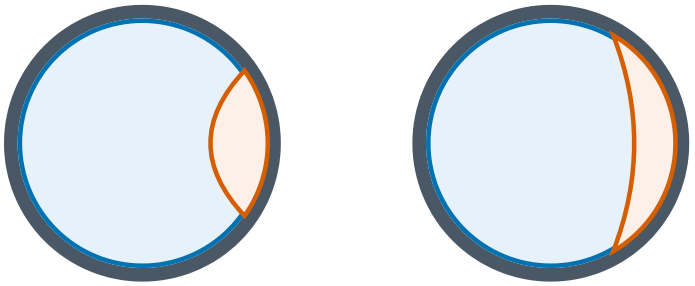
- **Acquired brain injury:** damage to the brain after birth that is not hereditary, congenital or degenerative: trauma, stroke, tumours, infection, lack of oxygen.
- **Traumatic brain injury (TBI):** an alteration in brain function, or other evidence of brain pathology, caused by an external force (a blow, acceleration–deceleration, a penetrating object or a blast).
- **Space-occupying lesion (SOL):** a mass inside the skull (tumour, abscess, haematoma, cyst) that compresses the brain and raises intracranial pressure.
- In India, road traffic accidents (often two-wheeler riders without helmets) and falls are the main causes of TBI, mostly in young men.

2 Classification and grading

Classification of head injury (3 marks in S24)

Basis	Types
Severity (GCS)	Mild 13–15; moderate 9–12; severe 3–8
Mechanism	Closed (blunt) or open (penetrating); acceleration–deceleration; blast
Skull	Linear, depressed or basal skull fracture (CSF leak from nose or ear, "raccoon eyes", bruising behind the ear)
Focal brain lesions	Extradural haematoma, subdural haematoma, intracerebral haemorrhage, contusion
Diffuse brain lesions	Concussion; diffuse axonal injury; diffuse swelling
Timing	**Primary injury** at impact; **secondary injury** afterwards from low oxygen, low blood pressure, raised intracranial pressure, seizures and infection

Extradural vs subdural haematoma on a CT slice



Extradural
lens-shaped, lucid interval

Subdural
thin crescent

Extradural: an arterial bleed (usually the middle meningeal artery after a temporal fracture) that strips the dura and stops at the sutures, so it is lens-shaped; the person may talk normally before collapsing (lucid interval). Subdural: torn bridging veins; the blood spreads along the brain surface as a crescent; acute after major trauma, chronic in older people after a minor fall.

Glasgow Coma Scale (3 marks in W16, S19, S20 and S26)

Response	Scores
Eye opening (E, 1–4)	4 spontaneous; 3 to sound (speech); 2 to pressure (pain); 1 none
Verbal response (V, 1–5)	5 oriented; 4 confused; 3 words (inappropriate); 2 sounds (incomprehensible); 1 none
Motor response (M, 1–6)	6 obeys commands; 5 localises pain; 4 normal flexion (withdraws); 3 abnormal flexion (decorticate); 2 extension (decerebrate); 1 none

- **Total 3–15.** Severe 3–8 (8 or less is coma, and the airway usually needs protection); moderate 9–12; mild 13–15.
- **Interpreting GCS (S26):** always report the parts, for example "E2 V2 M4 = 8"; the motor score predicts outcome best; record "T" for verbal when the person has a tube; use the best response, and the better side for motor.
- **Used in** (W16): head injury, stroke, meningitis and encephalitis, drug overdose, any patient with reduced consciousness.

Rancho Los Amigos Levels of Cognitive Functioning

Level	Behaviour	Physiotherapy focus
I No response	No response to any stimulus	Positioning, chest, range, splints, family
II Generalised response	Same non-specific response to any stimulus	As above; structured sensory input
III Localised response	Responds specifically (turns to a voice, pulls a tube); follows simple commands inconsistently	Simple one-step commands; sitting, tilt table
IV Confused, agitated	Heightened activity, aggression, poor attention, no short-term memory	Calm, safe, familiar activity; no demands for new learning
V Confused, inappropriate	Not agitated; follows simple commands; easily distracted	Structured, repeated practice of simple tasks
VI Confused, appropriate	Goal-directed with cues; memory still poor	Routine, cues, memory aids; functional training
VII Automatic, appropriate	Routine tasks automatic; poor judgement and insight	Community skills, dual tasks, safety awareness
VIII Purposeful, appropriate	Independent in familiar settings; subtle problems remain	Fitness, return to work, sport and driving assessment

The revised scale adds levels IX and X (purposeful, appropriate, needing stand-by help on request, then modified independence).

3 Causes and risk factors

- **TBI:** road traffic accidents, falls (children and older people), assault, sports, industrial accidents. Risk rises with alcohol, no helmet and no seat belt.
- **Space-occupying lesions:** tumours (gliomas, meningiomas, metastases, pituitary tumours), abscess, tuberculoma (common in India), chronic subdural haematoma, cysts.

4 Pathophysiology to impairments

1. **Primary injury:** contusions under the impact and opposite it (coup and contrecoup), haematomas, and shearing of axons in rotational injuries (**diffuse axonal injury**: coma from the moment of injury, often with a nearly normal CT; MRI shows small lesions in the corpus callosum, internal capsule and brainstem).
2. **Secondary injury:** swelling and bleeding raise intracranial pressure; because the skull cannot expand, cerebral perfusion pressure falls (mean arterial pressure minus intracranial pressure), so more brain becomes ischaemic. Low oxygen and low blood pressure make it worse.
3. If pressure keeps rising, the brain herniates: the uncus presses on the third cranial nerve (a fixed, dilated pupil), then the brainstem fails.
4. **Impairments** depend on where the damage is: reduced consciousness; motor problems (hemiplegia, spasticity, ataxia, rigidity); frontal lobe problems (poor attention, planning, judgement, disinhibition, apathy); memory loss; speech, swallowing and visual problems; seizures.
5. **Secondary complications:** contractures, heterotopic ossification, pressure sores, chest infection, deconditioning.

Raised intracranial pressure (3 marks in W22 and S25)

- **Symptoms:** headache (worse on waking, coughing and bending), vomiting (may be projectile), blurred vision.
- **Signs:** falling level of consciousness; papilloedema; a dilating pupil on the side of the mass; sixth nerve palsy; seizures; in infants a bulging fontanelle.
- **Cushing's triad** (late, dangerous): rising blood pressure with a wide pulse pressure, slowing heart rate and irregular breathing.
- **Numbers:** normal ICP is about 5–15 mmHg in adults; guidelines treat ICP above **22 mmHg** and aim for a cerebral perfusion pressure of **60–70 mmHg** (Brain Trauma Foundation, 2016).

Recall check. Why does a rising ICP reduce blood flow to the brain, and which physiotherapy actions can raise ICP?

5 Clinical features

- **Consciousness:** coma, then vegetative state (eyes open, no awareness) or minimally conscious state; post-traumatic amnesia while confused.
- **Motor:** hemiparesis, often on both sides; spasticity; abnormal postures (decorticate, decerebrate); ataxia; balance loss.
- **Cognitive and behavioural:** attention, memory, executive function, slowed processing, agitation, impulsivity, poor insight.
- **Frontal lobe injury** (S23, S24 cases): poor planning and judgement, disinhibition or apathy, poor safety awareness; if the motor cortex is involved, weakness on the opposite side.
- **Others:** visual field or acuity loss (S24 case), double vision, dizziness and benign positional vertigo, speech and swallowing problems, seizures, fatigue, headache.

- **Space-occupying lesions:** slowly progressive headache, seizures, focal weakness, personality change, signs of raised ICP.

6 Investigations

- **CT brain:** the first scan: fractures, haematomas, contusions, swelling, midline shift.
- **MRI:** diffuse axonal injury, tumours and smaller lesions.
- **ICP monitoring** in severe TBI; blood gases; electrolytes; EEG for seizures.
- **Bedside:** GCS and pupils at regular intervals; post-traumatic amnesia testing.

7 Medical and surgical management

- **Prevent secondary injury:** airway, oxygen and blood pressure; treat raised ICP (head-up position, sedation, osmotic therapy, drainage of CSF); prevent seizures and fever.
- **Surgery:** evacuation of extradural or acute subdural haematoma; elevation of a depressed fracture; decompressive craniectomy for uncontrolled ICP (a skull flap is left out, so the soft area must be protected); burr holes for chronic subdural haematoma; tumour removal.
- **Later:** tracheostomy and feeding tube if needed; treatment of spasticity (botulinum toxin, baclofen) and agitation; tumour therapy (radiotherapy, chemotherapy, steroids for swelling).

8 Complications and prevention

Complication	What to look for	Physio role
Chest infection, aspiration	Secretions, fever, falling oxygen; poor swallow	Positioning, airway clearance within ICP limits, early sitting
Contractures	Ankle plantarflexion, elbow and wrist flexion, hip and knee flexion with spasticity	Positioning, daily range, splints, serial casting
Heterotopic ossification	Warm, swollen joint with losing range (hips, elbows, shoulders)	Gentle range within comfort; report early
Pressure sores	Sacrum, heels, back of the head	Turning schedule, mattress, positioning
Seizures	Twitching, staring, collapse	Know the seizure plan; protect the head; record
Falls and injury in agitation	Restlessness, pulling at tubes, climbing out of bed	Low bed, calm setting, supervised mobility

9 PT assessment

- **Before treatment:** GCS, pupils, ICP and CPP if monitored, blood pressure, oxygen, sedation, ventilator, skull precautions (craniectomy), fractures elsewhere.
- **Level of cognition:** Rancho level; post-traumatic amnesia; Coma Recovery Scale-Revised in disorders of consciousness; Agitated Behaviour Scale.
- **Respiratory:** chest, secretions, cough, swallow.
- **Motor:** tone (Modified Ashworth), posture, range, voluntary movement, reflexes; balance and gait when able (Berg, High-level Mobility Assessment Tool later).
- **Others:** vision, hearing, sensation, communication, behaviour; family, job and home.

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case. Write the GCS or Rancho level in the health condition box.

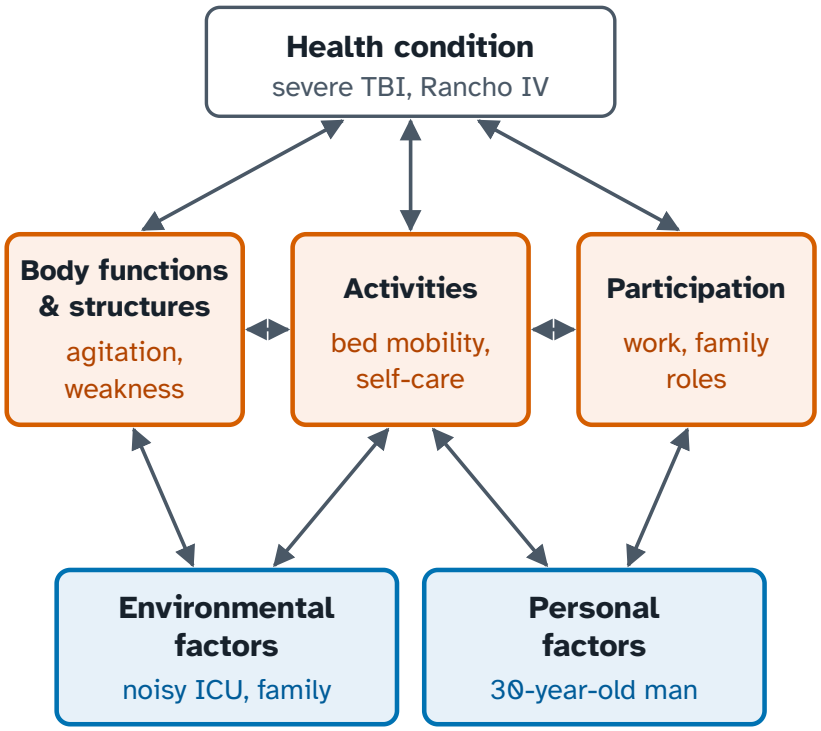
Worked example from a real paper (W24, 5 + 7 + 3 marks)

> "Discuss ICF with clinical reasoning for a 30 year old male who sustained multiple skull fractures after a road traffic accident 2 weeks back. Patient is currently in the ICU and is conscious with Rancho scale level IV. Formulate short term and long term goals of physiotherapy management."

Spot the ICF clues before you write

multiple skull fractures after a road accident means a severe TBI with possible contusions and fracture precautions. 2 weeks, ICU means the early phase. conscious, Rancho level IV means confused and agitated: no new learning yet; safety and a calm structure come first.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Severe TBI with multiple skull fractures, 2 weeks; Rancho level IV (confused, agitated); in the ICU	Contusions and diffuse injury disturb arousal, attention and behaviour
Body function and structure impairments	• Agitation, poor attention, no short-term memory, poor safety awareness • Weakness or hemiparesis; altered tone; poor balance • Possible secretions, poor swallow, fatigue • Risk of contractures, pressure sores, heterotopic ossification • Skull fractures: protect the head	• Frontal and diffuse damage cause the behavioural stage of recovery • Motor damage and immobility in the ICU weaken and stiffen the body
Activity limitations	Dependent for bed mobility, sitting, transfers, walking, feeding and hygiene; cannot follow complex instructions	Cognition, not only strength, limits activity at this level
Participation restrictions	Off work; family roles suspended	Long recovery expected

ICF component	From this case	Clinical reasoning
Environmental factors	**Facilitators:** ICU care, family visits. **Barriers:** noise, alarms, many staff, restraints (they increase agitation)	A calm, consistent setting reduces agitation
Personal factors	30-year-old man, likely an earner	Young age favours recovery

One-line functional diagnosis

A 30-year-old man 2 weeks after a severe TBI with multiple skull fractures, at Rancho level IV (confused and agitated), with weakness, poor balance and poor safety awareness, who is dependent for mobility and self-care in the ICU and cannot work or fulfil family roles.

Practice cases from the papers: S26 40-year-old with diffuse axonal injury, GCS 5, unresponsive (goals and techniques); S23 and S24 45-year-old men with a right frontal fracture and subdural haematoma, GCS 12 (S24 with impaired vision, a truck driver); S17 20-year-old student with a left intracranial bleed, conscious and oriented, with right-sided paralysis.

Recall check. Fill the ICF for the S24 truck driver at GCS 12 with impaired vision. What decides whether he can return to driving?

11 Problem list and goals

Goals for the 30-year-old at Rancho level IV

Problem	Short-term (2–4 weeks)	Long-term (3–6 months)
Agitation and safety	Calm during short sessions; no falls or injuries	Follows routines; safe in familiar settings
Chest and skin	Clear chest; no pressure sores	No respiratory complications
Range and tone	Full range; no new contractures	Range sufficient for walking and self-care
Mobility	Sits out of bed; stands with help	Walks independently indoors and outdoors
Daily life and work	Takes part in simple familiar tasks (washing the face)	Independent self-care; planned return to work

12 PT management

Principles: protect the injured brain (ICP and CPP limits); prevent secondary complications; match the session to the level of consciousness and cognition; use functional, familiar tasks; involve the family.

Acute stage in the ICU (7 marks in S18 and S20)

- **Positioning:** head of the bed raised about 30° and the head in midline as the unit advises; avoid neck flexion and rotation, which compress the jugular veins; side lying and supported positions that reduce abnormal postures; turn regularly.
- **Chest care:** positioning, gentle manual techniques and short, well-oxygenated suctioning as allowed; no head-down postural drainage while ICP is raised or unstable.
- **Range and splints:** daily passive range of all joints; splints or serial casts for ankles and wrists when tone rises.
- **Early mobilisation:** sitting over the bed edge and tilt table once ICP is controlled and the team agrees.
- **Sensory input:** structured, meaningful input (family voices, familiar music, touch) in short periods; evidence for "coma stimulation" is weak, so avoid overstimulation.
- Watch ICP, CPP, blood pressure and oxygen before, during and after each treatment.

By level of cognition

- **Unresponsive (GCS 3–8, Rancho I–II; S26 case):** positioning and turning; chest care; range, splints and casting; tilt table and sitting for arousal; structured sensory input; teach the family.

- **Confused and agitated (Rancho IV; W24 case):** reduce stimulation (quiet room, one therapist, short sessions); familiar, automatic tasks such as walking, sitting to standing and washing the face; let the person move safely (low bed, mats) instead of restraining; do not argue or test memory; stop and try later if agitation rises.
- **Confused, non-agitated (Rancho V–VI):** structured routine at the same time each day; one-step instructions; errorless learning (guide the right movement rather than letting mistakes happen); memory aids; repeated functional practice.
- **Automatic to purposeful (Rancho VII–VIII):** balance, dual tasks, community walking, fitness, vocational and driving assessment.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Raised ICP risk	Head up about 30°, head in midline; treat only within the unit's ICP and CPP limits (treatment threshold ICP **22 mmHg**; CPP target **60–70 mmHg**)	Protects brain perfusion during handling
Contracture risk with spasticity	Daily passive range; positioning; serial casting of ankles or wrists when needed	High tone and immobility shorten muscles within weeks
Low arousal	Upright sitting and tilt table; structured, meaningful stimuli in short periods	Upright positions increase alertness
Agitation	Low stimulation, consistent staff, familiar tasks, short sessions	Overload worsens confusion and aggression
Poor balance and gait	Task-specific balance and walking practice; later dual-task and community walking	Practice drives recovery of mobility
Low fitness and fatigue	Graded aerobic exercise once stable; pacing	Builds endurance for daily life and work

Recall check. How would your session differ for a patient at Rancho II, IV and VI?

13 Precautions, contraindications and stop criteria

Take care

- Stop if ICP rises and stays above the set limit, CPP falls, or the GCS drops; report a new dilated pupil at once.
- No head-down positions or tight neck collars that block venous drainage while ICP is raised; keep suctioning short with pre-oxygenation.
- After craniectomy, protect the skull defect (helmet when mobilising); no pressure on the soft area.
- Clear other fractures and spinal injuries before moving the patient.
- Seizure plan known; agitation: keep yourself and the patient safe, keep exits clear.
- Brain tumours: watch fatigue, seizures and steroid effects (weak muscles, high sugar); set goals with the patient's prognosis in mind.

14 Education, lifestyle and home programme

- Explain the stages of recovery to the family, especially that agitation is a stage and usually passes.
- Teach positioning, range, safe transfers and a calm routine at home.
- Memory aids, a daily timetable, rest breaks; avoid alcohol.
- Helmet and seat-belt use; driving only after medical clearance (vision, cognition, seizures).
- Graded return to work or study; support groups and counselling for the family.

15 Prognosis and discharge criteria

- Worse outcome with older age, a low GCS motor score, non-reacting pupils, low oxygen or blood pressure after injury, and large lesions or swelling on CT.
- A longer period of post-traumatic amnesia means a more severe injury and a longer recovery.
- Recovery is fastest in the first months but continues for years; cognitive and behavioural problems often limit return to work more than physical ones.
- Discharge from inpatient care when safe at home with supervision as needed, with outpatient or community rehabilitation in place.

16 Exam corner

Past questions (MUHS, 2016–2026)

S26 Q3(b)

40-year-old man, road accident a week ago, in a multispeciality hospital; MRI shows diffuse axonal injury in the left internal capsule; no voluntary movement of the limbs; GCS 5; completely unresponsive. Short- and long-term goals; treatment techniques in detail.

W24 Q3(b), 5 + 7 + 3 marks

The worked example in section 10.

S24 Q3(a)

45-year-old truck driver with a depressed right frontal fracture and subdural haematoma 2 months ago; GCS 12; affected vision. ICF; short- and long-term goals.

S23 Q4(a), 5 + 5 + 5 marks

45-year-old teacher with a right frontal fracture and subdural haematoma 15 days ago; GCS 12; in hospital. ICF; short- and long-term goals.

S17 Q4(a)

20-year-old college student, road accident a week ago, left intracranial bleed; stable, conscious and oriented; on the ward with right-sided paralysis. Assessment with ICF; short- and long-term management.

7-mark questions

Physiotherapy in the acute stage of TBI (S18, S20).

3-mark questions

Components of GCS and two conditions where it is used (W16); GCS (S19, S20); interpreting GCS (S26); features of raised ICP (W22); signs of raised ICP (S25); classification of head injury (S24).

Pages for this topic: stroke (p. 50), cognition and perception (p. 38), vestibular disorders (p. 116), scales (p. 21).

Keywords to underline

primary and secondary injury GCS Rancho level raised ICP CPP head up 30°, midline diffuse axonal injury agitation low stimulation serial casting family education

Answer skeleton: any brain injury long case

1. Restate: age, job, mechanism, lesion, days since injury, GCS and Rancho level, ICU or ward.
2. **[ICF, 5]** ICF diagram and table with clinical reasoning; one-line functional diagnosis.

3. **[Goals, 5–7]** Short-term goals for the present level (chest, skin, range, arousal or behaviour, sitting); long-term goals for walking, self-care, cognition and return to work.

4. **[Management, 3–8]** ICU care within ICP limits; management by level of cognition; motor and balance retraining; family education; precautions.

Draw these

- Extradural vs subdural haematoma shapes (section 2)
- The GCS table with the three parts
- The ICF model filled from the case

Viva questions

1. What are the three parts of the GCS, and what does a score of 8 mean?
2. What is Cushing's triad?
3. Why is head-down postural drainage avoided after TBI?
4. What is the lucid interval?
5. How do you treat a patient at Rancho level IV?
6. What is diffuse axonal injury?

Linked practical spots

- GCS chart; Rancho levels; serial cast; tilt table.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the components of the Glasgow Coma Scale.**

Answer
Eye opening 1–4 (spontaneous, to sound, to pressure, none); verbal 1–5 (oriented, confused, words, sounds, none); motor 1–6 (obeys, localises, normal flexion, abnormal flexion, extension, none). Total 3–15; severe 3–8, moderate 9–12, mild 13–15.

3 m **Enumerate the signs of raised intracranial pressure.**

Answer
Headache worse on waking and coughing; vomiting; papilloedema; falling consciousness; dilating pupil; sixth nerve palsy; Cushing's triad (rising blood pressure, slow pulse, irregular breathing); seizures; bulging fontanelle in infants.

7 m **Physiotherapy in the acute stage of traumatic brain injury.**

Answer
Assess GCS, pupils, ICP and CPP, vital signs, chest and tone. Position with the head of the bed up about 30° and the head in midline; avoid neck flexion and head-down positions. Chest care within ICP limits: positioning, gentle manual techniques, short suctioning with pre-oxygenation. Turning and pressure care. Daily passive range, splints and serial casting for rising tone. Early sitting and tilt table once ICP is controlled. Structured sensory input. Family education. Stop if ICP rises above the limit or GCS falls.

LAQ **A 40-year-old man a week after a road accident has diffuse axonal injury, GCS 5, no voluntary movement and no response to stimuli. Plan goals and treatment techniques. (S26)**

Answer

Goals: short-term (2–4 weeks): clear chest; no pressure sores or contractures; tolerates sitting and tilt table; shows more responses. Long-term (months): sits with support; follows commands; family able to manage positioning and range; progress to active rehabilitation as consciousness returns.

Techniques: positioning with head up 30° and midline; turning; chest care within ICP limits; passive range of all joints daily; resting splints and serial casts; tilt table and sitting for arousal; structured sensory input (familiar voices, touch, music) in short periods; monitoring of ICP, CPP and vital signs; family teaching.

MCQ **A patient opens the eyes to pressure, makes incomprehensible sounds and withdraws from pain. The GCS is:**

A. 6

B. 8

C. 10

D. 12

Answer

B. E2 + V2 + M4 = 8, a severe injury.

MCQ **A patient at Rancho level IV is best managed by:**

A. Teaching new complex tasks

B. A quiet setting, short sessions and familiar tasks

C. Restraining him for every session

D. Testing his memory repeatedly

Answer

B. Agitation is a stage of recovery; low stimulation and familiar activity calm it.

MCQ **The Brain Trauma Foundation (2016) recommends treating ICP above:**

A. 10 mmHg

B. 15 mmHg

C. 22 mmHg

D. 40 mmHg

Answer

C. With a cerebral perfusion pressure target of 60–70 mmHg.

Last recall check. Write the GCS, the eight Rancho levels with one physio action each, and five signs of raised ICP.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Teasdale G, Jennett B. Assessment of coma and impaired consciousness: a practical scale. *Lancet* 1974;2:81–4. Teasdale G, et al. The Glasgow Coma Scale at 40 years: standing the test of time. *Lancet Neurol* 2014;13:844–54.

3. Carney N, et al. Guidelines for the management of severe traumatic brain injury, fourth edition. Brain Trauma Foundation; 2016 (*Neurosurgery* 2017;80:6–15).
4. Hagen C, et al. Rancho Los Amigos Levels of Cognitive Functioning (original 1972; revised scale).
5. Structure and classical content: O'Sullivan and Schmitz, *Physical Rehabilitation*; Stokes, *Physical Management for Neurological Conditions*; Pryor and Prasad, *Physiotherapy for Respiratory and Cardiac Problems* (ICU precautions).

5A(c) Spinal cord injury: level and grade, complications, ICF and rehabilitation

> Spinal cord injury was a long case in 9 of the 20 papers (2016–2026), and autonomic dysreflexia alone was asked 8 times. Most cases are adults with a lesion near T10 (a fracture, tuberculosis or myelitis), asked for ICF, goals and the expected level of walking. Every answer starts from the level and grade of the injury, because they decide what the person can learn to do, what equipment they need and which complications to prevent.

Syllabus	Ch 5A(c) Spinal cord disorders, traumatic and non-traumatic, management including bladder training (bladder and bowel are on their own page)
Priority	**Tier 1.** In 18 of 20 papers (2016–2026): 9 long cases, 3 seven-mark and 15 three-mark questions. The 9 bladder questions are on the bladder and bowel page.
Asked as	LAQ S16 · S17 · W18 · S19 · S21 · S22 · W22 · S25 · S26 · 7m S18 · W18 · W24 · 3m S16 · S17 ×2 · W17 · S19 · W19 · W20 · S21 · W21 · S22 · W22 · S23 · S24 · W24 · W25
Usual split	15 (ICF with clinical reasoning 5, goals 3–5, management 7–10; or impairments 7 and ambulation 8); S16 asked level 2 and grade 1 first
Checked against	ISNCSCI (revised 2019; a 9th edition was released in July 2026); Consortium for Spinal Cord Medicine guidelines on autonomic dysreflexia (2020), neurogenic bowel (2020) and expected outcomes (2nd edition, 2026); international exercise guidelines for adults with SCI (2018); Ditunno four-phase model of spinal shock (2004); Indian SCI data (Jaipur, 2015). Full list under Sources (p. 93). Content checked 19 October 2026.

1 Definition

- **Spinal cord injury (SCI)** is damage to the spinal cord or the cauda equina that changes its motor, sensory or autonomic function, for a time or permanently. It may be **traumatic** (fracture or dislocation) or **non-traumatic** (tuberculosis, tumour, inflammation, disc, blood-supply loss).
- **Tetraplegia:** loss of function in the cervical segments, so the arms, trunk, legs and pelvic organs are affected.
- **Paraplegia:** loss of function in the thoracic, lumbar or sacral segments (or the cauda equina). The arms are spared; the trunk, legs and pelvic organs are affected, depending on the level.
- **Neurological level of injury (NLI):** the lowest segment where sensation and antigravity power (grade 3 or more) are both normal, with normal function above it.
- **Complete injury:** no sensory or motor function in the lowest sacral segments (S4–S5). **Incomplete injury:** some function is preserved there ("sacral sparing").

2 Classification and grading

The ISNCSCI examination (ASIA)

- **Motor:** 10 key muscles on each side graded 0–5. C5 elbow flexors, C6 wrist extensors, C7 elbow extensors, C8 finger flexors (middle finger), T1 little-finger abductors; L2 hip flexors, L3 knee extensors, L4 ankle dorsiflexors, L5 long toe extensors, S1 ankle plantar flexors.
- **Sensory:** 28 key points on each side (C2 to S4–S5), each tested for light touch and pinprick and scored 0 (absent), 1 (altered) or 2 (normal).
- **Anorectal examination:** voluntary anal contraction (VAC) and deep anal pressure (DAP).
- **Motor level:** the lowest key muscle with grade 3 or more, provided the muscles above are grade 5. **Sensory level:** the lowest dermatome with normal light touch and pinprick.

ASIA Impairment Scale (AIS) (3 marks in S19)

Grade	Meaning	Key test
A: complete	No sensory or motor function in S4–S5	No light touch, pinprick or deep anal pressure at S4–S5, and no voluntary anal contraction
B: sensory incomplete	Sensation, but no motor function, preserved below the level, including S4–S5	No motor function more than three levels below the motor level on either side
C: motor incomplete	Motor function preserved below the level, but less than half of the key muscles below the NLI reach grade 3	Voluntary anal contraction, or sacral sensory sparing with some motor function more than three levels below the motor level
D: motor incomplete	As C, but half or more of the key muscles below the NLI reach grade 3 or more	Count the key muscles below the NLI
E: normal	Sensation and power normal in all segments, in a person who had deficits before	Not used if the person never had an SCI

Sacral sparing (3 marks in S16) and complete vs incomplete (3 marks in W25)

Sacral sparing means any sensation at S4–S5 (light touch, pinprick or deep anal pressure) or voluntary anal contraction. It shows that some fibres cross the lesion, so the injury is **incomplete**. Sacral fibres lie on the outer edge of the long tracts, so they often survive a central lesion. No sacral sparing means a **complete** injury (AIS A).

Incomplete cord syndromes (Brown-Séquard: 3 marks in W17 and S23; cauda equina: 3 marks in W24)

Incomplete cord syndromes: which part of the cord is damaged



Each oval is a cross-section of the cord, back at the top and front at the bottom; the H shape is the grey matter. Hatching shows the damaged area.

Syndrome	Typical cause and features	Outlook
Central cord (the commonest)	Older person with a stiff, arthritic neck falls forward and hyperextends it. Arms much weaker than legs, hands worst; variable sensory loss; sacral sparing	Good for walking; the hands recover least
Anterior cord	Flexion injury or loss of the anterior spinal artery. Loss of power, pain and temperature below the level; position sense, vibration and fine touch kept (posterior columns intact)	Poor for motor recovery

Syndrome	Typical cause and features	Outlook
Brown-Séquard (hemisection)	Stab or bullet wound, tumour, disc. **Same side:** weakness with UMN signs and loss of position sense and vibration below the lesion. **Opposite side:** loss of pain and temperature from 1–2 segments below the lesion. A band of loss at the level on the same side	Best of all syndromes: most people walk again
Posterior cord (rare)	Loss of position sense, vibration and fine touch; power kept; gait is ataxic and worse with the eyes closed	Walking is possible, using vision for balance
Conus medullaris	Fracture at T12–L1 vertebrae damaging the sacral cord. Saddle anaesthesia, areflexic bladder and bowel, sexual dysfunction early and symmetrical; legs may show mixed UMN and LMN signs	Bladder and bowel recover least
Cauda equina	Fracture or disc below L1–L2 vertebrae damaging the nerve roots. LMN picture: flaccid, often asymmetrical leg weakness, lost knee and ankle reflexes, saddle anaesthesia, root pain, areflexic bladder and bowel	Better than cord lesions, because roots are peripheral nerves and can regrow

Paraplegia in extension and in flexion (3 marks in W22)

Feature	Paraplegia in extension	Paraplegia in flexion
Tracts damaged	Mainly the pyramidal (corticospinal) tracts; extrapyramidal tracts such as the vestibulospinal tract still work	Pyramidal and extrapyramidal tracts both damaged: a complete or near-complete lesion
Posture and tone	Legs extended and adducted; extensor tone; clonus	Hips and knees drawn up in flexion; flexor spasms; a touch on the foot may set off a mass reflex
Outlook	Better: an incomplete lesion	Poorer: a severe lesion

3 Causes and risk factors

Group	Causes
Traumatic	**Falls from height** (roofs, trees, buildings, trains), road traffic accidents, diving into shallow water, sports, assault, falls from standing in older people
Infection	**Tuberculosis of the spine (Pott's spine)**, epidural abscess, HIV and other viral myelitis
Inflammation	**Transverse myelitis**, multiple sclerosis, neuromyelitis optica
Compression and degeneration	Tumours (often spinal metastases), cervical spondylotic myelopathy, prolapsed disc
Vascular and others	Anterior spinal artery infarct, vascular malformations; vitamin B12 deficiency (subacute combined degeneration); syringomyelia; radiation

India: falls cause more injuries than road accidents. In 2,716 patients in Jaipur, falls from height caused 53% and road accidents 28%; most patients were young men (71% aged 20–49) and 79% came from rural areas (Mathur et al., 2015).

4 Pathophysiology to impairments

1. **Primary injury:** the force of the fracture, dislocation or compression tears and bruises the cord.
2. **Secondary injury** over hours to days: swelling, bleeding, poor blood flow and chemical damage spread the lesion. Early decompression, good blood pressure and oxygen limit it.
3. **Spinal shock:** reflexes below the lesion are lost or depressed at first, then return over days to weeks and become exaggerated.
4. **Below the lesion:** because the cord above the sacral segments is cut off from the brain, a cord lesion gives **UMN signs** (spasticity, brisk reflexes, reflex bladder) once spinal shock ends. Because a cauda equina lesion damages nerve roots, it gives **LMN signs** (flaccid, areflexic, atonic bladder).

5. **Autonomic:** because the sympathetic outflow (T1–L2) loses brain control, lesions at or above T6 cause low blood pressure, orthostatic hypotension and autonomic dysreflexia, and poor temperature control.
6. **Secondary problems:** immobility and lost sensation lead to pressure injuries, DVT, contractures, osteoporosis, chest infections and pain.

Vertebral level and cord segment in adults (approximate)

Fracture at	Cord segments at risk
Cervical vertebrae	Add 1 (C5 vertebra lies over the C6 segment)
T1–T6 vertebrae	Add 2
T7–T9 vertebrae	Add 3
T10 vertebra	L1–L2 segments
T11 vertebra	L3–L4 segments
T12 vertebra	L5 segment
L1 vertebra	Sacral segments (conus medullaris)
Below L1–L2	No cord: cauda equina (nerve roots) only

The neurological level comes from the examination, not the X-ray: the roots that leave just below a fracture are often damaged too.

? Why does a T10 fracture give a reflex (spastic) bladder but an L1 fracture often gives a flaccid one?

> The bladder reflex centre is in the sacral cord (S2–S4), which lies at the L1 vertebra. A T10 fracture damages the cord above that centre, so the reflex arc is intact but cut off from the brain, and the bladder empties by reflex. An L1 fracture can destroy the sacral centre itself, so the reflex arc is broken and the bladder becomes atonic.

Recall check. Which cord segments lie under a T10 vertebral fracture, and will the legs be spastic or flaccid?

5 Clinical features

Spinal shock (3 marks in S17 and W20)

- Starts at injury: flaccid paralysis and loss of all reflexes below the lesion, including the bulbocavernosus reflex; loss of sensation.
- Atonic bladder with retention; paralytic ileus; loss of sweating and temperature control below the lesion.
- **Four phases** (Ditunno et al., 2004): reflexes absent or depressed (day 0–1); first reflexes return, starting with the bulbocavernosus reflex (days 1–3); early hyperreflexia (1–4 weeks); late hyperreflexia and spasticity (1–12 months).
- Do not confuse it with **neurogenic shock**: low blood pressure and slow heart rate in injuries at or above T6, from loss of sympathetic tone.

By level

- **C1–C4:** all four limbs and the trunk paralysed; the diaphragm (C3–C5) is weak or paralysed, so breathing may need a ventilator.
- **C5–C8:** partial arm function by level; weak cough; orthostatic hypotension and autonomic dysreflexia; hand function decides independence.
- **T1–T12:** arms normal; trunk control falls the higher the lesion; lesions at or above T6 risk autonomic dysreflexia.
- **Conus and cauda equina:** LMN legs, saddle anaesthesia, areflexic bladder and bowel.

Non-traumatic conditions in the papers

- **Transverse myelitis** (S17, S21): inflammation across a few segments; weakness of both legs, a sensory level and bladder loss developing over hours to days. Treated with steroids or plasma exchange. Recovery begins within weeks and varies: roughly a third each recover well, partly or poorly.
- **Pott's spine** (W22): tuberculosis of the vertebrae, most often thoracic; back pain, kyphosis (a gibbus), weight loss, evening fever; paraplegia from pus, granulation tissue or bone pressing on the cord. Treated with anti-tubercular drugs for many months and a brace, with surgery when the deficit worsens or does not improve.

6 Investigations

- **X-ray and CT** of the spine: fracture, dislocation, stability.
- **MRI:** cord compression, bleeding, swelling; disc, abscess (Pott's spine) or tumour; inflammation in transverse myelitis.
- **ISNCSCI examination** within 72 hours and repeated: level and AIS grade.
- **Urodynamics** and kidney ultrasound: bladder pressure, detrusor overactivity, reflux.
- **Others:** Doppler for DVT; blood tests (ESR in tuberculosis); bone density scan later.

7 Medical and surgical management

- **At the scene and in casualty:** immobilise the spine; airway, breathing and blood pressure; transfer to a spinal unit.
- **Surgery:** decompression and stabilisation (screws and rods), ideally early, so the person can sit up and start rehabilitation within days. Stable fractures may be treated with a brace.
- **Prevention:** blood thinners and compression stockings for DVT; stomach protection; bowel and bladder care (catheter, then intermittent catheterisation).
- **Later:** drugs for spasticity (baclofen) and nerve pain; antimuscarinic drugs for the bladder; surgery for tendon transfers in tetraplegia.
- **Non-traumatic:** anti-tubercular treatment (Pott's spine); steroids or plasma exchange (transverse myelitis); radiotherapy or surgery (tumours).

Classic answer vs current practice: early care

Classic	Current	Write in the exam
Bed rest with postural reduction for 6–12 weeks before sitting up.	Early surgical stabilisation where possible, then sitting and rehabilitation within days, following the surgeon's precautions.	"After stabilisation, mobilise early within the precautions advised: chest care, turning, range of motion, then tilt table and sitting."

8 Complications and prevention

Autonomic dysreflexia (7 marks in W18 and W24; 3 marks in S17, W19, S21, W21, S22, S24)

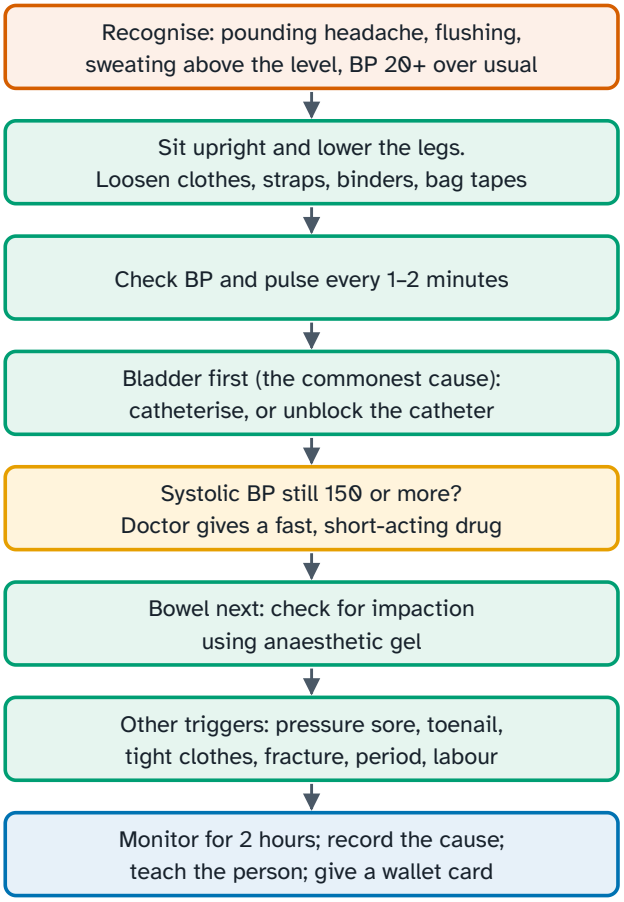
Definition: a sudden rise in systolic blood pressure of **more than 20 mmHg** above the person's usual level (more than 15 mmHg in children), in people with SCI **at or above T6**. It can cause a stroke, seizures or death, so it is an emergency.

1. A harmful stimulus below the lesion, most often a **full bladder**, sends impulses up the cord.
2. Because the large splanchnic blood vessels are controlled from T6–L2, below the lesion, a massive reflex sympathetic discharge narrows them, so the blood pressure rises sharply.

3. The brain senses the high pressure and slows the heart through the vagus nerve and widens vessels above the lesion (flushing, headache, blocked nose).
4. Because the calming signals cannot pass down through the lesion, the pressure stays high until the stimulus is removed.

Item	What to know
Causes	**Bladder distension** (the commonest: blocked or kinked catheter, missed catheterisation), bowel impaction, pressure injury, ingrown toenail, tight clothes or straps, fractures, menstruation and labour, sexual activity, procedures such as cystoscopy
Symptoms and signs	Sudden pounding headache; high blood pressure; slow (or fast) pulse; flushing and sweating above the lesion; goose bumps; blurred vision or spots; blocked nose; anxiety. Some people have few symptoms

Autonomic dysreflexia: what to do, in order



Order from the 2020 Consortium for Spinal Cord Medicine guideline. A physiotherapist starts the first four steps at once and calls the doctor.

Text version of the flowchart

1. Recognise it: pounding headache, flushing and sweating above the lesion, systolic BP more than 20 mmHg above the usual level.
2. Sit the person upright and lower the legs; loosen tight clothes, straps, binders and leg-bag tapes.
3. Check BP and pulse every 1–2 minutes until stable.
4. Check the bladder first: catheterise, or look for a kinked or blocked catheter and flush or change it.
5. If systolic BP stays at 150 mmHg or more, the doctor gives a fast, short-acting drug (for example nitrate paste or nifedipine; ask about sildenafil-type drugs before nitrates).
6. Then check the bowel for impaction, using anaesthetic gel.
7. Look for other triggers: pressure injury, ingrown toenail, tight clothing, fracture, menstruation or labour.
8. Monitor for at least 2 hours, record the cause, and teach the person and family; give a written card.

Prevent it in the physiotherapy department

- Ask about the bladder before every session; check that the catheter drains and the leg bag is not full.
- Avoid tight straps on the tilt table and in standing frames; check skin and toenails.
- Know the person's usual blood pressure: in tetraplegia it is often only 90–110 systolic, so 130 may already be high.
- Stop exercise or electrical stimulation at once if a headache starts, and check blood pressure.

Other complications

Complication	What to look for	Physio role
Orthostatic hypotension	Fall of 20 mmHg systolic or 10 mmHg diastolic on sitting up; dizziness, blurred vision, fainting; worst in tetraplegia	Tilt table with slow progression; abdominal binder and stockings; sit up in stages; tilt back and raise the legs if symptoms come
Pressure injury	Sacrum and heels in lying; ischial tuberosities in sitting; trochanters in side lying	Turning every 2–6 hours by skin tolerance; wheelchair pressure relief **every 15–30 minutes** for **30–90 seconds**; cushion; skin checks **twice a day** with a mirror
DVT and pulmonary embolism	Swollen, warm leg (no pain is felt below the lesion); sudden breathlessness	Early mobilisation; stockings; passive movements; stop leg exercise and report if a DVT is suspected
Chest infection	Weak cough, retained secretions, low vital capacity in tetraplegia and high paraplegia	Breathing exercises, assisted cough, positioning, binder, inspiratory muscle training
Spasticity and contractures	Spasms, tight hip flexors, adductors, hamstrings and calves	Daily stretching, prone lying, standing, positioning; look for triggers (bladder, skin)
Heterotopic ossification	New bone around a joint, most often the hip, in the first months: swelling, warmth, loss of range	Gentle range within comfort; no forceful stretching; report early
Osteoporosis and fractures	Swelling or deformity after a minor injury below the lesion	Gentle handling and stretching; safe transfers; standing programme
Shoulder pain	Overuse from transfers and wheeling	Strengthen the shoulder and scapular muscles; level transfers; long, smooth push strokes
Pain and low mood	Burning nerve pain below the level; depression	Activity, exercise and goals; refer for medicines and counselling

9 PT assessment

- **History:** mechanism and date, surgery and spinal precautions, bladder and bowel method, autonomic dysreflexia episodes, skin, pain, home (floor, stairs, toilet type), job, family, goals.
- **ISNCSCI examination:** key muscles, key sensory points, anorectal examination; level and AIS grade.
- **Respiratory:** breathing pattern, chest expansion, vital capacity, cough strength.
- **Tone and range:** Modified Ashworth Scale, spasm frequency, range (hamstrings, hip flexors, ankles, shoulders).
- **Skin:** pressure areas; risk score.
- **Function:** bed mobility, sitting balance (long sitting and short sitting), transfers, wheelchair skills; **Spinal Cord Independence Measure** (SCIM III); **Walking Index for SCI** (WISCI II) and 10-metre walk test in walkers.
- **Others:** blood pressure lying and sitting; mood; hand function in tetraplegia.

Worked example: level and grade (S16, 2 + 1 marks)

> "A 21 year old male suffered from traumatic cervical spinal cord injury… bilateral muscle power: biceps 5/5, wrist extensors 5/5, triceps 4/5 and no active contraction below C7. Intact pinprick and light touch C2–T4, absent below T4. Intact anal sensation… Write his neurological level of injury. What is the patient's ASIA impairment classification?"

1. **Motor level:** C5 and C6 are 5/5 and C7 (triceps) is 4/5, which is grade 3 or more with normal muscles above, so the motor level is **C7**.
2. **Sensory level:** light touch and pinprick are normal to T4, so the sensory level is **T4**.
3. **Neurological level:** the higher of the two, so the NLI is **C7**.

4. **Grade:** anal sensation (deep anal pressure) is intact, so the injury is incomplete; there is no motor function below C7, so it is **AIS B** (sensory incomplete).

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case. Write the level and grade in the health condition box.

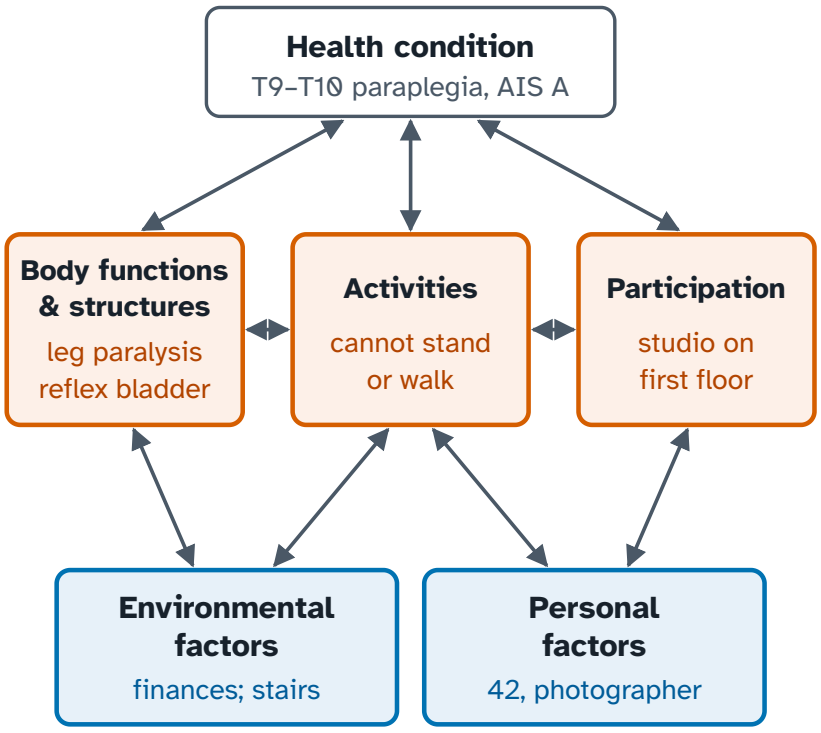
Worked example from a real paper (S22, 7 + 8 marks)

> "A 42 year old photographer had RTA resulting in fracture of T9–T10 vertebrae with complete spinal cord injury. A decompression-stabilization surgery was done 1 month back. He can sit without support, and can transfer himself in wheel chair. He has a spastic bladder. His financial condition is good, his studio is in his own home on first floor. Describe his impairments with clinical reasoning. Describe the PT management for expected type and mode of ambulation."

Spot the ICF clues before you write

complete T9–T10 means AIS A with no function below the level. sits without support, transfers himself are activities he already has: build on them. spastic bladder means a UMN (reflex) bladder. financial condition is good is a facilitator for orthoses and home changes. studio on first floor is a barrier (stairs) to his work.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Traumatic complete paraplegia (AIS A) after T9–T10 fracture; decompression and stabilisation 1 month ago	The cord above the conus is damaged, so UMN signs below the level

ICF component	From this case	Clinical reasoning
Body function and structure impairments	• Paralysis of both legs and the lower trunk • Loss of all sensation below the level • Spasticity and spasms developing • Reflex (spastic) bladder; neurogenic bowel; sexual dysfunction • Weak lower abdominal muscles, so trunk control is reduced • Risks: pressure injury, DVT, contractures, osteoporosis • Pain at the fixation site	• Complete lesion, so no voluntary power or feeling below it • Spinal shock is over, so tone and reflexes return and become exaggerated • The sacral reflex centre is intact but cut off from the brain • Insensate skin cannot warn of pressure • Below T6, so autonomic dysreflexia is unlikely
Activity limitations	Cannot stand or walk; cannot climb stairs; needs a routine for bladder and bowel; difficulty dressing the lower body and bathing. Already sits without support and transfers to the wheelchair	Arms and upper trunk are normal, so full wheelchair independence is expected
Participation restrictions	Cannot reach his studio on the first floor; outdoor photography assignments; travel and social life	Stairs, not his skills, are the main barrier to work
Environmental factors	**Facilitators:** good finances (orthoses, light wheelchair, a lift or moving the studio); studio within his own home; family. **Barriers:** stairs to the studio; possibly an Indian-style toilet; rough roads	Finances make home changes and equipment possible
Personal factors	42-year-old man; self-employed photographer; coping with a sudden disability	A job that can be done from a wheelchair supports return to work

One-line functional diagnosis

A 42-year-old photographer with complete (AIS A) T9–T10 paraplegia, 1 month after stabilisation, with paralysis and loss of sensation below the level and a reflex bladder, who sits and transfers independently but cannot stand, walk or climb to his first-floor studio. Good finances are a facilitator; the stairs are the main barrier.

Practice cases from the papers: W18 24-year-old accountant with a T10 fracture, spasticity grade 2 and loss of bladder control; S25 25-year-old labourer with a T10 fracture after a fall from the second floor; S21 30-year-old woman with transverse myelitis at T10; W22 29-year-old executive with Pott's spine T10–T12 after anti-tubercular treatment; S19 36-year-old computer operator with an L1 fracture.

Recall check. Fill the ICF for the S25 labourer. What is different about his participation compared with the photographer?

11 Problem list and goals

Goals for the photographer with complete T9–T10 paraplegia

Problem	Short-term (4–6 weeks)	Long-term (3–6 months)
Skin	Does his own pressure relief every 15–30 minutes and checks his skin twice a day	No pressure injury
Upper limb and trunk strength	Lifts his body clear of the seat in push-ups	Floor-to-wheelchair transfer on his own
Wheelchair mobility	Wheels independently on ramps and turns in small spaces	Manages kerbs and rough ground; balances on the back wheels

Problem	Short-term (4–6 weeks)	Long-term (3–6 months)
Standing and walking	Stands in a frame daily; stands in KAFOs in the parallel bars	Walks household distances with KAFOs and elbow crutches, swing-through
Bladder and bowel	Self-catheterises on schedule with volumes under 400–500 mL	Independent bladder and bowel routine
Work and home	Plan for a lift or a ground-floor studio	Back at work; drives with hand controls

12 PT management

Principles: start early within spinal precautions; prevent complications; strengthen what works; teach new ways to move; set goals from the level and grade; plan for home, work and lifelong fitness.

Expected outcomes by level (complete injury, AIS A or B)

Level	Mobility	Self-care
C1–C4	Power wheelchair with head, chin or sip-and-puff control; hoist transfers; C1–C3 need a ventilator	Dependent; directs own care
C5 (elbow flexors)	Power wheelchair; manual wheelchair on level ground, independent to some help; board transfers with help, or a hoist	Feeding and grooming with set-up and adapted equipment; bladder and bowel dependent
C6 (wrist extensors: tenodesis grasp)	Manual wheelchair on level ground with coated rims, help outdoors; level transfers independent to some help	Feeds and grooms with adapted equipment; bladder and bowel usually need help; may drive an adapted van
C7–C8 (triceps, finger flexors)	Manual wheelchair independent at home and in the community, help with kerbs and ramps; level transfers independent	Independent with adapted equipment, including bladder and bowel; drives with hand controls
T1–T12	Independent in a light manual wheelchair, including ramps and kerbs; floor transfers; stands with orthoses; walks household distances with KAFOs or HKAFOs and a walker or elbow crutches, with less help the lower the level	Independent; help with heavy housework
L1–L3 (hip flexors, quadriceps)	Walks with orthoses (KAFO or AFO by strength) and crutches; wheelchair for long distances	Independent
L4–S1 (ankle muscles, hip abductors and extensors)	Walks with AFOs and crutches or canes; wheelchair for long distances	Independent

From the Consortium for Spinal Cord Medicine outcomes guideline (2nd edition, 2026). Incomplete injuries usually do better. Walking with KAFOs costs far more energy than wheeling, so most people with thoracic paraplegia use the wheelchair for daily mobility and walk for exercise or short distances at home.

By phase

- **Acute (bed and ICU, first days to weeks):** chest care (deep breathing, assisted cough, positioning); turning and pressure care; daily range of motion within the surgeon's precautions; resting splints to keep ankles at 90°; strengthening of all working muscles; DVT prevention; sitting up in stages once the spine is stable, watching blood pressure.
- **Rehabilitation (weeks to months):** tilt table and wheelchair sitting; mat programme (rolling, prone on elbows, long sitting, sitting balance, push-ups); transfers (sliding board, then lifting transfers, then floor transfers); wheelchair skills; standing programme; gait training where the level allows; hand training in tetraplegia; bladder and bowel routines with the nurse.

- **Community and lifelong:** fitness and sport; home and workplace changes; driving with hand controls; yearly review of skin, shoulders, bladder and equipment.

Mat and transfer skills, step by step

1. Rolling using momentum: swing the arms across the body, with the legs crossed in the direction of the roll.
2. Prone on elbows, then shifting weight between the elbows.
3. Long sitting: needs a hamstring length of about **100–110°** straight-leg raise; do not overstretch, because a little tightness helps sitting balance.
4. Sitting balance: hold, then reach out, then catch and throw a ball, first with a mirror, then without.
5. Push-ups on blocks: lift the buttocks clear using shoulder depressors, latissimus dorsi and triceps.
6. Transfers: sliding board first, then a lifting (depression) transfer, to level and then to uneven surfaces; finally floor to wheelchair.

Ambulation with orthoses (the S22 and S26 long cases)

- **Before walking:** strong shoulder depressors, latissimus dorsi and triceps; full hip extension and ankle dorsiflexion to neutral; standing tolerance in a frame without dizziness; good skin.
- **Orthoses:** bilateral KAFOs with drop-lock knees and the ankles set in slight dorsiflexion; for higher thoracic levels, a hip-knee-ankle-foot orthosis or a reciprocating gait orthosis (RGO).
- **In the parallel bars:** putting on the orthoses; sit-to-stand with the knees locked; standing balance with the hips pushed forward (the body hangs on the Y-ligaments) and the shoulders behind the hips; push-ups; swing-to gait, then **swing-through**; turning.
- **Progress:** walker, then elbow crutches; four-point gait with an RGO; sitting down, falling safely and getting up from the floor; kerbs; stairs backwards with a rail and a crutch.
- **Mode:** household or exercise walking with KAFOs at T9–T12; the wheelchair stays the main way to get about.

Ambulation after an L3 vertebral fracture (7 marks in S18)

An L3 fracture lies below the end of the cord, so it damages the cauda equina: an LMN, often incomplete and uneven, weakness of the legs. Strengthen the hip extensors, abductors and quadriceps; train standing balance; fit **AFOs** for weak ankles (KAFOs only if the quadriceps are weak); walk with elbow crutches using a four-point, then two-point gait, then canes; practise stairs, kerbs and slopes. Expect community walking. Protect the insensate feet, and follow the brace and lifting precautions.

Tetraplegia: what is different

- **Breathing:** breathing exercises; assisted cough (hands push in and up under the ribs as the person coughs); abdominal binder when sitting; inspiratory muscle training.
- **Tenodesis grasp (C6–C7):** wrist extension makes the fingers close. Stretch the finger flexors only with the wrist flexed, and never stretch the fingers and wrist into extension together.
- **Upper limb:** strengthen every working muscle; splints to hold the wrist and thumb; practise feeding and writing with the occupational therapist.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Low fitness	Arm cycling, wheelchair pushing, swimming: at least **20 minutes** at moderate-to-vigorous intensity **2 times a week**; for heart and metabolic health, **30 minutes** **3 times a week**	SCI raises the risk of heart disease and diabetes, and fitness makes wheeling and transfers easier
Weak working muscles	Resistance training of shoulder depressors, latissimus dorsi, triceps, pectorals and scapular muscles: **3 sets** for each major working muscle group, **2 times a week**	These muscles lift the body for transfers, push-ups and crutch walking

Impairment	Technique and dose	Why
Poor sitting balance	Sitting on a mat and on the edge of a plinth: hold, reach beyond arm's length, catch a ball; mirror first, then without	Vision and the remaining trunk and shoulder muscles take over from the lost trunk sensation
Orthostatic hypotension	Tilt table, raising the angle in small steps as blood pressure allows; abdominal binder; stockings	Gradual upright time trains blood pressure control; binders reduce pooling of blood
Pressure injury risk	Push-ups, side leans or forward leans **every 15–30 minutes** for **30–90 seconds**; turning in bed every 2–6 hours by skin tolerance; skin checks twice a day	Insensate skin cannot warn of pressure, and relief restores blood flow
Spasticity and tight muscles	Daily stretching of hip flexors, adductors, hamstrings and calves; prone lying; standing in a frame	Keeps the range needed for sitting, transfers and orthoses and reduces spasms
Weak cough	Assisted cough; breathing exercises; inspiratory muscle training; binder in sitting	The abdominal muscles that power a cough are paralysed in high lesions
Cannot walk (incomplete injury)	Overground walking and treadmill training, with body-weight support if needed; strengthening of the muscles that work	Walking practice helps people with AIS C and D injuries; in chronic AIS A and B it does not restore walking

Fitness and strength doses: international exercise guidelines for adults with SCI (Martin Ginis et al., 2018).

Recall check. What can a person with complete C6 tetraplegia expect to do, and how do you protect the tenodesis grasp?

13 Precautions, contraindications and stop criteria

Take care

- Follow the surgeon's spinal precautions: brace, log rolling, limits on bending, twisting and lifting.
- Stop and check blood pressure if a headache, flushing or sweating starts (autonomic dysreflexia) or if dizziness starts (orthostatic hypotension).
- No heat packs or ice on insensate skin; check skin under orthoses and splints.
- Handle osteoporotic limbs gently; no forceful stretching, especially with heterotopic ossification.
- Protect the tenodesis grasp in tetraplegia and the shoulders in all wheelchair users.
- No leg exercise or massage if a DVT is suspected until it is treated.

14 Education, lifestyle and home programme

- Pressure relief, skin checks with a mirror, cushion care and nutrition.
- Autonomic dysreflexia: signs, first steps and a wallet card (lesions at or above T6).
- Bladder and bowel routine; enough fluids; signs of urine infection.
- Home: ramps, wide doors, a western toilet or commode chair, a bathroom with a shower chair, ground-floor living or a lift.
- Daily stretching, standing and the exercise doses above; safe transfers; wheelchair maintenance.
- Work, driving with hand controls, sport and sexuality; disability certificate and benefits; peer support.

15 Prognosis and discharge criteria

- Most motor recovery happens in the first 6–9 months, fastest in the first 3 months.

- Most injuries graded AIS A stay AIS A; about 20–30% improve, half of them to AIS B and half to AIS C or D. Conversion is more likely in tetraplegia than paraplegia.
- Walking is more likely with a lower AIS grade (D best), younger age, stronger quadriceps (L3) and calf muscles (S1), and preserved pinprick sensation in the legs and sacral area.
- Discharge from inpatient rehabilitation when the person reaches the goals expected for the level, can direct their own care, and has the equipment, home changes and follow-up in place.

16 Exam corner

Past questions (MUHS, 2016–2026)

S26 Q4(a)

55-year-old man, fall from the first floor, T9–T10 fracture, decompression and stabilisation 2 weeks ago; senses a full bladder but cannot void without stimulation; lives with his wife and son on the second floor. Short- and long-term goals; expected ambulation level and rehabilitation outcomes; type of bladder.

S25 Q3(b), 5 + 10 marks

25-year-old labourer, fall from the second floor, T10 vertebral fracture a month ago, cannot move both legs. ICF with clinical reasoning; goals with physiotherapy.

W22 Q3(b), 5 + 7 + 3 marks

29-year-old unmarried executive, Pott's spine at T10–T12, weak legs, cannot walk, anti-tubercular treatment completed. ICF; physiotherapy; goals.

S22 Q3(a), 7 + 8 marks

The worked example in section 10.

S21 Q3(b), 5 + 10 marks

30-year-old woman with acute transverse myelitis at T10. ICF with clinical reasoning; physiotherapy.

S19 Q4(a), 5 + 10 marks

36-year-old computer operator, road accident 6 months ago, traumatic L1 fracture. ICF; goals and physiotherapy.

W18 Q3(a), 5 + 5 + 5 marks

24-year-old accountant, road accident 2 months ago, T10 fracture treated operatively; leg weakness, spasticity grade 2 and loss of bladder control. ICF; goals; physiotherapy.

S17 Q4(b)

45-year-old shopkeeper, the sole earner, with transverse myelitis (D8–D11) for 2 weeks. Assessment with ICF; short- and long-term management.

S16 Q3(a), 2 + 1 + 5 + 7 marks

21-year-old with traumatic cervical SCI: level and grade (worked in section 9), goals, physiotherapy.

7-mark questions

Autonomic dysreflexia and its management (W18); autonomic dysreflexia: causes, symptoms and intervention (W24); ambulation training after an L3 vertebral fracture (S18).

3-mark questions

Sacral sparing (S16); spinal shock (S17, W20); autonomic dysreflexia (S17, W19, S21, W21, S22, S24); Brown-Séquard syndrome (W17, S23); ASIA scale (S19); paraplegia in flexion and extension (W22); cauda equina injury (W24);

complete vs incomplete paraplegia (W25).

Pages for this topic: bladder and bowel training (p. 238) (types of bladder, asked in 9 papers), transfers and gait training (p. 232), orthoses (p. 249), wheelchairs (p. 257), autonomic disorders (p. 193).

Keywords to underline

neurological level AIS grade sacral sparing spinal shock UMN or LMN autonomic dysreflexia pressure relief tenodesis KAFO swing-through expected outcome for the level home modification

Answer skeleton: any SCI long case

1. Restate: age, job, level (vertebral and likely neurological), complete or incomplete, time since injury and surgery, bladder, home.
2. **[ICF, 5]** ICF diagram and table with clinical reasoning; one-line functional diagnosis.
3. **[Goals, 3–5]** Short-term and long-term goals that match the expected outcome for the level.
4. **[Management, 7–10]** Phase; chest and skin care; strengthening; mat skills and transfers; wheelchair skills; standing and ambulation with orthoses (or locomotor training if incomplete); bladder and bowel; autonomic dysreflexia if at or above T6; home, work and fitness; precautions.

Answer: type of bladder in the S26 case

A T9–T10 fracture damages the cord above the sacral micturition centre (S2–S4, at the L1 vertebra), so the reflex arc is intact but cut off from the brain: a **reflex (UMN, spastic) neurogenic bladder**. It empties by reflex when stimulated, often against a sphincter that does not relax (detrusor-sphincter dyssynergia). His sense of fullness suggests that some sensory fibres are spared. Manage it with clean intermittent catheterisation 4–6 times a day and drugs to calm the bladder; avoid tapping and pressing to empty it unless urodynamics show safe pressures. Full detail is on the bladder and bowel page (p. 238).

Draw these

- The four incomplete cord syndromes (section 2)
- The autonomic dysreflexia flowchart (section 8)
- The ICF model filled from the case

Viva questions

1. What is the difference between spinal shock and neurogenic shock?
2. How do you find the neurological level of injury?
3. Why is sacral sparing important?
4. Why does autonomic dysreflexia occur only at or above T6?
5. What can a person with C6 tetraplegia do, and what is tenodesis?
6. Which orthosis and gait pattern would you choose for T10 complete paraplegia?

Linked practical spots

- KAFO, RGO and AFO; sliding board; wheelchair cushion; tilt table; ISNCSCI chart.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the clinical features of Brown-Séquard syndrome.**

Answer
Hemisection of the cord. Same side: UMN weakness and loss of position sense and vibration below the lesion; a band of sensory loss at the level. Opposite side: loss of pain and temperature from 1–2 segments below the lesion. Best outlook of the cord syndromes.

3 m **What is spinal shock? Write its features.**

Answer
Temporary loss or depression of all reflexes below the lesion after SCI. Features: flaccid paralysis, areflexia including the bulbocavernosus reflex, sensory loss, urinary retention, paralytic ileus. Reflexes return over days to weeks (the bulbocavernosus first) and then become exaggerated.

7 m **Explain autonomic dysreflexia: causes, symptoms and intervention.**

Answer
Definition: systolic BP more than 20 mmHg above the usual level in SCI at or above T6. Mechanism: a stimulus below the lesion triggers a reflex sympathetic discharge that narrows splanchnic vessels; the brain cannot send calming signals down. Causes: full bladder (commonest), bowel impaction, pressure injury, ingrown toenail, tight clothing. Symptoms: pounding headache, flushing and sweating above the lesion, blocked nose, blurred vision, slow pulse. Intervention: sit up and lower the legs; loosen clothing; BP every 1–2 minutes; check the bladder; drug if systolic stays 150 or more; check the bowel; other triggers; monitor 2 hours; educate with a wallet card.

LAQ **A 25-year-old labourer fell from the second floor and had a T10 vertebral fracture a month ago; he cannot move both legs. Discuss the ICF with clinical reasoning, and short- and long-term goals with physiotherapy. (5 + 10, S25)**

Answer
ICF (5): health condition: traumatic paraplegia after T10 fracture (examine for level and grade). Impairments: paralysis and sensory loss of both legs, spasticity developing, neurogenic bladder and bowel, risks of pressure injury and DVT. Activities: cannot stand or walk; bed mobility, transfers and self-care to be learnt. Participation: cannot do manual labour; family income lost. Environment: likely a poor family, stairs, Indian toilet, rough roads (barriers); free government schemes and family (facilitators). Personal: young man, manual worker.

Goals and physiotherapy (10): short-term: chest care, pressure relief and skin checks, full range, strong arms and shoulders, sitting balance, independent transfers and wheelchair skills. Long-term: independent in a wheelchair at home and outside; standing and household walking with KAFOs and crutches; bladder and bowel routine; a new seated job (training, disability schemes). Management by phase as in section 12, with low-cost equipment and a home programme.

MCQ **Biceps and wrist extensors are 5/5, triceps 3/5, nothing below; sensation normal to C8; no sacral sensation or anal contraction. The classification is:**

A. C6 AIS A

B. C7 AIS A

C. C7 AIS B

D. C8 AIS A

Answer
B. Motor level C7 (grade 3 with normal muscles above), sensory level C8, so the NLI is C7; no sacral sparing means AIS A.

MCQ **The commonest cause of autonomic dysreflexia is:**

A. Pressure injury

B. Bladder distension

C. Ingrown toenail

D. Tight clothing

Answer

B. A full bladder or a blocked catheter, so check the bladder first.

MCQ **How often should a wheelchair user with paraplegia do pressure relief?**

A. Every 15–30 minutes

B. Every 2 hours

C. Every 4 hours

D. Only when the skin is red

Answer

A. Every 15–30 minutes, for at least 30–90 seconds each time.

Last recall check. Draw the four cord syndromes, write the AIS grades, and list the eight steps for autonomic dysreflexia.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. American Spinal Injury Association. International Standards for Neurological Classification of Spinal Cord Injury (ISNCSCI), revised 2019; 9th edition released July 2026.
3. Consortium for Spinal Cord Medicine. Evaluation and management of autonomic dysreflexia and other autonomic dysfunctions: preventing the highest immediate threats to life. Paralyzed Veterans of America; 2020 (*Top Spinal Cord Inj Rehabil* 2021).
4. Consortium for Spinal Cord Medicine. Outcomes following traumatic spinal cord injury, 2nd edition. Paralyzed Veterans of America; 2026.
5. Consortium for Spinal Cord Medicine. Management of neurogenic bowel dysfunction in adults after spinal cord injury. Paralyzed Veterans of America; 2020.
6. Martin Ginis KA, et al. Evidence-based scientific exercise guidelines for adults with spinal cord injury: an update and a new guideline. *Spinal Cord* 2018;56:308–21.
7. Ditunno JF, et al. Spinal shock revisited: a four-phase model. *Spinal Cord* 2004;42:383–95.
8. Mathur N, et al. Spinal cord injury: scenario in an Indian state. *Spinal Cord* 2015;53:349–52.
9. Model Systems Knowledge Translation Center. Skin care and pressure injuries: prevention (fact sheet).
10. EAU Guidelines on Neuro-Urology, 2026 limited update.
11. Structure and classical content: Somers, *Spinal Cord Injury: Functional Rehabilitation*; Bromley, *Tetraplegia and Paraplegia*; O'Sullivan and Schmitz, *Physical Rehabilitation*; Snell, *Clinical Neuroanatomy* (vertebral and segmental levels).

5A(d) Peripheral nerve injuries and the brachial plexus: classification, deformities, splints and physiotherapy

> Nerve injuries appeared in 14 of the 20 papers: the classification of nerve injury was a 3-mark question five times, splints for radial palsy three times, and Erb's palsy or a plexus case often enough to expect one. Every answer follows the nerve's recovery: protect and splint while the muscle is denervated, then re-educate movement and sensation as the regrowing nerve (about 1 mm a day) reaches each muscle.

Syllabus	Ch 5A(d) Peripheral neuropathies, traumatic: upper limb and lower limb, brachial plexus (non-traumatic neuropathies and obstetric plexus palsy have their own pages)
Priority	**Tier 1.** In 14 of 20 papers (2016–2026): 3 long cases, 11 seven-mark and 8 three-mark questions.
Asked as	LAQ W16 · S22 · S23 · 7m S16 · S17 · S18 · W18 · W19 · S21 · W21 · S22 · W24 · S25 · W25 · 3m W17 · S21 · W21 · S22 · S23 · S25 ×2 · W25
Usual split	15 (ICF with clinical reasoning 7, goals and techniques 8; S22 asked the part of the plexus 2, goals 6, management with orthoses 7)
Checked against	APTA clinical practice guideline on carpal tunnel syndrome (2019; revised 2026); systematic review of timing of brachial plexus surgery (Martin et al., 2018); Seddon (1943) and Sunderland (1951) classifications. Full list under Sources (p. 106). Content checked 20 October 2026.

1 Definition

- A **peripheral nerve injury** is damage to a peripheral nerve that causes a lower motor neuron (LMN) loss in its area: weakness and wasting of the muscles it supplies, loss of reflexes, sensory loss and loss of sweating.
- A **brachial plexus injury** damages the C5–T1 roots, trunks or cords that supply the upper limb, usually by traction.
- Because a peripheral nerve can regrow, many injuries recover, slowly, from the injury site downwards.

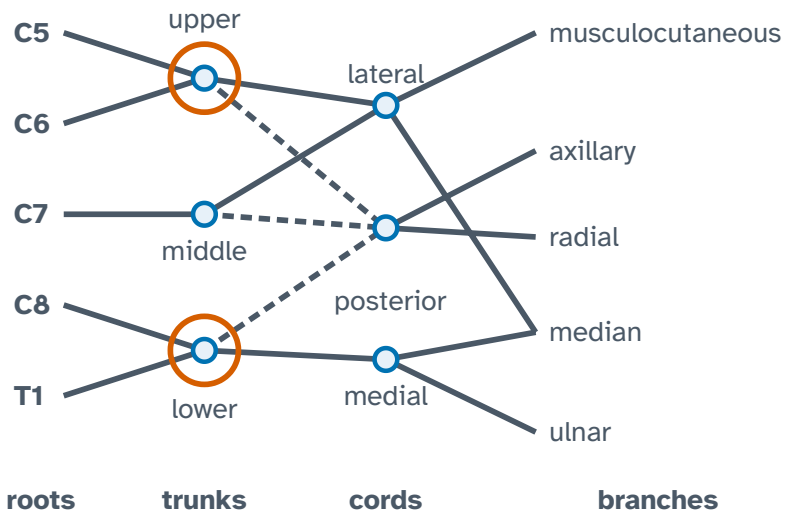
2 Classification and grading

Asked for 3 marks in S21, W21, S22, S23 and S25, and with foot drop for 7 marks in S16.

Sunderland degree (Seddon type)	What is damaged	Recovery
I (neurapraxia)	Local myelin damage; conduction block; the axon is intact; no Wallerian degeneration	Full, in days to about 3 months; no Tinel's sign moving down the limb
II (axonotmesis)	Axon cut; endoneurium intact; Wallerian degeneration below the injury	Full, at about 1 mm a day, because each axon regrows in its own tube
III	Axon and endoneurium cut; perineurium intact	Incomplete: axons enter the wrong tubes (misdirection)
IV	Axon, endoneurium and perineurium cut; only the epineurium holds the nerve together (a neuroma in continuity)	Poor without surgery: excise the neuroma and repair or graft
V (neurotmesis)	The whole nerve is divided	None without surgical repair

Seddon's three types: **neurapraxia** (conduction block), **axonotmesis** (axon cut, sheaths intact) and **neurotmesis** (nerve divided). Sunderland split axonotmesis into degrees II–IV by how much connective tissue is cut.

The brachial plexus (simplified)



Dashed lines are the posterior divisions. Circles mark the upper trunk (Erb's palsy, C5–C6) and the lower trunk (Klumpke's palsy, C8–T1).

3 Causes and risk factors

Nerve	Common causes
Brachial plexus	Two-wheeler accidents and falls: head and shoulder forced apart (upper plexus) or the arm pulled overhead (lower plexus); stab and gunshot wounds
Radial	Fracture of the shaft of the humerus (spiral groove); "Saturday night palsy" (arm hung over a chair); crutch pressure in the axilla
Ulnar	Elbow fractures and old deformity (tardy ulnar palsy); cuts at the wrist; leprosy
Median	Cuts at the wrist (glass, knife); carpal tunnel compression; supracondylar fracture
Axillary	Shoulder dislocation; fracture of the surgical neck of the humerus
Common peroneal	Fracture of the fibular neck; tight plaster; prolonged squatting or crossed legs; knee dislocation
Sciatic	Wrongly placed gluteal injections (still seen in India, often in children); hip dislocation and hip surgery

4 Pathophysiology to impairments

1. When an axon is cut, the part below the injury breaks down (**Wallerian degeneration**) over days, so the muscles lose their nerve supply.
2. Denervated muscle becomes flaccid and wastes; reflexes are lost; the skin loses sensation and sweating (dry, shiny, cool skin).
3. Because the nerve regrows from the injury site at about **1 mm a day** (about 2.5 cm a month), muscles close to the injury recover first and distant muscles last.
4. A **Tinel's sign** (tingling when the nerve is tapped) moves down the limb as the regrowing axons advance.
5. If reinnervation takes too long, muscle fibres and motor end-plates degenerate, so proximal injuries far from the hand recover less.
6. Axons that regrow into the wrong tubes cause poor localisation of touch and imprecise movement.

Recall check. A radial nerve is injured 25 cm above the wrist extensors. Roughly when would you expect the first flicker of wrist extension?

5 Clinical features

Nerve by nerve

Nerve and level	Motor loss and deformity	Sensory loss and tests
Radial, at the spiral groove (7 marks in S18 and W18)	**Wrist drop**, finger drop and thumb drop; triceps usually spared; weak grip because the wrist cannot be held extended	Small area on the back of the first web space; ask the person to extend the wrist and fingers
Posterior interosseous (radial in the forearm)	Finger and thumb drop; the wrist still extends, but deviates to the radial side	No sensory loss
Ulnar, at the wrist	**Claw hand** of the ring and little fingers (hyperextended knuckles, bent finger joints); wasting of the hypothenar and interossei; no finger spreading	Little finger and half the ring finger; **Froment's sign** (the thumb bends at its tip when pinching paper), card test
Ulnar, at the elbow	As above, but **less clawing**, because the long flexors of the ring and little fingers are also weak (the "ulnar paradox")	As above, plus the back of the hand on the ulnar side
Median, at the wrist	**Ape-thumb**: thenar wasting, the thumb lies flat with the palm and cannot oppose or abduct	Palm side of the thumb, index, middle and half the ring finger; pen test for abductor pollicis brevis
Median, at the elbow	As above, plus weak finger flexors: the index (and middle) finger stays straight on making a fist ("hand of benediction")	As above; Ochsner's clasp test
Median and ulnar together	Claw of all fingers with an ape-thumb; all intrinsic muscles lost	Most of the palm
Axillary	Weak shoulder abduction (deltoid); flattened shoulder	Patch over the lower deltoid ("regimental badge" area)
Common peroneal (7 marks in W25)	**Foot drop**: weak dorsiflexion, eversion and toe extension; high-stepping gait; the foot slaps down	Outer leg and the back of the foot
Sciatic	Foot drop with weak hamstrings and calf; flail foot in complete lesions	Below the knee except the inner leg

Claw hand (7 marks in W19)

Claw hand is hyperextension of the knuckle (MCP) joints with flexion of the finger (IP) joints, because the lumbricals and interossei are paralysed: the long extensors pull the knuckles back and the long flexors bend the fingers, unopposed. An ulnar lesion alone claws the ring and little fingers; median and ulnar lesions together claw all four fingers. Other causes: Klumpke's palsy, leprosy, Volkmann's contracture.

Brachial plexus injuries (adult)

Type	Mechanism and lost function	Posture and sensory loss
Erb's (upper plexus, C5–C6) (7 marks in S17, S21, W21, W24)	Head and shoulder forced apart, as in a fall from a two-wheeler. Lost: shoulder abduction (deltoid, supraspinatus) and external rotation (infraspinatus), elbow flexion (biceps, brachialis, brachioradialis), supination	**"Waiter's tip"**: arm adducted and internally rotated, elbow extended, forearm pronated. Sensory loss over the outer arm and forearm; biceps reflex lost

Type	Mechanism and lost function	Posture and sensory loss
Klumpke's (lower plexus, C8–T1) (7 marks in S25)	Arm pulled forcibly overhead, as when grabbing a branch while falling. Lost: hand intrinsics, finger and wrist flexors	**Claw hand**; sensory loss on the inner forearm and hand; **Horner's syndrome** (drooping eyelid, small pupil, dry face) if the T1 root is torn out
Total plexus (C5–T1)	High-energy accident	Flail, numb arm; often root avulsions and severe burning pain

Root avulsion (preganglionic) vs injury beyond the root (postganglionic)

Feature	Preganglionic (avulsion)	Postganglionic
Clues	Horner's syndrome; winged scapula; weak rhomboids; raised hemidiaphragm; severe burning pain	Tinel's sign over the plexus in the neck
Sensory nerve action potential	Preserved in the numb area, because the sensory cell bodies in the dorsal root ganglion are still connected to the nerve	Absent
Recovery	None by itself: needs nerve transfer	Possible: repair or graft if it does not recover

6 Investigations

- **Nerve conduction studies and EMG:** show the site and severity; denervation potentials (fibrillations) appear about 2–3 weeks after injury, so studies are most useful from then on, and repeated studies show reinnervation before it is visible.
- **Imaging:** X-rays for fractures; ultrasound or MRI of the nerve; MRI or CT myelography of the neck for root avulsion (pseudomeningocele) in plexus injuries.
- **Bedside:** sensory mapping, Tinel's sign tracking, sweat (ninhydrin) test.

7 Medical and surgical management

- **Neurapraxia and axonotmesis:** conservative care with splints and exercise; watch for recovery with Tinel's sign and repeated nerve studies.
- **Clean cuts:** early primary repair; **gaps:** nerve grafts (often the sural nerve).
- **Brachial plexus:** exploration with repair, grafting or **nerve transfers** (for example a branch of the spinal accessory nerve to the suprascapular nerve for shoulder movement, or an ulnar nerve fascicle to the biceps branch for elbow flexion). Results are best when surgery is done within 6 months of injury, and about 3 months is usually appropriate (Martin et al., 2018). Late cases are treated with muscle or tendon transfers.
- **Tendon transfers** when a nerve does not recover: for example for wrist drop, foot drop or claw hand.
- **Carpal tunnel syndrome:** splint and education first; steroid injection; surgical release for severe or persistent cases (thenar wasting, constant numbness).
- **Pain:** medicines for nerve pain.

8 Complications and prevention

Complication	What to look for	Physio role
Contractures and deformity	Tight thumb web, clawing, wrist flexion, equinus, frozen shoulder	Splints in a functional position; daily passive range
Overstretching of paralysed muscles	Wrist extensors in radial palsy, dorsiflexors in foot drop, deltoid in shoulder subluxation	Support them with splints, a cock-up splint, an AFO or a sling
Burns, cuts and ulcers	Numb skin cannot warn of heat or injury	Teach daily skin checks and protection; no heat over numb skin

Complication	What to look for	Physio role
Oedema	Swollen, stiff hand	Elevation, active movement, compression
Neuropathic pain and CRPS	Burning pain, hypersensitivity, colour and temperature change	Desensitisation, graded activity, refer for medicines
Learned non-use and job loss	The person stops using the arm	Task practice, work planning, adaptive devices

9 PT assessment

- **History:** mechanism, date, surgery, dominant hand, job, pain.
- **Inspection:** posture of the limb, deformity, wasting, skin and nails, sweating, swelling.
- **Motor:** manual muscle testing of every muscle the nerve supplies, in order from proximal to distal; trick movements.
- **Sensory:** map light touch and pinprick; two-point discrimination; monofilaments; Tinel's sign and how far it has moved.
- **Special tests:** Froment's sign, card test, pen test, Ochsner's clasp test; for carpal tunnel: Phalen's test, Tinel's sign at the wrist, carpal compression test.
- **Function:** grip and pinch strength, Nine-Hole Peg Test, the DASH questionnaire; gait for foot drop.

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case. Name the nerve or plexus level in the health condition box.

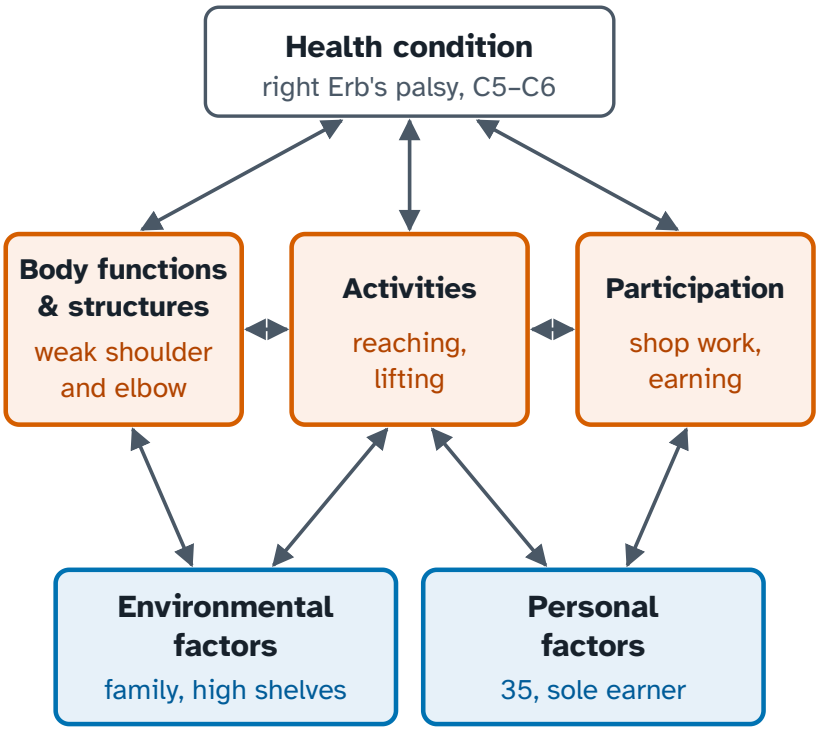
Worked example from a real paper (W16, 7 + 8 marks)

> "Discuss the ICF with clinical reasoning and short term rehabilitation goals and techniques of a 35 years old shopkeeper, who suffered right sided Erb's palsy during a road traffic accident 5 months back. He complains of difficulty in using his right upper extremity. He has a dependent wife and 2 children."

Spot the ICF clues before you write

Erb's palsy means C5–C6 (upper trunk): shoulder abduction, external rotation and elbow flexion are lost. 5 months back means some recovery may have started, and the surgical window (about 6 months) is closing: check for recovery and refer if there is none. shopkeeper lifting and reaching to shelves is his work. dependent wife and 2 children means he is the sole earner.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Right Erb's palsy (upper trunk, C5–C6) after a road accident 5 months ago; dominant side likely	Traction injury of the upper plexus; LMN lesion
Body function and structure impairments	• Paralysis or weakness of deltoid, supraspinatus, infraspinatus, biceps, brachialis, brachioradialis, supinator • "Waiter's tip" posture; risk of inferior shoulder subluxation • Tight shoulder internal rotators and adductors; stiff shoulder • Sensory loss over the outer arm and forearm; biceps reflex lost • Wasting of shoulder and arm muscles; possible pain	• C5–C6 roots supply these muscles through the suprascapular, axillary and musculocutaneous nerves • Unopposed adductors and internal rotators pull the arm in • Five months of disuse adds stiffness and wasting
Activity limitations	Cannot lift the arm to shelves, carry goods, comb his hair, bring food to his mouth with the right hand, or dress easily	Shoulder elevation and elbow flexion are needed for these tasks; the hand itself works
Participation restrictions	Difficulty running his shop; family income at risk; social roles reduced	Shop work needs repeated reaching and lifting
Environmental factors	**Facilitators:** family support; his own shop (he can rearrange it). **Barriers:** high shelves; heavy goods; cost of surgery and travel	Rearranging the shop lets him keep working while he recovers
Personal factors	35-year-old man, sole earner with a wife and 2 children; worried about income	Young age favours nerve recovery; financial pressure may push an early return to work

One-line functional diagnosis

A 35-year-old shopkeeper with right Erb's palsy (C5–C6), 5 months after a road accident, with paralysis of shoulder abductors, external rotators and elbow flexors, a stiffening shoulder and sensory loss on the outer arm, who cannot reach or lift at work, putting his family's income at risk.

Practice cases from the papers: S22 21-year-old with a lower plexus traction injury after a fall from a tree (identify the level, splints); S23 23-year-old who fell from a train, with a clavicle fracture and a paralysed right arm.

Recall check. Fill the ICF for the S23 man who cannot move his right arm after a fall from a train. Which test tells you if the roots are torn out?

11 Problem list and goals

Goals for the shopkeeper with Erb's palsy

Problem	Short-term (4–6 weeks)	Long-term (6–12 months)
Shoulder range and subluxation	Full passive shoulder range; no pain; supported arm	No contracture; stable shoulder
Shoulder and elbow strength	Flicker progresses to movement with gravity eliminated (if recovering)	Biceps and deltoid grade 4; lifts goods to chest height
Sensation	Protects numb skin; no injuries	Useful protective sensation
Daily tasks and work	Uses the right hand for tasks at waist height; shop rearranged	Runs the shop independently
Surgical decision	Referred to a plexus surgeon now if there is no sign of recovery	Rehabilitation after nerve or muscle transfer if done

12 PT management

Principles: protect and support the denervated part; keep joints mobile; keep the skin safe; re-educate each muscle as it is reinnervated; retrain sensation; return the person to work.

By phase

- **Denervation (protective) phase:** splints in a functional position (cock-up splint, AFO, sling or shoulder support); daily passive range of motion, gently, to every joint; prevent overstretching of paralysed muscles; oedema control; skin protection; strengthening of muscles that work; education about the expected recovery time.
- **Reinnervation phase** (Tinel's sign advancing, first flicker of movement): muscle re-education one muscle at a time, with EMG biofeedback or electrical stimulation as a cue; gravity-eliminated active-assisted, then active, then resisted exercise; short sessions to avoid fatigue; stop trick movements.
- **Functional phase:** grip and pinch strengthening; task practice; work simulation; sensory re-education; endurance.
- **After nerve transfers:** start by contracting the donor muscle to fire the new target (for example squeezing the fingers to bend the elbow after an ulnar fascicle transfer), then practise the new movement on its own.

Sensory re-education

1. **Protective stage:** teach the person to look at the hand while using it, test bath water with the other hand, and avoid sharp and hot objects.
2. **Desensitisation** for hypersensitive skin: graded textures (cotton, then towel, then rougher), tapping, vibration.
3. **Discriminative stage:** localise touch with the eyes open, then closed; recognise textures, shapes and everyday objects without looking.

Splints for nerve injuries (3 marks in W17, S25 and W25)

Splint	Used for	How it works
Static cock-up splint	Radial palsy (rest and night); carpal tunnel syndrome (in neutral); wrist fractures and arthritis	Holds the wrist in about 20–30° of extension (neutral for carpal tunnel), leaving the fingers free
Dynamic cock-up (lively) splint	Radial palsy, for daytime function	Supports the wrist, and an outrigger with elastic slings lifts the fingers and thumb at the knuckles; the person bends the fingers actively and the elastic straightens them again, so grasp and release are possible
Knuckle-bender (anti-claw) splint	Ulnar or median-ulnar claw hand; Klumpke's palsy	Blocks knuckle hyperextension, so the long extensors can straighten the finger joints
Thumb web spacer or opponens splint	Median palsy	Keeps the thumb in opposition and the first web space open
AFO (posterior leaf spring)	Foot drop from a peroneal or sciatic lesion	Lifts the foot in swing and prevents the toes catching
Shoulder sling or support	Flail arm in plexus injury	Reduces subluxation and traction on the plexus

Dynamic cock-up splint: uses and principle (S25)

Uses: radial nerve palsy (wrist, finger and thumb drop), after extensor tendon repair, and after tendon transfers for radial palsy. **Principle:** the splint replaces the paralysed extensors. A static wrist support keeps the wrist extended so the flexors grip strongly, and elastic slings on an outrigger return the fingers and thumb to extension after each active grasp. It allows function, prevents overstretching of the extensors and stops the flexors shortening.

Specific lesions

- **Radial palsy (7 marks in S18 and W18):** static cock-up at night and a dynamic splint by day; range of motion of the wrist and fingers; stop the flexors shortening; re-educate wrist and finger extensors as recovery comes (brachioradialis and wrist extensors first, then finger and thumb extensors); grip activities with the wrist supported.
- **Claw hand (7 marks in W19):** knuckle-bender splint; passive stretching of the knuckle joints into flexion; intrinsic re-education (finger spreading, "table-top" position); protect the numb little finger; tendon transfer if no recovery.
- **Foot drop (3 marks in S16) and peroneal nerve injury (7 marks in W25):** AFO; daily calf stretching; dorsiflexor and evertor re-education as recovery comes, with functional electrical stimulation if available; gait training to replace the high-stepping pattern; protect the numb top of the foot; tendon transfer if there is no recovery.
- **Erb's palsy (7 marks in S17, S21, W21 and W24):** support the arm; keep full shoulder range, especially external rotation and abduction, and supination; prevent subluxation; re-educate deltoid, spinati, biceps and supinator in gravity-eliminated positions first (lying, a powder board, a sling suspension); biofeedback; functional reaching; after surgery, follow the surgeon's protocol.
- **Klumpke's palsy (7 marks in S25):** anti-claw splint; range of the fingers and wrist; intrinsic and long flexor re-education; sensory protection and re-education of the inner hand; check for Horner's syndrome.

Carpal tunnel syndrome (7 marks in S22)

- **Neutral wrist splint at night** for short-term relief (moderate evidence); day or full-time wear if night wear alone fails.
- Education: pathology, risk factors, avoiding sustained wrist bending and forceful gripping, keyboard and mouse habits.

- Manual therapy to the neck and upper limb, a trial of superficial heat, interferential current or diathermy, and splinting with stretching may help in mild to moderate cases (weak evidence).
- **Do not use** low-level laser, iontophoresis or magnets (moderate evidence) or thermal ultrasound (weak evidence). Nerve gliding has conflicting evidence.
- Refer for injection or surgery if there is thenar wasting, constant numbness or no response.

Recommendations from the APTA carpal tunnel guideline (2019). The guideline was revised in 2026; check the new edition for any changes.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Joint stiffness	Passive range of motion of every affected joint, gently, **every day**; splints in a functional position	Paralysed limbs stiffen within weeks
Overstretched paralysed muscles	Cock-up splint, AFO or sling, worn as advised	An overstretched muscle recovers less well when the nerve returns
Early reinnervation (flicker)	Isolated contractions with biofeedback; gravity-eliminated, few repetitions, short sessions	Newly reinnervated muscles tire quickly and need exact practice
Weakness (grade 3 or more)	Progressive resistance and functional tasks	Builds strength for work and daily tasks
Numb or hypersensitive skin	Protective education; desensitisation; discriminative sensory training	The brain must learn to read the new sensory signals
Foot drop	AFO; calf stretching; dorsiflexor re-education; gait training	Prevents tripping and equinus contracture

Recall check. Which splint would you give for wrist drop, claw hand and foot drop, and why?

13 Precautions, contraindications and stop criteria

Take care

- No heat, ice or electrical treatment over numb skin without close checks; teach skin inspection.
- After nerve repair, protect the repair as the surgeon advises (no stretch on the nerve for the first weeks).
- Check splints for pressure points over bony areas and numb skin.
- Do not overstretch paralysed muscles or a subluxing shoulder; no traction on a flail arm.
- Avoid overworking newly reinnervated muscles.
- Watch for signs of CRPS (burning pain, swelling, colour change).

14 Education, lifestyle and home programme

- Explain how slowly nerves regrow (about 2.5 cm a month) so the person keeps exercising for months.
- Daily home range-of-motion and strengthening programme; splint wearing and care.
- Skin protection: check the skin, use gloves and long handles in the kitchen, test water with the other hand.
- Work: modify tasks and tools; rearrange the workplace; plan a staged return.
- Carpal tunnel: posture at work, breaks, tool and keyboard changes.

15 Prognosis and discharge criteria

- **Better outcome:** neurapraxia or low-degree injury; young age; a distal injury close to the muscles; a clean cut repaired early; a pure motor or pure sensory nerve.
- **Worse outcome:** crush or traction injuries over a long segment; root avulsion; old age; long delay before repair; a proximal injury far from the hand muscles.
- In brachial plexus injuries, the upper plexus recovers better than the lower, and avulsed roots do not recover without transfer.
- Discharge when recovery has plateaued, the person uses the limb in daily life and work, and knows the home programme and skin care.

16 Exam corner

Past questions (MUHS, 2016–2026)

S23 Q3(a), 7 + 8 marks

23-year-old man who fell while hanging out of a train a month ago, with a clavicle fracture treated conservatively; since then he cannot move his right arm and has sensory loss. ICF and physiotherapy.

S22 Q4(b), 2 + 6 + 7 marks

21-year-old man who fell from a tree onto an outstretched hand 3 weeks ago (traction injury); cannot make a fist, straighten his elbow or move his thumb; sensory loss on the inner arm and forearm; power below grade 2. Which part of the plexus; goals; physiotherapy with orthoses.

W16 Q3(a), 7 + 8 marks

The worked example in section 10.

7-mark questions

Sunderland's classification and physiotherapy for foot drop (S16); motor re-education in Erb's palsy (S17); radial nerve injury (S18); radial nerve palsy: features and physiotherapy (W18); claw hand (W19); Erb's palsy: features and physiotherapy (S21), management (W21), physiotherapy (W24); carpal tunnel syndrome (S22); Klumpke's paralysis (S25); common peroneal nerve injury (W25).

3-mark questions

Orthosis for radial nerve injury (W17); classification of nerve injuries (S21, S23, S25); Sunderland's classification (W21, S22); dynamic cock-up splint: uses and principle (S25); cock-up splint (W25).

Pages for this topic: obstetric brachial plexus injury (p. 309), neuropathies and nerve root lesions (p. 107), orthoses and splints (p. 249), electrotherapy (p. 204).

Keywords to underline

Wallerian degeneration neurapraxia, axonotmesis, neurotmesis Sunderland degree 1 mm a day Tinel's sign wrist drop claw hand foot drop waiter's tip root avulsion dynamic splint sensory re-education

Answer skeleton: a nerve or plexus long case

1. Restate: age, job, dominant side, mechanism, time since injury; name the nerve or plexus level and the likely degree.
2. **[ICF, 5–7]** ICF diagram and table with clinical reasoning; one-line functional diagnosis.
3. **[Goals, 3–6]** Short-term: range, protection, splints, first active movement; long-term: strength, hand use, return to work; refer for surgery on time.

4. **[Management, 7–8]** Phase-wise: protection and splints; passive range; re-education with biofeedback; strengthening; sensory re-education; function and work; precautions.

Answer: which part of the plexus in the S22 case (2 marks)

The **lower plexus, C7–T1** (middle and lower trunks): cannot make a fist (finger flexors, C8), cannot straighten the elbow (triceps, C7) or move the thumb (thumb muscles, C8–T1), with sensory loss on the inner arm (T1) and inner forearm (C8). A fall from a tree onto an outstretched, raised arm pulls on the lower roots. Check for Horner's syndrome, which suggests T1 root avulsion.

Draw these

- The brachial plexus with Erb's and Klumpke's levels (section 2)
- The ICF model filled from the case
- A dynamic cock-up splint (wrist support with an outrigger and finger slings)

Viva questions

1. Give Sunderland's classification.
2. How fast do peripheral nerves regrow?
3. What is the ulnar paradox?
4. Why is the sensory action potential preserved in root avulsion?
5. What is "waiter's tip" posture?
6. Which splint do you give for wrist drop, and how does it work?

Linked practical spots

- Static and dynamic cock-up splints, knuckle-bender, AFO; Froment's sign, card test, Phalen's test.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write Sunderland's classification of nerve injury.**

Answer

I: conduction block, myelin damage (neurapraxia). II: axon cut, endoneurium intact. III: axon and endoneurium cut, perineurium intact. IV: only the epineurium intact (neuroma in continuity). V: complete division (neurotmesis). Degrees I and II recover fully; III partly; IV and V need surgery.

3 m **Describe the orthosis used for radial nerve injury.**

Answer

A static cock-up splint holding the wrist in about 20–30° of extension (rest and night), and a dynamic (lively) cock-up splint with an outrigger and elastic finger slings for daytime function: the person grips actively and the elastic straightens the fingers and thumb again.

7 m **Describe Erb's palsy and its physiotherapy.**

Answer

Features: C5–C6 upper trunk traction injury; loss of shoulder abduction and external rotation, elbow flexion and supination; "waiter's tip" posture; sensory loss over the outer arm and forearm; biceps reflex lost.

Physiotherapy: support the arm and prevent subluxation; daily full passive range, especially external rotation, abduction and supination; protect numb skin; re-educate deltoid, spinati, biceps and supinator in gravity-eliminated positions with biofeedback as recovery comes, then active and resisted; functional reaching and lifting; refer for nerve transfer if no recovery by about 3–6 months; rehabilitation after surgery.

LAQ **A 21-year-old fell from a tree onto an outstretched hand 3 weeks ago. He cannot make a fist, straighten his elbow or move his thumb, with sensory loss on the inner arm and forearm. Identify the part of the plexus, write goals and physiotherapy with orthoses. (2 + 6 + 7, S22)**

Answer

Level (2): lower plexus, C7–T1 (middle and lower trunks).

Goals (6): short-term: full range of fingers, wrist and elbow; no claw contracture; protected numb skin; splints worn correctly. Long-term: active grip and release; elbow extension against gravity; useful hand for daily tasks and study or work.

Management (7): anti-claw (knuckle-bender) splint and a resting hand splint at night; passive range of motion daily; oedema control; sensory protection; check Horner's sign and refer for nerve studies and surgical opinion; as recovery comes, re-educate triceps, finger flexors and intrinsics with biofeedback; grip and pinch training; sensory re-education.

MCQ **Wrist drop after a fracture of the shaft of the humerus is due to injury of the:**

A. Median nerve

B. Ulnar nerve

C. Radial nerve

D. Axillary nerve

Answer

C. The radial nerve runs in the spiral groove of the humerus.

MCQ **Froment's sign tests the:**

A. Adductor pollicis (ulnar nerve)

B. Abductor pollicis brevis (median nerve)

C. Extensor pollicis longus (radial nerve)

D. Deltoid (axillary nerve)

Answer

A. With the adductor weak, the person bends the thumb tip (flexor pollicis longus) to hold the paper.

MCQ **For carpal tunnel syndrome, which treatment has moderate evidence for short-term relief?**

A. Low-level laser

B. Magnets

C. Neutral wrist splint at night

D. Iontophoresis

Answer

C. The others should not be used (APTA 2019).

Last recall check. Draw the brachial plexus, write Sunderland's degrees, and list the deformity and splint for each major nerve.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Erickson M, et al. Hand pain and sensory deficits: carpal tunnel syndrome. Clinical practice guidelines from the APTA. *J Orthop Sports Phys Ther* 2019;49:CPG1–85 (revised 2026).
3. Martin E, et al. Timing of surgery in traumatic brachial plexus injury: a systematic review. *J Neurosurg* 2018 (online ahead of print).
4. Seddon HJ. Three types of nerve injury. *Brain* 1943;66:237–88. Sunderland S. A classification of peripheral nerve injuries producing loss of function. *Brain* 1951;74:491–516.
5. Structure and classical content: Lundborg, *Nerve Injury and Repair*; Stokes, *Physical Management for Neurological Conditions*; Kisner and Colby; standard orthopaedic texts for nerve lesions.

5A(d) Peripheral neuropathies: diabetic foot, leprosy, alcoholic neuropathy, neuropathic pain and CRPS

> These neuropathies came up in 9 of the 20 papers, mostly as 7-mark questions on the diabetic foot (four times), plus long cases on diabetic neuropathy and leprosy. Every answer protects a foot or hand that can no longer feel: test sensation, prevent ulcers and deformity, keep the person active safely, and treat pain with movement as well as medicine.

Syllabus	Ch 5A(d) Peripheral neuropathies, non-traumatic: metabolic and endocrine; nerve root lesions (traumatic nerve injuries have their own page)
Priority	**Tier 2.** In 9 of 20 papers (2016–2026): 2 long cases and 8 seven-mark questions.
Asked as	LAQ S16 · W25 · 7m W17 · W19 · S21 · W21 · W22 ×2 · S23 · S25
Usual split	15 (investigations 3, goals 5, management 7 in S16; diagnosis with reasoning 5 and goals with management 10 in W25); 7-mark classification with diabetic foot care
Checked against	IWGDF practical guidelines on the diabetic foot (2023); ADA position statement on diabetic neuropathy (2017); ICMR-INDIAB study (2023); WHO leprosy guidelines (2018) and 2023 leprosy data; reviews of CRPS rehabilitation and the Budapest criteria. Full list under Sources (p. 115). Content checked 20 October 2026.

1 Diabetic neuropathy and the diabetic foot

Asked for 7 marks in W17, W19, W22 and S23 (classification and diabetic foot care) and as a long case in S16. India has about **101 million** adults with diabetes (11.4%; ICMR-INDIAB, 2023), so this is a daily problem in practice.

Definition and types (ADA, 2017)

Group	Types
Diffuse neuropathy	**Distal symmetric polyneuropathy** (the commonest: "glove and stocking"; small-fibre, large-fibre or mixed); **autonomic neuropathy** (heart rate and blood pressure, stomach and bowel, bladder and sexual function, sweating)
Mononeuropathy	A single cranial or limb nerve (for example the third cranial nerve, median, ulnar or peroneal nerve); several nerves (mononeuritis multiplex)
Radiculopathy	Lumbosacral radiculoplexus neuropathy (diabetic amyotrophy: painful thigh weakness and wasting); thoracic radiculopathy

Causes and key features

- Long-standing, poorly controlled blood sugar damages nerves and their blood supply; the longest nerves fail first, so symptoms start in the toes and rise.
- **Sensory:** numbness, tingling, burning pain, "walking on cotton wool"; loss of protective sensation; poor position sense, so balance worsens in the dark.
- **Motor:** weak intrinsic foot muscles, so the toes claw, slippers slip off unnoticed and pressure rises under the metatarsal heads.
- **Autonomic:** dry, cracked skin (no sweating); orthostatic hypotension; resting tachycardia.
- **The diabetic foot:** neuropathy plus peripheral artery disease plus deformity leads to ulcers, infection and amputation. A hot, swollen foot without pain may be a **Charcot foot** (neuropathic joint destruction).

Medical point that matters for physiotherapy

- Good control of blood sugar, blood pressure and lipids; stop smoking.

- Investigations (S16, 3 marks): blood sugar and HbA1c; the **10 g monofilament** test and a 128 Hz tuning fork; nerve conduction studies; foot pulses and the ankle-brachial index; vitamin B12 (low on metformin), kidney and thyroid tests.
- Ulcers: offloading with a total contact cast or a non-removable knee-high walker for plantar ulcers; infection and artery disease need urgent specialist care.
- Ulcer grading: Wagner grades 0–5 (0 at-risk foot, 1 superficial ulcer, 2 deep ulcer, 3 with abscess or bone infection, 4 gangrene of part of the foot, 5 gangrene of the whole foot).

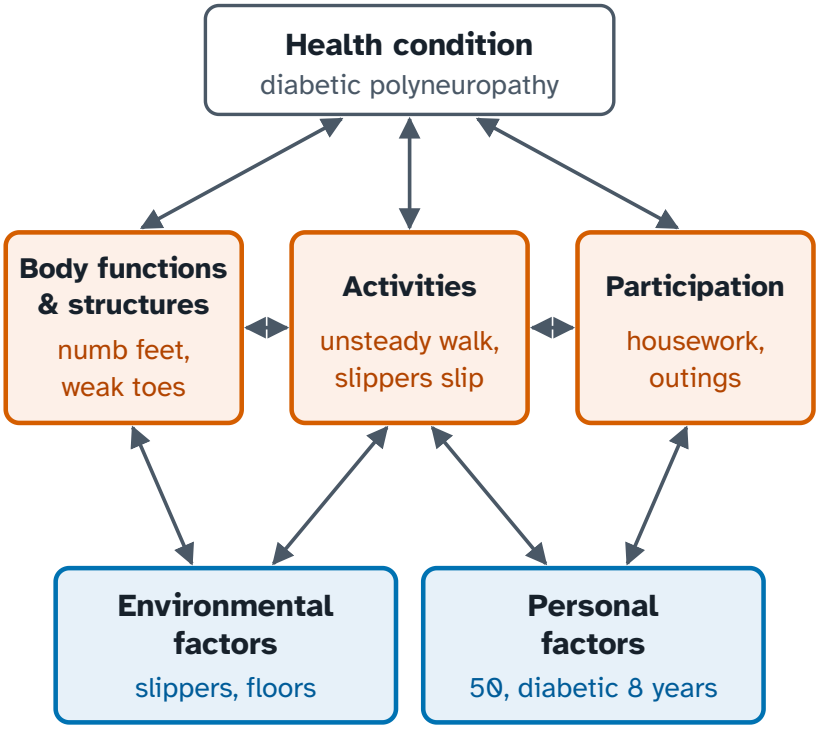
Foot risk categories and screening (IWGDF, 2023)

Category	Findings	Screen
0, very low	No loss of protective sensation (LOPS), no peripheral artery disease (PAD)	Once a year
1, low	LOPS or PAD	Every 6–12 months
2, moderate	LOPS and PAD, or either with foot deformity	Every 3–6 months
3, high	LOPS or PAD with a past ulcer, an amputation or end-stage kidney disease	Every 1–3 months

ICF summary: the S16 case

> "A 50 year old female is a known case of diabetes since last 8 years. She complains of numbness in bilateral lower extremities and while walking she feels as if she is walking on cotton. She also complains of inability to hold her footwear (slippers) while walking. Symptoms are gradually progressing. Plan investigations. Plan short term goals and long term goals. Write physiotherapy management for her." (3 + 5 + 7)

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

- **Impairments:** loss of protective sensation and position sense in both feet; weak intrinsic and toe muscles (cannot grip slippers); possible clawing and high-pressure areas; poor balance.
- **Activities:** unsteady walking, worse in the dark and on uneven ground; footwear falls off; risk of falls.
- **Participation:** housework, market and religious outings, family roles.
- **Environment:** loose slippers and barefoot walking at home are barriers; family and diabetes clinic are facilitators.

- **Functional diagnosis:** a 50-year-old woman with progressive diabetic distal symmetric polyneuropathy, with loss of protective sensation and weak foot muscles, causing unsteady walking and a high risk of falls and foot ulcers.

Physiotherapy by key problem

Problem	Technique and dose	Why
Ulcer risk	Daily foot inspection with a mirror; proper-fitting protective footwear (no barefoot walking, socks only or thin slippers); callus and nail care; screening by risk category	The person cannot feel injuries or high pressure
Weak foot and ankle, stiff joints	Foot-ankle exercise programme (toe flexion and extension, ankle range and strength), preferably supervised	Improves range and strength and may lower plantar pressure (IWGDF: may be considered)
Poor balance	Static and dynamic balance training with vision, then on varied surfaces; safe home lighting	Vision and training compensate for lost position sense; reduces falls
Low activity and poor glucose control	Walking and other weight-bearing activity in proper footwear, building up to an **extra 1000 steps a day**; non-weight-bearing exercise (cycling, swimming) if an ulcer is present	Foot risk is not a barrier to training if footwear is right (IWGDF 2023); activity improves glucose control
Burning pain	Graded activity, education; TENS may be tried; medicines from the doctor	See neuropathic pain below
Ulcer present	Offloading (total contact cast or non-removable walker), crutches or a wheelchair; keep the rest of the body active	Pressure relief is the key to healing

Precautions

- No heat, ice or electrical treatment over numb feet without close checks.
- Check the feet after every exercise session; stop weight-bearing exercise if a new red area or blister appears.
- Check blood sugar before exercise; carry sugar for hypoglycaemia; watch orthostatic hypotension.
- A hot, swollen foot: suspect Charcot foot or infection and refer the same day.

Exam points

- Classify (diffuse, mononeuropathy, radiculopathy), then give diabetic foot care: inspection, footwear, skin and nail care, exercise, offloading, education.
- Name the 10 g monofilament and the IWGDF risk categories.

2 Leprosy neuropathy

Asked as a long case in W25 and for 7 marks (leprous neuritis and lepromatous leprosy) in W22. India reported about 59% of the world's new cases in 2023 (107,851 of 182,815).

Definition and types

- A chronic infection with *Mycobacterium leprae* that affects the skin and peripheral nerves.
- **WHO groups for treatment:** paucibacillary (1–5 skin lesions, smear negative) and multibacillary (more than 5 lesions, nerve involvement or smear positive).
- **Ridley–Jopling spectrum:** tuberculoid, borderline tuberculoid, mid-borderline, borderline lepromatous, lepromatous.

Causes and key features

- Hypopigmented or reddish skin patches with loss of sensation; **thickened nerves** (ulnar at the elbow, common peroneal at the fibular neck, posterior tibial, median, facial, great auricular).
- **Deformities:** claw hand (ulnar and median), wrist drop (rarely radial), foot drop (peroneal), clawed toes and plantar ulcers (posterior tibial), lagophthalmos (facial).

- **Lepromatous leprosy** (W22): many symmetrical lesions and nodules, thickened facial skin ("leonine face"), loss of eyebrows, a collapsed nose, glove-and-stocking anaesthesia, symmetrical nerve thickening, eye damage; skin smears strongly positive.
- **Leprous neuritis** (W22): a painful, tender, swollen nerve with new loss of sensation or power, often during a reaction (type 1, reversal reaction; or type 2, erythema nodosum leprosum); "silent neuritis" causes loss without pain.
- **WHO disability grade:** 0, no anaesthesia or visible damage; 1, anaesthesia without visible deformity; 2, visible deformity or damage (or severe eye problems).

Medical point that matters for physiotherapy

- WHO (2018) multidrug therapy for all patients: rifampicin, dapsone and clofazimine, for **6 months** (paucibacillary) or **12 months** (multibacillary). Contacts may receive single-dose rifampicin.
- Neuritis is treated with a course of oral steroids; the limb is rested during acute neuritis.
- Reconstructive surgery (tendon transfers for claw hand, foot drop and lagophthalmos) once the disease and neuritis are controlled.

ICF summary: the W25 case

> "A 42 year old labourer comes to the PT department with complaints of loss of sensation and power in both hands since past one month. On observation some red coloured patches were noted on his skin. He stays in a slum area with his wife and two children and is the only earning member. Explain the diagnosis with clinical reasoning. Plan short term and long term goals with physiotherapy management." (5 + 10)

- **Diagnosis with reasoning:** leprosy, probably multibacillary (nerve involvement in both hands), with ulnar and median neuropathy. Reasoning: skin patches with sensory loss; weakness and numbness of both hands; nerve thickening to be checked at the elbows and wrists; living in a crowded area. Confirm with skin smear and nerve examination. Also rule out other neuropathies (diabetic, hereditary) and cervical radiculopathy.
- **Impairments:** hand intrinsic weakness, early clawing, sensory loss in both hands, dry skin; possible acute neuritis.
- **Activities and participation:** difficulty gripping tools and lifting; risk of burns and cuts; cannot work as a labourer; family income at risk; stigma.
- **Environment:** free multidrug therapy and government leprosy services (facilitators); slum living, manual work with hot or sharp tools, stigma (barriers).

Physiotherapy by key problem

Problem	Technique	Why
Monitoring nerve function	Voluntary muscle testing and sensory testing (monofilaments) of hands, feet and eyes at diagnosis and at every visit	Detects new nerve damage early so steroids can be started
Acute neuritis	Rest the limb in a splint in a functional position until pain settles; no exercise of the inflamed nerve; then gentle range and graded exercise	Movement over an inflamed nerve can increase damage
Claw hand	Knuckle-bender splint; passive stretching of the fingers; intrinsic exercises; prepare for and rehabilitate after tendon transfer	Prevents fixed contractures and keeps the hand useful
Dry, numb skin	Self-care: soak the hands and feet in water, scrape thick skin, then oil; daily inspection; padded tool handles and gloves	Prevents cracks, burns and ulcers
Foot drop and plantar anaesthesia	AFO; protective microcellular rubber footwear; walking short distances; ulcer care	Prevents trips and plantar ulcers
Work and stigma	Tool changes, work planning, vocational training, family counselling	Keeps the sole earner working; reduces isolation

Precautions

- Do not exercise an acutely painful, tender nerve; refer for steroids.
- Protect numb skin from heat (cooking, smoking, hot tools) and from splint pressure.
- Leprosy is not spread by touching; treated patients quickly become non-infectious.

Exam points

- Skin patch with lost sensation plus thickened nerves equals leprosy until proven otherwise.
- Physiotherapy answers: nerve function monitoring, neuritis rest, splints, self-care (soak, scrape, oil), footwear, surgery rehabilitation, stigma.

3 Alcoholic and other neuropathies

Asked for 7 marks in S25.

Definition and key features

- A length-dependent, mainly axonal sensorimotor polyneuropathy from long-term heavy drinking, made worse by poor nutrition (especially thiamine deficiency).
- Burning, painful feet; numbness in a glove-and-stocking pattern; lost ankle reflexes; sensory ataxia (worse in the dark); distal weakness and foot drop in severe cases; muscle wasting; autonomic problems.
- Other causes of polyneuropathy: vitamin B12 deficiency, chronic kidney disease, hypothyroidism, drugs (isoniazid without pyridoxine, some chemotherapy), HIV, hereditary neuropathy (Charcot–Marie–Tooth).

Medical point

Stop alcohol; thiamine and other vitamins; a good diet; medicines for nerve pain. Recovery is slow and often partial.

Physiotherapy by key problem

- **Sensory ataxia and falls:** balance training with visual feedback; a stick; good lighting; Frenkel's exercises.
- **Distal weakness:** progressive strengthening of ankle and hand muscles; AFO for foot drop.
- **Pain:** graded activity; TENS trial; education.
- **Foot protection:** daily inspection and footwear, as for the diabetic foot.
- **General fitness** and support to stay off alcohol.

Precautions

- Watch for withdrawal, low blood sugar and liver disease; protect numb skin.

4 Neuropathic pain

Asked for 7 marks in W21.

- **Definition (IASP):** pain caused by a lesion or disease of the somatosensory nervous system.
- **Features:** burning, shooting or electric pain; pins and needles; pain from light touch (allodynia); more pain from a mildly painful stimulus (hyperalgesia); numbness in the same area.
- **Causes:** diabetic neuropathy, post-herpetic neuralgia, trigeminal neuralgia, nerve injury, radiculopathy, spinal cord injury, stroke, multiple sclerosis, phantom limb pain.
- **Medicines** (prescribed by the doctor): first-line gabapentin or pregabalin, duloxetine, or amitriptyline.

Physiotherapy

- Explain pain in plain words: hurt does not always mean harm; nerves can become over-sensitive.
- Graded activity and regular exercise, with pacing to avoid boom-and-bust.
- Desensitisation for allodynia: graded textures and pressure, from tolerable to less tolerable.

- Sensory discrimination training, graded motor imagery and mirror therapy for CRPS and phantom pain.
- TENS may be tried for short-term relief; relaxation and sleep advice.

5 Complex regional pain syndrome (reflex sympathetic dystrophy)

Asked for 7 marks in S21.

Definition and types

- A painful condition of a limb, usually after an injury, surgery or immobilisation (often a wrist fracture), in which the pain is out of proportion to the injury and the limb shows sensory, vasomotor, sweating, swelling, motor and trophic changes.
- **Type I** (formerly reflex sympathetic dystrophy): no definite nerve injury. **Type II** (formerly causalgia): with a nerve injury.

Key features and diagnosis (Budapest criteria)

- Continuing pain disproportionate to the injury.
- At least one **symptom** in three of four groups, and at least one **sign** in two of four groups: sensory (allodynia, hyperalgesia); vasomotor (temperature or colour change); sudomotor and swelling (oedema, sweating change); motor and trophic (stiffness, weakness, tremor, hair, nail or skin changes).
- No other diagnosis explains the picture.

Physiotherapy

- Encourage gentle movement and functional use of the limb as early as possible; avoid prolonged immobilisation.
- Graded motor imagery (left/right recognition, imagined movement, then mirror therapy); evidence suggests possible benefit.
- Desensitisation and tactile discrimination training.
- Oedema control: elevation, active pumping exercises, retrograde massage or compression if tolerated.
- Graded exercise and return to normal activities and work; pacing.
- Work with the team: medicines, psychology, occupational therapy.

Precautions

- Do not force painful passive stretching; it can flare symptoms.
- Never splint for long periods without movement.

6 Nerve root lesions (radiculopathy)

- **Causes:** disc prolapse, spondylosis with foraminal narrowing, tumours, herpes zoster; diabetic radiculopathy.
- **Features:** pain along a dermatome (sciatica, brachialgia), sensory change in the dermatome, weakness of muscles in the myotome, and reduced reflexes at that level (for example C6: biceps and brachioradialis reflexes; L5: big toe extension; S1: ankle reflex).
- **Peripheral nerve vs root:** a root lesion follows a dermatome and myotome; a nerve lesion follows the nerve's own area. A root lesion may weaken muscles supplied by different nerves from the same root.
- **Red flags:** bladder or bowel loss and saddle numbness (cauda equina), progressive weakness: refer urgently.
- **Physiotherapy:** education, staying active, directional preference exercises, neural mobilisation, traction where appropriate, strengthening, ergonomic advice.

7 Exam corner

Past questions (MUHS, 2016–2026)

W25 Q4(b), 5 + 10 marks

42-year-old labourer with loss of sensation and power in both hands for a month and red patches on the skin; lives in a slum with his wife and two children and is the only earner. Diagnosis with clinical reasoning; goals with physiotherapy (section 2).

S16 Q4(b), 3 + 5 + 7 marks

50-year-old diabetic woman with numb feet, "walking on cotton" and slippers that slip off (section 1).

7-mark questions

Physiotherapy of the diabetic foot (W17); types of diabetic neuropathy and diabetic foot care (W19); diabetic neuropathy classification and physiotherapy for the diabetic foot (W22, S23); leprous neuritis and features of lepromatous leprosy (W22); physiotherapy for reflex sympathetic dystrophy (S21); physiotherapy for neuropathic pain (W21); alcoholic neuropathy and its physiotherapy (S25).

Pages for this topic: nerve injuries and splints (p. 94), Guillain-Barré syndrome (p. 144), orthoses (p. 249), electrotherapy (p. 204).

Keywords to underline

glove and stocking · loss of protective sensation · 10 g monofilament · IWGDF risk category · offloading · protective footwear · thickened nerves · neuritis · multidrug therapy · soak, scrape, oil · allodynia · graded motor imagery

Answer skeleton: diabetic neuropathy and diabetic foot care (7 marks)

1. **[Classification, 2]** Diffuse (distal symmetric, autonomic), mononeuropathy, radiculopathy.
2. **[Assessment, 1]** Monofilament, tuning fork, pulses, deformity, skin; IWGDF risk category.
3. **[Physiotherapy, 4]** Daily inspection and foot hygiene; protective footwear and no barefoot walking; foot-ankle exercises and balance training; graded walking in proper footwear (extra 1000 steps a day); offloading for ulcers; education and screening by risk; precautions with heat and blood sugar.

Draw these

- The ICF model filled from the S16 case (section 1)
- A foot showing monofilament test sites on the sole and common ulcer sites (metatarsal heads, heel, toe tips)

Viva questions

1. Why does diabetic neuropathy start in the feet?
2. What does the 10 g monofilament test?
3. What is a Charcot foot?
4. Which nerves are thickened in leprosy?
5. What is "soak, scrape and oil"?
6. What are the Budapest criteria for CRPS?

Linked practical spots

- Monofilament and tuning fork; microcellular rubber footwear; knuckle-bender splint; AFO.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **List three types of diabetic neuropathy.**

Answer
Distal symmetric polyneuropathy; autonomic neuropathy; mononeuropathy (single or multiple nerves); radiculopathy such as diabetic amyotrophy.

3 m **Write three features of lepromatous leprosy.**

Answer
Many symmetrical skin lesions and nodules with a thickened "leonine" face; loss of eyebrows and a collapsed nose; glove-and-stocking anaesthesia with symmetrical nerve thickening; strongly positive skin smears.

7 m **Physiotherapy management of the diabetic foot.**

Answer
Assess sensation (10 g monofilament, tuning fork), pulses, deformity, skin, footwear and the IWGDF risk category. Teach daily inspection with a mirror, washing and drying between the toes, moisturising, nail care. Prescribe protective footwear; no barefoot walking, socks only or thin slippers. Foot-ankle exercises and balance training. Graded walking in proper footwear, adding about 1000 steps a day; non-weight-bearing exercise if an ulcer is present. Offloading (total contact cast or non-removable walker) for plantar ulcers. Precautions: no heat on numb feet, check blood sugar, refer a hot swollen foot.

LAQ **A 50-year-old diabetic woman has numb feet, feels she walks on cotton and cannot keep her slippers on. Plan investigations, goals and physiotherapy. (3 + 5 + 7, S16)**

Answer
Investigations (3): fasting sugar and HbA1c; monofilament and tuning fork; nerve conduction studies; foot pulses and ankle-brachial index; vitamin B12, kidney and thyroid tests.

Goals (5): short-term: daily foot care routine, proper footwear, safer walking with balance training, no new skin damage. Long-term: no ulcers or falls, active lifestyle with walking every day, good sugar control.

Physiotherapy (7): education and self-inspection; protective footwear with a back strap; foot-ankle strengthening (toe grip, ankle range); balance training; graded walking programme; home safety and lighting; pain management if needed; screening every 6–12 months (IWGDF risk 1).

MCQ **The standard screening tool for loss of protective sensation in the diabetic foot is:**

A. Pinwheel

B. 10 g Semmes–Weinstein monofilament

C. Two-point discriminator

D. Reflex hammer

Answer
B. A 128 Hz tuning fork or the Ipswich touch test can be used when it is not available.

MCQ **The WHO (2018) treatment for multibacillary leprosy is:**

A. Dapsone alone for 6 months

B. Rifampicin, dapsone and clofazimine for 12 months

C. Rifampicin and dapsone for 6 months

D. Steroids only

Answer

B. Paucibacillary leprosy gets the same three drugs for 6 months.

MCQ **In acute leprous neuritis, the first physiotherapy step is:**

A. Vigorous stretching

B. Resting the limb in a splint and referring for steroids

C. Resisted exercise of the affected muscles

D. Heat over the nerve

Answer

B. Exercise over an inflamed nerve can increase damage.

Last recall check. Write the IWGDF risk categories, the leprosy self-care routine and the Budapest criteria groups.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. International Working Group on the Diabetic Foot. IWGDF practical guidelines on the prevention and management of diabetes-related foot disease (2023 update).
3. Pop-Busui R, et al. Diabetic neuropathy: a position statement by the American Diabetes Association. *Diabetes Care* 2017;40:136–54.
4. Anjana RM, et al. Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17). *Lancet Diabetes Endocrinol* 2023;11:474–89.
5. World Health Organization. Guidelines for the diagnosis, treatment and prevention of leprosy (2018); global leprosy update for 2023.
6. Harden RN, et al. Validation of proposed diagnostic criteria (the "Budapest Criteria") for complex regional pain syndrome. *Pain* 2010;150:268–74. Reviews of CRPS rehabilitation (graded motor imagery, mirror therapy).
7. Structure and classical content: ILEP and national leprosy programme manuals on prevention of disability; standard neurology texts on polyneuropathy.

5A(e) Vestibular disorders: tests, BPPV manoeuvres and vestibular rehabilitation

> Vestibular disorders were a 7-mark question in 8 papers: BPPV and its manoeuvres four times, vestibular rehabilitation twice, plus vestibular tests and the exercise types. The answers rest on one idea: the brain can recalibrate the vestibular system if the person moves the head and challenges balance, so treatment is exercise, not rest or medicines.

Syllabus	Ch 5A(e) Vestibular disorders: central and peripheral
Priority	**High yield.** In 8 of 20 papers (2016–2026): 8 seven-mark questions.
Asked as	7m S17 · S20 · W21 · S22 · S24 · W24 · W25 · S26
Usual split	7 (BPPV test 3 and one manoeuvre 4; principles 3 and two manoeuvres 4; or two tests, or two exercise types)
Checked against	Vestibular rehabilitation for peripheral vestibular hypofunction, APTA Academy of Neurologic Physical Therapy guideline (Hall et al., 2022); BPPV clinical practice guideline update (Bhattacharyya et al., 2017); HINTS examination (Kattah et al., 2009). Full list under Sources (p. 123). Content checked 20 October 2026.

1 The vestibular system

- **Parts:** three semicircular canals on each side (anterior, posterior, horizontal) sense head rotation; two otolith organs (utricle and saccule) sense linear movement and gravity; the vestibular nerve; the vestibular nuclei; and the cerebellum.
- **Vestibulo-ocular reflex (VOR):** moves the eyes opposite to the head so vision stays clear during head movement.
- **Vestibulospinal reflexes:** keep the body upright and steady.
- Balance comes from three inputs, vision, somatosensation and the vestibular system, integrated by the brain.

2 Vestibular lesions: signs, symptoms and types

Signs and symptoms of a vestibular lesion (7 marks with adaptation exercises in S17)

- **Vertigo:** a false sense of spinning or movement; dizziness and light-headedness.
- **Nystagmus:** in peripheral lesions, horizontal-torsional, beating away from the damaged side, reduced by fixing the gaze.
- **Imbalance:** unsteady standing and walking, veering or falling towards the damaged side; falls.
- **Oscillopsia:** the world seems to jump during head movement (VOR failure), worst in bilateral loss.
- **Nausea and vomiting**, sweating, motion sensitivity.
- **Hearing loss, tinnitus, ear fullness** if the cochlea is involved (Ménière's disease, labyrinthitis).
- **Anxiety and avoidance** of head movement, which delays recovery.

Peripheral vs central vertigo

Feature	Peripheral (inner ear, nerve)	Central (brainstem, cerebellum)
Examples	BPPV, vestibular neuritis, labyrinthitis, Ménière's disease	Stroke (vertebrobasilar), MS, tumour, vestibular migraine
Nystagmus	One direction, horizontal-torsional, suppressed by fixation	May change direction or be purely vertical or torsional; not suppressed
Head impulse test	Abnormal (catch-up saccade)	Usually normal

Feature	Peripheral (inner ear, nerve)	Central (brainstem, cerebellum)
Balance	Unsteady but can walk	Often cannot stand or walk
Other signs	Hearing symptoms possible	Double vision, slurred speech, weakness, numbness, ataxia, skew deviation

Red flag: HINTS in acute vertigo

In a person with continuous vertigo and nystagmus, a **normal head impulse test**, **direction-changing nystagmus** or **skew deviation** (vertical eye misalignment on the cover test) points to a stroke. Refer as an emergency.

3 Assessment and vestibular tests

Asked for 7 marks in W24 (enumerate the tests, explain any two).

Test	How it is done	Positive result
Head impulse test	The person fixes on your nose; you turn the head quickly and unpredictably about 10–20° to each side	A catch-up saccade back to the target shows a weak VOR on the side the head was turned towards
Dynamic visual acuity	Read an eye chart with the head still, then while the head is moved side to side at about 2 per second	A drop of more than two lines suggests vestibular hypofunction
Dix-Hallpike test	Sitting, head turned 45° to one side; lie back quickly with the neck extended about 20° below horizontal	Vertigo with upbeating, torsional nystagmus after a short delay, lasting under a minute: posterior canal BPPV on the lower side
Supine roll test	Lying supine, head raised 30°; turn the head quickly 90° to each side	Horizontal nystagmus: horizontal canal BPPV
Head-shaking test	Shake the head side to side for about 20 seconds with the eyes closed, then look for nystagmus	Nystagmus after shaking suggests a one-sided loss
Romberg and sharpened Romberg	Stand with feet together (or heel to toe), eyes open then closed	Loss of balance with the eyes closed
Fukuda stepping test	March on the spot with the arms out and eyes closed for 50 steps	Turning of more than about 45° towards one side suggests a lesion on that side
Modified Clinical Test of Sensory Interaction in Balance	Stand on a firm surface and on foam, with eyes open and closed (four conditions), 30 seconds each	Falls on foam with the eyes closed show reliance on vision and poor vestibular use

- **Questionnaires and function:** Dizziness Handicap Inventory; Motion Sensitivity Quotient (which movements provoke symptoms); Dynamic Gait Index or Functional Gait Assessment; Timed Up and Go.
- **Laboratory:** videonystagmography with caloric testing, video head impulse test, rotary chair, vestibular evoked myogenic potentials, posturography; hearing tests.

4 Benign paroxysmal positional vertigo (BPPV)

Asked for 7 marks in S20 and W21 (physiotherapy for BPPV), S24 (two manoeuvres) and S26 (diagnostic test for posterior canal BPPV and one treatment).

- **What it is:** otoconia (calcium crystals) come loose from the utricle and move into a semicircular canal, most often the **posterior canal**. Head movement shifts the crystals and stimulates the canal, causing brief vertigo.
- **Features:** brief spinning (seconds to under a minute) on lying down, turning in bed, looking up or bending; worse in the morning; no hearing loss.

- **Common causes:** idiopathic in older people; after head injury; after inner-ear disease; prolonged bed rest.

Diagnosis (S26, 3 marks)

Posterior canal BPPV is diagnosed when the **Dix-Hallpike test** provokes vertigo with **torsional, upbeating nystagmus**: the person is brought from sitting to lying with the head turned 45° to one side and the neck extended 20°, affected ear down. If it is negative, test the other side. If the history fits but the test shows horizontal or no nystagmus, do a **supine roll test** for horizontal canal BPPV (BPPV guideline, 2017).

Canalith repositioning: the Epley manoeuvre (right posterior canal)

1. Sit the person on the couch with the head turned 45° to the right.
2. Lie them back quickly with the neck extended about 20° (the right Dix-Hallpike position); wait until the vertigo and nystagmus stop, about 30 seconds or more.
3. Turn the head 90° to the left, keeping the neck extended; wait about 30 seconds.
4. Roll the person onto the left side while turning the head a further 90°, so the nose points down towards the floor; wait about 30 seconds.
5. Bring them up to sitting with the head still turned and the chin slightly down.
6. Repeat the Dix-Hallpike test; repeat the manoeuvre if it is still positive.

Semont (liberatory) manoeuvre (right posterior canal)

1. Sit the person on the side of the couch with the head turned 45° to the left (away from the affected ear).
2. Move them quickly onto the right side, keeping the head turned, so the back of the head rests on the couch; wait about 1 minute.
3. Swing them quickly in one movement through sitting onto the left side, head still turned, so the nose points down; wait about 1 minute.
4. Return slowly to sitting.

- **Horizontal canal BPPV:** barbecue (Lempert) roll or the Gufoni manoeuvre.
- **Brandt–Daroff exercises:** a home habituation option (repeated lying to each side from sitting); less effective than repositioning, useful for residual symptoms.

What the BPPV guideline (2017) says

- Treat posterior canal BPPV with a canalith repositioning procedure; observation with follow-up is also an option.
- **Do not** order imaging or vestibular testing if the diagnosis is clear, and do not routinely give vestibular suppressant medicines.
- Vestibular rehabilitation may be offered; reassess within **1 month**; teach about falls and recurrence.

Classic answer vs current guidance: after the Epley manoeuvre

Classic	Current	Write in the exam
Sleep propped upright for 48 hours, wear a soft collar and avoid bending or turning the head after the manoeuvre.	The 2017 guideline advises against post-procedure postural restrictions: they add nothing to the success of the manoeuvre.	"Epley manoeuvre, then repeat the Dix-Hallpike test; no postural restrictions are needed; reassess within a month."

Recall check. Write the Epley steps for the left posterior canal, and the nystagmus that confirms posterior canal BPPV.

5 Vestibular rehabilitation

Asked for 7 marks in S22 (vestibular rehabilitation), S24 (principles and two manoeuvres), S17 (adaptation exercises) and W25 (substitution and habituation exercises).

Principles

- Offer exercise-based vestibular rehabilitation to adults with one-sided or two-sided vestibular hypofunction; start early, within the first 2 weeks of an acute loss (Hall et al., 2022).
- The brain recovers by **adaptation** (recalibrating the VOR), **habituation** (reducing the response to provoking movements) and **substitution** (using other senses and strategies).
- Movement provokes mild symptoms during exercise; that is expected and helps recovery. Vestibular suppressant medicines slow compensation and should be stopped as soon as possible.
- Supervised sessions plus a home programme; progress difficulty step by step; combine with balance and gait training.

The three exercise types

Type	How it works and examples	Used for
Adaptation (gaze stabilisation)	Image movement on the retina during head movement (retinal slip) drives the cerebellum to increase VOR gain. **VOR ×1:** keep the eyes on a fixed target while moving the head side to side, then up and down. **VOR ×2:** the target moves opposite to the head. Progress speed, duration, background and position (sitting, standing, walking)	One-sided loss (vestibular neuritis, after surgery)
Habituation	Repeated, controlled exposure to the movements or visual scenes that provoke mild dizziness lessens the response over time. Choose movements from the Motion Sensitivity Quotient test; repeat each a few times, twice a day, letting symptoms settle between repetitions; Brandt–Daroff exercises are one example	Motion-provoked dizziness; visual motion sensitivity; persistent dizziness
Substitution	Replaces lost vestibular function with other systems: vision and touch for balance; planned eye–head movements (look at a target, then turn the head); a stick, night lights, avoiding dark or uneven ground	Bilateral or complete loss

Doses (Hall et al., 2022)

Condition	Gaze stabilisation	Balance and gait
Acute or subacute one-sided loss	**3 times a day**, at least **12 minutes** in total daily	Static and dynamic balance and walking, progressed
Chronic one-sided loss	**3–5 times a day**, at least **20 minutes** daily, for **4–6 weeks**	At least **20 minutes** daily for **4–6 weeks**
Bilateral loss	**3–5 times a day**, **20–40 minutes** daily, for about **5–7 weeks**	At least 20 minutes daily for 6–9 weeks

Weekly clinic visits plus the home programme. Stop when goals are met, symptoms resolve, or progress plateaus.

Classic answer vs current guidance: eye exercises

Classic	Current	Write in the exam
Cawthorne–Cooksey exercises begin with eye movements alone (looking up and down, side to side) with the head still.	Voluntary saccades or smooth pursuit without head movement should not be used to improve gaze stability: the evidence shows harm outweighs benefit (Hall et al., 2022). Use gaze stabilisation with head movement.	"Gaze stabilisation exercises with head movement (VOR ×1, VOR ×2), habituation, balance and gait training, as a supervised and home programme."

Balance and gait training

- Standing with a narrowing base, on firm then soft surfaces, eyes open then closed.
- Weight shifts, reaching, stepping, tandem walking; walking with head turns and pivots; dual tasks; varying speed.
- Community walking: crowds, stairs, uneven ground; graded return to driving and work.

Recall check. Define adaptation, habituation and substitution, with one exercise for each.

6 Other vestibular disorders

Disorder	Key features	Physiotherapy
Vestibular neuritis	Sudden, continuous vertigo for days after a viral illness; no hearing loss; abnormal head impulse test	Early gaze stabilisation and balance training
Labyrinthitis	As neuritis, with hearing loss	As above
Ménière's disease	Attacks of vertigo lasting hours, with fluctuating hearing loss, tinnitus and ear fullness	Not during attacks; rehabilitation if there is lasting loss between attacks
Bilateral vestibular loss	Oscillopsia, imbalance worse in the dark; often after ototoxic drugs (gentamicin)	Substitution, gaze stabilisation, balance; fall prevention
Central disorders	Brainstem or cerebellar stroke, MS, concussion, vestibular migraine	Longer, slower programmes with balance and gait training
Persistent postural-perceptual dizziness	Daily non-spinning dizziness for 3 months or more, worse upright and with busy visual scenes	Habituation and graded exposure, with psychological support

7 Precautions and red flags

Take care

- Refer urgently if vertigo comes with a new headache, double vision, slurred speech, weakness, numbness, inability to walk, or a central HINTS finding.
- Before the Dix-Hallpike test, check for neck problems, carotid or vertebral artery disease and recent neck injury; use a modified side-lying test if neck extension is unsafe.
- Guard closely during manoeuvres and balance work; keep a vomit bowl ready.
- Advise the person not to drive until symptoms settle.

8 Exam corner

Past questions (MUHS, 2016–2026)

S26 Q2

Explain the diagnostic test for posterior semicircular canal BPPV and describe any one physiotherapy intervention for it.

W25 Q2(e)

Explain in detail substitution and habituation exercises.

W24 Q2(c)

Enumerate the tests for assessing vestibular disorders; explain any two.

S24 Q2(b)

Describe the principles of vestibular rehabilitation; write any two manoeuvres.

S22 Q2(e)

Write a note on vestibular rehabilitation.

S20 Q2(f), W21 Q2(c)

Physiotherapy management of BPPV.

S17 Q2(e)

Describe the signs and symptoms of a vestibular lesion; explain adaptation exercises in detail.

Pages for this topic: balance and coordination (p. 223), cerebellar disease and ataxia (p. 155), brain injury (p. 66).

Keywords to underline

VOR vertigo nystagmus oscillopsia head impulse test Dix-Hallpike torsional upbeating nystagmus canalith repositioning Epley adaptation habituation substitution

Answer skeleton: BPPV (7 marks)

1. **[What it is, 1]** Loose otoconia in a canal (posterior most often); brief positional vertigo.
2. **[Diagnosis, 2]** Dix-Hallpike test with the result; supine roll test for the horizontal canal.
3. **[Treatment, 3]** Epley (or Semont) steps; repeat the test; Brandt–Daroff for residual symptoms; habituation and balance if needed.
4. **[Advice, 1]** No postural restrictions; no routine suppressant medicines; reassess within a month; recurrence and fall risk.

Draw these

- The Dix-Hallpike position (head turned 45°, neck extended 20°)
- VOR ×1 and VOR ×2: arrows for head and target movement

Viva questions

1. What is the VOR, and how do you test it at the bedside?
2. Which canal is most often involved in BPPV, and why?
3. Describe the nystagmus in posterior canal BPPV.
4. What is the difference between adaptation and habituation?
5. Which findings in acute vertigo suggest a stroke?
6. Why should vestibular suppressants be stopped early?

Linked practical spots

- Dix-Hallpike test, head impulse test, Epley manoeuvre demonstration; foam pad for mCTSIB.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Describe the Dix-Hallpike test.**

Answer
Sitting, head turned 45° to one side; lie back quickly with the neck extended 20° below horizontal (that ear down). Positive for posterior canal BPPV: vertigo with torsional, upbeating nystagmus after a short delay, lasting under a minute. Repeat on the other side if negative.

3 m **What are substitution exercises?**

Answer
Exercises that replace lost vestibular function with other senses and strategies: balance training using vision and touch, planned eye–head movements (look at the target first, then turn the head), assistive devices and environmental changes (night lights). Used mainly for bilateral loss.

7 m **Describe the principles of vestibular rehabilitation and any two manoeuvres.**

Answer
Principles: early, exercise-based, supervised plus home programme; adaptation (VOR ×1, ×2), habituation (repeated exposure to provoking movements) and substitution (vision, touch, strategies); balance and gait training; progress step by step; mild symptoms during exercise are expected; stop suppressant medicines early. Doses: gaze exercises 3 times a day for at least 12 minutes in acute loss, 3–5 times a day for 20 minutes in chronic loss for 4–6 weeks.

Manoeuvres: Epley (sitting with the head turned 45° to the affected side, lie back with the neck extended, turn the head 90° to the other side, roll onto that side nose down, sit up) and Semont (head turned away, lie quickly onto the affected side, swing through to the other side nose down, sit up).

LAQ **A 58-year-old woman has had brief spinning attacks for a week whenever she lies down or turns in bed. Plan the assessment and physiotherapy. (Practice case)**

Answer
Assessment: history (brief positional vertigo, no hearing loss, no neurological symptoms); red flags; neck and vascular screen; Dix-Hallpike both sides (torsional upbeating nystagmus = posterior canal BPPV); supine roll test if needed; balance (Romberg, mCTSIB), falls risk, Dizziness Handicap Inventory.

Physiotherapy: Epley manoeuvre on the affected side, repeat the test, repeat the manoeuvre if still positive; no postural restrictions; teach about recurrence and falls; Brandt–Daroff or habituation exercises if symptoms remain; balance training if unsteady; review within a month.

MCQ **Torsional, upbeating nystagmus on the Dix-Hallpike test indicates:**

A. Horizontal canal BPPV

B. Posterior canal BPPV

C. Vestibular neuritis

D. Cerebellar stroke

Answer
B. Horizontal canal BPPV gives horizontal nystagmus on the roll test.

MCQ **VOR ×2 exercise means:**

A. Eyes on a still target while the head moves

B. Target and head move in opposite directions

C. Eye movements with the head still

D. Walking with eyes closed

Answer

B. It demands twice the VOR gain of VOR ×1.

MCQ **After the Epley manoeuvre, the 2017 BPPV guideline advises:**

A. Sleeping upright for 48 hours

B. A soft collar for a week

C. No postural restrictions

D. Bed rest

Answer

C. Restrictions add nothing to success.

Last recall check. List six vestibular tests, write the Epley steps, and the gaze stabilisation doses.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Hall CD, et al. Vestibular rehabilitation for peripheral vestibular hypofunction: an updated clinical practice guideline from the Academy of Neurologic Physical Therapy of the APTA. *J Neurol Phys Ther* 2022;46:118–77.
3. Bhattacharyya N, et al. Clinical practice guideline: benign paroxysmal positional vertigo (update). *Otolaryngol Head Neck Surg* 2017;156(3 Suppl):S1–47.
4. Kattah JC, et al. HINTS to diagnose stroke in the acute vestibular syndrome. *Stroke* 2009;40:3504–10.
5. Structure and classical content: Herdman and Clendaniel, *Vestibular Rehabilitation*; O'Sullivan and Schmitz, *Physical Rehabilitation*.

5A(f) Facial (VII) nerve palsy, Bell's palsy, trigeminal nerve and trigeminal neuralgia

> The facial and trigeminal nerves came up in 9 papers: a long case on Bell's palsy, physiotherapy for Bell's palsy for 7 marks three times, trigeminal neuralgia twice, and short notes on Bell's phenomenon, synkinesis and nerve assessment. Most people with Bell's palsy recover, especially with early steroids; physiotherapy protects the eye, keeps the face moving symmetrically and limits the unwanted movements that can follow regrowth.

Syllabus	Ch 5A(f) VII cranial nerve (trigeminal neuralgia and nerve assessment included)
Priority	**Tier 2.** In 9 of 20 papers (2016–2026): 1 long case, 5 seven-mark and 5 three-mark questions.
Asked as	LAQ W19 · 7m W16 · W17 · S20 · W22 · S24 · 3m S16 · W18 ×3 · W24
Usual split	15 (ICF with goals); 7 (features 2–3 and physiotherapy 4–5); 3 (assessment steps or a sign)
Checked against	AAO-HNS clinical practice guideline on Bell's palsy (2013); Sullivan et al., prednisolone and acyclovir trial (2007); Cochrane review of physical therapy for Bell's palsy (2011); House-Brackmann grading. Full list under Sources (p. 132). Content checked 20 October 2026.

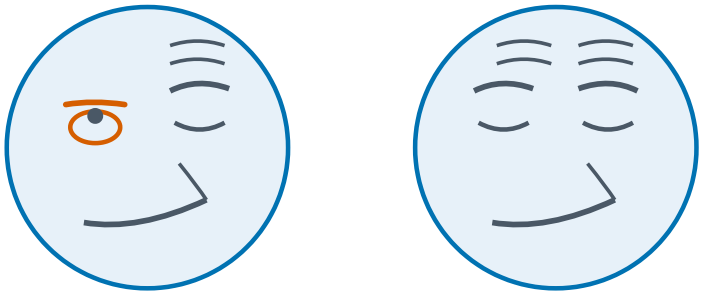
1 Definition

- **Bell's palsy** is an acute, one-sided, lower motor neuron (LMN) facial nerve palsy with no identifiable cause, developing over less than 72 hours. It is the commonest cause of facial palsy.
- It is thought to follow viral inflammation (often herpes simplex) that makes the nerve swell inside its narrow bony canal.

2 Classification and grading

UMN vs LMN facial weakness

Right facial weakness: LMN vs UMN, on trying to close the eyes and smile



LMN (Bell's palsy)
whole half of the face

UMN (stroke)
lower face only

The faces are drawn as you see the patient, so the patient's right side is on your left. The forehead is spared in UMN palsy because the upper face receives nerve supply from both sides of the brain.

House-Brackmann grading

Grade	Description
I	Normal

Grade	Description
II	Mild: slight weakness on close look; symmetry at rest; eye closes fully with little effort
III	Moderate: obvious but not disfiguring weakness; eye closes fully with effort; synkinesis may be present
IV	Moderately severe: disfiguring asymmetry; eye does not close fully
V	Severe: barely perceptible movement
VI	Total paralysis

The **Sunnybrook Facial Grading System** (resting symmetry, voluntary movement and synkinesis, 0–100) is more detailed and sensitive to change.

3 Causes of facial palsy

- **LMN:** Bell's palsy (idiopathic); Ramsay Hunt syndrome (herpes zoster: painful vesicles in the ear, worse outlook); trauma (temporal bone fracture, facial wounds); middle ear infection and cholesteatoma; parotid tumours or surgery; Guillain-Barré syndrome (often both sides); leprosy; diabetes; Lyme disease.
- **UMN:** stroke, tumour, MS affecting the corticobulbar fibres.
- Risk of Bell's palsy is higher in pregnancy, diabetes and after viral infections.

4 Pathophysiology to impairments

1. Inflammation makes the nerve swell inside the narrow facial canal of the temporal bone, so the nerve is compressed: mostly conduction block (neurapraxia), sometimes axon loss.
2. Because all the facial muscles on one side lose their supply, the face droops, the eye cannot close and the mouth cannot seal.
3. Because branches inside the canal can be involved, taste (front of the tongue), tear and saliva production and the stapedius reflex (loud-sound sensitivity) can be affected.
4. If axons are lost, regrowing fibres can reach the wrong muscles, causing **synkinesis**, muscle tightness (contracture) and "crocodile tears".

5 Clinical features

Asked for 3 marks in W24 and with management for 7 marks in S24.

- Sudden weakness of one whole side of the face over hours to 3 days; pain behind the ear may come first.
- Cannot raise the eyebrow or wrinkle the forehead; eye does not close fully; flattened nasolabial fold; mouth droops and pulls to the normal side when smiling.
- Food collects in the cheek; drinks dribble; speech of lip sounds (p, b, m) is unclear.
- Watering or dry eye; loss of taste on the front two-thirds of the tongue; sensitivity to loud sounds.
- **Bell's phenomenon** (3 marks in W18): when the patient tries to close the eyes, the eyeball on the weak side rolls upwards and outwards. This normal reflex becomes visible because the eyelid does not close.
- **Facial synkinesis** (3 marks in W18): an involuntary movement of one part of the face during a voluntary movement of another part (for example the eye narrows when the person smiles, or the corner of the mouth lifts when blinking), due to misdirected regrowth of nerve fibres; it appears months after severe palsy. "Crocodile tears" (tears while eating) are a similar misdirection.

Clinical assessment of the facial nerve (3 marks in S16)

Test	Muscle or function
Raise the eyebrows, wrinkle the forehead	Frontalis (spared in UMN palsy)

Test	Muscle or function
Close the eyes tightly against resistance	Orbicularis oculi; watch Bell's phenomenon
Show the teeth, smile	Zygomaticus and levator muscles of the lip
Puff out the cheeks, whistle	Buccinator and orbicularis oris
Tense the neck	Platysma
Taste, tears, hearing	Taste on the front two-thirds of the tongue; tear production (Schirmer's test); sensitivity to sound (stapedius)

6 Investigations

- Bell's palsy is a clinical diagnosis; routine blood tests and imaging are not advised unless the picture is atypical (AAO-HNS, 2013).
- Nerve conduction studies (electroneurography) and EMG may be offered in complete paralysis to predict recovery, not in incomplete paralysis.
- Ear examination for Ramsay Hunt syndrome and middle ear disease; blood sugar in diabetics.

7 Medical and surgical management

- **Oral steroids within 72 hours** of onset for people aged 16 and over (strong recommendation). In a large trial, prednisolone raised complete recovery from 64% to 83% at 3 months and from 82% to 94% at 9 months (Sullivan et al., 2007).
- **Antivirals alone are not advised;** they may be added to steroids.
- **Eye protection** when the eye does not close: lubricating drops by day, ointment and taping or a patch at night, glasses outdoors.
- Refer to a facial nerve specialist if new neurological signs, eye problems, or incomplete recovery at **3 months**.
- Botulinum toxin for synkinesis; surgery (nerve transfer, muscle transfer, gold weight in the eyelid) in long-standing paralysis.

8 Complications and prevention

Complication	Prevention and physio role
Corneal exposure and ulcer	Eye drops and ointment, taping at night, glasses; report a red or painful eye at once
Synkinesis and facial tightness	Slow, small, controlled symmetrical movements with a mirror; avoid forceful mass movements; massage and stretching of tight muscles
Drooling, food pocketing	Chew on the strong side; cheek support; lip seal exercises
Low mood and social withdrawal	Reassurance about recovery; counselling

9 PT assessment

- History: onset, pain, ear symptoms, taste, hearing, eye symptoms, steroid start date, occupation (speaking, teaching).
- Observation at rest: symmetry of eyebrow, eye, nasolabial fold and mouth.
- Voluntary movement of each muscle group (section 5) and grading: House-Brackmann or Sunnybrook; photographs for comparison.
- Eye closure and Bell's phenomenon; synkinesis in later stages.

- Function: eating, drinking, speaking, blowing; quality of life (Facial Disability Index).

10 PT diagnosis by ICF

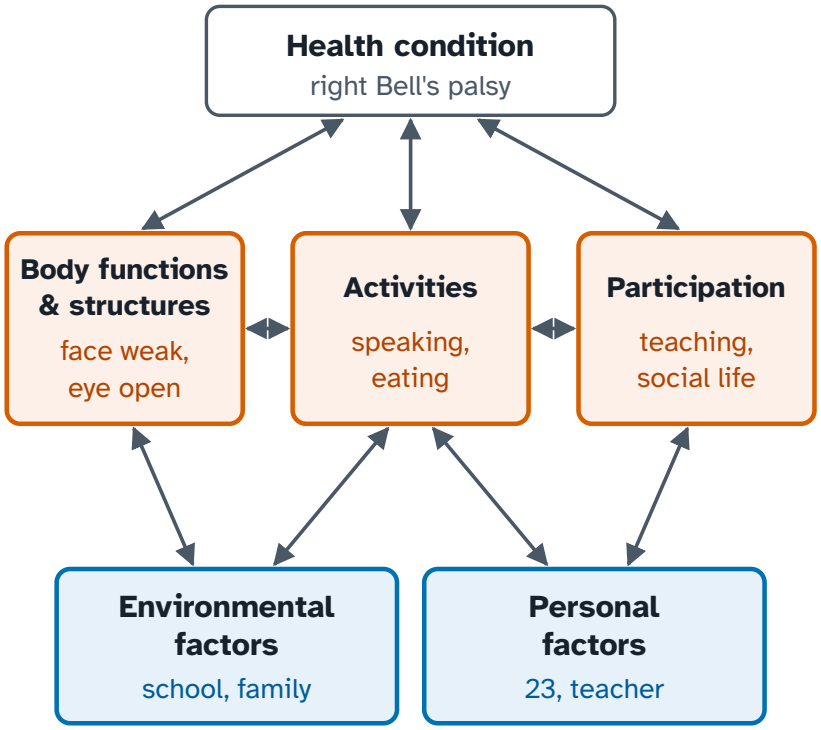
The long case (W19) asked for ICF with goals.

> "Give ICF and short term and long term goals of physiotherapy management for a 23 years old male teacher who has been diagnosed as right Bell's palsy."

Spot the ICF clues before you write

right Bell's palsy is an acute LMN palsy of the whole right face. 23 years: young, good outlook. teacher: clear speech and facial expression are central to his work, and appearance affects confidence in class.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Right Bell's palsy (acute idiopathic LMN facial palsy)	Swollen nerve compressed in the facial canal
Body function and structure impairments	Weakness of all right facial muscles; incomplete eye closure (risk to the cornea); drooping mouth; possible loss of taste and sound sensitivity; later risk of synkinesis	All branches of the right facial nerve are affected
Activity limitations	Unclear speech of lip sounds; difficulty eating, drinking, whistling, blowing; facial expression	Lip seal and cheek control are lost
Participation restrictions	Teaching (speaking for long periods, being seen by students); social outings	Appearance and speech affect confidence
Environmental factors	**Facilitators:** early medical care, family, supportive school. **Barriers:** dust and wind (eye), stigma	Eye protection outdoors and in class
Personal factors	23-year-old man; anxious about appearance	Young age favours full recovery

One-line functional diagnosis

A 23-year-old teacher with right Bell's palsy, with weakness of the whole right side of the face and incomplete eye closure, causing unclear speech and difficulty eating, which affects his teaching and social life.

11 Problem list and goals

Goals for the 23-year-old teacher

Problem	Short-term (2–4 weeks)	Long-term (2–3 months)
Eye	No corneal damage; protects the eye day and night	Full eye closure
Facial movement	Flicker to visible movement in each muscle group; symmetry at rest	House-Brackmann grade I or II
Speech and eating	Lip seal for drinking; clear lip sounds with effort	Normal speech for a full teaching day
Synkinesis	Understands slow, controlled, symmetrical practice	No troublesome synkinesis

12 PT management

Asked for 7 marks in W16 (with education and advice), W17 and S24.

By stage

- **Acute (paralysed):** reassurance (most recover); eye care; gentle massage of the face (upward strokes towards the eye and ear); moist warmth if comfortable, protecting the eye; supporting the cheek with taping; chewing on the strong side.
- **Recovering (movement returning):** active-assisted, then active exercises in front of a mirror for each muscle group: raise eyebrows, gently close the eyes, smile, purse the lips, puff the cheeks, flare the nostrils. Small, slow, symmetrical movements; hold the strong side back with the fingers so it does not overpower.
- **Late (synkinesis risk):** facial neuromuscular retraining: slow, isolated movements with mirror or EMG biofeedback; relax the unwanted muscle while moving the wanted one; stretching and massage of tight muscles; botulinum toxin by the doctor.

Education and advice (W16)

- Most people recover within weeks to months; start steroids early (doctor).
- Eye: drops, ointment and taping at night, glasses outdoors; report pain or redness.
- Eat on the strong side; clean food from the cheek after meals; sip through a straw.
- Do exercises gently in front of a mirror several times a day; avoid forceful grimacing and gum chewing that encourage mass movements.

Electrical stimulation

Traditionally used for the paralysed face, but trials show no benefit over placebo for recovery, and many clinicians avoid it in case it encourages mass movement (Cochrane, 2011).

Classic answer vs current evidence: physiotherapy for Bell's palsy		
Classic Faradic or galvanic stimulation of the facial muscles, infrared, massage and vigorous facial exercises.	**Current** The 2013 guideline makes no recommendation for or against physiotherapy because evidence is weak; electrical stimulation showed no benefit over placebo. Tailored facial exercises and mirror or EMG biofeedback may help, especially in moderate and chronic cases (low-quality evidence).	**Write in the exam** "Eye care, reassurance, gentle massage, mirror-guided slow symmetrical facial exercises, neuromuscular retraining for synkinesis; electrical stimulation has no proven benefit."

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Incomplete eye closure	Drops by day, ointment and taping at night; gentle eye-closure practice	Protects the cornea
Weak muscles	Mirror-guided active-assisted and active exercises, a few repetitions of each, **several times a day**	Practice with visual feedback restores controlled, symmetrical movement
Synkinesis	Slow, isolated movements with mirror or EMG biofeedback; relax unwanted muscles	Retrains selective control after misdirected regrowth
Tight muscles	Massage and gentle stretching	Eases contracture and discomfort

Recall check. Write the education and advice you would give on day 1 to a person with Bell's palsy.

13 Trigeminal nerve and trigeminal neuralgia

Clinical assessment of the trigeminal nerve (3 marks in W18)

- **Sensory:** light touch and pinprick over the forehead (V1, ophthalmic), cheek (V2, maxillary) and jaw (V3, mandibular), comparing sides.
- **Motor:** clench the teeth and feel the masseter and temporalis; open the mouth against resistance: the jaw deviates to the weak side (pterygoids).
- **Reflexes:** corneal reflex (V1 carries the sensation; the facial nerve closes the eye); jaw jerk (brisk in UMN lesions above the pons).

Trigeminal neuralgia (7 marks in S20 and W22)

- **Features:** sudden, severe, electric shock-like pain on one side of the face, usually in the V2 or V3 area, lasting seconds to about 2 minutes, in repeated attacks; triggered by light touch, talking, chewing, brushing the teeth, shaving or a cold wind ("trigger zones"); normal sensation between attacks.
- **Causes:** usually a blood vessel pressing on the nerve root; sometimes MS or a tumour (suspect in young people or with sensory loss).
- **Medical:** carbamazepine (first choice) or oxcarbazepine; microvascular decompression or other procedures if drugs fail.
- **Physiotherapy (supportive):** education on avoiding triggers (scarf in cold wind, lukewarm water, soft toothbrush, chewing on the other side); relaxation and stress management; posture and jaw relaxation; TENS may be tried by some patients away from trigger zones; encourage nutrition and oral hygiene, which patients often avoid. Do not stimulate trigger zones directly.

14 Precautions

Take care

- Protect the eye during heat or light treatment; never apply heat with the eye open.
- Avoid forceful, mass facial exercises and vigorous gum chewing in the recovery phase.
- Refer if weakness spreads, other cranial nerves are involved, the ear has vesicles, or there is no recovery by 3 months.
- Trigeminal neuralgia: do not touch trigger zones during treatment.

15 Prognosis

- Most people recover fully; with early prednisolone, about 94% recover completely by 9 months (Sullivan et al., 2007).
- Worse outlook: complete paralysis, older age, Ramsay Hunt syndrome, diabetes, no recovery by 3 weeks, and signs of axon loss on nerve tests.
- Synkinesis and tightness may remain after severe palsy.

16 Exam corner

Past questions (MUHS, 2016–2026)

W19 Q4(a)

The ICF case in section 10 (23-year-old teacher with right Bell's palsy): ICF with short- and long-term goals.

7-mark questions

Physiotherapy of facial palsy with education and advice (W16); physiotherapy of Bell's palsy (W17); Bell's palsy: features and management (S24); trigeminal neuralgia: features and physiotherapy (S20, W22).

3-mark questions

Clinical assessment of the facial nerve (S16); clinical assessment of the trigeminal nerve, facial synkinesis and Bell's phenomenon (W18); signs and symptoms of Bell's palsy (W24).

Pages for this topic: stroke (UMN facial palsy) (p. 50), Guillain-Barré syndrome (bilateral facial palsy) (p. 144), leprosy (lagophthalmos) (p. 107).

Keywords to underline

LMN, whole half of face forehead spared in UMN Bell's phenomenon eye care steroids within 72 hours House-Brackmann mirror exercises synkinesis neuromuscular retraining trigger zones

Answer skeleton: Bell's palsy, features and management (7 marks)

1. **[Features, 2–3]** Sudden LMN weakness of the whole half face; eye closure, Bell's phenomenon, mouth droop, taste and hearing.
2. **[Medical, 1]** Steroids within 72 hours, eye protection, refer at 3 months if incomplete.
3. **[Physiotherapy, 3–4]** Education and eye care; massage; mirror-guided slow symmetrical exercises by muscle group; taping; neuromuscular retraining for synkinesis; evidence note on electrical stimulation.

Draw these

- LMN vs UMN facial palsy faces (section 2)

- The ICF model filled from the case

Viva questions

1. Why is the forehead spared in UMN facial palsy?
2. What is Bell's phenomenon?
3. What is synkinesis, and how do you prevent it?
4. Within what time should steroids start?
5. How do you test the trigeminal nerve?

Linked practical spots

- Demonstrate facial nerve testing; mirror exercise; taping of the cheek.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **What is Bell's phenomenon?**

Answer
On attempted eye closure, the eyeball on the paralysed side rolls upward and outward; it is visible because the eyelid cannot close in LMN facial palsy.

3 m **Write the signs and symptoms of Bell's palsy.**

Answer
Sudden one-sided weakness of the whole face; cannot wrinkle the forehead or close the eye; Bell's phenomenon; flattened nasolabial fold and drooping mouth; drooling and food pocketing; pain behind the ear; loss of taste on the front of the tongue; sensitivity to loud sound; watery or dry eye.

7 m **Physiotherapy management of Bell's palsy with education and advice.**

Answer
Assess and grade (House-Brackmann). Education: good outlook, early steroids, eye care, eating on the strong side, gentle practice. Acute: massage, warmth with eye protection, cheek taping. Recovery: mirror-guided active-assisted then active exercises of each muscle group, slow and symmetrical, holding the strong side. Late: neuromuscular retraining with biofeedback for synkinesis, stretching tight muscles. Evidence note: electrical stimulation shows no benefit. Refer if incomplete at 3 months.

LAQ **Give ICF and short- and long-term goals for a 23-year-old teacher with right Bell's palsy. (W19)**

Answer
Use the ICF table and goals in sections 10 and 11; add the management outline from section 12.

MCQ **Weakness of the lower face with normal forehead wrinkling suggests:**

A. Bell's palsy
B. UMN lesion such as stroke
C. Ramsay Hunt syndrome
D. Parotid tumour

Answer
B. The upper face has bilateral cortical supply.

MCQ **The 2013 Bell's palsy guideline recommends:**

A. Antivirals alone

B. Oral steroids within 72 hours for adults

C. Routine MRI

D. Electrical stimulation for all

Answer
B.

MCQ **The first-choice drug for trigeminal neuralgia is:**

A. Prednisolone

B. Carbamazepine

C. Baclofen

D. Paracetamol

Answer
B.

Last recall check. Draw LMN vs UMN facial palsy, list the facial nerve tests, and write the physiotherapy by stage.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Baugh RF, et al. Clinical practice guideline: Bell's palsy. *Otolaryngol Head Neck Surg* 2013;149(3 Suppl):S1–27.
3. Sullivan FM, et al. Early treatment with prednisolone or acyclovir in Bell's palsy. *N Engl J Med* 2007;357:1598–607.
4. Teixeira LJ, et al. Physical therapy for Bell's palsy (idiopathic facial paralysis). *Cochrane Database Syst Rev* 2011;(12):CD006283.
5. House JW, Brackmann DE. Facial nerve grading system. *Otolaryngol Head Neck Surg* 1985;93:146–7.

5A(g) Multiple sclerosis: types, features, EDSS, ICF and physiotherapy

> Multiple sclerosis was a long case in 6 papers, and the same 29-year-old executive after her second relapse was asked three times. Short questions ask for features, Uhthoff's phenomenon, Charcot's triad and the stages of the disease. Every answer turns on two facts: symptoms come and go with relapses and heat, and exercise is safe and helpful at every level of disability if it is paced around fatigue.

Syllabus	Ch 5A(g) Demyelinating diseases: multiple sclerosis and G.B. syndrome (GBS has its own page)
Priority	**Tier 1.** In 14 of 20 papers (2016–2026): 6 long cases, 4 seven-mark and 4 three-mark questions.
Asked as	LAQ W17 · S18 · W19 · S23 · S24 · S25 · 7m W18 · S19 · S20 · W20 · 3m W16 · S22 · W22 · W25
Usual split	15 (ICF with clinical reasoning 7, management with goals 8; or functional diagnosis 3, goals 5, management 7)
Checked against	Exercise and lifestyle physical activity recommendations for people with MS (Kalb et al., 2020); 2024 revisions of the McDonald criteria (published 2025); Atlas of MS, third edition (2020); MS phenotype definitions (Lublin et al., 2014); Kurtzke EDSS. Full list under Sources (p. 143). Content checked 20 October 2026.

1 Definition

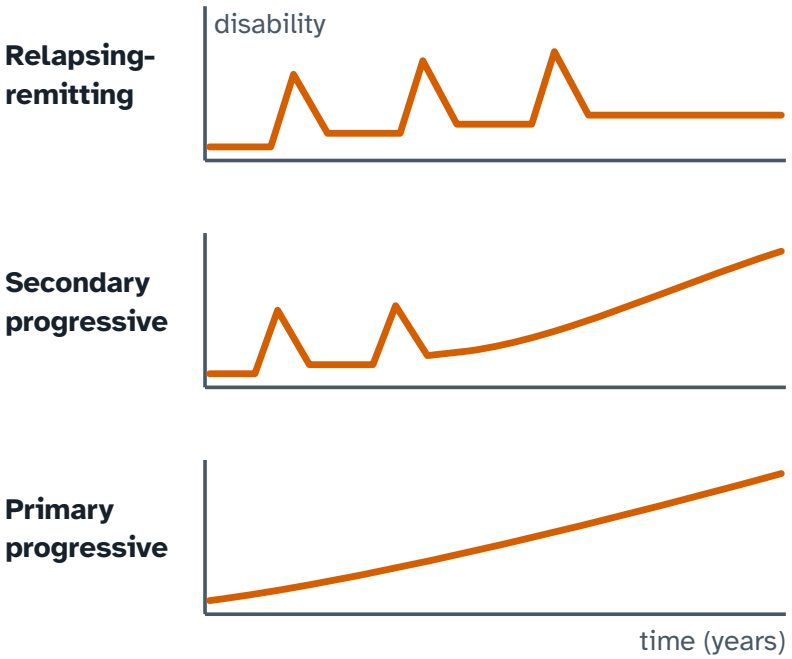
Multiple sclerosis (MS) is a chronic immune-mediated disease of the central nervous system in which patches of inflammation strip the myelin from nerve fibres in the brain, spinal cord and optic nerves, leaving scars (plaques) and later axonal loss. The lesions are scattered in **space** (different parts of the CNS) and in **time** (new lesions at different times).

- About **2.8 million** people live with MS worldwide; the average age at diagnosis is 32, and women are affected about twice as often as men (Atlas of MS, 2020).
- It is less common in India than in Europe and North America, where it is the commonest non-traumatic cause of disability in young adults.

2 Classification and grading

Disease courses ("stages of MS": 7 marks in S20)

Courses of MS: disability over time



About 85% start as relapsing-remitting MS. After years, many move into a secondary progressive course. A minority are progressive from the start.

Course	What happens
Clinically isolated syndrome (CIS)	A first attack (for example optic neuritis) that may become MS
Relapsing-remitting (RRMS)	Relapses lasting days to weeks, followed by full or partial recovery; stable between relapses
Secondary progressive (SPMS)	After an initial relapsing course, disability increases steadily, with or without relapses
Primary progressive (PPMS)	Steady worsening from the start, often a slowly progressive walking difficulty

Each course is also described as active or not active (new relapses or MRI lesions) and progressing or not (Lublin et al., 2014).

Expanded Disability Status Scale (EDSS, Kurtzke)

Score	Meaning
0	Normal neurological examination
1.0–3.5	Mild to moderate signs; walks fully
4.0–5.5	Walks without aid, but the distance falls: about 500 m at 4.0, 100 m at 5.5
6.0	Needs a stick or one-sided support to walk 100 m
6.5	Needs support on both sides to walk 20 m
7.0–7.5	Cannot walk more than about 5 m even with aid; uses a wheelchair
8.0–9.5	Bed- or chair-bound; from some self-care with the arms to totally dependent
10	Death due to MS

3 Causes and risk factors

- **Epstein–Barr virus infection** (the strongest known environmental risk factor).
- Low vitamin D and little sunlight (more MS far from the equator); smoking; obesity in adolescence.
- Genes (for example HLA-DRB1*15:01); female sex.
- Relapses may be triggered by infections and the period after childbirth.

4 Pathophysiology to impairments

1. Immune cells cross the blood–brain barrier and attack myelin, causing inflammation and plaques, mainly near the ventricles, in the brainstem and cerebellum, the spinal cord and the optic nerves.
2. Because demyelinated fibres conduct slowly or not at all, symptoms appear according to the site: weakness, numbness, visual loss, ataxia, bladder problems.
3. Because inflammation settles and some remyelination occurs, symptoms improve after a relapse (remission).
4. Because axons are lost over time, disability gradually builds up and does not recover (progression).
5. Because heat slows conduction further in demyelinated fibres, symptoms worsen temporarily with a rise in body temperature (Uhthoff's phenomenon).

Recall check. Why do symptoms improve after a relapse but disability still builds up over years?

5 Clinical features

Asked for 3 marks in W16 and S22 and with physiotherapy for 7 marks in W20.

- **Sensory:** numbness, tingling, band-like feelings; **Lhermitte's sign** (an electric shock down the spine on bending the neck).
- **Visual:** optic neuritis (painful eye movement, blurred central vision, faded colours); double vision; nystagmus.
- **Motor:** weakness and spasticity, usually in the legs; UMN signs.
- **Cerebellar:** ataxia, intention tremor, dysarthria.
- **Fatigue:** very common and often the most disabling symptom.
- **Bladder and bowel:** urgency, frequency, incomplete emptying, constipation; sexual dysfunction.
- **Cognitive and emotional:** slowed thinking, poor memory and attention; depression, anxiety, emotional lability.
- **Pain** (including trigeminal neuralgia) and heat sensitivity.

Uhthoff's phenomenon (3 marks in W22) and Charcot's triad (3 marks in W25)

Uhthoff's phenomenon: a temporary worsening of symptoms (often blurred vision or weakness) when body temperature rises, as with exercise, a hot bath, hot weather or fever. Heat slows conduction in demyelinated fibres; symptoms settle when the person cools down. It is not a relapse.

Charcot's triad in MS: nystagmus, intention tremor and scanning (staccato) speech. These are cerebellar and brainstem signs.

Visual, cognitive and behavioural disturbances (7 marks in W18)

Area	Features	What the physiotherapist does
Visual	Optic neuritis, double vision, nystagmus, blurred or jumping vision, poor colour vision	Good lighting, high-contrast markings on steps, uncluttered spaces; refer for eye care; gaze stability exercises if appropriate
Cognitive	Slow processing, poor memory, attention and planning	Short, simple instructions; written programmes; practise in quiet surroundings; avoid dual tasks at first

Area	Features	What the physiotherapist does
Behavioural and emotional	Depression, anxiety, irritability, sudden laughing or crying, low motivation	Agree on goals together; encourage; exercise helps mood; refer for counselling

6 Investigations

- **MRI** of brain and spine: plaques near the ventricles, in the cortex, brainstem, cerebellum and cord.
- **Diagnosis (McDonald criteria, 2024 revision):** typical lesions in at least two of five regions (around the ventricles, cortex or just beneath it, brainstem and cerebellum, spinal cord, and now the **optic nerve**). Proof of new lesions over time is no longer always needed. CSF oligoclonal bands, or a raised kappa free light chain index, support the diagnosis.
- **Visual evoked potentials:** slowed in optic nerve involvement.
- Blood tests exclude mimics (B12 deficiency, neuromyelitis optica antibodies).

7 Medical and surgical management

- **Relapse:** high-dose methylprednisolone for 3–5 days speeds recovery; it does not change the long-term course.
- **Disease-modifying therapy** to reduce relapses and new lesions: interferon beta, glatiramer acetate, teriflunomide, dimethyl fumarate, fingolimod, natalizumab, ocrelizumab and others. Ocrelizumab is also used in primary progressive MS.
- **Symptoms:** baclofen or tizanidine for spasticity; fampridine for walking speed in some; drugs for bladder, pain and depression; botulinum toxin for focal spasticity.

8 Complications and prevention

Complication	What to look for	Physio role
Falls	Weakness, spasticity, ataxia, poor vision, fatigue, heat	Balance and strength training, aids, home safety
Deconditioning	Activity falls because of fatigue and fear of worsening	Graded aerobic and resistance exercise
Contractures and spasms	Calf, hamstrings, hip flexors and adductors	Stretching, standing, positioning; check for triggers such as a full bladder or infection
Urine infection	Worse spasticity, fever, new weakness (a pseudo-relapse)	Recognise and refer; bladder programme
Pressure sores, osteoporosis	Wheelchair users; steroid courses	Pressure relief, seating, standing programme

9 PT assessment

- **History:** course of MS, date of last relapse, medicines, fatigue pattern through the day, heat sensitivity, bladder, falls, work and home.
- **Disability:** EDSS.
- **Motor:** strength, tone (Modified Ashworth Scale), range, coordination (finger-nose, heel-shin).
- **Sensory, vision and cognition** screening.
- **Balance and gait:** Berg Balance Scale, Timed 25-Foot Walk, Timed Up and Go, 6-minute walk, 12-item MS Walking Scale.
- **Fatigue:** Modified Fatigue Impact Scale or Fatigue Severity Scale.
- **Upper limb:** Nine-Hole Peg Test.

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case. Write the course and the time since the relapse in the health condition box.

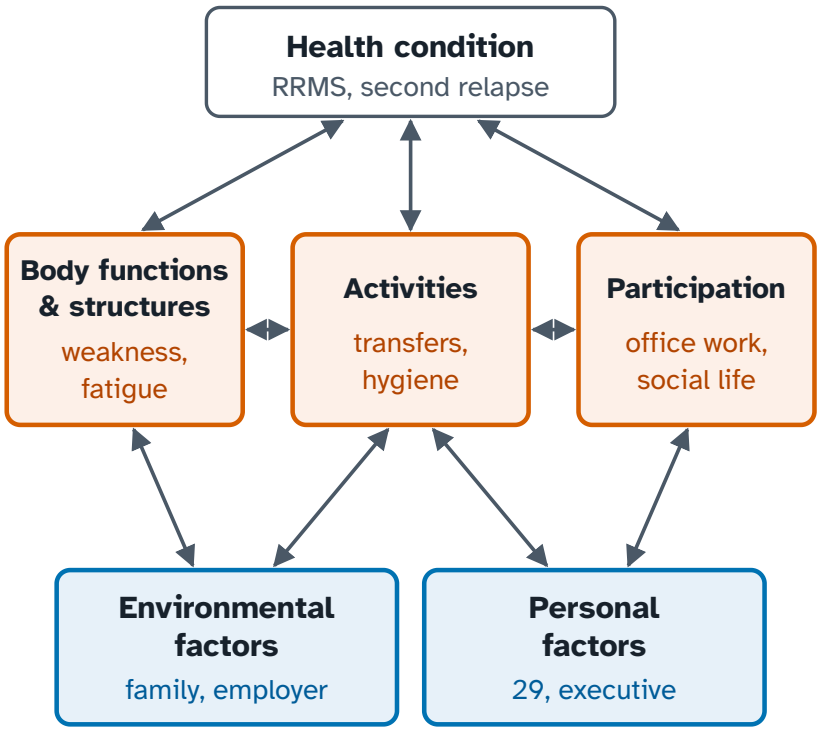
Worked example from a real paper (W19, S24 and S25; 7 + 8 marks)

> "Write ICF and physiotherapy management with long and short term goals for a 29 year old unmarried female executive working in a multinational company, diagnosed as a case of multiple sclerosis 2 years back. She is suffering from a second relapse. She was discharged from the hospital 1 week back. She is moderately functionally dependent in transfers and hygiene."

Spot the ICF clues before you write

second relapse means relapsing-remitting MS, recovering now. discharged 1 week back means early recovery after steroids: start gently and grade up. moderately dependent in transfers and hygiene is the activity limitation. executive in a multinational company means a desk job with travel and long hours: plan fatigue management and return to work.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Relapsing-remitting MS, 2 years; recovering from a second relapse; discharged 1 week ago	Inflammation is settling; partial recovery is expected over weeks
Body function and structure impairments	• Weakness and spasticity, mainly in the legs • Impaired balance and possible ataxia • Fatigue; heat sensitivity • Possible sensory loss, visual disturbance, bladder urgency • Reduced endurance after the hospital stay • Possible low mood and slowed thinking	• Demyelination in the cord and brain blocks or slows conduction • Fatigue comes from the disease itself and from deconditioning • Heat slows conduction further (Uhthoff's phenomenon)

ICF component	From this case	Clinical reasoning
Activity limitations	Needs moderate help for transfers, bathing and toileting; reduced walking distance; stairs and travel difficult	Weak legs, poor balance and fatigue
Participation restrictions	Off work as an executive; reduced social life; worry about career and marriage	A demanding job with long hours and travel needs stamina
Environmental factors	**Facilitators:** family; a large employer that may allow flexible or home working; access to medicines. **Barriers:** commuting, hot weather, long working hours	Workplace adjustments help return to work
Personal factors	29-year-old unmarried, educated woman	Young and educated: able to self-manage fatigue and exercise

One-line functional diagnosis

A 29-year-old executive with relapsing-remitting MS, 1 week after discharge from her second relapse, with leg weakness, spasticity, poor balance and fatigue, who needs moderate help with transfers and hygiene and cannot yet return to her office job. Her family and employer are possible facilitators; heat, commuting and long hours are barriers.

Practice cases from the papers: W17 30-year-old teacher with double vision and an ataxic gait (functional diagnosis); S18 23-year-old engineering student with relapsing-remitting MS who walks with support and has poor fine motor control; S23 35-year-old medical representative, newly diagnosed, with visual disturbance and imbalance, who must travel for work.

Recall check. Fill the ICF for the S23 medical representative. What is her main participation restriction?

11 Problem list and goals

Goals for the 29-year-old executive

Problem	Short-term (2–4 weeks)	Long-term (2–3 months)
Transfers and hygiene	Independent bed-to-chair and toilet transfers; bathes with a shower chair	Independent in all self-care
Strength and spasticity	Stronger hip and knee muscles; full calf and hamstring length	Climbs stairs at home and at work
Balance and walking	Walks indoors with a stick	Walks outdoors without an aid; no falls
Fatigue	Uses pacing and rest breaks; exercises without next-day exhaustion	Manages a full working day
Work	Plans a phased return with her employer	Back at work with adjustments; exercises at least 150 minutes a week

12 PT management

Principles: exercise is safe and helpful at every level; grade it to the person's EDSS and fatigue; keep cool; train balance and function; pace activity; support work and family life (7 marks in S19 and W20).

By phase and stage

- **During a relapse:** rest the affected function, keep range, position well, and maintain gentle activity as tolerated; avoid overheating.
- **Recovery after a relapse:** grade back up: transfers, balance, walking, strength and endurance; short sessions with rests.

- **Mild to moderate disability (EDSS 0–6.5):** aerobic and resistance training at the doses below; balance and gait training; fatigue management; work support.
- **Severe disability (EDSS 7–9):** short daily sessions; upper-body exercise; standing frame; transfers; seating and pressure care; breathing; carer training.

Exercise doses (Kalb et al., 2020)

Disability	Aerobic	Resistance
EDSS 0–6.5	**2–3 times a week**, **10–30 minutes** at moderate intensity (**40–60% of maximum heart rate**, or perceived exertion 11–13 on the 6–20 scale)	**2–3 times a week**; **1–3 sets** of **8–15 repetitions**; 5–10 exercises
EDSS 7.0–9.0	Up to **20 minutes a day** (less at higher scores), **3–7 days a week**, with rests; arm exercise in short intervals	Upper body **3 times a week**, **3 sets of 10** as able

Overall target for everyone: at least **150 minutes a week** of exercise or of everyday physical activity, reached gradually.

Classic answer vs current evidence: exercise in MS

Classic	Current	Write in the exam
Avoid exertion and heat because exercise worsens MS and fatigue.	Exercise and lifestyle physical activity are safe for people with MS and should be promoted at every disability level. Heat may briefly worsen symptoms in heat-sensitive people, so cool them rather than stop exercise (Kalb et al., 2020).	"Graded aerobic and resistance exercise 2–3 times a week, with pacing and cooling, aiming at 150 minutes of activity a week."

Other techniques

- **Spasticity:** daily stretching, standing, positioning; check triggers (bladder, skin, infection).
- **Balance and gait:** static and dynamic balance, stepping, turning, walking with a second task when safe; AFO for foot drop; walking aids as needed.
- **Ataxia and intention tremor:** proximal stability, Frenkel's exercises, slow controlled practice with visual feedback; weighted cuffs on the wrist may reduce tremor for some tasks.
- **Fine motor control** (S18 case): grip and pinch strength, task practice (writing, buttoning, eating) with adapted pens and cutlery.
- **Fatigue:** energy conservation (plan, prioritise, pace, sit for tasks), rest breaks, exercise at the coolest time of day.
- **Heat:** pre-cooling (cool drink, cool shower, fan), cool room, cooling vest; avoid hot packs and hot baths.
- **Bladder:** timed voiding and pelvic floor training (see the bladder page (p. 238)).

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Low fitness and fatigue	Cycling, walking, swimming in cool water: **2–3 times a week**, **10–30 minutes**, moderate intensity	Aerobic training improves fitness and can reduce fatigue
Weak legs	Resistance training **2–3 times a week**, **1–3 sets of 8–15**	Stronger legs improve walking, stairs and transfers
Poor balance	Progressive balance and stepping practice; vision and surface challenges as able	Reduces falls from weakness, ataxia and sensory loss
Spasticity	Daily stretching and standing; positioning	Keeps range and eases spasms

Impairment	Technique and dose	Why
Heat sensitivity	Pre-cooling, cool environment, cooling vest, breaks	Prevents Uhthoff's worsening so the person can complete exercise
Severe disability (EDSS 7 or more)	Up to **20 minutes a day**, **3–7 days a week**; arm exercise; standing frame	Keeps strength for transfers, prevents complications

Recall check. Plan a week of exercise for a person with MS at EDSS 4, including how you manage heat and fatigue.

13 Precautions, contraindications and stop criteria

Take care

- Avoid overheating: cool room, cool water, breaks; no hot packs on numb skin.
- Fatigue: stop before exhaustion; if the person is worse the next day, reduce the dose.
- A new symptom lasting more than 24 hours without fever or heat may be a relapse: refer.
- Fever or urine infection can cause a pseudo-relapse; treat the cause.
- Falls: guard during balance work; check vision and footwear.
- After steroid courses: bones may be weaker; watch blood sugar and mood.

14 Education, lifestyle and home programme

- MS is not contagious or inherited in a simple way; most people continue to work and live full lives.
- Daily home exercise plus everyday activity, building towards 150 minutes a week.
- Energy conservation at home and at work: sit for tasks, plan the day, rest before tiredness, keep things within reach.
- Keep cool in summer; avoid very hot baths.
- Workplace adjustments: flexible hours, home working, a seat near the toilet, a cool office, less travel.
- Stop smoking; healthy weight; take disease-modifying medicines regularly; report new symptoms.

15 Prognosis and discharge criteria

- The course varies widely. Better outlook: female, young onset, sensory or optic onset, good recovery from the first relapse, few early relapses.
- Poorer outlook: older onset, motor or cerebellar onset, frequent early relapses, a progressive course from the start.
- Disease-modifying medicines have reduced relapses and slowed disability for many people.
- Physiotherapy is given in blocks after relapses and when function changes; discharge from a block when goals are met and the person exercises on their own.

16 Exam corner

Past questions (MUHS, 2016–2026)

W19 Q4(b), S24 Q4(b), S25 Q4(b) (7 + 8 marks)

The worked example in section 10 (asked three times).

S23 Q3(b), 7 + 8 marks

35-year-old medical representative recently diagnosed with MS, with visual disturbance and loss of balance; her job needs a lot of travelling. ICF and physiotherapy.

S18 Q4(a), 6 + 9 marks

23-year-old engineering student with relapsing-remitting MS for 3 years; walks with support, uses public transport, attends college, has difficulty with grooming, writing and using a spoon. Goals; physiotherapy for his impairments.

W17 Q3(a), 3 + 5 + 7 marks

30-year-old primary school teacher with difficulty walking and double vision for 5 months; wide-based ataxic gait, mild slurred speech, tremor on finger-nose test, MRI with many white areas; a few steps to enter her house. Functional diagnosis with rationale; goals; physiotherapy.

7-mark questions

Visual, cognitive and behavioural disturbances in MS (W18); physiotherapy for MS (S19); stages of MS and physiotherapy for any one stage (S20); clinical features of MS and physiotherapy (W20).

3-mark questions

Three clinical symptoms of MS (W16); three clinical features of MS (S22); Uhthoff's phenomenon (W22); Charcot's triad in MS (W25).

Pages for this topic: Guillain-Barré syndrome (p. 144), cerebellar disease and ataxia (p. 155), bladder and bowel training (p. 238), tone management (p. 216).

Keywords to underline

demyelination relapsing-remitting dissemination in space and time Uhthoff's phenomenon Charcot's triad fatigue EDSS energy conservation cooling graded aerobic and resistance exercise 150 minutes a week

Answer skeleton: any MS long case

1. Restate: age, job, course of MS, time since the last relapse, function now (EDSS if possible).
2. **[ICF, 5–7]** ICF diagram and table with clinical reasoning; one-line functional diagnosis.
3. **[Goals, 3–5]** Short-term goals for transfers, self-care, balance and fatigue; long-term goals for walking, work and lifelong exercise.
4. **[Management, 7–8]** Phase (relapse, recovery or stable); aerobic and resistance doses by EDSS; balance and gait; spasticity; ataxia and fine motor; fatigue and heat; bladder; work and home; precautions.

Answer: functional diagnosis with rationale in the W17 case (3 marks)

MS with cerebellar and brainstem involvement: a 30-year-old woman with double vision and gait difficulty for 5 months, a wide-based ataxic gait, slurred speech and intention tremor (cerebellar signs, close to Charcot's triad), and multiple white-matter lesions on MRI (dissemination in space). She has difficulty walking and climbing the steps into her house.

Draw these

- The three disease courses (section 2)
- The ICF model filled from the case

Viva questions

1. What is Uhthoff's phenomenon, and how do you manage it?
2. What is Charcot's triad?
3. Name the courses of MS.
4. What does EDSS 6.0 mean?
5. How would you dose aerobic exercise for a person with MS?

6. What is energy conservation?

Linked practical spots

- Cooling vest; Nine-Hole Peg Test; EDSS and fatigue scales; Frenkel's exercises.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **What is Uhthoff's phenomenon?**

Answer

A temporary worsening of MS symptoms with a rise in body temperature (exercise, hot bath, hot weather, fever), because heat slows conduction in demyelinated fibres. It settles on cooling and is not a relapse. Manage with pre-cooling, a cool room and breaks.

3 m **Describe Charcot's triad in multiple sclerosis.**

Answer

Nystagmus, intention tremor and scanning (staccato) speech, from cerebellar and brainstem lesions.

7 m **Describe the stages of multiple sclerosis and physiotherapy for any one stage.**

Answer

Stages: clinically isolated syndrome; relapsing-remitting (relapses with recovery); secondary progressive (steady worsening after a relapsing phase); primary progressive (worsening from the start). Disability stages by EDSS: mild (0–4), moderate (4.5–6.5), severe (7 or more).

Physiotherapy for relapsing-remitting MS with mild to moderate disability: aerobic exercise 2–3 times a week for 10–30 minutes at moderate intensity; resistance training 2–3 times a week, 1–3 sets of 8–15; balance and gait training; stretching for spasticity; energy conservation and cooling; a home programme towards 150 minutes a week; graded return after relapses.

LAQ **A 23-year-old engineering student with relapsing-remitting MS for 3 years walks with support and has difficulty grooming, writing and using a spoon. Write short- and long-term goals and physiotherapy for his impairments. (6 + 9, S18)**

Answer

Goals (6): short-term: steadier walking with a stick; better grip and control for writing and eating with adapted tools; manage fatigue during college hours. Long-term: walks on campus without an aid or with one stick; writes for a full exam; independent grooming; exercises 150 minutes a week.

Physiotherapy (9): aerobic and resistance training at the Kalb doses; balance and gait training; proximal stability and Frenkel's exercises for ataxia; fine motor practice with thick pens, weighted cutlery and task repetition; stretching for spasticity; energy conservation and cooling; travel planning on public transport; a home programme and review after any relapse.

MCQ **A person with MS develops blurred vision after a hot shower that clears in an hour. This is:**

A. A new relapse

B. Uhthoff's phenomenon

C. Lhermitte's sign

D. Optic atrophy

Answer
B. Heat-related, temporary and not a relapse.

MCQ **The most common course of MS at onset is:**

A. Primary progressive

B. Secondary progressive

C. Relapsing-remitting

D. Progressive-relapsing

Answer
C. About 85% start as relapsing-remitting MS.

MCQ **For a person with MS at EDSS 3, a suitable aerobic dose is:**

A. 10–30 minutes, 2–3 times a week, moderate intensity

B. 60 minutes daily at maximal effort

C. No aerobic exercise

D. Only passive movements

Answer
A. Kalb et al. (2020) for EDSS 0–6.5.

Last recall check. Draw the three courses of MS, define Uhthoff's phenomenon and Charcot's triad, and write the aerobic and resistance doses.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Kalb R, et al. Exercise and lifestyle physical activity recommendations for people with multiple sclerosis throughout the disease course. *Mult Scler* 2020;26:1459–69.
3. Montalban X, et al. Diagnosis of multiple sclerosis: 2024 revisions of the McDonald criteria. *Lancet Neurol* 2025;24:850–65.
4. Walton C, et al. Rising prevalence of multiple sclerosis worldwide: insights from the Atlas of MS, third edition. *Mult Scler* 2020;26:1816–21.
5. Lublin FD, et al. Defining the clinical course of multiple sclerosis: the 2013 revisions. *Neurology* 2014;83:278–86.
6. Kurtzke JF. Rating neurologic impairment in multiple sclerosis: an expanded disability status scale (EDSS). *Neurology* 1983;33:1444–52.
7. Structure and classical content: O'Sullivan and Schmitz, *Physical Rehabilitation*; Stokes, *Physical Management for Neurological Conditions*.

5A(g) Guillain-Barré syndrome: ICU to recovery, ICF and physiotherapy

> Guillain-Barré syndrome was a long case in 9 of the 20 papers (2016–2026), and most of those patients were in the ICU or had just left it. The examiner wants three things: watch the breathing, prevent the complications of lying still, and rebuild strength without overworking muscles whose nerves are still recovering.

Syllabus	Ch 5A(g) Demyelinating diseases: multiple sclerosis and G.B. syndrome (multiple sclerosis has its own page)
Priority	**Tier 1.** In 10 of 20 papers (2016–2026): 9 long cases and 2 seven-mark questions.
Asked as	LAQ W16 · S17 · W17 · S18 · S20 · W21 · S24 · S25 · W25 · 7m S19 · W21
Usual split	15 (ICF with clinical reasoning 5 or 7, goals 3–5, management 7–10); W17 asked the nerve conduction picture for 3 marks and S18 the diagnosis with rationale
Checked against	Leonhard et al., diagnosis and management of GBS in ten steps (2019); EAN/PNS guideline on diagnosis and treatment of GBS (2023); Garssen et al., exercise training after GBS (2004); Hughes GBS disability scale. Full list under Sources (p. 154). Content checked 20 October 2026.

1 Definition

Guillain-Barré syndrome (GBS) is an acute, immune-mediated disease of the peripheral nerves and nerve roots (a polyradiculoneuropathy). It causes rapidly progressive, usually symmetrical weakness of the limbs with absent or reduced tendon reflexes, reaching its worst within 4 weeks.

- **How common:** about **1–2 per 100,000** people a year, at any age.
- **A trigger:** about two-thirds have an infection in the 6 weeks before, often diarrhoea or a chest infection.
- It is a lower motor neuron (LMN) disease: the brain and cord are normal.

2 Classification and grading

Subtype	What is damaged	Points
AIDP (acute inflammatory demyelinating polyradiculoneuropathy)	The myelin sheath	Sensory and motor; the usual form in Europe and the Americas
AMAN (acute motor axonal neuropathy)	Motor axons	Pure motor, often after *Campylobacter* diarrhoea; common in Asia
AMSAN (acute motor and sensory axonal neuropathy)	Motor and sensory axons	Severe; slower recovery
Miller Fisher syndrome	Cranial nerves and sensory pathways	Eye-muscle weakness, ataxia and areflexia

The sensorimotor form makes up about 70% of cases in Europe and the Americas but only 30–40% in Asia, where pure motor forms are more common.

GBS disability scale (Hughes)

Score	Meaning
0	Healthy
1	Minor symptoms; can run
2	Walks 10 m or more without help but cannot run
3	Walks 10 m across an open space with help

Score	Meaning
4	Bedridden or chair-bound
5	Needs a ventilator for at least part of the day
6	Dead

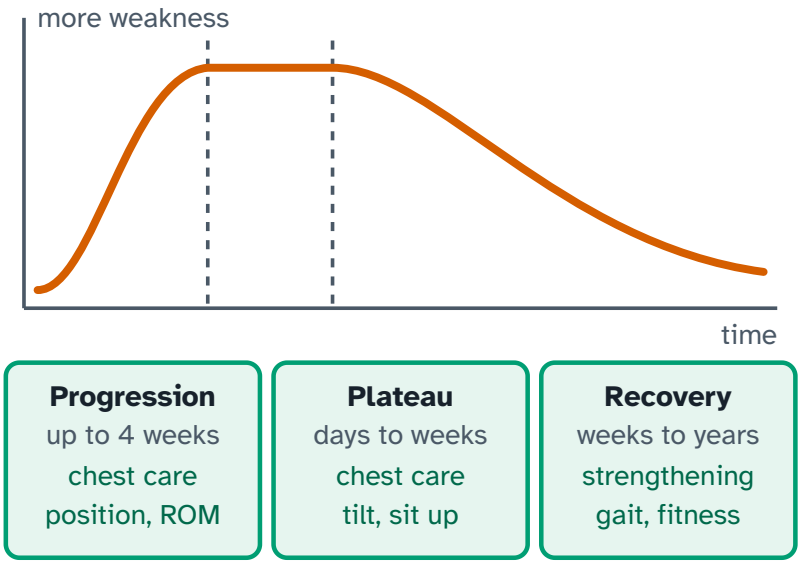
3 Causes and risk factors

- **Infections linked to GBS:** *Campylobacter jejuni* (diarrhoea), cytomegalovirus, hepatitis E, *Mycoplasma pneumoniae*, Epstein–Barr virus and Zika virus.
- Other triggers reported: influenza and other viral illnesses, surgery, and rarely vaccines.
- **Mechanism:** the immune response to the infection cross-reacts with nerve molecules (molecular mimicry), so antibodies and immune cells attack the myelin or the axon.

4 Pathophysiology to impairments

1. An immune attack on myelin (AIDP) slows or blocks conduction; an attack on the axon (AMAN) stops it.
2. Because the motor nerves fail, muscles become weak and flaccid with lost reflexes, usually starting in the legs and rising (ascending paralysis).
3. Because the phrenic and intercostal nerves can be involved, breathing muscles weaken: up to 22% need a ventilator in the first week.
4. Because cranial nerves can be involved, facial weakness and swallowing problems appear.
5. Because autonomic nerves are affected, blood pressure and heart rate swing and arrhythmias can occur.
6. Because sensory nerves are affected, there is tingling, loss of position sense and nerve pain.
7. Recovery comes from remyelination (weeks) and from axonal regrowth (months), so demyelinating forms recover faster than axonal ones.

The course of GBS and the physiotherapy focus in each phase



Most people are at their weakest within 2 weeks. Recovery is fastest in the first year but can continue for more than 5 years.

Recall check. Why does a person with GBS need their vital capacity checked, and why do axonal forms recover more slowly?

5 Clinical features

- **Weakness:** rapidly progressive, symmetrical, flaccid; usually starts in the legs and ascends to the arms, trunk, breathing and face. Proximal muscles are often as weak as distal ones.
- **Reflexes:** absent or reduced in the weak limbs.
- **Sensory:** tingling in the hands and feet; loss of position sense; often **pain** in the back and legs.
- **Cranial nerves:** bilateral facial weakness; swallowing and speech problems.
- **Breathing:** breathlessness, weak cough, rapid shallow breathing, use of accessory muscles.
- **Autonomic:** swings in blood pressure and heart rate, arrhythmias, sweating, urinary retention, constipation.
- Fatigue is very common and can persist after strength returns.

Diagnosis

- **Required:** progressive weakness of both arms and legs (it may start in the legs only), and absent or reduced tendon reflexes in the weak limbs.
- **Strongly support it:** progression over days to 4 weeks (usually under 2); relative symmetry; mild sensory signs; cranial nerve involvement, especially bilateral facial weakness; autonomic dysfunction; pain.
- **Make you doubt it:** a sensory level, early bladder or bowel loss, clear asymmetry, or more than 50 cells in the CSF; these point to a cord lesion or infection.

6 Investigations

- **CSF (lumbar puncture):** raised protein with a normal cell count (albuminocytological dissociation). Protein is normal in 30–50% in the first week, so a normal result does not rule GBS out.
- **Nerve conduction studies and EMG** (3 marks in W17): see the table.
- **Breathing tests:** vital capacity, maximum inspiratory and expiratory pressure, single breath count, arterial blood gases.
- **Others:** ECG; stool tests for *Campylobacter*; antibodies (anti-GQ1b in Miller Fisher syndrome); MRI of the spine if a cord lesion is possible.

Nerve conduction and EMG picture

Finding	Demyelinating (AIDP)	Axonal (AMAN, AMSAN)
Conduction velocity	Slowed	Normal or nearly normal
Distal latency	Prolonged	Normal
F-waves	Prolonged or absent: often the earliest sign, because the roots are hit first	Reduced or absent
Amplitude	Conduction block and temporal dispersion	Low motor (and in AMSAN sensory) amplitudes
Sensory nerves	Abnormal in the arms with a normal sural response ("sural sparing")	Normal in AMAN
EMG	Reduced recruitment	Denervation (fibrillations) after 2–3 weeks

7 Medical and surgical management

- **Immunotherapy** (EAN/PNS 2023) for people who cannot walk unaided: **IVIg** **0.4 g/kg a day for 5 days**, within 2 weeks of onset (up to 4 weeks as good practice); or **plasma exchange** **12–15 L** over 4–5 exchanges within 4 weeks. Both work equally well; plasma exchange followed straight away by IVIg is not advised.
- **Not advised:** corticosteroids (oral steroids are recommended against); a second IVIg course.

- **ICU care:** admit if breathing is failing, swallowing is unsafe or the autonomic system is unstable. Intubate early when the **20/30/40 rule** is met: vital capacity under **20 mL/kg**, maximum inspiratory pressure under **30 cmH2O**, or maximum expiratory pressure under **40 cmH2O**. A single breath count of 19 or less predicts the need for ventilation. Tracheostomy if ventilation is prolonged.
- **Supportive:** DVT prophylaxis; monitoring of ECG, heart rate, blood pressure, bladder and bowel; nutrition (tube feeding if swallowing is unsafe); eye care in facial weakness; medicines for pain (gabapentinoids, tricyclic antidepressants or carbamazepine).

8 Complications and prevention

Complication	What to look for	Physio role
Respiratory failure	Falling vital capacity or single breath count; rapid shallow breathing; weak cough; mucus plugs	Measure and report; chest clearance; positioning; assisted cough
Pneumonia and aspiration	Fever, secretions; coughing with food when the bulbar muscles are weak	Airway clearance; upright positioning for meals; report swallowing problems
Autonomic instability	Sudden high or low blood pressure; arrhythmia, especially when turned or suctioned	Monitor during treatment; slow position changes; stop and report
DVT and pulmonary embolism	Swollen leg; sudden breathlessness	Passive and active leg movements; early mobilisation; stockings as prescribed
Pressure ulcers and pressure palsies	Sacrum and heels; ulnar nerve at the elbow and peroneal nerve at the fibular head	Regular turning; pad and position the elbows and knees
Contractures	Ankle plantarflexion, shoulder, fingers	Daily range of motion; resting splints for wrist and ankle
Pain, anxiety, hallucinations and depression	Back and limb pain; fear while on a ventilator	Gentle handling, positioning, explanation and reassurance; refer

9 PT assessment

- **History:** preceding infection, date of onset, speed of progression, immunotherapy given, ventilation, swallowing, pain, job, home and family.
- **Vital signs and monitors:** heart rate and rhythm, blood pressure (lying and sitting), oxygen saturation, ventilator settings.
- **Respiratory:** vital capacity, single breath count, cough strength, chest expansion, auscultation, secretions.
- **Motor:** manual muscle testing; **MRC sum score** (6 muscle groups on each side, total 0–60); reflexes; facial and bulbar muscles.
- **Sensory:** light touch, position sense; pain score.
- **Range of motion** and skin.
- **Function:** GBS disability scale; bed mobility, sitting balance, transfers, standing, walking (10-metre walk, 6-minute walk later); hand function.
- **Fatigue:** Fatigue Severity Scale.

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case. In an ICU case, breathing goes first in the impairments.

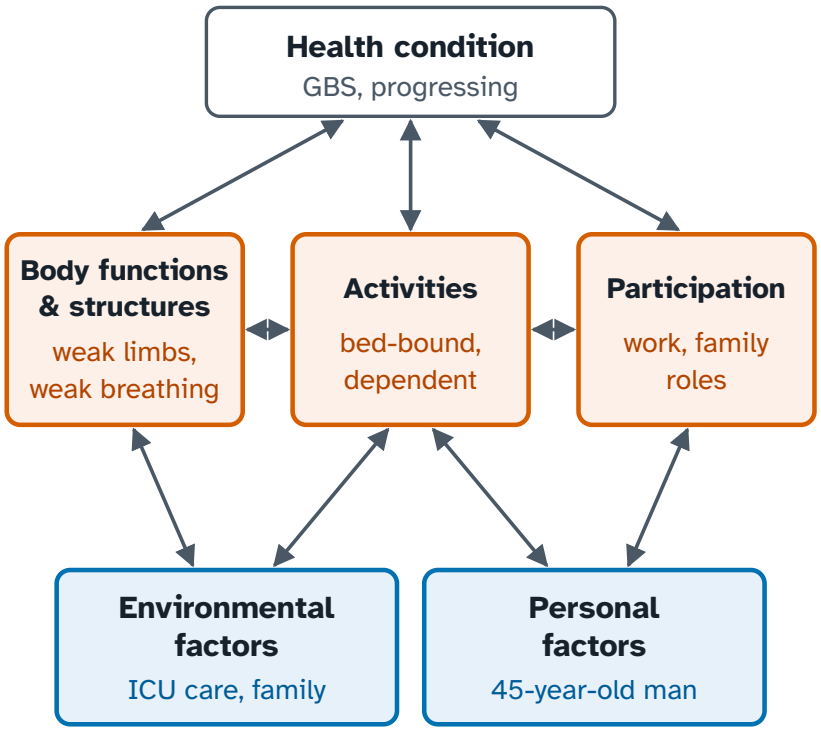
Worked example from a real paper (S25, 5 + 7 + 3 marks)

> "Write down ICF with clinical reasoning for a 45 year old male having complaints of abdominal pain and viral fever followed by weakness of upper limb and lower limb since 10 days. Presently the patient has also started feeling breathing difficulties. The patient is admitted in the ICU and diagnosed as Guillain-Barré syndrome. Write the short term and long term goals with physiotherapy management."

Spot the ICF clues before you write

abdominal pain and viral fever is the preceding infection. weakness since 10 days means he is still in the progression phase, near his worst. breathing difficulties and ICU mean respiratory muscle weakness: monitor vital capacity and keep the chest clear. A 45-year-old man is likely a working earner: plan the return to work.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	GBS, day 10, still progressing, with respiratory involvement, after a viral illness	Immune attack on peripheral nerves and roots; most people are at their worst within 2 weeks
Body function and structure impairments	• Weak breathing muscles: low vital capacity, weak cough, risk of respiratory failure • Symmetrical flaccid weakness of all four limbs and the trunk; absent reflexes • Possible facial and swallowing weakness • Tingling, nerve pain • Unstable blood pressure and heart rate • Risks: chest infection, DVT, pressure ulcers, contractures, pressure palsies • Anxiety	• Phrenic and intercostal nerves are involved • LMN lesion of motor nerves and roots • Cranial nerves can be involved • Sensory and autonomic fibres are involved too • Immobility in bed causes the secondary risks

ICF component	From this case	Clinical reasoning
Activity limitations	Cannot turn in bed, sit, stand or walk; dependent for feeding, washing and toileting; cannot clear secretions; cannot speak if intubated	Weak limbs, trunk and breathing muscles
Participation restrictions	Off work; cannot fulfil family and social roles	Hospital stay of weeks; recovery over months
Environmental factors	**Facilitators:** ICU care, immunotherapy, family support. **Barriers:** cost of treatment; an unfamiliar, noisy ICU	Early immunotherapy and good supportive care improve the outcome
Personal factors	45-year-old man, likely the earner; fear of the ventilator	Older age and severe weakness lower the chance of a quick recovery

> **One-line functional diagnosis**
>
> A 45-year-old man with GBS, 10 days from onset and still progressing, with weak breathing muscles and flaccid weakness of all four limbs, who is bed-bound in the ICU, dependent for all care and unable to work.

Practice cases from the papers: W16 20-year-old student out of the ICU after plasma exchange who rolls and sits but cannot stand; W17 20-year-old woman with a tracheostomy, now on the ward, who stands with difficulty; W25 10-year-old boy out of the ICU for 2 days who must cycle to school again.

Recall check. Fill the ICF for the W25 10-year-old boy. Which facilitators and goals are different for a child?

11 Problem list and goals

Goals for the 45-year-old man in the ICU

Problem	Short-term (1–3 weeks)	Long-term (3–6 months)
Breathing	Chest clear; no collapse or pneumonia; vital capacity monitored and kept stable	Normal breathing and cough
Range and skin	Full range in all joints; no pressure ulcers or pressure palsies	No contractures
Strength and mobility	Sits out of bed with support; tolerates the tilt table	Walks independently indoors and outdoors, with an AFO if foot drop remains
Self-care	Feeds himself with set-up once the arms recover	Independent in all daily activities
Fatigue and fitness	Short sessions without next-day exhaustion	Endurance to return to work

12 PT management

Principles: watch the breathing and the autonomic system; prevent the complications of immobility; exercise within the person's strength; never overwork recovering muscles; plan the return to home and work.

By phase

- **Progression and plateau (ICU, 7 marks in W21):**
  - Chest: positioning (side lying, upright when allowed), deep breathing and incentive spirometry if the person can, airway clearance and suctioning, assisted cough; measure vital capacity or single breath count and report a fall.
  - Ventilated patients: positioning, secretion clearance as per the unit protocol, support during weaning.
  - Limbs: neutral positioning; resting splints for the wrist and hand and an AFO or foot support; gentle passive range of motion once or twice a day; pad the elbows and fibular heads; turn regularly.
  - Pain: gentle handling and positioning; explain everything before touching.

- Start sitting up and the tilt table as soon as breathing and blood pressure are stable, watching for postural hypotension.

- **Early recovery (ward):** active-assisted, then active exercise, first with gravity eliminated; sitting balance; rolling, bridging, sit-to-stand; standing in the parallel bars; short sessions with rests.
- **Recovery and rehabilitation:** progressive strengthening; gait training (parallel bars, walker, then a stick), with an AFO for foot drop; balance training with vision when position sense is poor; hand function with the occupational therapist; endurance training; fatigue management (pacing, planning the day); return to school, work and sport.

Avoid overwork

Supervised exercise improves fitness and independence, but its intensity must be watched closely, because overwork causes fatigue (Leonhard et al., 2019). If weakness or tiredness is worse the next day, the session was too hard: shorten it or lower the load.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Weak breathing muscles	Deep breathing, incentive spirometry, positioning; vital capacity or single breath count each shift; report vital capacity under **20 mL/kg**	Falling vital capacity warns of respiratory failure before blood gases change
Retained secretions	Positioning, manual techniques, suctioning, assisted cough; watch the monitor during suctioning	A weak cough lets mucus plug the airways; suctioning can trigger arrhythmias
Risk of contracture	Passive range of motion **once or twice a day**, gently; resting splints for the wrist and ankle	Flaccid muscles shorten when joints rest in one position
Weak limbs (recovering)	Active-assisted, then active, then progressive resistance; few repetitions, rests between sets; stop before fatigue	Recovering nerves supply fewer muscle fibres, which tire quickly and can be overworked
Foot drop	AFO for walking; strengthen dorsiflexors as they recover	Prevents tripping and ankle sprains
Fatigue and low fitness (walking again)	Cycling: **30 minutes** at **70–90% of maximum heart rate**, **3 times a week** for **12 weeks**, with a 5-minute warm-up and a cool-down	This programme reduced fatigue and improved fitness after GBS (Garssen et al., 2004)
Poor balance from lost position sense	Balance training with visual feedback: mirror, then eyes closed as it improves	Vision replaces the missing position sense

Recall check. What would you do in the ICU on day 10, and how would the plan change once the person can stand?

13 Precautions, contraindications and stop criteria

Take care

- Stop and report if breathing worsens: vital capacity or single breath count falling, rising respiratory rate, a weaker cough.
- Watch heart rate, rhythm and blood pressure during turning, suctioning and sitting up; tilt back if blood pressure drops.
- Do not overstretch flaccid joints; support the shoulders.
- Avoid fatigue: short sessions, rests, and less work if the person is weaker the next day.
- Upright positioning for meals if swallowing is weak; protect the eyes if the face is weak.

14 Education, lifestyle and home programme

- Explain that GBS is not contagious, that most people recover, and that recovery is slow.
- Teach the family positioning, range of motion and safe transfers.
- A home exercise programme that builds slowly; pacing and energy conservation for fatigue.
- Return to school or work in stages; hand controls or an AFO if needed.
- Seek help at once if weakness returns: treatment-related fluctuations happen in 6–10% and recurrence in 2–5%.

15 Prognosis and discharge criteria

- About **80%** walk independently 6 months after onset. Recovery is fastest in the first year but may continue for more than 5 years.
- Death occurs in 3–10%, even with the best care, mostly in older people with severe disease.
- Older age, diarrhoea before the illness and severe weakness are linked to a poorer outcome; the modified Erasmus GBS outcome score (mEGOS) estimates the chance of walking.
- Severe pain still affects at least a third at 1 year, and fatigue affects 60–80%.
- Worsening 8 weeks or more after onset, or three or more relapses after treatment, suggests acute-onset CIDP rather than GBS.
- Discharge when the person is safe at home, independent or with planned help, with a home programme and follow-up.

16 Exam corner

Past questions (MUHS, 2016–2026)

W25 Q4(a), 5 + 10 marks

10-year-old boy with ascending, symmetrical, progressive weakness (acute demyelinating disorder); legs grade 2, arms grade 3, hands grade 2; weak breathing muscles; out of the ICU 2 days ago; sensation and cranial nerves intact; studies in 5th standard and needs to ride a bicycle to school; family of four in an apartment. Short- and long-term goals; management in detail.

S25 Q3(a), 5 + 7 + 3 marks

The worked example in section 10.

S24 Q3(b)

15-year-old girl with GBS in the ICU, with difficulty in breathing. ICF with physiotherapy.

W21 Q4(b), 5 + 10 marks

19-year-old college student with leg weakness spreading to the arms, 2 weeks after a stomach infection; lives on the third floor with a lift. ICF with reasoning; detailed physiotherapy.

S20 Q3(a), 5 + 10 marks

30-year-old woman with fever 2 weeks ago, then progressive weakness of all four limbs, breathless on exertion, in hospital for a week. ICF; detailed physiotherapy.

S18 Q3(b)

44-year-old clerk with a gut infection 3 weeks ago and 2 days of weakness of all four limbs, proximal more than distal; severe difficulty grooming, bathing, eating and walking; normal sensation; delayed F-waves. Diagnosis with rationale; goals; physiotherapy for the impairments.

W17 Q4(a), 3 + 5 + 7 marks

20-year-old woman in the ICU with sudden weakness of all limbs and the trunk, tracheostomy for respiratory distress; after 15 days, recovering on the ward; stands with difficulty and walks with full support. EMG and NCV picture; short-term goals; physiotherapy.

S17 Q3(a)

25-year-old man with leg weakness for a week after a viral infection, spreading on both sides, with sensory loss and areflexia; difficulty sitting and standing. Assessment with ICF; goals with treatment.

W16 Q3(b), 7 + 8 marks

20-year-old student with GBS, in the ICU for 5 days for respiratory involvement, improved after plasma exchange and now on the ward; rolls and sits but cannot stand. ICF with reasoning; short-term goals with management.

7-mark questions

Physiotherapy management of GBS (S19); physiotherapy in acute-stage GBS (W21).

Pages for this topic: neuropathies (p. 107), orthoses (p. 249), transfers and gait training (p. 232).

Keywords to underline

ascending symmetrical flaccid weakness areflexia preceding infection albuminocytological dissociation vital capacity 20/30/40 rule autonomic instability IVIg or plasma exchange avoid overwork fatigue

Answer skeleton: any GBS long case

1. Restate: age, days since onset (progression, plateau or recovery), breathing and ICU status, current mobility, home and role.
2. **[ICF, 5–7]** ICF diagram and table with clinical reasoning, breathing first; one-line functional diagnosis.
3. **[Goals, 3–5]** Short-term goals for the present phase; long-term goals for walking, self-care, fatigue and return to role.
4. **[Management, 7–10]** By phase: ICU chest care, monitoring, positioning, splints and range; then graded strengthening, sitting, standing and gait with an AFO; balance; endurance; fatigue; education; precautions (autonomic instability, overwork).

Answer: diagnosis with rationale in the S18 case

Guillain-Barré syndrome, probably a pure motor (axonal) form. Rationale: a gut infection 3 weeks before; acute weakness of all four limbs over 2 days; symmetrical, with proximal muscles as weak as distal ones; sensation normal; delayed F-waves, which show early nerve root involvement. Rule out myasthenia gravis (fatigable weakness, eye signs, normal F-waves), an acute cord lesion (sensory level, bladder loss, later spasticity), hypokalaemic paralysis (low potassium) and polymyositis (raised muscle enzymes, normal nerve conduction).

Draw these

- The course of GBS with the physiotherapy focus (section 4)
- The ICF model filled from the case

Viva questions

1. What is albuminocytological dissociation?
2. What is the 20/30/40 rule?
3. Why are F-waves often the first nerve conduction sign?
4. Why should you avoid overworking muscles in GBS?
5. How do you tell GBS from acute myelitis at the bedside?

6. What is the GBS disability scale?

Linked practical spots

- Incentive spirometer; resting hand splint and AFO; MRC grading; peak cough flow meter.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Describe the EMG and NCV picture in GBS.**

Answer

Demyelinating form: slowed conduction velocity, prolonged distal latency, prolonged or absent F-waves (often the earliest sign), conduction block and temporal dispersion, abnormal arm sensory responses with a normal sural response. Axonal form: low motor amplitudes with near-normal velocity; denervation on EMG after 2–3 weeks.

3 m **List three features that support a diagnosis of GBS.**

Answer

Progression over days to 4 weeks (usually under 2); relative symmetry; bilateral facial weakness; autonomic dysfunction; raised CSF protein with a normal cell count; nerve conduction showing demyelination or axonal loss.

7 m **Physiotherapy in acute-stage GBS.**

Answer

Assess breathing (vital capacity, single breath count, cough) and report a fall; chest care: positioning, deep breathing, incentive spirometry, airway clearance, suctioning, assisted cough; monitor heart rate and blood pressure during turning and suctioning. Neutral positioning, regular turning, padding of elbows and knees; resting splints; gentle passive range once or twice a day. DVT prevention with leg movements. Pain relief by positioning and gentle handling. Sitting up and tilt table once stable. Explain and reassure; teach the family.

LAQ **A 10-year-old boy is out of the ICU for 2 days after GBS; legs grade 2, arms grade 3, hands grade 2, weak breathing muscles. He needs to ride a bicycle to school. Write short- and long-term goals and the management. (5 + 10, W25)**

Answer

Goals (5): short-term (2–4 weeks): clear chest and stronger cough; sits unsupported; stands with support; grips a spoon and pencil; no contractures. Long-term (3–6 months): walks independently; climbs stairs; writes in class; rides his bicycle to school; full school day without exhaustion.

Management (10): breathing exercises with incentive spirometry and games (blowing bubbles); positioning and range of motion; graded strengthening through play, few repetitions with rests; sitting balance, sit-to-stand, standing frame, gait training with AFOs if foot drop; hand activities (putty, building blocks, writing); balance and coordination; static cycling before road cycling; fatigue pacing at school; teach parents a home programme; watch for overwork.

MCQ **Which finding means a patient with GBS needs ventilation?**

A. Vital capacity 35 mL/kg

B. Vital capacity under 20 mL/kg

C. Single breath count of 30

D. Absent ankle reflexes

Answer
B. The 20/30/40 rule: vital capacity under 20 mL/kg, inspiratory pressure under 30 or expiratory pressure under 40 cmH2O; a single breath count of 19 or less also predicts ventilation.

MCQ **The classic CSF finding in GBS is:**

A. High cells, normal protein

B. High protein, normal cells

C. Low glucose

D. Blood-stained fluid

Answer
B. Albuminocytological dissociation; it may be absent in the first week.

MCQ **Which treatment is not recommended for GBS?**

A. IVIg

B. Plasma exchange

C. Oral corticosteroids

D. DVT prophylaxis

Answer
C. The 2023 EAN/PNS guideline recommends against oral corticosteroids.

Last recall check. Draw the course of GBS, write the 20/30/40 rule, and list five ICU physiotherapy actions.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Leonhard SE, et al. Diagnosis and management of Guillain–Barré syndrome in ten steps. *Nat Rev Neurol* 2019;15:671–83.
3. van Doorn PA, et al. European Academy of Neurology/Peripheral Nerve Society guideline on diagnosis and treatment of Guillain–Barré syndrome. *Eur J Neurol* 2023;30:3646–74.
4. Garssen MPJ, et al. Physical training and fatigue, fitness, and quality of life in Guillain–Barré syndrome and CIDP. *Neurology* 2004;63:2393–5.
5. Hughes RAC, et al. Controlled trial of prednisolone in acute polyneuropathy. *Lancet* 1978;2:750–3 (GBS disability scale).
6. Structure and classical content: Lundy-Ekman, *Neuroscience: Fundamentals for Rehabilitation*; O'Sullivan and Schmitz, *Physical Rehabilitation*; Stokes, *Physical Management for Neurological Conditions*.

5A(h) Cerebellar disease and ataxia: types, tests, Friedreich's ataxia and physiotherapy

> Ataxia came up in 8 papers: three long cases (two young women and men with Friedreich's ataxia, and an older man after a cerebellar stroke), physiotherapy of cerebellar ataxia for 7 marks three times, and the rebound test and sensory ataxia for 3 marks. Every answer separates cerebellar from sensory ataxia first, because sensory ataxia can be helped by vision while cerebellar ataxia needs intensive, repeated coordination and balance practice.

Syllabus	Ch 5A(h) Cerebellar diseases and ataxia
Priority	**Tier 1.** In 8 of 20 papers (2016–2026): 3 long cases, 3 seven-mark and 2 three-mark questions.
Asked as	LAQ W18 · W22 · W24 · 7m S16 · W16 · W19 · 3m W17 · S19
Usual split	15 (ICF 5–7, goals 3, management 7–8; W18 asked structural impairments 2, assessment 5, management 8)
Checked against	Intensive coordinative training in degenerative ataxia (Ilg et al., 2009 and 2010); Scale for the Assessment and Rating of Ataxia (Schmitz-Hübsch et al., 2006); exercise to prevent falls (Sherrington et al., 2017). Full list under Sources (p. 163). Content checked 20 October 2026.

1 Definition

- **Ataxia** is uncoordinated movement (errors in the direction, range, timing and force of movement) that is not due to weakness.
- The cerebellum compares the intended movement with what is happening and corrects it; when it fails, movements overshoot, break into parts and tremble as they near the target.

2 Classification and grading

Type	Lesion	Key difference
Cerebellar ataxia	Cerebellum or its connections	Present with eyes open and closed; nystagmus, dysarthria, intention tremor; Romberg's test usually negative (unsteady even with eyes open)
Sensory ataxia	Posterior columns, dorsal roots or peripheral nerves	Much worse with eyes closed or in the dark; loss of position and vibration sense; Romberg's test positive; stamping gait while watching the feet
Vestibular ataxia	Inner ear, vestibular nerve or nuclei	Vertigo, nystagmus; limb coordination normal; veers to one side

- **Cerebellar hemisphere lesions** cause limb ataxia on the **same side**; **midline (vermis)** lesions cause truncal and gait ataxia.
- **Scales:** Scale for the Assessment and Rating of Ataxia (SARA: 8 items, 0–40, higher is worse); International Cooperative Ataxia Rating Scale (ICARS).

3 Causes and risk factors

- **Vascular:** cerebellar infarct or haemorrhage (posterior circulation stroke).
- **Inherited:** Friedreich's ataxia, spinocerebellar ataxias, ataxia telangiectasia.
- **Acquired:** multiple sclerosis; alcohol; drugs (phenytoin); tumours; head injury; infections and post-infectious cerebellitis in children; hypothyroidism; vitamin E and B12 deficiency.
- **Congenital:** ataxic cerebral palsy.

4 Pathophysiology to impairments

1. The cerebellum receives the motor plan and sensory feedback and corrects errors in timing and force; it also adapts and learns movements.
2. Because error correction fails, movements overshoot or undershoot (**dysmetria**), break into separate parts (**decomposition**), and tremble near the target (**intention tremor**).
3. Because postural control is poor, the trunk sways and the gait is wide-based and staggering; falls are common.
4. Because motor learning depends on the cerebellum, progress is slower, so training needs more repetitions.
5. Secondary problems: reduced activity and fitness, fear of falling, contractures and scoliosis in progressive disease.

Recall check. Why does vision help a patient with sensory ataxia but not much one with cerebellar ataxia?

5 Clinical features

- **Limbs:** dysmetria (past-pointing), dysdiadochokinesia (slow, irregular rapid alternating movements), intention tremor, decomposition of movement, rebound phenomenon, hypotonia.
- **Trunk and gait:** truncal sway (titubation), wide-based, staggering gait with irregular steps; cannot walk heel to toe.
- **Eyes and speech:** nystagmus; dysarthria with scanning or slurred speech.

Rebound test (3 marks in W17)

The patient pulls the forearm towards the shoulder against the examiner's resistance (elbow flexion); the examiner suddenly lets go while guarding the patient's face. **Normal:** the triceps checks the movement at once. **Cerebellar lesion:** the arm flies towards the face or shoulder (impaired check, Holmes' rebound phenomenon). A variant: tap the outstretched arms downward and watch for excessive swing.

Assessment of sensory ataxia (3 marks in S19)

- **Romberg's test:** steady with feet together and eyes open, sways or falls with eyes closed.
- Joint position sense and vibration (128 Hz tuning fork) reduced, starting in the toes.
- Finger-nose and heel-shin tests much worse with the eyes closed.
- Pseudoathetosis: outstretched fingers wander with the eyes closed.
- High-stepping, stamping gait; the person watches the feet and is worse in the dark.

Friedreich's ataxia (W18 and W24 cases)

- The commonest inherited ataxia; autosomal recessive (a GAA repeat in the frataxin gene); onset usually in childhood or adolescence, before 25.
- Progressive gait and limb ataxia, dysarthria, loss of knee and ankle reflexes with extensor plantar responses, loss of position and vibration sense (mixed cerebellar and sensory ataxia), nystagmus.
- **Pes cavus, kyphoscoliosis**, cardiomyopathy (the main cause of death), diabetes, sometimes hearing and visual loss.
- Most people need a wheelchair within 10–20 years of onset.

6 Investigations

- **MRI brain:** stroke, MS plaques, tumours, cerebellar atrophy; spinal cord atrophy in Friedreich's ataxia.
- **Genetic testing** for inherited ataxias.
- Nerve conduction (sensory neuropathy), ECG and echocardiogram (cardiomyopathy), blood sugar, vitamin levels, thyroid function.

7 Medical and surgical management

- Treat the cause: stroke care, MS treatment, stopping alcohol or the causative drug, vitamin replacement.
- **Friedreich's ataxia:** omaveloxolone, the first approved drug (2023, in some countries, for ages 16 and over); treatment of cardiomyopathy and diabetes; scoliosis and foot surgery when needed.
- Few drugs help ataxia itself, so rehabilitation is the main treatment.

8 Complications and prevention

Complication	What to look for	Physio role
Falls and injuries	Falls history, fear of falling, Berg score	Balance training, walking aids, home safety
Deconditioning	Less walking and activity	Aerobic exercise (cycling is safe and useful)
Scoliosis and foot deformity	Kyphoscoliosis, pes cavus (Friedreich's)	Posture, stretching, footwear, orthoses
Cardiac strain	Breathlessness, palpitations (Friedreich's cardiomyopathy)	Moderate intensity; monitor pulse and symptoms
Social isolation	Slurred speech mistaken for drunkenness; stopping outings	Mobility aids, community activity, counselling

9 PT assessment

W18 asked for assessment and tests (5 marks).

- **History:** onset and progression, family history, falls, function, work, home.
- **Coordination tests:** finger-nose, finger-finger, heel-shin, rapid alternating movements, rebound test, drawing circles; eyes open and closed.
- **Sensation:** position sense, vibration, Romberg's test.
- **Balance and gait:** sitting and standing balance, tandem walk, Berg Balance Scale, Timed Up and Go, 10-metre walk.
- **Ataxia scale:** SARA.
- **Others:** tone, strength (often near normal), range, posture and spine, feet, speech, upper limb function (Nine-Hole Peg Test), fitness.

10 PT diagnosis by ICF

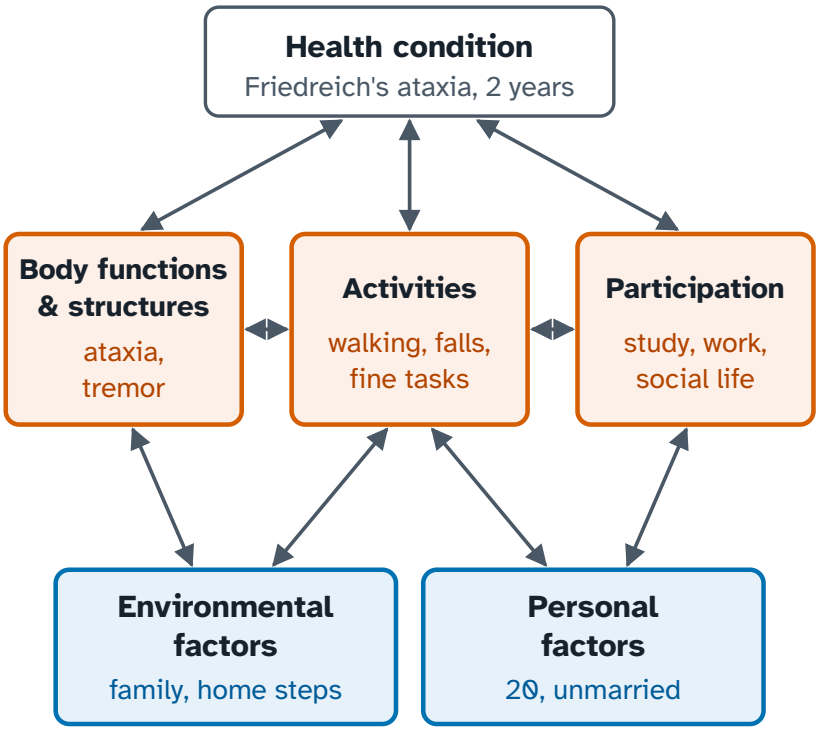
Draw the six-box ICF model, then fill it from the case.

Worked example from a real paper (W24, 5 + 7 + 3 marks)

> "Write ICF and physiotherapy management with short term and long term goals for a 20 year old unmarried female diagnosed with Friedreich's ataxia since 2 years. She complains of ataxic gait and tremors. She is also having difficulty in fine motor activities and complains of repeated falls."

> **Spot the ICF clues before you write**
>
> Friedreich's ataxia is a progressive, mixed cerebellar and sensory ataxia. ataxic gait and tremors are cerebellar signs. fine motor difficulty limits writing and self-care. repeated falls make safety the first goal. 20 years, unmarried: study, work and social life matter.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Friedreich's ataxia, diagnosed 2 years ago; progressive	Degeneration of spinocerebellar tracts, posterior columns, peripheral sensory nerves and cerebellar connections
Body function and structure impairments	• Gait and limb ataxia; intention tremor; dysmetria • Loss of position and vibration sense in the legs • Poor sitting and standing balance • Dysarthria; nystagmus • Possible pes cavus, scoliosis, cardiomyopathy	• Cerebellar and sensory pathways both fail, so vision helps only partly • Progressive disease: maintain function and prevent complications
Activity limitations	Unsteady walking with repeated falls; difficulty with stairs, writing, buttoning, eating, typing	Ataxia and tremor disturb precise and balanced movement
Participation restrictions	College or work, travel, social outings, future marriage plans	Falls and speech change cause withdrawal
Environmental factors	**Facilitators:** family support; assistive devices. **Barriers:** stairs, crowded transport, stigma	Home and transport adaptations reduce falls
Personal factors	20-year-old educated woman; anxiety about the future	Young and motivated: able to follow an intensive home programme

One-line functional diagnosis

A 20-year-old woman with Friedreich's ataxia for 2 years, with gait and limb ataxia, intention tremor and loss of position sense, who walks unsteadily with repeated falls and struggles with fine hand tasks, limiting her study, work and social life.

Practice cases from the papers: W18 22-year-old man with 10 years of clumsy hands, unsteady gait, slurred speech, intention tremor, nystagmus, kyphoscoliosis and pes cavus; W22 65-year-old diabetic man 3 months after a right cerebellar stroke, Berg score 38.

Recall check. Fill the ICF for the W22 cerebellar stroke patient with a Berg score of 38. What does that score mean for falls?

11 Problem list and goals

Goals for the 20-year-old with Friedreich's ataxia

Problem	Short-term (4–6 weeks)	Long-term (3–6 months)
Falls	Knows her fall triggers; uses a walking aid outdoors	Fewer or no falls
Balance and gait	Better Berg and SARA scores; walks indoors with supervision only	Walks safely at home and college with an aid
Fine motor control	Writes a short paragraph with an adapted pen	Independent in self-care and study tasks
Fitness and posture	Cycles 20 minutes; daily posture and stretching routine	Maintains fitness; spine and feet monitored

12 PT management

Asked for 7 marks in S16 and W16 (physiotherapy of cerebellar ataxia) and W19 (rationale of balance training).

Principles

- Intensive, repeated coordination and balance practice: in degenerative ataxia, 4 weeks of intensive coordinative training reduced ataxia and improved gait, and the gains lasted a year with home practice (Ilg et al., 2009 and 2010).
- Train the person's own goals as whole tasks; increase difficulty gradually.
- Separate cerebellar from sensory ataxia: use vision and feedback to compensate in sensory ataxia.
- In progressive disease, keep training for life to maintain function; add aids before falls happen.

Rationale of balance training in cerebellar ataxia (W19)

The cerebellum adapts movements through practice and error. Balance training repeatedly challenges postural control, so the remaining cerebellar circuits and other brain areas learn to reduce sway and correct errors; it also strengthens postural muscles, builds confidence and reduces falls. Exercise that challenges balance for more than 3 hours a week reduces falls most in older people (Sherrington et al., 2017).

Techniques

- **Proximal stability:** weight-bearing in prone on elbows, four-point kneeling, kneeling, half-kneeling and standing; approximation; PNF rhythmic stabilisation and stabilising reversals.
- **Balance training:** sitting, then standing with a reducing base, eyes open and closed (for sensory ataxia start with vision), reaching, weight shifts, stepping, perturbations (see the balance page (p. 223)).
- **Coordination training:** whole-body tasks (kneeling to standing, stepping onto targets), slow, controlled limb movements to targets, gradually faster; video or mirror feedback; Frenkel's exercises for the sensory component.
- **Gait training:** walking with a slightly wider base, then narrowing; walking along lines and over obstacles; turning; treadmill with or without support; a rollator or weighted walker if needed.
- **Upper limb:** task practice of writing, eating, buttoning; proximal stabilisation (elbows on the table); weighted cutlery and pens, wrist weights for some tasks; adaptive devices.
- **Fitness:** stationary cycling or a recumbent bike; strength for proximal muscles.

- **Friedreich's ataxia extras:** posture and spinal care, foot orthoses for pes cavus, cardiac precautions, wheelchair and seating when walking becomes unsafe.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Truncal and gait ataxia	Intensive coordinative balance and walking practice, daily, for at least **4 weeks**, then a home programme	Improved ataxia and gait in degenerative ataxia (Ilg 2009)
Falls	Challenging balance exercise for more than **3 hours a week**; walking aid; home safety	Higher-dose balance programmes prevent most falls
Limb dysmetria and tremor	Slow, accurate movements to targets with feedback; proximal stabilisation; weighted utensils	Practice and added stability reduce errors
Sensory ataxia	Frenkel's exercises; balance with vision, then reduced vision	Vision replaces lost position sense
Low fitness	Stationary or recumbent cycling at moderate intensity, building towards **150 minutes a week** (WHO activity guidance for adults)	Safe aerobic work without balance risk

Recall check. List five treatment methods for cerebellar ataxia and the reason for each.

13 Precautions, contraindications and stop criteria

Take care

- Guard closely; use a gait belt and parallel bars at first.
- Friedreich's ataxia: watch for breathlessness, palpitations and chest pain (cardiomyopathy); check blood sugar (diabetes).
- Cerebellar stroke: watch for new headache, vomiting or drowsiness (swelling can compress the brainstem early on).
- Avoid fatigue, which worsens ataxia; rest between sets.

14 Education, lifestyle and home programme

- A daily home programme of balance, coordination and walking practice; exercise is the main treatment.
- Home safety: grab bars, rails on stairs, non-slip floors, good lighting, sitting for dressing.
- Adaptive equipment: weighted cutlery, thick pens, rollator, later a wheelchair for distances.
- Avoid alcohol and sedatives, which worsen ataxia; genetic counselling for inherited ataxias.
- Support to stay in education, work and social life.

15 Prognosis and discharge criteria

- Cerebellar stroke often recovers well over months with training.
- Degenerative ataxias progress; training slows functional decline and keeps people safer.
- Friedreich's ataxia: most people need a wheelchair within 10–20 years of onset; cardiomyopathy limits life expectancy.
- Review regularly; discharge from a block when goals are met and the home programme is established.

16 Exam corner

Past questions (MUHS, 2016–2026)

W24 Q4(a), 5 + 7 + 3 marks

The worked example in section 10.

W22 Q4(a), 7 + 8 marks

65-year-old retired man 3 months after a right cerebellar stroke, with weakness and loss of balance; diabetic for 10 years; Berg Balance score 38. ICF and physiotherapy.

W18 Q3(b), 2 + 5 + 8 marks

22-year-old man with difficulty walking, clumsy hands and unclear speech for 10 years (insidious onset); abnormal finger-nose test with intention tremor, nystagmus, kyphoscoliosis and pes cavus. Structural impairments; assessment and tests; physiotherapy.

7-mark questions

Physiotherapy of cerebellar ataxia (S16); treatment of cerebellar ataxia (W16); cerebellar ataxia and the rationale of balance training (W19).

3-mark questions

Rebound test (W17); assessment of sensory ataxia (S19).

Pages for this topic: balance and coordination (p. 223), multiple sclerosis (p. 133), vestibular disorders (p. 116), cerebral palsy (ataxic type) (p. 275).

Keywords to underline

dysmetria dysdiadochokinesia intention tremor rebound phenomenon Romberg positive in sensory ataxia Friedreich's ataxia SARA proximal stability intensive coordinative training Frenkel's exercises falls

Answer skeleton: any ataxia long case

1. Restate: age, type and cause of ataxia, duration, progression, falls, function, roles.
2. **[ICF or impairments, 2–7]** ICF diagram and table with reasoning; separate cerebellar from sensory signs.
3. **[Assessment, 0–5]** Coordination tests, Romberg, balance and gait scales, SARA.
4. **[Goals and management, 7–11]** Goals; intensive balance and coordination training; proximal stability; gait training and aids; upper limb task practice; fitness; Frenkel's for the sensory part; home programme; precautions.

Answer: structural impairments in the W18 case (2 marks)

Degeneration of the spinocerebellar tracts, posterior columns and dorsal root ganglia, corticospinal tracts and cerebellar connections (Friedreich's ataxia), with skeletal deformities: kyphoscoliosis and pes cavus.

Draw these

- The ICF model filled from the case
- A cerebellar gait: wide base, irregular steps, arms out

Viva questions

1. How do you differentiate cerebellar from sensory ataxia?
2. How is the rebound test done?
3. What are the features of Friedreich's ataxia?

4. Why do Frenkel's exercises suit sensory ataxia?
5. What is SARA?

Linked practical spots

- Coordination tests; Romberg's test; tuning fork; weighted cuffs and cutlery.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Describe the rebound test.**

Answer
The patient flexes the elbow against the examiner's resistance; the examiner suddenly releases while guarding the face. Normally the movement is checked at once; in cerebellar disease the hand flies towards the face (Holmes' rebound phenomenon).

3 m **How do you assess sensory ataxia?**

Answer
Romberg's test (steady with eyes open, sways with eyes closed); position and vibration sense; finger-nose and heel-shin worse with eyes closed; pseudoathetosis; stamping gait, worse in the dark.

7 m **Physiotherapy management of cerebellar ataxia.**

Answer
Assess (coordination, balance, gait, SARA). Intensive coordination and balance training with progression; proximal stability (weight-bearing, approximation, rhythmic stabilisation); gait training with a wider base, then narrowing, treadmill, aids; upper limb task practice with stabilisation and weighted utensils; fitness (cycling); home programme; fall prevention; precautions. Evidence: 4 weeks of intensive coordinative training improves ataxia and gait, maintained with home practice.

LAQ **A 65-year-old diabetic man 3 months after a right cerebellar stroke has weakness and poor balance, with a Berg score of 38. Discuss ICF and physiotherapy. (7 + 8, W22)**

Answer
ICF (7): right cerebellar stroke; impairments: right-sided limb ataxia, truncal ataxia, mild weakness, poor balance (Berg 38, below 45 means high fall risk), possible nystagmus and dysarthria, diabetic neuropathy adding sensory loss. Activities: unsteady walking, stairs, dressing and bathing standing. Participation: outings, community roles. Environment: family, home layout. Personal: retired, older, diabetic.

Physiotherapy (8): balance training more than 3 hours a week with progression; coordination training of the right limbs; gait training with a stick or rollator; strengthening; sensory component (vision, footwear, foot care); fall prevention at home; aerobic cycling; check blood sugar before exercise; home programme.

MCQ **A cerebellar hemisphere lesion causes limb ataxia on:**

A. The same side

B. The opposite side

C. Both sides equally

D. The legs only

Answer
A.

MCQ **A positive Romberg's test suggests:**

A. Cerebellar ataxia

B. Sensory ataxia

C. Parkinson's disease

D. Spasticity

Answer
B.

MCQ **Which feature is typical of Friedreich's ataxia?**

A. Brisk ankle reflexes

B. Pes cavus and kyphoscoliosis

C. Onset after 60

D. Rigidity

Answer
B. Ankle reflexes are lost, with extensor plantar responses.

Last recall check. Write the cerebellar signs, cerebellar vs sensory ataxia, and five treatment methods.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Ilg W, et al. Intensive coordinative training improves motor performance in degenerative cerebellar disease. *Neurology* 2009;73:1823–30. Ilg W, et al. Long-term effects of coordinative training in degenerative cerebellar disease. *Mov Disord* 2010.
3. Schmitz-Hübsch T, et al. Scale for the assessment and rating of ataxia: development of a new clinical scale. *Neurology* 2006;66:1717–20.
4. Sherrington C, et al. Exercise to prevent falls in older adults. *Br J Sports Med* 2017;51:1750–8.
5. World Health Organization. Guidelines on physical activity and sedentary behaviour (2020).
6. Structure and classical content: O'Sullivan and Schmitz, *Physical Rehabilitation* (coordination); Stokes, *Physical Management for Neurological Conditions*.

5A(i) Parkinson's disease: features, Hoehn and Yahr, gait, falls and physiotherapy

> Parkinson's disease was a long case in 7 papers, usually a retired person at Hoehn and Yahr stage III, and parkinsonian gait was a 7-mark question four times. Every answer rests on one idea: the basal ganglia no longer scale and automate movement, so the person must move big and on purpose, use cues, and keep exercising to stay strong, fit and safe from falls.

Syllabus	Ch 5A(i) Extrapyramidal diseases, with emphasis on Parkinson's disease
Priority	**Tier 1.** In 15 of 20 papers (2016–2026): 7 long cases, 6 seven-mark and 6 three-mark questions.
Asked as	LAQ W18 · S19 · W19 · S21 · W21 · S22 · W22 · 7m S16 · W16 · S17 · W17 · S22 · S26 · 3m W17 · W19 · W20 · W21 · S24 · S25
Usual split	15 (ICF with clinical reasoning 5–7, goals 3–5, management 7–10); S19 added home management for daily activities and S22 fall prevention
Checked against	APTA clinical practice guideline for physical therapist management of Parkinson disease (2022); MDS clinical diagnostic criteria (2015); Allen et al. review of falls (2013); LSVT BIG trial (Ebersbach et al., 2010); SPARX trial (Schenkman et al., 2018). Full list under Sources (p. 176). Content checked 20 October 2026.

1 Definition

Parkinson's disease (PD) is a slowly progressive neurodegenerative disease in which dopamine-producing neurons of the **substantia nigra** die, so the basal ganglia cannot control movement normally. It causes **parkinsonism**: bradykinesia together with rest tremor, rigidity or both (Movement Disorder Society criteria, 2015), plus many non-motor problems.

- It affects about 1% of people over 60 and becomes more common with age; men are affected a little more often.
- **Parkinsonism** is the clinical picture; PD is its commonest cause.

2 Classification and grading

Causes of parkinsonism

- **Idiopathic PD** (the commonest), including young-onset and genetic forms.
- **Drug-induced:** antipsychotics such as haloperidol, metoclopramide; often reversible when the drug is stopped.
- **Vascular parkinsonism:** small-vessel disease; legs more than arms ("lower-body parkinsonism"), little tremor.
- **Parkinson-plus syndromes:** progressive supranuclear palsy (early falls, poor vertical gaze), multiple system atrophy (early autonomic failure, ataxia), corticobasal degeneration. They respond poorly to levodopa.
- **Others:** after encephalitis or repeated head injury; toxins (manganese); Wilson's disease in young people.

Hoehn and Yahr stages (3 marks in W20)

Stage	Original scale	Modified scale adds
1	One side only; little or no disability	1.5: one side plus the trunk (axial) involved
2	Both sides, balance not impaired	2.5: both sides, recovers on the pull test
3	Both sides, mild to moderate; some postural instability; physically independent	No change
4	Severe disability; can still walk or stand without help	No change
5	Wheelchair-bound or bedridden unless helped	No change

Stage III is the turning point that examiners like: balance is now impaired (falls begin), but the person is still independent.

Clinical subtypes: **tremor-dominant** and **postural instability and gait difficulty** (PIGD). The PIGD type has more falls and tends to progress faster.

3 Causes and risk factors

- **Age:** the strongest risk factor.
- **Genes:** about 5–10% of cases (for example LRRK2, SNCA, PINK1, parkin, GBA), more in young-onset disease.
- **Environment:** pesticides, rural living and well water have been linked with PD; head injury.
- Most cases come from a mix of ageing, genes and environment; no single cause is found.

4 Pathophysiology to impairments

What the basal ganglia do (3 marks in W21)

- Select and start the wanted movement and suppress unwanted ones.
- Scale the size, speed and force of movement.
- Run well-learned movements automatically and in sequence (walking, writing, turning in bed).
- Help motor learning and habits; loops also serve thinking, motivation and mood.

What goes wrong in PD

1. Dopamine neurons of the substantia nigra (pars compacta) die, and clumps of the protein alpha-synuclein (Lewy bodies) form.
2. Because striatal dopamine falls, the output of the basal ganglia over-inhibits the thalamus and motor cortex, so movements become **slow and small** (bradykinesia, hypokinesia).
3. Because automatic control is lost, routine movements need conscious attention, so walking deteriorates during a second task, and steps freeze in doorways and turns.
4. Rigidity and rest tremor arise from abnormal basal ganglia rhythms.
5. Because other brain systems are affected too, non-motor problems (sleep, mood, memory, bowel, blood pressure) appear, some years before the motor signs.
6. Symptoms appear only after a large part of the nigral neurons is lost, so the disease begins years before diagnosis.

Recall check. Why does a person with PD walk better when you put stripes on the floor or count a rhythm?

5 Clinical features

Cardinal features (3 marks in S25)

Feature	Description
Tremor	At rest, 4–6 per second, "pill-rolling" of thumb and fingers; often starts in one hand; less with movement and absent in sleep; worse with stress
Rigidity	Stiffness through the whole range in flexors and extensors alike (lead-pipe), or jerky (cogwheel) when tremor is added
Akinesia and bradykinesia	Slowness, difficulty starting, and movements that get smaller as they repeat; mask-like face, soft voice, small handwriting
Postural instability	Poor balance reactions, a later feature; falls backwards on the pull test

Memory hook

TRAP

- **T**remor at rest
- **R**igidity
- **A**kinesia and bradykinesia
- **P**ostural instability

Checked against the MDS 2015 criteria: diagnosis needs bradykinesia plus rest tremor or rigidity; postural instability is a later feature, not a diagnostic one.

Rigidity and its types (3 marks in W19 and S24)

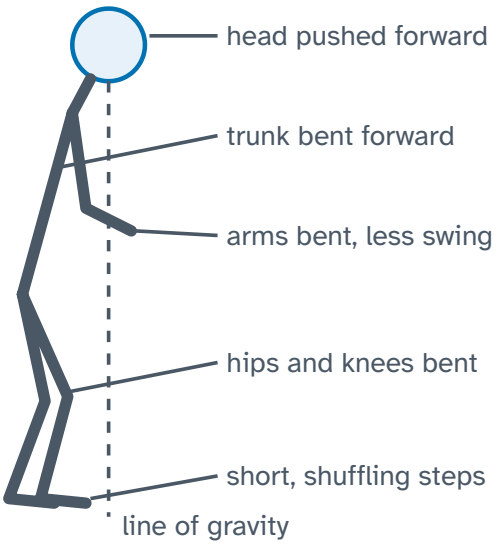
- **Lead-pipe rigidity:** smooth, even resistance through the whole range, in both directions, at any speed (PD).
- **Cogwheel rigidity:** jerky, ratchet-like resistance, because tremor is added to lead-pipe rigidity (PD).
- **Decorticate rigidity:** arms flexed, legs extended; lesion above the red nucleus (severe brain injury).
- **Decerebrate rigidity:** arms and legs extended, arms internally rotated; brainstem lesion (worse sign).

Spasticity vs rigidity (3 marks in W17)

Feature	Spasticity	Rigidity
Lesion	Upper motor neuron (pyramidal)	Extrapyramidal (basal ganglia)
Speed	Depends on speed: worse with fast stretch	Same at any speed
Pattern	Clasp-knife: resistance then sudden give	Lead-pipe or cogwheel through the whole range
Muscles	Mainly antigravity muscles (arm flexors, leg extensors)	Flexors and extensors equally
Reflexes	Brisk; clonus; Babinski sign	Normal; no Babinski sign

Gait abnormalities (7 marks in S16 and W17; gait training in W16 and S17)

Parkinsonian posture and gait



The dashed line of gravity falls in front of the feet, so the person takes faster and faster short steps to keep up with it (festination).

- **Flexed posture:** head forward, rounded shoulders, trunk, hips and knees bent.
- **Short, shuffling steps** with a flat foot (no heel strike) and little foot clearance.

- **Festination:** steps become shorter and faster as if chasing the centre of gravity; propulsion forwards, retropulsion backwards.
- **Reduced arm swing** (often one side first) and **reduced trunk rotation**.
- **Turning en bloc** in many small steps.
- **Start hesitation and freezing**: feet "glued" to the floor at the start, in doorways, narrow spaces, turns, or when talking while walking.
- Walking worsens with a second task and in the "off" period of medicines.

Other features

- **Motor:** mask face, soft voice, small handwriting, drooling, swallowing difficulty; stooped or sideways-leaning trunk (camptocormia, Pisa syndrome).
- **Non-motor:** loss of smell, constipation, acting out dreams in sleep, depression and anxiety, dizziness on standing, urinary urgency, pain, fatigue, and memory problems or dementia in later years.

6 Investigations

- The diagnosis is **clinical**; a clear, lasting response to levodopa supports it.
- **Dopamine transporter scan (DaTscan):** shows loss of nigral nerve endings when the diagnosis is uncertain (for example, PD vs essential tremor).
- **MRI:** excludes vascular parkinsonism, tumours and normal-pressure hydrocephalus.

7 Medical and surgical management

- **Levodopa with carbidopa:** the most effective drug. After some years, its effect wears off before the next dose ("off" periods) and involuntary movements (dyskinesia) appear at peak dose.
- **Others:** dopamine agonists (pramipexole, ropinirole); MAO-B inhibitors (selegiline, rasagiline); COMT inhibitors (entacapone); amantadine for dyskinesia; anticholinergics (trihexyphenidyl) for tremor in young people, avoided in the elderly because of confusion.
- **Deep brain stimulation** of the subthalamic nucleus or globus pallidus for motor fluctuations, dyskinesia and tremor in selected people.
- Speech and swallowing therapy, occupational therapy, and treatment of depression, constipation and low blood pressure.

> **For the physiotherapist**
>
> Plan exercise and gait training in the "on" period, when the medicine works; teach safety strategies for the "off" period. Record the time of the last dose with every assessment.

8 Complications and prevention

Complication	What to know	Physio role
Falls and fractures	About 60% of people with PD fall each year and 39% fall repeatedly (Allen et al., 2013). Risks: previous falls, freezing, severity, poor mobility, fear of falling, cognitive problems	Balance and gait training, cueing, home safety, footwear, strength
Kyphosis and contractures	Hip and knee flexors, pectorals, neck flexors; reduced chest expansion	Extension and rotation exercises, posture training, flexibility
Deconditioning	Less activity as movement becomes effortful	Aerobic and resistance exercise; community classes
Orthostatic hypotension	Dizziness on standing, from the disease and its drugs	Check lying and standing blood pressure; stand up slowly

Complication	What to know	Physio role
Aspiration pneumonia	Swallowing difficulty in later stages	Upright posture for meals; breathing exercises; refer to speech therapy
Pressure sores (stage 5)	Bed-bound or chair-bound	Positioning, turning, carer training

9 PT assessment

- **History:** onset and side, medicines and timing (on and off periods), falls and near-falls, freezing, daily activities, home, job, goals.
- **Staging:** Hoehn and Yahr; MDS-UPDRS (part III is the motor examination).
- **Posture, range and strength:** kyphosis, hip and knee flexion contractures, chest expansion, sit-to-stand (Five Times Sit to Stand).
- **Balance:** Mini-BESTest or Berg Balance Scale; the pull (retropulsion) test.
- **Gait:** 10-metre walk (speed, step length, cadence); Timed Up and Go, also with a second task; 6-minute walk; freezing questionnaire; observation of turns and doorways.
- **Others:** cognition, mood, fatigue, lying and standing blood pressure; quality of life (PDQ-39).

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case. Note the stage and the medicine timing in the health condition box.

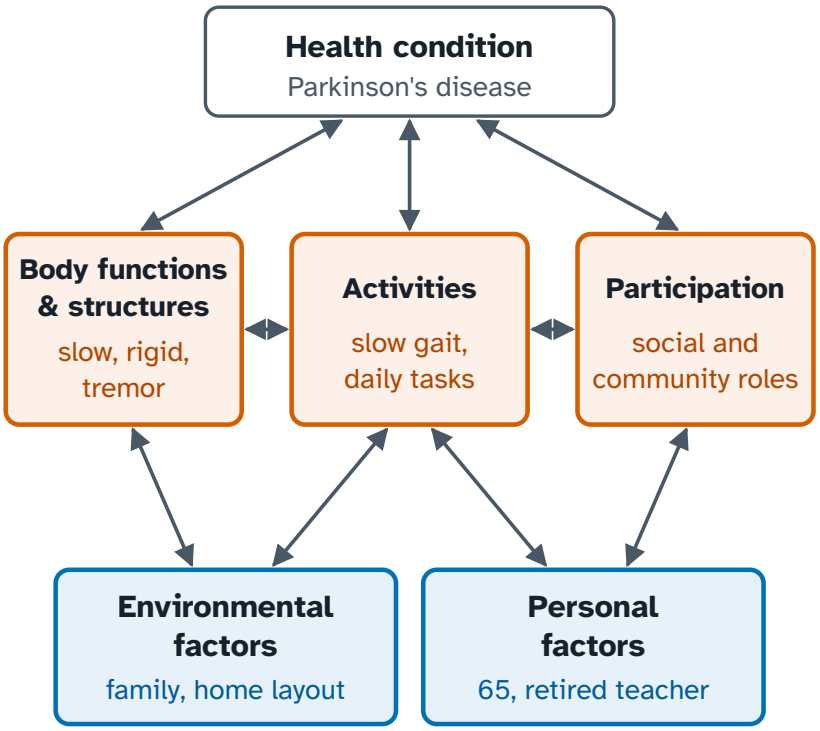
Worked example from a real paper (W21, 5 + 10 marks)

> "A 65 years old male, a retired teacher, has difficulty in initiating movement, tremor, rigidity and slowness of gait. Patient diagnosed as Parkinson disease and has difficulty in activities of daily living. ICF with clinical reasoning. Physiotherapy short term and long term goals and treatment."

Spot the ICF clues before you write

difficulty initiating movement means akinesia and start hesitation. tremor, rigidity are cardinal features. slowness of gait means bradykinesia with short steps. difficulty in activities of daily living is the activity limitation. retired teacher suggests roles in family and community: ask what he still wants to do.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Idiopathic Parkinson's disease; stage to be confirmed (likely Hoehn and Yahr II–III); on medicines	Loss of nigral dopamine neurons; basal ganglia fail to scale and automate movement
Body function and structure impairments	• Akinesia: difficulty starting movement • Bradykinesia and hypokinesia: slow, small movements • Lead-pipe or cogwheel rigidity • Rest tremor • Flexed posture; reduced trunk rotation and chest expansion • Impaired balance reactions (if stage III) • Reduced fitness and strength	• Dopamine loss over-inhibits the motor thalamus and cortex • Automatic movement needs conscious effort • Rigidity in flexors and extensors stiffens the trunk • Inactivity adds deconditioning to the disease
Activity limitations	Slow gait with short steps; difficulty getting up from a chair, turning in bed, dressing, buttoning, writing, eating with a tremor	Tasks that need sequencing and fine control are hit first
Participation restrictions	Less social, religious and community activity; outings; tutoring or teaching if he still does it	Slowness, fear of falling and embarrassment about tremor
Environmental factors	**Facilitators:** family, medicines, community exercise groups. **Barriers:** loose rugs, poor lighting, low seats, an Indian-style toilet, crowded roads	Home changes reduce falls and effort
Personal factors	65-year-old retired teacher; educated, can follow strategies	Good understanding helps cueing and cognitive strategies

One-line functional diagnosis

A 65-year-old retired teacher with Parkinson's disease, with akinesia, bradykinesia, rigidity and rest tremor causing slow, short-stepped gait and difficulty with daily activities, which limit his social and community roles. Family support and medicines are facilitators; home hazards are barriers.

Practice cases from the papers: Hoehn and Yahr III cases in W18, W19, S21 and W22; S19 68-year-old man with a stooped posture who walks only indoors (add home management); S22 68-year-old retired teacher with falls on turning and freezing who still teaches slum children (add fall prevention).

Recall check. Fill the ICF for the S22 teacher who falls on turning. Which impairment explains the falls?

11 Problem list and goals

Goals for the 65-year-old retired teacher

Problem	Short-term (4–6 weeks)	Long-term (3–6 months)
Starting and slowness	Gets up from a chair in one attempt using a movement strategy	Moves through daily tasks without help
Gait	Longer steps with a cue; faster 10-metre walk	Walks outdoors safely without cues for short distances
Posture and rigidity	More trunk rotation and extension; better chest expansion	Upright posture held in daily activity
Balance and falls	Better Mini-BESTest score; knows his fall triggers	No falls; turns safely
Fitness and activity	Exercises 3 days a week	Attends a community exercise group; keeps his social roles

12 PT management

Principles: exercise early and for life; train big, fast and on purpose; use cues and movement strategies to bypass the damaged automatic control; practise the real tasks; prevent falls; time sessions to the "on" period.

What the evidence supports (APTA guideline, 2022)

Strong recommendations ("should"): aerobic exercise, resistance training, balance training, gait training, task-specific training, external cueing, community-based exercise, behaviour-change approaches and integrated (team) care. Weak ("may"): flexibility exercises and telerehabilitation. Most evidence comes from Hoehn and Yahr stages 1–3.

By stage (Hoehn and Yahr)

- **Stages 1–2 (early):** stop inactivity and fear of moving; aerobic, resistance and balance exercise; join a community class (dance, tai chi, walking groups).
- **Stages 2–4 (middle, 7 marks for stage III in S26):** keep the above, and add gait training with cues, cognitive movement strategies for transfers and turning, balance and fall prevention, posture, and home safety.
- **Stage 5 (late):** keep the person safe and comfortable: positioning, pressure care, breathing, assisted transfers, and carer training.

Gait training (7 marks in S16, W16, S17 and W17)

- **External cues:** visual stripes on the floor at the target step length; a metronome or music beat; a verbal cue ("big step"). Cues bypass the faulty basal ganglia timing.
- **Attention:** think about long steps and heel strike; avoid talking or carrying things while walking in the early sessions.

- **Practise:** walking with long steps and arm swing; treadmill walking; walking with Nordic poles; turning in a wide arc rather than pivoting; doorways and narrow spaces; stops and starts; walking backwards and sideways.
- **Festination:** stop, stand tall, and restart with a long step; widen the base.
- **Dose:** gait training **20–60 minutes**, **3–5 days a week**, for **4–12 weeks**; cueing training **2–5 times a week** (APTA 2022). Gains fade when training stops, so keep it going.

Freezing of gait

1. Stop trying to push forward; stand tall and shift weight to one leg.
2. Step over a real or imagined line, or onto a mark on the floor.
3. Use a rhythm: count aloud, march on the spot, or a metronome.
4. Turn in a wide arc; walk through doorways with a target beyond them.

Cognitive movement strategies (for transfers and daily tasks)

Break a complex movement into separate steps, rehearse it in the mind, then perform each step consciously in order. Example, rising from a chair: move to the front of the seat, feet back, lean well forward ("nose over toes"), push up and stand tall. The same approach works for turning in bed and getting up from the floor.

Bradykinesia and rigidity (7 marks in S22)

- **Bradykinesia:** large-amplitude training such as LSVT BIG (**16 one-hour sessions**, **4 days a week for 4 weeks**) using whole-body movements at maximal size; speed and power training; cues and cognitive strategies; aerobic exercise.
- **Rigidity:** slow rhythmic trunk and limb rotation; rhythmic initiation (passive, then active-assisted, then active); large reciprocal movements; flexibility exercises for pectorals, hip flexors and hamstrings; relaxation; warm-up before exercise.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Low fitness and motor severity	Treadmill, cycling, brisk walking: at least **3 days a week**, **30–40 minutes**, moderate (**60–75% of maximum heart rate**) to high (**75–85%**) intensity; use perceived exertion too	Improves fitness and lowers motor severity; heart rate response can be blunted in PD
Weakness and low power	Resistance training **1–2 days a week** for **30–60 minutes**; about **80% of 1RM** for strength and **40% of 1RM** fast for power; add load once 3 sets of 15 are done well	Leg strength and power are needed for sit-to-stand, stairs and recovering balance
Poor balance	Multimodal balance training (reactive steps, weight shifts, reaching, turning, dual tasks when ready): **2–3 times a week**, **16–30 hours** in total over 5–10 weeks	Balance training improves balance, gait and confidence and helps reduce falls
Short, slow steps and freezing	Gait training with visual and auditory cues (doses above)	External cues supply the timing and size the basal ganglia no longer provide
Small, slow movements	LSVT BIG or other large-amplitude task practice	Recalibrates the person's sense of how big a movement should be
Flexed posture and rigidity	Trunk extension and rotation, chest expansion, flexibility; prone lying if tolerated	Counters the flexed posture and keeps range for walking and breathing
Inactivity over years	Community classes (dance, tai chi, walking groups) and behaviour-change support: goals, diaries, follow-up	Benefits fade when exercise stops, so lifelong activity is the aim

Classic answer vs current evidence: high-intensity exercise

Classic	Current	Write in the exam
Gentle range-of-motion and relaxation exercises; avoid fatigue.	Moderate- to high-intensity aerobic and resistance exercise is recommended (APTA 2022). In early PD, treadmill exercise at 80–85% of maximum heart rate was safe, and motor scores stayed stable over 6 months while usual care worsened (SPARX, 2018).	"Aerobic exercise at moderate to high intensity 3 days a week for 30–40 minutes, with resistance and balance training, cueing and task practice."

Recall check. Plan one week of exercise for a person at Hoehn and Yahr stage III, with doses.

13 Precautions, contraindications and stop criteria

Take care

- Train in the "on" period; expect slower, riskier movement in "off" periods.
- Guard closely during balance and turning practice; use a gait belt.
- Check lying and standing blood pressure; stop if dizzy.
- Use perceived exertion as well as heart rate, because the heart rate response may be blunted.
- Keep dual tasks for when single-task walking is safe; watch for confusion in people with cognitive decline.
- Swallowing difficulty: no exercise straight after meals in a reclined position.

14 Education, lifestyle and home management

S19 asked for home management that helps the person perform better (5 marks).

- **Home safety:** remove loose rugs and clutter; good lighting with switches at the bed; grab bars in the bathroom; a raised western toilet or commode; firm chairs with armrests; clear wide paths for turning; floor stripes at freezing spots such as doorways.
- **Daily tasks:** do demanding tasks in the "on" period; sit to dress; loose clothes with Velcro or large buttons, elastic laces; large-handled or weighted cutlery and a non-slip mat for tremor; a bed rope or rail for turning.
- **Walking out:** a stick or walker if advised; avoid crowds at first; good footwear.
- **Exercise:** a daily home programme plus a community class; keep it up for life.
- **Other:** take medicines on time; fluids and fibre for constipation; speech and swallowing exercises from the speech therapist; family education and support groups.

15 Prognosis and discharge criteria

- PD progresses slowly over many years; the speed varies. The tremor-dominant type usually progresses more slowly than the postural instability and gait difficulty type.
- Falls, swallowing problems and dementia mark the later stages and raise the risk of death, often from pneumonia.
- Physiotherapy is given in blocks with regular review, and the person keeps exercising in between; discharge from a block when its goals are met and a home programme is in place.

16 Exam corner

Past questions (MUHS, 2016–2026)

S22 Q3(b), 4 + 4 + 7 marks

68-year-old woman with finger tremor, falls while turning and slowness for 2 years, now freezing while speaking and walking; retired teacher who still teaches slum children; on medicines. Short- and long-term goals; fall prevention strategies; physiotherapy in detail.

W22 Q4(b), 7 + 8 marks

64-year-old retired housewife with PD for 3 years, Yahr stage 3. ICF and physiotherapy.

W21 Q3(a), 5 + 10 marks

The worked example in section 10.

S21 Q4(b), 5 + 5 + 5 marks

65-year-old retired clerk with PD for 3 years, on medicines, Yahr stage 3. ICF with clinical reasoning; short- and long-term goals.

W19 Q3(a)

65-year-old retired clerk with PD for 2 years, Yahr stage III. ICF; short- and long-term goals.

S19 Q4(b), 5 + 5 + 5 marks

68-year-old man with PD for 3 years who walks indoors with a stooped posture and has difficulty outdoors. ICF; physiotherapy; home management for better performance.

W18 Q4(a), 3 + 7 + 5 marks

60-year-old woman with PD for 7 years, difficulty walking, Hoehn and Yahr stage 3. ICF; physiotherapy with goals.

7-mark questions

Gait abnormalities and their physiotherapy (S16, W17); gait training (W16); festinating gait and principles of gait training (S17); bradykinesia and rigidity (S22); physiotherapy at Hoehn and Yahr stage III (S26).

3-mark questions

Spasticity vs rigidity (W17); rigidity and its types (W19, S24); Hoehn and Yahr scale (W20); three functions of the basal ganglia (W21); cardinal features (S25).

Pages for this topic: balance and coordination (p. 223), transfers and gait training (p. 232), paediatric extrapyramidal disorders (p. 350), scales (p. 21).

Keywords to underline

dopamine loss basal ganglia bradykinesia lead-pipe and cogwheel rigidity festination freezing Hoehn and Yahr external cues cognitive movement strategies "on" period falls aerobic exercise

Answer skeleton: any PD long case

1. Restate: age, years since diagnosis, Hoehn and Yahr stage, medicines, falls and freezing, home and roles.
2. **[ICF, 5–7]** ICF diagram and table with clinical reasoning; one-line functional diagnosis.
3. **[Goals, 3–5]** Short-term goals for gait, transfers, posture and balance; long-term goals for safe independent activity, no falls, lifelong exercise and roles.

4 **[Management, 7–10]** Principles; aerobic, resistance and balance training with doses; gait training with cues; freezing strategies; cognitive movement strategies; posture and rigidity; fall prevention and home changes; "on" period timing; precautions.

Answer: fall prevention strategies (S22, 4 marks)

- Find her risks: previous falls, falls on turning, freezing, dual-task walking, medicine timing, low blood pressure, home hazards.
- Balance training 2–3 times a week (reactive steps, turning, weight shifts), with leg strengthening.
- Turning in a wide arc; freezing strategies and floor cues; no talking while walking in risky places until trained.
- Home safety (rugs, lighting, grab bars, raised toilet); safe footwear; a walking aid if needed.
- Activities in the "on" period; stand up slowly; teach getting up from the floor.

Draw these

- Parkinsonian posture with the line of gravity (section 5)
- The ICF model filled from the case

Viva questions

1. What are the cardinal features of PD?
2. How is rigidity different from spasticity?
3. What is festination, and why does it happen?
4. How do external cues help gait in PD?
5. What happens at Hoehn and Yahr stage III?
6. When in the day would you schedule physiotherapy, and why?

Linked practical spots

- Metronome; floor stripes; Nordic walking poles; Hoehn and Yahr and Mini-BESTest as scale spots.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the Hoehn and Yahr scale.**

Answer
1: one side only. 2: both sides, balance normal. 3: both sides, mild to moderate, some postural instability, physically independent. 4: severe disability, can still walk or stand unaided. 5: wheelchair-bound or bedridden unless helped. (The modified scale adds 1.5 and 2.5.)

3 m **Write three functions of the basal ganglia.**

Answer
Selecting and starting wanted movements while suppressing unwanted ones; scaling the size, speed and force of movement; running learned movement sequences automatically; also motor learning and habit formation.

7 m **Describe the common gait abnormalities in Parkinson's disease and their physiotherapy. (2 + 5)**

Answer

Abnormalities (2): flexed posture; short shuffling steps without heel strike; festination; reduced arm swing and trunk rotation; en bloc turning; start hesitation and freezing; worse with dual tasks and in "off" periods.

Physiotherapy (5): visual stripes and auditory rhythm cues; attention to long steps and heel strike; treadmill and overground practice with arm swing; wide turns, doorways, stops and starts; freezing strategies; trunk rotation and extension; balance and leg strength; gait training 20–60 minutes, 3–5 days a week; train in the "on" period.

LAQ **A 68-year-old man with PD for 3 years walks indoors with a stooped posture and has difficulty outdoors. Discuss ICF, physiotherapy, and home management for better performance. (5 + 5 + 5, S19)**

Answer

ICF (5): impairments: bradykinesia, rigidity, flexed posture, short steps, poor balance, low fitness. Activities: limited outdoor walking, slow transfers and dressing. Participation: social outings, shopping, religious places. Environment: crowded roads and uneven ground (barriers); family (facilitator). Personal: 68-year-old man.

Physiotherapy (5): aerobic walking or cycling 3 days a week for 30–40 minutes; resistance and balance training; gait training with cues and turning practice; trunk extension and rotation; graded outdoor walking in the "on" period.

Home management (5): remove rugs and clutter; lighting; grab bars and a raised toilet; firm chairs with armrests; floor stripes at doorways; dressing aids and adapted cutlery; tasks in the "on" period; a daily home programme with family support.

MCQ **Rigidity in Parkinson's disease is best described as:**

A. Velocity-dependent, clasp-knife

B. Uniform resistance in flexors and extensors at any speed

C. Present only in antigravity muscles

D. Associated with a Babinski sign

Answer

B. Lead-pipe rigidity; the others describe spasticity.

MCQ **Which strategy best helps a patient who freezes in a doorway?**

A. Pull the patient forward

B. Tell the patient to hurry

C. Step over a line on the floor with a counted rhythm

D. Stop all walking practice

Answer

C. Visual and auditory cues bypass the faulty automatic control.

MCQ **The APTA 2022 guideline gives a strong recommendation to all of these except:**

A. Aerobic exercise

B. Balance training

C. External cueing

D. Flexibility exercises

Answer

D. Flexibility exercises get a weak recommendation ("may").

Last recall check. Draw the parkinsonian posture, write TRAP and the Hoehn and Yahr stages, and list four freezing strategies.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Osborne JA, et al. Physical therapist management of Parkinson disease: a clinical practice guideline from the American Physical Therapy Association. *Phys Ther* 2022;102:pzab302.
3. Postuma RB, et al. MDS clinical diagnostic criteria for Parkinson's disease. *Mov Disord* 2015;30:1591–601.
4. Allen NE, Schwarzel AK, Canning CG. Recurrent falls in Parkinson's disease: a systematic review. *Parkinsons Dis* 2013;2013:906274.
5. Ebersbach G, et al. Comparing exercise in Parkinson's disease: the Berlin LSVT BIG study. *Mov Disord* 2010;25:1902–8.
6. Schenkman M, et al. Effect of high-intensity treadmill exercise on motor symptoms in patients with de novo Parkinson disease (SPARX). *JAMA Neurol* 2018;75:219–26.
7. Hoehn MM, Yahr MD. Parkinsonism: onset, progression and mortality. *Neurology* 1967;17:427–42.
8. Structure and classical content: Keus SHJ, et al. European Physiotherapy Guideline for Parkinson's Disease (KNGF/ParkinsonNet, 2014); O'Sullivan and Schmitz, *Physical Rehabilitation*; Morris, movement strategies in PD.

5A(j) Anterior horn cell diseases: motor neuron disease, poliomyelitis and SMA

> Motor neuron disease came up in 8 papers: two long cases of young men with ALS, the types and diagnostic criteria of MND, pseudobulbar palsy, and physiotherapy for a terminally ill patient. MND progresses steadily, so physiotherapy aims to keep the person independent, comfortable and breathing well for as long as possible, staying a step ahead with equipment and never exercising to exhaustion.

Syllabus	Ch 5A(j) Anterior horn cell diseases, hereditary and acquired: MND, PMA, SMA, poliomyelitis (post-polio residual paralysis and paediatric SMA have their own pages)
Priority	**Tier 1.** In 8 of 20 papers (2016–2026): 2 long cases, 2 seven-mark and 4 three-mark questions.
Asked as	LAQ S16 · W16 · 7m S23 · W25 · 3m S18 · S19 · S21 · W24
Usual split	15 (impairments with reasoning 3–7, goals 4–5, management 4–7)
Checked against	Gold Coast diagnostic criteria for ALS (Shefner et al., 2020); Cochrane reviews of riluzole and of therapeutic exercise in ALS; the randomised trial of non-invasive ventilation (Bourke et al., 2006); NICE guideline on motor neurone disease (NG42). Full list under Sources (p. 184). Content checked 20 October 2026.

1 Definition

- **Motor neuron disease (MND)** is a progressive degeneration of motor neurons in the cortex (upper motor neurons), brainstem and spinal cord anterior horns (lower motor neurons), causing increasing weakness, wasting and spasticity, and eventually breathing failure. Sensation is usually spared.
- It is rare (about 1–3 new cases per 100,000 people a year), usually starting after 50; about 1 in 10 cases is familial.

2 Classification and grading

Asked for 3 marks in S21 and W24 (types of MND) and for 7 marks in S23 (classify MND with features).

Type	Neurons affected	Features
Amyotrophic lateral sclerosis (ALS, the commonest)	UMN and LMN	Wasting, fasciculations and weakness together with spasticity, brisk reflexes and Babinski signs; often starts in one hand or foot
Progressive bulbar palsy	Brainstem motor nuclei (with UMN signs)	Slurred speech, swallowing difficulty, wasted fasciculating tongue; worse outlook
Progressive muscular atrophy	LMN only	Wasting, fasciculations, weakness, lost reflexes
Primary lateral sclerosis	UMN only	Slowly progressive spasticity; the best outlook

Diagnostic criteria (3 marks in S18)

- **Gold Coast criteria (2020):** (1) progressive motor impairment, after previously normal function, shown by history or repeated examination; (2) UMN and LMN dysfunction in at least one body region, or LMN dysfunction in at least two regions; (3) investigations that exclude other diseases.
- The older **revised El Escorial criteria** graded ALS as definite, probable or possible according to the number of regions (bulbar, cervical, thoracic, lumbosacral) with UMN and LMN signs.
- EMG (fibrillations, fasciculations, chronic neurogenic changes) shows LMN involvement even in muscles that look normal.

Function scale: ALS Functional Rating Scale-Revised (ALSFRS-R, 0–48: bulbar, fine motor, gross motor and breathing).

3 Causes and risk factors

- Cause unknown in most; about 10% familial (genes such as SOD1 and C9orf72); age and male sex increase risk.
- Mechanisms include protein misfolding, glutamate excitotoxicity, oxidative stress and inflammation.

4 Pathophysiology to impairments

1. Because LMNs die, muscles lose their supply: wasting, fasciculations (twitches), cramps and weakness.
2. Because UMNs die, there is spasticity, brisk reflexes and Babinski signs, even in wasted limbs.
3. Because brainstem neurons fail, speech, swallowing and chewing weaken (risk of aspiration and weight loss).
4. Because the diaphragm and intercostals weaken, breathing fails, first at night (morning headache, poor sleep).
5. Eye movements, sensation and bladder and bowel control are usually spared; some people develop frontotemporal dementia.

Recall check. Why can a patient with ALS have wasted, fasciculating muscles and brisk reflexes in the same arm?

5 Clinical features

- **Limb onset:** weakness and wasting of the small hand muscles (difficulty buttoning or opening the fist), foot drop, cramps, fasciculations; the thenar muscles waste more than the hypothenar ("split hand").
- **Bulbar onset:** slurred speech, choking, drooling, wasted tongue with fasciculations.
- **Later:** neck extensor weakness (head drop), breathing weakness, weight loss, fatigue, emotional lability.

Pseudobulbar palsy vs bulbar palsy (3 marks in S19)

Feature	Pseudobulbar palsy (UMN)	Bulbar palsy (LMN)
Lesion	Both corticobulbar tracts (bilateral strokes, MND, MS)	Cranial nerve nuclei IX–XII or their nerves (MND, GBS, polio)
Tongue	Small, stiff, no wasting or fasciculation	Wasted, fasciculating
Speech	Spastic, slow, strained	Nasal, slurred
Jaw jerk and gag	Brisk jaw jerk	Absent or normal jaw jerk; weak gag
Emotion	Emotional lability: sudden laughing or crying	Normal

Poliomyelitis and spinal muscular atrophy (brief)

- **Acute poliomyelitis:** a virus that destroys anterior horn cells, causing sudden, asymmetrical, flaccid paralysis after a feverish illness, with normal sensation. India has been polio-free since 2014; the residual effects are on the post-polio residual paralysis page (p. 322).
- **Spinal muscular atrophy:** an inherited loss of anterior horn cells, usually in childhood: symmetrical, proximal weakness with hypotonia; see the SMA and HSMN page (p. 345).

6 Investigations

- **EMG and nerve conduction:** widespread denervation with normal sensory studies.
- **MRI** of brain and spine to exclude mimics (cervical myelopathy, tumours).
- Blood tests (thyroid, B12, calcium, CK); genetic tests in familial cases.
- **Breathing:** forced vital capacity (FVC), sniff nasal pressure, oxygen saturation overnight, blood gases.

7 Medical and surgical management

- **Riluzole** prolongs median survival by about 2–3 months; other drugs (edaravone; tofersen for SOD1-related ALS) are available in some countries.
- **Non-invasive ventilation** for breathing weakness improves sleep, symptoms and survival, most clearly when bulbar weakness is not severe.
- **Nutrition:** gastrostomy feeding when swallowing is unsafe or weight falls.
- Symptom control: cramps, spasticity (baclofen), saliva, emotional lability, pain, low mood.
- A multidisciplinary team, with early palliative care and advance care planning.

8 Complications and prevention

Complication	Physio role
Respiratory failure and chest infection	Monitor FVC and cough; airway clearance; assisted cough or cough machine; support NIV
Falls	AFOs, walking aids, home changes
Contractures and painful joints	Daily range of motion, positioning, shoulder care
Head drop and neck pain	Supportive neck collar
Pressure sores, DVT	Positioning, pressure-relieving equipment, leg movements
Aspiration	Upright positioning for meals; refer to speech therapy

9 PT assessment

- History: onset region, speed of progression, falls, fatigue, cramps, breathing at night, swallowing, work and home.
- Muscle strength (MRC grades or hand-held dynamometry), wasting, fasciculations, tone, reflexes.
- Function: ALSFRS-R; grip and pinch; timed walk; Berg; transfers.
- Breathing: FVC (sitting and lying), cough peak flow, breathlessness lying flat.
- Range of motion, posture, neck control, pain, fatigue, mood and cognition.

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case.

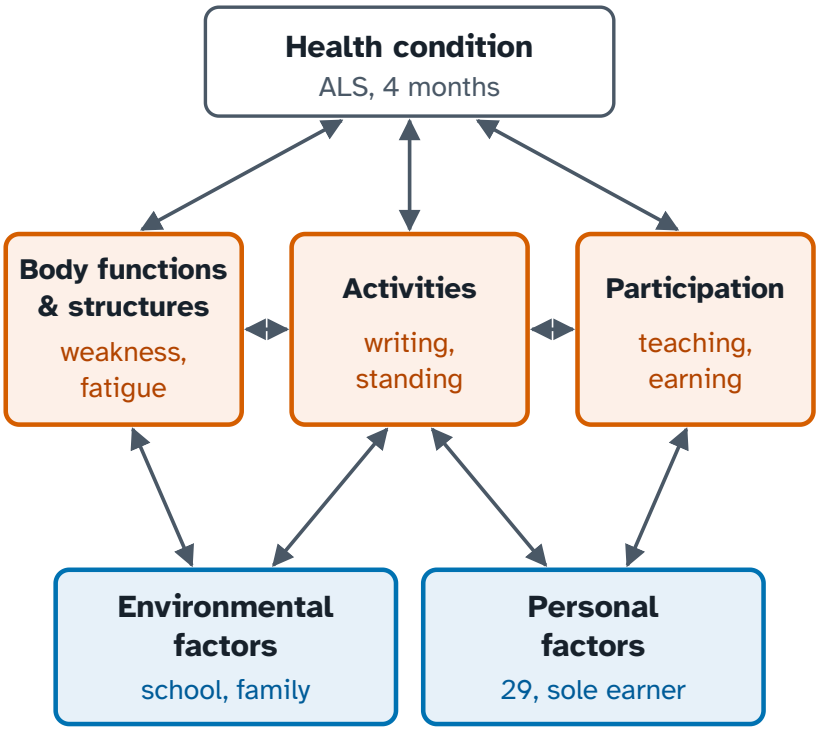
Worked example from a real paper (W16, 7 + 4 + 4 marks)

> "A 29 years old male teacher, who belongs to the middle class and has one child and a non working spouse is diagnosed with amyotrophic lateral sclerosis 4 months back. He complains of difficulty in writing on the board and fatigue along with difficulty in standing for prolonged period of time. Describe functional impairments with clinical reasoning short and long term PT management."

Spot the ICF clues before you write

ALS 4 months: early stage, mixed UMN and LMN. difficulty writing on the board: shoulder and hand weakness. fatigue, difficulty standing for long: leg weakness and reduced endurance. teacher, one child, non working spouse: he is the only earner, so keep him working as long as possible with energy conservation and adaptations, and plan ahead for the family.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	ALS, 4 months since diagnosis; early stage	Progressive UMN and LMN loss
Body function and structure impairments	Shoulder and hand weakness; leg weakness; fatigue and low endurance; possible cramps, fasciculations, spasticity; breathing to be checked	Fewer motor units, each overworked, so muscles tire quickly
Activity limitations	Writing on the board (arm held up), standing for a whole lesson, walking long distances, buttoning	Sustained overhead and standing tasks demand endurance of weak muscles
Participation restrictions	Teaching full days; family income at risk; social life	Fatigue limits the working day
Environmental factors	**Facilitators:** family; a school that may adapt the timetable. **Barriers:** blackboard teaching, stairs, cost of equipment, middle-class income with a non-working spouse	Workplace changes keep him working longer
Personal factors	29-year-old man, sole earner; coping with a serious diagnosis	Psychological support is part of care

One-line functional diagnosis

A 29-year-old teacher with early ALS, with shoulder, hand and leg weakness and marked fatigue, who struggles to write on the board and stand through lessons, putting his job and family income at risk.

Practice case: S16 36-year-old dentist with ALS, weakness and wasting of the small hand muscles, fasciculations in the forearm and upper arm, brisk reflexes in the arms, reduced reflexes in the legs and bilateral Babinski signs (structural and functional impairments, goals, physiotherapy).

11 Problem list and goals

Goals for the 29-year-old teacher

Problem	Short-term (4–6 weeks)	Long-term (3–6 months)
Fatigue	Uses energy conservation; rests between lessons	Manages the working day with planned rests
Strength and function	Maintains strength of unaffected and mildly affected muscles	Independent in self-care as long as possible
Teaching	Writes using a whiteboard at a lower height, or a projector; sits to teach	Continues teaching with adaptations
Safety and breathing	No falls; breathing monitored	Equipment and NIV in place before they are urgently needed

12 PT management

By stage

- **Early (independent):** moderate, non-fatiguing strengthening of muscles at grade 3 or more and gentle aerobic activity (walking, cycling, swimming); daily range of motion and stretching; energy conservation and pacing; ergonomic changes at work; education.
- **Middle (some help):** orthoses (AFOs for foot drop, wrist and thumb splints, a supportive neck collar), walking aids, then a wheelchair for distances; transfer training; breathing exercises and airway clearance; manage cramps and spasticity with stretching.
- **Late (dependent):** positioning and pressure care, passive range of motion to prevent painful stiffness, chest care with assisted cough and suction, support of NIV, equipment (hoist, profiling bed, tilt-in-space chair), carer training.

Two small trials (43 people) found no harm from moderate exercise and slightly better function at 3 months, but they are too small to be sure (Cochrane 2013). Never exercise to exhaustion.

Physiotherapy for the terminally ill patient (7 marks in W25)

- **Comfort and positioning:** supported positions, pillows, a profiling bed, pressure-relieving mattress; regular turning.
- **Range of motion:** gentle passive movements to prevent painful stiff joints, especially shoulders.
- **Breathlessness:** upright or forward-lean positions, a hand-held fan, breathing control and relaxation; support NIV use and mask comfort.
- **Secretions:** positioning, gentle airway clearance, manually assisted cough or a cough-assist machine, suction as needed.
- **Equipment:** hoist, slide sheets, tilt-in-space wheelchair, head support.
- **Carers and patient:** teach safe handling and positioning; respect the person's wishes and advance care plan; support emotionally; work with the palliative care team.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Weakness (grade 3 or more)	Moderate resistance, few repetitions, rests; stop well before fatigue	Maintains strength without overworking the remaining motor units
Fatigue	Energy conservation: plan, prioritise, pace, sit for tasks	Saves energy for valued activities such as work
Foot drop, wrist weakness, head drop	AFO, wrist and thumb splints, neck collar	Keeps function and prevents falls and pain
Stiffness and cramps	Daily stretching and range of motion	Prevents painful contractures
Weak cough	Assisted cough, cough machine, airway clearance	Prevents chest infection

Recall check. How does physiotherapy change between early, middle and late MND?

13 Precautions, contraindications and stop criteria

Take care

- Avoid exercise to fatigue, heavy eccentric work and strengthening of muscles below grade 3.
- Stop if cramps, fasciculations or weakness increase the next day.
- Watch for breathlessness lying flat, morning headache and poor sleep (night hypoventilation): refer.
- Aspiration risk: upright for meals and for 30 minutes after.

14 Education, lifestyle and home programme

- Honest information and time; plan ahead for equipment, work and finances.
- Daily range of motion and gentle exercise; energy conservation.
- Home changes: ground-floor living, rails, a commode, bathroom changes.
- Communication aids early, before speech fails.
- Support for the spouse and children; patient associations.

15 Prognosis

- Median survival is about 3 years from the first symptoms (2–4 years in most studies), mostly limited by breathing failure; up to 1 in 10 people live more than 10 years.
- Bulbar onset and older age progress faster; primary lateral sclerosis progresses slowest.
- NIV, nutrition and riluzole lengthen survival; physiotherapy maintains function and comfort.

16 Exam corner

Past questions (MUHS, 2016–2026)

W16 Q4(a), 7 + 4 + 4 marks

The worked example in section 10.

S16 Q4(a), 3 + 5 + 7 marks

36-year-old dentist recently diagnosed with ALS: difficulty opening the fist, weakness and wasting of the small hand muscles, painless twitching in the forearm and upper arm, hyperreflexia in both arms, hyporeflexia in both legs, bilateral Babinski signs. Structural and functional impairments; goals; physiotherapy.

7-mark questions

Classify MND with features (S23); physiotherapy for a terminally ill patient with MND (W25).

3-mark questions

Diagnostic criteria of MND (S18); pseudobulbar palsy (S19); types of MND (S21, W24).

Pages for this topic: neurological assessment (UMN vs LMN) (p. 29), post-polio residual paralysis (p. 322), SMA and HSMN (p. 345).

Keywords to underline

UMN and LMN signs together · fasciculations · split hand · sensation spared · Gold Coast criteria · ALSFRS-R · energy conservation · non-fatiguing exercise · orthoses · NIV · palliative care

Answer skeleton: an MND long case

1. Restate: age, job, type and duration of MND, main problems, family.
2. **[Impairments or ICF, 3–7]** UMN and LMN impairments with reasons; activities; participation; one-line diagnosis.
3. **[Goals, 4–5]** Maintain function and independence, prevent complications, keep working, plan equipment.
4. **[Management, 4–7]** Stage-based plan: non-fatiguing exercise, range, energy conservation, orthoses and aids, breathing care, work adaptations, carer training; precautions.

Draw these

- The motor pathway with the UMN and LMN levels marked, showing where ALS acts on both
- The ICF model filled from the case

Viva questions

1. Name the four types of MND.
2. Which functions are usually spared in MND?
3. How do pseudobulbar and bulbar palsy differ?
4. Why should exercise in MND never be exhausting?
5. What is the role of NIV in MND?

Linked practical spots

- AFO, wrist and thumb splint, neck collar; peak cough flow meter.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the types of MND.**

Answer
Amyotrophic lateral sclerosis (UMN and LMN); progressive bulbar palsy; progressive muscular atrophy (LMN only); primary lateral sclerosis (UMN only).

3 m **What is pseudobulbar palsy?**

Answer
Weakness of bulbar muscles from bilateral UMN (corticobulbar) lesions, as in bilateral strokes, MND or MS: small stiff tongue without wasting, spastic speech, dysphagia, brisk jaw jerk and emotional lability.

7 m **Physiotherapy for a terminally ill patient with MND.**

Answer
Comfort positioning and pressure care; gentle passive range of motion; breathlessness management (positions, fan, breathing control, NIV support); secretion clearance (positioning, assisted cough, cough machine, suction); equipment (hoist, tilt-in-space chair, head support); carer training; respect for the patient's wishes and advance care plan; teamwork with palliative care.

LAQ **A 36-year-old dentist with ALS has wasting of the small hand muscles, fasciculations, brisk arm reflexes and bilateral Babinski signs. Write structural and functional impairments, goals and physiotherapy. (3 + 5 + 7, S16)**

Answer

Impairments (3): structural: degeneration of anterior horn cells and corticospinal tracts; functional: hand weakness and wasting (cannot open the fist, poor fine grip for dental work), fasciculations, spasticity and brisk reflexes, possible leg weakness, fatigue. **Goals (5):** keep hand function for work as long as possible with adapted instruments; maintain strength and range; manage fatigue; prevent falls; plan for work changes and equipment. **Physiotherapy (7):** non-fatiguing strengthening of muscles at grade 3 or more; range and stretching; splints for the thumb and wrist; adapted, thicker instrument handles; energy conservation; aerobic walking or cycling at moderate intensity; breathing monitoring; education and stage-based planning.

MCQ **Which is usually spared in ALS?**

A. Hand muscles

B. Bulbar muscles

C. Sensation and eye movements

D. Respiratory muscles

Answer

C.

MCQ **Primary lateral sclerosis affects:**

A. UMN only

B. LMN only

C. Sensory neurons

D. Neuromuscular junction

Answer

A. It has the slowest course.

MCQ **In early MND, exercise should be:**

A. Maximal and to exhaustion

B. Moderate and non-fatiguing

C. Avoided completely

D. Only eccentric

Answer

B.

Last recall check. Write the MND types, the Gold Coast criteria and the stage-wise physiotherapy plan.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Shefner JM, et al. A proposal for new diagnostic criteria for ALS (Gold Coast criteria). *Clin Neurophysiol* 2020;131:1975–8.

3. Miller RG, et al. Riluzole for amyotrophic lateral sclerosis (ALS)/motor neuron disease (MND). *Cochrane Database Syst Rev* 2012;(3):CD001447.
4. Dal Bello-Haas V, Florence JM. Therapeutic exercise for people with amyotrophic lateral sclerosis or motor neuron disease. *Cochrane Database Syst Rev* 2013;(5):CD005229.
5. Bourke SC, et al. Effects of non-invasive ventilation on survival and quality of life in patients with amyotrophic lateral sclerosis: a randomised controlled trial. *Lancet Neurol* 2006;5:140–7.
6. Talbott EO, Malek AM, Lacomis D. The epidemiology of amyotrophic lateral sclerosis. *Handb Clin Neurol* 2016;138:225–38.
7. National Institute for Health and Care Excellence. Motor neurone disease: assessment and management (NG42).

5A(k) Myopathies and myasthenia gravis: features, crises and physiotherapy

> Myasthenia gravis came up twice: physiotherapy for 7 marks (W19) and three features for 3 marks (W24). Its hallmark is weakness that grows with use and recovers with rest, so physiotherapy is timed to the medicine, paced with rests and built up gradually; the other muscle diseases on this page follow the same rule of moderate exercise without exhaustion.

Syllabus	Ch 5A(k) Myopathies (adult); myasthenia gravis and other neuromuscular junction disorders (Duchenne muscular dystrophy is on its own page)
Priority	**Tier 3.** In 2 of 20 papers (2016–2026): 1 seven-mark and 1 three-mark question, both on myasthenia gravis.
Asked as	7m W19 · 3m W24
Usual split	7 (features 2, physiotherapy 5); 3 (three features, one line each)
Checked against	MGFA clinical classification (2000); review of physical training in myasthenia gravis (Gilhus, 2021); randomised trial of inspiratory muscle training in myasthenia gravis (Fregonezi et al., 2005); reviews of exercise in myositis (Alexanderson) and of training in muscle disease (Cochrane, Voet et al., 2019). Full list under Sources (p. 192). Content checked 20 October 2026.

1 Myasthenia gravis

Definition and types

- **Myasthenia gravis (MG)** is an autoimmune disease of the neuromuscular junction: antibodies block and destroy acetylcholine (ACh) receptors on the muscle, so muscles weaken with repeated use and recover with rest.
- **By antibody:** ACh receptor antibodies in about 85% of generalised MG (about half of purely ocular MG); MuSK antibodies in some of the rest; a few have none detectable.
- **By extent:** ocular MG (eyes only) or generalised MG. Ocular weakness is the commonest first symptom, and when it generalises it usually does so within 2 years.

MGFA clinical classification

Class	Severity
I	Eye muscles only (may include weak eye closure); all other muscles normal
II	Mild weakness of other muscles
III	Moderate weakness of other muscles
IV	Severe weakness of other muscles (a feeding tube without intubation is IVb)
V	Intubated, with or without ventilation (not routine after surgery)

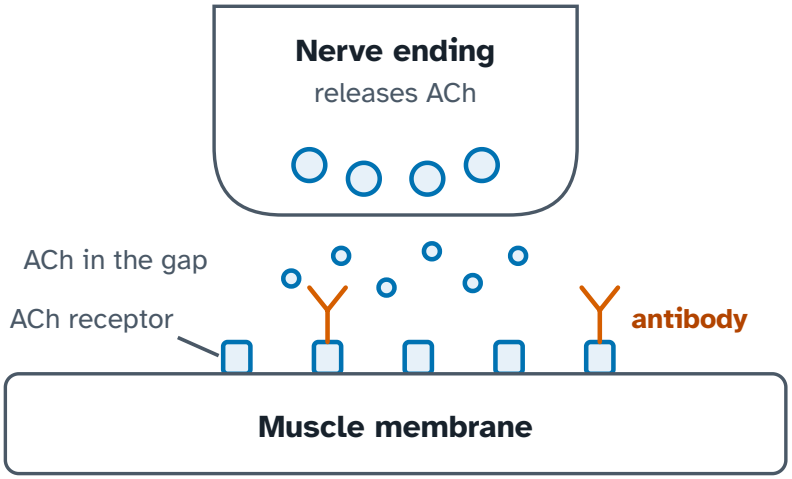
Classes II–IV are split into **a** (mainly limb and trunk) and **b** (mainly swallowing and breathing muscles).

Causes and key features (3 marks in W24)

- **Who:** any age; peaks in younger women and in older men. About 10–15% have a thymoma, and many younger patients have an enlarged (hyperplastic) thymus.
- **Fatigable weakness:** worse with repeated use and towards evening, better after rest.
- **Eyes:** drooping eyelids (ptosis) and double vision (diplopia).
- **Bulbar:** slurred or nasal speech, difficulty chewing and swallowing, a weak "snarling" smile.
- **Neck and limbs:** neck weakness (head drop), proximal arm and leg weakness (combing hair, climbing stairs).

- **Breathing:** breathlessness, especially lying flat; it can fail suddenly (myasthenic crisis).
- Reflexes, sensation and bladder are normal; wasting is unusual.
- **Triggers:** infection, overexertion, heat, emotional stress, surgery, pregnancy, and some drugs (certain antibiotics, magnesium, beta-blockers, botulinum toxin).

What goes wrong at the neuromuscular junction



In MG, antibodies block and destroy receptors, so each signal opens fewer of them: weakness that worsens with repeated use.

Pyridostigmine slows the breakdown of ACh.

Draw this for the features question: it explains why the weakness fatigues and why pyridostigmine helps.

Medical point that matters for physiotherapy

- **Diagnosis:** antibody tests; repetitive nerve stimulation (falling responses) and single-fibre EMG; the ice-pack test (ptosis improves with cold); CT of the chest for thymoma.
- **Pyridostigmine** (an anticholinesterase) improves strength for a few hours; its effect peaks about 1–2 hours after a dose, so plan exercise then.
- **Long-term:** steroids and other immunosuppressants; newer antibody-based drugs; thymectomy (sternotomy precautions after surgery).
- **Crisis:** IVIG or plasma exchange, and ventilation if needed.

Feature	Myasthenic crisis	Cholinergic crisis
Cause	Disease flare: infection, a missed dose, a trigger drug	Too much anticholinesterase
Signs	Rapidly increasing weakness, breathing and swallowing failure	Weakness with sweating, saliva, tears, cramps, diarrhoea, small pupils, slow pulse, twitching

In either crisis: stop the session, sit the person upright, call the medical team at once, and support breathing and the airway.

ICF summary for a typical patient with generalised MG

ICF component	Typical findings	Clinical reasoning
Body functions and structures	Fatigable proximal and neck weakness, ptosis, diplopia, weak chewing and swallowing, reduced breathing strength, fatigue	Fewer working receptors, so strength drops with each repetition
Activities	Stairs, rising from a chair, lifting the arms (hair, shelves), long walks, reading and driving with double vision	Worse late in the day and after effort

ICF component	Typical findings	Clinical reasoning
Participation	Full working day, household work, social life	Unpredictable fatigue limits plans
Environmental factors	Heat, stairs, long travel; supportive family and medical care	Cool rooms and planned rests help
Personal factors	Often a young woman or an older man; worry about crises	Education builds confidence to stay active

Physiotherapy by key problem (7 marks in W19)

Problem	Technique and dose	Why
Fatigable limb weakness	Moderate resistance and aerobic training in stable patients, started low and built up; short sets with rests; aim for **150 min a week** of activity in mild to moderate MG	Training is safe and improves strength and function (Gilhus, 2021)
Timing	Exercise about 1–2 hours after pyridostigmine, earlier in the day, in a cool room	Strength is best then
Weak breathing muscles	Inspiratory muscle training with breathing retraining, **3 times a week** for 8 weeks, load 20% rising to 60% of maximal inspiratory pressure	Improved respiratory muscle strength and breathing pattern in a trial (Fregonezi, 2005)
Weak cough, secretions	Breathing control, huffing, assisted cough, positioning	Prevents chest infection
Fatigue	Energy conservation: plan, prioritise, pace; sit for tasks; rest before tiredness	Saves strength for valued activities
Head drop, ptosis, diplopia	Soft collar; eyelid support or alternating eye patch (with the eye doctor); fall-proof home	Safer walking and reading
After thymectomy	Breathing exercises, early walking, sternal precautions	Prevents chest complications

Classic answer vs current guidance: exercise in myasthenia

Classic	Current	Write in the exam
Rest and avoid exercise, because effort makes the weakness worse.	In stable MG, regular aerobic and strength training is safe and improves strength and daily function; tailor the intensity and keep training to keep the gains (Gilhus, 2021).	"Graded, moderate aerobic and strength training when the disease is stable, timed after pyridostigmine, with rests and no exhaustion."

Precautions

- No exercise during a flare, an infection or a crisis; stop if breathing, speech or swallowing worsen.
- Avoid heat (hot packs, hot pools, hot rooms) and exhausting sessions.
- Watch for crisis: breathlessness lying flat, a weak cough, choking, a falling vital capacity.
- Tell the team before any injection of botulinum toxin for spasticity.

Exam points

- Three features: fatigable weakness, ptosis and diplopia, bulbar weakness (with normal reflexes and sensation).
- For 7 marks: aims, then exercise (type, timing, dose, rests), breathing care, energy conservation, aids, education about crisis and triggers.

2 Lambert–Eaton myasthenic syndrome

- Antibodies against calcium channels on the nerve ending reduce ACh release; about half are linked to small-cell lung cancer.

- Proximal leg weakness, reduced reflexes that briefly return after exercise, a dry mouth and other autonomic symptoms.
- Unlike MG, strength and reflexes improve for a short time with repeated effort.
- **Physiotherapy:** as for MG, plus care around cancer treatment.

3 Inflammatory myopathies (myositis)

Definition and key features

- **Dermatomyositis:** proximal weakness with a rash (violet eyelids, scaly patches over the knuckles); in adults, screen for cancer.
- **Polymyositis** and **immune-mediated necrotising myopathy** (sometimes after statins): symmetrical proximal weakness, high CK.
- **Inclusion body myositis:** over 50; slow, often asymmetrical weakness of the quadriceps and finger flexors; falls; poor response to drugs.
- Swallowing difficulty and lung fibrosis can occur.

Medical point

Diagnosed by CK, myopathic EMG, MRI and muscle biopsy; treated with steroids and immunosuppressants (steroids themselves can weaken muscles).

Physiotherapy

- Exercise has been found safe in all types and stages studied; start at low intensity under supervision and progress with regular review (Alexanderson, 2018).
- Active phase: range of motion, positioning, gentle active and isometric work, breathing exercises.
- Stable phase: progressive resistance and aerobic training; functional training (sit to stand, stairs); falls prevention (inclusion body myositis).

4 Adult muscular dystrophies

Type	Key features	Physiotherapy points
Myotonic dystrophy type 1	Autosomal dominant; distal weakness (hands, foot drop), facial and neck weakness, ptosis, myotonia (slow release of grip, better with repeated movement), cataracts, heart conduction problems, daytime sleepiness	Warm-up movements ease myotonia; AFO and hand aids; moderate training; check heart symptoms and ECG follow-up; breathing and swallowing care
Facioscapulohumeral (FSHD)	Face (cannot whistle), winged scapulae, upper arms, often asymmetrical; later foot drop and abdominal weakness	Moderate aerobic and strength work, scapular support for function, AFO
Limb-girdle	Weakness of the hip and shoulder girdles; some types affect the heart and breathing	Functional strengthening, contracture prevention, mobility aids
Becker	Milder dystrophin disorder than Duchenne; later onset, calf enlargement, cramps on exercise, heart muscle disease	Submaximal exercise, no eccentric or exhausting work; cardiac review (see DMD (p. 334))

Other causes of muscle weakness in adults: steroid and thyroid myopathy (proximal weakness), statin-related myopathy, and weakness acquired in intensive care.

5 Exercise rules for muscle disease

- Moderate aerobic and strength training appears safe in slowly progressive muscle diseases, but the trials are small and their quality is low (Cochrane, 2019).
- Avoid eccentric, high-intensity and exhausting exercise.

- Stop and review if pain, cramps or weakness last into the next day, or if the urine turns dark after exercise (muscle breakdown).
- Screen the heart and breathing (vital capacity sitting and lying) where the disease can affect them.
- Prevent contractures and falls; use orthoses and aids early; teach energy conservation.

Recall check. Why should exercise be timed after pyridostigmine, and what are the signs that a session was too hard for a patient with muscle disease?

6 Exam corner

Past questions (MUHS, 2016–2026)

W19 Q2(a), 7 marks
Write physiotherapy management for myasthenia gravis.

W24 Q1(a), 3 marks
Enlist three features of myasthenia gravis.

Pages for this topic: Duchenne muscular dystrophy (p. 334), neurological assessment (EMG patterns) (p. 29), motor neuron disease (p. 177).

Keywords to underline

neuromuscular junction, acetylcholine receptor antibodies, fatigable weakness, ptosis and diplopia, bulbar weakness, pyridostigmine timing, energy conservation, inspiratory muscle training, myasthenic crisis, no exhaustion

Answer skeleton: physiotherapy for MG (7 marks)

1. **[Intro, 1]** One line on the disease and fatigable weakness.
2. **[Assessment and aims, 1]** Strength before and after repetition, breathing, swallowing; aims.
3. **[Treatment, 4]** Graded aerobic and strength training timed after pyridostigmine, with rests; breathing exercises and IMT; airway clearance; energy conservation; aids (collar, eye patch).
4. **[Precautions, 1]** Heat, overexertion, infection, crisis signs, trigger drugs.

Draw these

- The neuromuscular junction with receptors blocked by antibodies (section 1)

Viva questions

1. Why does the weakness of MG get worse with repeated use?
2. Which muscles are usually affected first?
3. When in the day should a patient with MG exercise?
4. How do you tell a myasthenic crisis from a cholinergic crisis?
5. How does Lambert–Eaton syndrome differ from MG?
6. What is myotonia, and what eases it?

Linked practical spots

- Inspiratory muscle trainer; soft cervical collar; AFO; an EMG report showing a decrement on repetitive stimulation.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Enlist three features of myasthenia gravis. (W24)**

Answer

(1) Fatigable weakness that worsens with use and towards evening and improves with rest. (2) Ptosis and diplopia, usually the first symptoms. (3) Bulbar weakness: nasal speech, difficulty chewing and swallowing. Also proximal limb and neck weakness and breathing weakness, with normal reflexes and sensation.

3 m **Differentiate myasthenic crisis from cholinergic crisis.**

Answer

Myasthenic crisis is a flare of the disease (infection, missed dose, trigger drug) with rapidly increasing weakness and breathing failure. Cholinergic crisis is an overdose of anticholinesterase, with weakness plus sweating, salivation, cramps, diarrhoea, small pupils and twitching. Both are emergencies.

7 m **Write physiotherapy management for myasthenia gravis. (W19)**

Answer

MG is an autoimmune disease of the neuromuscular junction with fatigable weakness. **Assess** strength before and after repetition, endurance, breathing (vital capacity, cough), swallowing and function. **Aims:** maintain and improve strength and endurance, protect breathing, manage fatigue, keep independence. **Treatment:** graded moderate aerobic and resistance training in stable disease, timed 1–2 hours after pyridostigmine, in a cool room, with short sets and rests; inspiratory muscle training and breathing retraining; airway clearance and assisted cough; energy conservation and pacing; collar for head drop, eye patch for diplopia, falls prevention; after thymectomy, chest physiotherapy and sternal precautions. **Precautions:** no exercise in a flare or infection, avoid heat and exhaustion, recognise crisis and refer, avoid trigger drugs. **Education** on triggers and warning signs.

LAQ **Practice: a 32-year-old teacher with generalised myasthenia gravis (MGFA class III) has ptosis, double vision, and arm and leg weakness that is worse by evening, and is breathless climbing stairs. Write ICF with clinical reasoning, goals and physiotherapy. (5 + 3 + 7)**

Answer

ICF (5): body functions: fatigable proximal weakness, ptosis, diplopia, reduced breathing strength, fatigue; activities: stairs, writing on the board, standing to teach, reading; participation: full teaching day; environment: heat in classrooms, stairs; personal: young, worried about crises. **Goals (3):** short term: knows pacing and timing of exercise, starts an exercise and breathing programme; long term: teaches a full day with planned rests, climbs stairs without breathlessness. **Physiotherapy (7):** as in the 7-mark answer, adapted to work (sit to teach, lessons with the hardest tasks in the morning after a dose).

MCQ **In myasthenia gravis the antibodies most often act against:**

A. Calcium channels on the nerve ending

B. Acetylcholine receptors on the muscle

C. Myelin

D. Dystrophin

Answer

B. Calcium channels are the target in Lambert–Eaton syndrome.

MCQ **The best time for a patient with MG to exercise is:**

A. Late evening

B. Just before the next dose is due

C. About 1–2 hours after a dose of pyridostigmine

D. After a hot shower

Answer
C.

MCQ **Slow release of a handshake grip suggests:**

A. Myotonic dystrophy

B. Myasthenia gravis

C. Polymyositis

D. Becker muscular dystrophy

Answer
A. This is myotonia.

Last recall check. Write three features of MG, the crisis comparison and a seven-point physiotherapy plan.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Jaretzki A, et al. Myasthenia gravis: recommendations for clinical research standards (MGFA clinical classification). *Neurology* 2000;55:16–23.
3. Gilhus NE. Physical training and exercise in myasthenia gravis. *Neuromuscul Disord* 2021;31:169–73.
4. Fregonezi GA, et al. Effects of 8-week, interval-based inspiratory muscle training and breathing retraining in patients with generalized myasthenia gravis. *Chest* 2005;128:1524–30.
5. Alexanderson H. Exercise in myositis. *Curr Treat Options Rheumatol* 2018.
6. Voet NB, et al. Strength training and aerobic exercise training for muscle disease. *Cochrane Database Syst Rev* 2019: CD003907.

5A(l) Disorders of the autonomic nervous system: Horner's syndrome, blood pressure problems and autonomic dysreflexia

> Horner's syndrome was a 3-mark question in S23, and autonomic dysreflexia, asked in most papers, is covered fully on the spinal cord page. The autonomic system matters in every session with a neurological patient, because blood pressure, heart rate and temperature decide whether it is safe to sit up, stand and exercise.

Syllabus	Ch 5A(l) Disorders of the autonomic nervous system: Horner's syndrome, hypotension and hypertension, autonomic dysreflexia
Priority	**Tier 3.** In 1 of 20 papers (2016–2026): Horner's syndrome for 3 marks (S23). Autonomic dysreflexia questions are counted on the spinal cord page.
Asked as	3m S23
Usual split	3 (the triad with one line on the pathway and causes)
Checked against	Consensus definition of orthostatic hypotension (Freeman et al., 2011); autonomic dysreflexia guidance as checked on the spinal cord page (Consortium for Spinal Cord Medicine, 2020). Full list under Sources (p. 196). Content checked 20 October 2026.

1 The autonomic nervous system in brief

Feature	Sympathetic	Parasympathetic
Outflow	Thoracolumbar (T1–L2)	Craniosacral (cranial nerves III, VII, IX, X and S2–S4)
Main actions	"Fight or flight": raises heart rate and blood pressure, narrows skin and gut vessels, sweating, widens pupils	"Rest and digest": slows the heart, gut movement, bladder emptying, narrows pupils
Physio relevance	Lost below a cord lesion at or above T6: low blood pressure, autonomic dysreflexia, no sweating	Vagus intact in cord injury: slow heart rate on suction or turning in acute tetraplegia

2 Horner's syndrome (3 marks in S23)

- **Definition:** loss of the sympathetic supply to one side of the face and eye.
- **Clinical features (same side):** mild ptosis (drooping upper lid), miosis (small pupil, most obvious in dim light), anhidrosis (no sweating of the face), and an apparently sunken eye (enophthalmos); the pupil still reacts to light.

The three-neuron pathway and causes

Neuron	Course	Causes
First order (central)	Hypothalamus down the brainstem and cervical cord to T1	Lateral medullary (brainstem) stroke, syringomyelia, cervical cord injury, MS
Second order (preganglionic)	T1 root over the lung apex to the superior cervical ganglion	T1 root avulsion in a lower brachial plexus (Klumpke's) injury, lung apex (Pancoast) tumour, neck trauma or surgery
Third order (postganglionic)	Along the internal carotid artery to the eye	Carotid artery dissection, cluster headache, cavernous sinus lesions

Why physiotherapists care: in a brachial plexus injury, Horner's syndrome means the T1 root is torn from the cord (preganglionic), which carries a poor outlook for natural recovery of the hand (see obstetric plexus (p. 309) and nerve injuries (p. 94)).

3 Orthostatic (postural) hypotension

- **Definition:** a sustained fall in systolic blood pressure of at least 20 mmHg, or diastolic of at least 10 mmHg, within 3 minutes of standing or head-up tilt (consensus, 2011).
- **Neurological causes:** spinal cord injury at or above T6, Parkinson's disease and multiple system atrophy, diabetic and other autonomic neuropathies, GBS, long bed rest, and drugs (blood-pressure tablets, levodopa, diuretics).
- **Symptoms:** dizziness, blurred or greying vision, weakness, yawning, nausea, fainting; in tetraplegia symptoms may be few despite large falls.

Physiotherapy

- Measure blood pressure lying and after sitting or standing; record symptoms.
- **Gradual uprighting:** head-of-bed elevation, then tilt table or sitting, raising the angle in small steps as blood pressure allows (see spinal cord (p. 77)).
- **Reduce pooling:** abdominal binder, compression stockings, leg exercises (ankle pumps) before rising.
- **Slow position changes,** sitting on the bed edge before standing; good fluid intake (as the medical team advises); avoid heat and large meals before exercise.
- **Physical counter-manoeuvres:** crossing the legs, tensing the leg and buttock muscles, or squatting raise blood pressure for a short time.
- **If symptoms start:** stop, recline or lower the tilt, raise the legs, and check blood pressure.

4 Hypertension: autonomic dysreflexia

A sudden, dangerous rise in blood pressure in people with spinal cord injury at or above T6, triggered by a stimulus below the lesion (most often a full bladder): pounding headache, flushing and sweating above the lesion, a slow pulse. Sit up, loosen clothing, check the bladder and bowel, and monitor blood pressure. The full answer with the flowchart is on the spinal cord page (p. 77).

5 Other autonomic problems

- **Temperature control:** after high cord injury the person cannot sweat or shiver below the lesion, so body temperature follows the room (poikilothermia): avoid heat and cold, cool rooms for exercise, dress for the weather.
- **Heart rate:** slow heart rate in acute tetraplegia, especially with suction or turning; irregular heart rhythm and blood pressure swings in GBS (dysautonomia).
- **Paroxysmal sympathetic hyperactivity** after severe brain injury: episodes of fast heart rate, high blood pressure, fever, fast breathing, sweating and posturing; calm handling, and treatment by the medical team.
- **Bladder, bowel and sexual function:** see bladder and bowel (p. 238).
- **Complex regional pain syndrome** (sympathetic features of pain, swelling, colour and sweating change): see neuropathies (p. 107).

Recall check. Give the triad of Horner's syndrome and one cause at each of the three neuron levels.

6 Exam corner

Past questions (MUHS, 2016–2026)

S23 Q1(e), 3 marks

Clinical features of Horner's syndrome.

Most papers

Autonomic dysreflexia (see the spinal cord page).

Keywords to underline

ptosis, miosis, anhidrosis three-neuron pathway T1 root avulsion orthostatic hypotension: 20/10 within 3 minutes tilt table abdominal binder counter-manoeuvres autonomic dysreflexia poikilothermia

Answer skeleton: Horner's syndrome (3 marks)

1. Definition: loss of sympathetic supply to the eye and face on one side.
2. Features: ptosis, miosis, anhidrosis, apparent enophthalmos.
3. Causes by level, with the brachial plexus (T1 avulsion) link.

Draw these

- The three-neuron sympathetic pathway to the eye: hypothalamus, T1, superior cervical ganglion, carotid artery, eye

Viva questions

1. What is the triad of Horner's syndrome?
2. Why does Horner's syndrome in a plexus injury suggest a poor prognosis?
3. Define orthostatic hypotension.
4. How do you progress a tetraplegic patient on a tilt table?
5. Why can a person with high cord injury not control body temperature?

Linked practical spots

- Tilt table, abdominal binder, compression stockings, blood pressure measurement lying and standing.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Clinical features of Horner's syndrome. (S23)**

Answer
Same-side mild ptosis, miosis (small pupil, reacting to light), anhidrosis of the face and apparent enophthalmos, from loss of the sympathetic supply. Causes: central (lateral medullary stroke, syringomyelia, cord injury), preganglionic (T1 root avulsion in Klumpke's injury, Pancoast tumour), postganglionic (carotid dissection).

3 m **Define orthostatic hypotension and list three ways to manage it in physiotherapy.**

Answer
A sustained fall of at least 20 mmHg systolic or 10 mmHg diastolic within 3 minutes of standing or head-up tilt. Management: gradual uprighting (tilt table in small steps), abdominal binder and stockings, leg exercises before rising, slow position changes, counter-manoeuvres; stop and recline if symptoms start.

7 m **Physiotherapy considerations for autonomic problems in a patient with C6 tetraplegia.**

Answer
Orthostatic hypotension: binder and stockings, gradual tilt, blood pressure monitoring, counter-measures. **Autonomic dysreflexia:** know the signs and the steps (sit up, loosen, bladder, bowel, other triggers), avoid tight straps, check the catheter before sessions. **Temperature:** avoid overheating and cold, cool room, fluids. **Heart:** slow heart rate with suction or turning in the acute stage. **Bladder and bowel** routines before sessions. Educate the patient and carers to recognise and act.

LAQ **Practice: a 68-year-old man with Parkinson's disease faints twice when getting out of bed in the morning. Assess, explain the cause, and plan physiotherapy and advice. (5 + 3 + 7)**

Answer

Assess (5): blood pressure and pulse lying and at 1 and 3 minutes standing, symptoms, timing with meals and levodopa, fluid intake, falls and injuries, other autonomic signs (constipation, urinary urgency). **Cause (3):** orthostatic hypotension from autonomic failure in PD, worsened by levodopa, morning dehydration and heat. **Plan (7):** sit on the bed edge and do ankle pumps before standing; stand slowly with support; counter-manoeuvres; compression stockings or an abdominal binder; fluids as advised; avoid hot baths and big meals before activity; exercise lying or seated at first; report to the doctor for a medicine review; teach the family what to do if he feels faint.

MCQ **Which is not a feature of Horner's syndrome?**

A. Ptosis

B. Miosis

C. Anhidrosis

D. Dilated pupil

Answer

D.

MCQ **Autonomic dysreflexia occurs in cord lesions at or above:**

A. T6

B. T12

C. L2

D. S2

Answer

A.

MCQ **Orthostatic hypotension is a fall in systolic pressure of at least:**

A. 5 mmHg

B. 10 mmHg

C. 20 mmHg

D. 40 mmHg

Answer

C. Or 10 mmHg diastolic, within 3 minutes.

Last recall check. Write Horner's syndrome with causes, the definition and physiotherapy of orthostatic hypotension, and the steps for autonomic dysreflexia.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Freeman R, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Clin Auton Res* 2011;21:69–72.

3. Consortium for Spinal Cord Medicine. Evaluation and management of autonomic dysreflexia and other autonomic dysfunctions (clinical practice guideline), 2020.

5A(m) Psychosomatic pain and paralysis: functional neurological disorder and chronic pain

> The syllabus calls it psychosomatic pain and paralysis; today it is called functional neurological disorder (FND), and it was not asked on its own in 2016–2026. It matters because these patients are common, their symptoms are real and involuntary, and physiotherapy that explains the diagnosis and retrains movement with attention directed away from the limb can help them recover.

Syllabus	Ch 5A(m) Psychosomatic pain and paralysis
Priority	**Syllabus.** Not asked on its own in 2016–2026.
Asked as	Not asked on its own in 2016–2026
Usual split	Likely 3 marks (Hoover's sign; features of functional weakness) or 7 marks (physiotherapy for functional paralysis)
Checked against	Consensus recommendations for physiotherapy in functional motor disorders (Nielsen et al., 2015); Physio4FMD randomised trial (Nielsen et al., 2024); reviews of positive signs of functional limb weakness. Full list under Sources (p. 202). Content checked 20 October 2026.

1 What it is

- **Functional neurological disorder (FND)** (older terms: conversion disorder, hysteria, psychosomatic paralysis) is a problem of how the nervous system works, not of its structure: real, involuntary symptoms such as weakness, abnormal movements, seizures or numbness, without the damage seen in stroke or MS.
- It is common in neurology clinics and often follows a trigger: an injury, an illness, pain, a panic attack or stress (not always psychological trauma).
- **Not the same as malingering** (deliberately faking for gain) or factitious disorder (deliberately producing symptoms to be a patient). People with FND are not pretending.
- It can coexist with other neurological disease (for example functional seizures with epilepsy).

2 Types

- **Functional weakness or paralysis:** one limb, one side, or both legs.
- **Functional movement disorders:** tremor, dystonia (often a fixed clenched fist or inverted ankle), jerks, gait disorders.
- **Functional (dissociative) seizures.**
- **Functional sensory symptoms,** dizziness and cognitive symptoms; fatigue and pain are frequent companions.

3 Diagnosis by positive signs

FND is diagnosed from **positive clinical signs** showing that the system can work, not only by excluding other disease.

Sign	How it is done	Functional finding
Hoover's sign	Test hip extension of the weak leg directly, then while the patient flexes the other hip against resistance	Weak hip extension on direct testing returns to normal during the other hip's flexion (automatic movement)
Hip abductor sign	Compare voluntary abduction with the automatic abduction that occurs when the other leg abducts	The "weak" leg abducts strongly when it acts automatically
Give-way weakness	Manual muscle testing	Normal power at first, then sudden collapse

Sign	How it is done	Functional finding
Drift without pronation	Arms held out, palms up, eyes closed	The arm drifts down without the pronation seen in a pyramidal lesion
Dragging gait	Watch walking	The weak leg is dragged behind with the hip turned in or out, rather than circumducted
Entrainment test (tremor)	Ask the patient to copy a rhythm with the other hand	The tremor changes to the new rhythm, pauses or becomes irregular
Distractibility and variability	Observe during other tasks	Symptoms lessen when attention is elsewhere

Reflexes, tone and plantar responses are normal in pure functional weakness, and there is no wasting early on.

4 Physiotherapy for functional motor disorders

Principles (consensus, Nielsen et al., 2015)

1. **Explain** the diagnosis clearly and positively: the symptoms are real, the nervous system is not damaged, and it can relearn; show the patient their positive signs (for example normal movement during Hoover's test).
2. **Set expectations of recovery** and agree goals with the patient.
3. **Retrain movement with attention directed away from the limb:** use automatic movements, rhythm, dual tasks, music, imagery and distraction; start with what the patient can do (for example walking backwards or sideways, or on a treadmill) and build towards normal patterns.
4. **Avoid reinforcing abnormal patterns:** minimise aids, braces and hands-on support where safe; withdraw them gradually when used.
5. **Self-management plan:** daily practice, pacing for fatigue, handling setbacks, return to activities and work.
6. **Team work:** psychology (CBT) and psychiatry where needed, especially for anxiety, depression or trauma.

Classic answer vs current guidance: "psychosomatic paralysis"

Classic	Current	Write in the exam
A diagnosis of exclusion; reassure that "nothing is wrong" and refer to psychiatry; physiotherapy has no role.	A positive diagnosis with specific signs; physiotherapy based on explanation and movement retraining is a core treatment. In the Physio4FMD trial (2024), specialist physiotherapy did not improve the main physical-function score at 12 months more than usual community physiotherapy, but patients more often felt improved and it was likely cost-effective.	"Positive signs (Hoover's), clear explanation, movement retraining with redirected attention, minimal aids, self-management and team care."

5 Chronic pain with psychological factors

- Long-lasting pain often involves an over-sensitive nervous system (central sensitisation) and is worsened by fear, low mood, poor sleep and inactivity; the pain is real.
- **Physiotherapy:** pain education (how pain works), graded activity and graded exposure to feared movements, regular exercise, pacing, sleep and stress strategies, and a cognitive-behavioural approach; reduce reliance on passive treatments.
- Screen for red flags first (see ICF page (p. 4)), and refer for psychological care when distress is high.

Recall check. Describe Hoover's sign and three principles of physiotherapy for functional weakness.

6 Exam corner

Past questions (MUHS, 2016–2026)

Not asked on its own

Expect a 3-mark question on Hoover's sign or features of functional paralysis, or a 7-mark question on physiotherapy.

Keywords to underline

functional neurological disorder, real and involuntary, not malingering, positive signs, Hoover's sign, give-way weakness, entrainment, explanation, redirected attention, self-management

Answer skeleton: physiotherapy for functional paralysis (7 marks)

1. Define FND and its difference from malingering (1).
2. Positive signs used in assessment (1–2).
3. Explanation, goals, movement retraining with redirected attention, minimal aids, graded progression, self-management, team care (4).

Draw these

- Hoover's test: the examiner's hand under the heel of the "weak" leg while the patient lifts the other leg

Viva questions

1. How does FND differ from malingering?
2. Describe Hoover's sign.
3. Why should aids and braces be limited in FND?
4. What does "redirected attention" mean in treatment?
5. Name two other positive signs of functional weakness.

Linked practical spots

- Demonstrate Hoover's sign and the hip abductor sign on a model.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **What is Hoover's sign?**

Answer

A positive sign of functional leg weakness: hip extension of the "weak" leg is weak on direct testing but returns to normal when the patient flexes the other hip against resistance, because hip extension then happens automatically. The examiner feels the heel press down into the hand.

3 m **Differentiate functional weakness from malingering.**

Answer

Functional weakness is real and involuntary, arising from altered nervous system function, often after a trigger; the person is not pretending. Malingering is deliberate feigning of symptoms for external gain (money, leave, avoiding duty).

7 m **Physiotherapy for a patient with functional paralysis of one leg.**

Answer

Assessment including positive signs (Hoover's, hip abductor, give-way, dragging gait) and normal reflexes and tone. Explain the diagnosis positively and show the patient that the leg can move. Agree goals. Retrain movement with attention directed away from the leg: automatic tasks (sit to stand, walking backwards, treadmill, rhythm, dual tasks), then normal walking; minimal aids and hands-on help, withdrawn gradually; graded progression; daily self-management plan; manage fatigue and pain; involve psychology when needed; plan return to work.

LAQ **Practice: a 28-year-old bank clerk developed weakness of the right leg after a minor fall 3 months ago; reflexes and tone are normal and Hoover's sign is positive; she walks dragging the leg with a stick and has stopped working. Write assessment, goals and physiotherapy. (5 + 3 + 7)**

Answer

Assessment (5): history and triggers, fears and beliefs, fatigue and pain; positive signs, normal reflexes and plantar; gait pattern; function and work; screen mood. **Goals (3):** short term: walks indoors without the stick; long term: returns to work and normal walking outdoors. **Physiotherapy (7):** explanation with demonstration of normal movement; retraining walking with redirected attention (treadmill, rhythm, walking backwards, dual tasks); gradual removal of the stick; graded activity and fatigue pacing; home practice plan and setback plan; liaison with psychology; graded return-to-work plan.

MCQ **In pure functional weakness, the plantar response is usually:**

A. Extensor

B. Flexor (normal)

C. Absent

D. Variable with clonus

Answer

B.

MCQ **Which approach best fits physiotherapy for FND?**

A. Long-term bracing of the weak limb

B. Movement retraining with attention directed away from the limb

C. Telling the patient nothing is wrong

D. Bed rest

Answer

B.

MCQ **A tremor that changes to match a rhythm tapped by the other hand suggests:**

A. Parkinson's disease

B. Essential tremor

C. Functional tremor

D. Cerebellar tremor

Answer

C. The entrainment test.

Last recall check. Write five positive signs and six principles of physiotherapy for functional motor disorders.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Nielsen G, et al. Physiotherapy for functional motor disorders: a consensus recommendation. *J Neurol Neurosurg Psychiatry* 2015;86:1113–9.
3. Nielsen G, et al. Specialist physiotherapy for functional motor disorder in England and Scotland (Physio4FMD): a pragmatic, multicentre, phase 3 randomised controlled trial. *Lancet Neurol* 2024.
4. Reviews of positive signs in functional limb weakness and functional movement disorders (Stone, Edwards and colleagues).

Part TP

Physiotherapy management: the adult treatment programme

Treatment programme items 1 to 9

TP1	Electrotherapy in neurological rehabilitation	204
TP2	Neurotherapeutic approaches: Brunnstrom, Rood, Bobath, MRP, PNF and CIMT	209
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TP1 Electrotherapy in neurological rehabilitation: TENS, NMES, FES, biofeedback and electrodiagnosis

> The syllabus asks for electrotherapy for pain relief and functional re-education with clinical reasoning; it was not asked on its own in 2016–2026 but appears inside management answers. The modern rule is simple: use electrical stimulation to make the patient's own practice of a task possible or more effective, not as a passive treatment on its own.

Syllabus	Treatment programme 1: application of appropriate electrotherapeutic modes for relief of pain and functional re-education with clinical reasoning
Priority	**Syllabus.** Not asked on its own in 2016–2026.
Asked as	Not asked on its own in 2016–2026
Usual split	Likely 3 marks (FES, TENS, strength-duration curve) or 7 marks (electrotherapy in stroke or nerve injury)
Checked against	Cochrane review of electrical stimulation for post-stroke shoulder pain (Price and Pandyan); UK National Clinical Guideline for Stroke (2023) on NMES for subluxation; peroneal FES versus AFO trial (Bethoux et al., 2014); Bell's palsy guideline on electrical stimulation (as on the facial nerve page). Full list under Sources (p. 208). Content checked 20 October 2026.

1 Uses of electrotherapy in neurology

- **Pain relief:** TENS.
- **Muscle activation and re-education:** neuromuscular electrical stimulation (NMES) and functional electrical stimulation (FES).
- **Feedback:** EMG biofeedback.
- **Denervated muscle:** long-pulse stimulation (limited evidence).
- **Electrodiagnosis:** the strength-duration curve (now largely replaced by EMG and nerve conduction; see assessment (p. 29)).

2 TENS for pain

Type	Settings	How it works
Conventional (high frequency)	About 80–100 Hz, short pulses, a strong but comfortable tingling, no muscle twitch	Gate control at the spinal cord; works while it is on
Acupuncture-like (low frequency)	About 2–4 Hz, longer pulses, strong enough for visible twitches	Release of endorphins; effect may outlast the session
Burst or brief intense	Bursts of high-frequency pulses; brief intense for procedures	Mixed mechanisms

- **Neuro uses:** painful hemiplegic shoulder (alongside positioning and movement), musculoskeletal pain secondary to abnormal posture, some neuropathic pain (evidence is limited).
- Electrodes over or around the painful area or along the nerve; the patient must be able to feel and report the tingling.

3 NMES and FES

- **NMES:** stimulation of an innervated nerve to produce muscle contraction, for strengthening or re-education.
- **FES:** stimulation timed to a functional task, for example the dorsiflexors during swing, or wrist and finger extensors during grasp and release.

- **Typical settings:** about 30–50 Hz for a smooth contraction; pulse width about 200–300 µs (longer for large muscles); a ramp up and down; on–off cycles with long rests at first (about 1:3 to 1:5) to avoid fatigue; intensity for a strong, comfortable contraction; sessions of about 20–30 minutes.
- **Best used** with active effort and task practice (for example the patient tries to open the hand while the stimulator assists, then grasps an object).

Use	How	Evidence
Shoulder subluxation after stroke	Supraspinatus and posterior deltoid, early after stroke	Reduces subluxation and improves pain-free lateral rotation, without a clear effect on pain (Cochrane); UK guideline: consider it for subluxation within 6 months (2023)
Foot drop (stroke, MS)	Peroneal nerve stimulation triggered by a heel switch or tilt sensor	As good as an AFO for walking (Bethoux et al., 2014)
Arm and hand recovery after stroke	Wrist and finger extensors with task practice	Helpful as an add-on to practice, mainly in people with some movement
Spinal cord injury	FES cycling, standing and grasp systems	Builds muscle and fitness; useful for exercise in paralysed legs
Disuse weakness	Quadriceps NMES in bed-bound or ICU patients	Can maintain or build strength when active exercise is not possible

4 EMG biofeedback

- Surface electrodes pick up muscle activity and show it as a sound or visual signal, so the patient can learn to increase activity in a weak muscle or reduce it in an overactive one.
- **Uses:** early muscle re-education after nerve injury or stroke (a flicker shows up before visible movement), relaxing overactive muscles, retraining after tendon transfers.
- Evidence from small trials suggests it may help as an add-on to exercise; it does not replace practice.

5 Denervated muscle and the strength-duration curve

- **Denervated muscle** responds only to long pulses (tens to hundreds of milliseconds, for example interrupted direct or triangular currents). Stimulation is sometimes used to slow wasting until the nerve regrows, but evidence in people is limited; for Bell's palsy, electrical stimulation has shown no benefit (see facial nerve (p. 124)).
- **Strength-duration curve:** the current needed to produce a minimal contraction plotted against pulse duration.
  - **Rheobase:** the smallest current that causes a contraction with a long pulse.
  - **Chronaxie:** the pulse duration needed at twice the rheobase; under about 1 ms in normally innervated muscle and much longer in denervated muscle.
  - Denervation shifts the curve upwards and to the right; a kink suggests partial denervation; serial curves can show reinnervation.

6 Precautions and contraindications

Take care

- **Spinal cord injury at or above T6:** stimulation below the lesion can trigger autonomic dysreflexia; monitor blood pressure and stop at once if symptoms start.
- Pacemakers and implanted defibrillators (avoid the chest and seek advice); over the carotid sinus or the front of the neck; over the eyes.
- Over known malignancy, active infection, deep vein thrombosis, or broken or fragile skin.
- The abdomen or low back in pregnancy; around the head and neck in uncontrolled epilepsy.
- Numb skin or poor understanding: lower intensities, frequent skin checks, close supervision.
- Always check skin before and after, and explain the sensation.

Recall check. Give the settings for NMES and two neurological uses with their evidence.

7 Exam corner

Past questions (MUHS, 2016–2026)

Not asked on its own

Expect 3 marks on FES, TENS types or the strength-duration curve, or 7 marks on electrotherapy in stroke or nerve injury.

Keywords to underline

gate control conventional and acupuncture-like TENS NMES FES timed to the task 30–50 Hz shoulder subluxation peroneal stimulator EMG biofeedback rheobase and chronaxie autonomic dysreflexia

Answer skeleton: FES (3 marks)

1. Definition: stimulation timed to a functional task.
2. Settings and an example (peroneal stimulator for foot drop).
3. Evidence or one precaution.

Draw these

- A strength-duration curve for normal and denervated muscle, with rheobase and chronaxie marked

Viva questions

1. What is the difference between NMES and FES?
2. Where do you place electrodes for shoulder subluxation?
3. What is chronaxie, and how does it change after denervation?
4. Why is electrical stimulation risky in high spinal cord injury?
5. How does conventional TENS relieve pain?

Linked practical spots

- Set up TENS, NMES for shoulder subluxation, and a foot-drop stimulator; plot a strength-duration curve.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Functional electrical stimulation.**

Answer

Electrical stimulation of nerves to produce contractions timed to a functional activity, for example peroneal nerve stimulation during swing for foot drop or wrist and finger extensor stimulation during grasp. About 30–50 Hz, 200–300 µs, triggered by a switch or sensor. Peroneal FES is as good as an AFO after stroke; check skin and watch for autonomic dysreflexia in high cord injury.

3 m **Strength-duration curve.**

Answer

A plot of the current needed for a minimal contraction against pulse duration. Rheobase is the smallest current with a long pulse; chronaxie is the pulse duration at twice the rheobase (under about 1 ms normally, much longer when denervated). Denervation shifts the curve up and to the right; a kink suggests partial denervation; serial curves show reinnervation.

7 m **Electrotherapy in the management of hemiplegia.**

Answer

Use with task practice, not alone. NMES of supraspinatus and posterior deltoid for early shoulder subluxation (reduces subluxation, pain-free range improves). FES for wrist and finger extension during grasp practice. Peroneal FES for foot drop as an alternative to an AFO. EMG biofeedback for early muscle re-education. TENS for shoulder or musculoskeletal pain. Settings (30–50 Hz, 200–300 µs, ramps, rests). Precautions: sensation, cognition, skin, pacemakers, DVT.

LAQ **Practice: a 50-year-old woman 6 weeks after a right MCA stroke has a flaccid left arm with a two-finger-breadth shoulder subluxation and left foot drop. Plan electrotherapy and how it fits with the rest of her treatment. (15)**

Answer

Assess sensation, skin, cognition, pain and contraindications. NMES to supraspinatus and posterior deltoid daily, with positioning and arm support, aiming to reduce subluxation and protect the shoulder; combine with active-assisted shoulder movement. As movement returns, FES for wrist and finger extension during grasp-and-release tasks. Peroneal FES or an AFO for foot drop during gait training. Settings 30–50 Hz, 200–300 µs, ramps, rest periods, building tolerance; skin checks; family taught to apply at home. Measure subluxation, pain-free range, Fugl-Meyer arm score and walking speed.

MCQ **Conventional TENS uses a frequency of about:**

A. 2–4 Hz

B. 10 Hz

C. 80–100 Hz

D. 1000 Hz

Answer

C.

MCQ **In denervated muscle, chronaxie:**

A. Falls below 0.1 ms

B. Stays the same

C. Becomes much longer

D. Cannot be measured

Answer
C.

MCQ **Electrical stimulation below the lesion in T4 paraplegia may cause:**

A. Autonomic dysreflexia

B. Hypoglycaemia

C. Spinal shock

D. Bell's palsy

Answer
A.

Last recall check. Write the TENS types, NMES settings, three neuro uses with evidence, the strength-duration curve and five precautions.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Price CI, Pandyan AD. Electrical stimulation for preventing and treating post-stroke shoulder pain. *Cochrane Database Syst Rev* 2000 (updated); CD001698.
3. Intercollegiate Stroke Working Party. National Clinical Guideline for Stroke for the UK and Ireland. 2023.
4. Bethoux F, et al. The effects of peroneal nerve functional electrical stimulation versus ankle-foot orthosis in patients with chronic stroke: a randomized controlled trial. *Neurorehabil Neural Repair* 2014;28:688–97.
5. Standard electrotherapy texts for TENS and NMES parameters and the strength-duration curve.

TP2 Neurotherapeutic approaches: Rood, Bobath, Brunnstrom, MRP, PNF, imagery and CIMT

The named approaches came up in 8 papers, as 3-mark lists of principles (Rood, Bobath, MRP) and 7-mark questions on Rood, MRP and PNF. Learn each approach's principles and two or three techniques, because examiners ask for them by name. Then add one line every time: no single approach is better than the others, and the best results come from active, task-specific practice that borrows from all of them.

Syllabus	Treatment programme 2: neurotherapeutic approaches (Brunnstrom, Rood, Bobath, NDT, MRP, mental imagery, CIMT, learning transfers)
Priority	**High yield.** In 8 of 20 papers (2016–2026): 3 seven-mark and 5 three-mark questions.
Asked as	7m S17 · W18 · W25 · 3m S16 · W16 · S18 · W19 · S22
Usual split	7 (principles 3–4 and techniques 3–4); 3 (three principles, or three techniques, or one definition with an example)
Checked against	Cochrane review of physical rehabilitation approaches after stroke (Pollock et al., 2014); Cochrane review of mental practice for the upper limb after stroke (2020); state of the evidence for cerebral palsy (Novak et al., 2020). Full list under Sources (p. 215). Content checked 20 October 2026.

1 What the approaches are, and what the evidence says

- The **neurophysiological approaches** (Rood, Bobath, Brunnstrom, PNF) were developed from the 1940s to the 1970s, using reflexes, sensory stimulation and handling to change tone and bring out movement.
- The **motor learning approaches** (Motor Relearning Programme, task-oriented training, CIMT) treat the patient as an active learner who practises real tasks.
- **Evidence:** physical rehabilitation that combines components from different approaches is effective after stroke, but no one named approach is more or less effective than another, so treatment should not be limited to compartmentalised approaches (Pollock et al., Cochrane, 2014).

Classic answer vs current evidence: choosing an approach

Classic	Current	Write in the exam
Choose one approach (for example Bobath) and follow its techniques for the whole programme.	Use evidence-based components from any approach, centred on active, repetitive, task-specific practice with enough dose; passive handling alone does not improve function.	Give the approach's principles and techniques as asked, then add: "Used as part of task-oriented practice; no single approach is superior (Cochrane, 2014)."

2 Rood's approach

Asked for 3 marks in S16 (three Rood techniques for spasticity) and S22 (three principles), and for 7 marks in S17 (principles and techniques).

Principles

1. **Normalise tone and evoke the wanted response with the right sensory stimulus:** some stimuli facilitate, others inhibit.
2. **Sensorimotor control develops in sequence** (the ontogenetic sequence), so treatment follows the developmental patterns.
3. **Movement is purposeful:** use activities that call out the response automatically.
4. **Repetition** of sensorimotor responses is needed for learning.

Ontogenetic motor patterns: supine withdrawal, roll over, pivot prone, neck co-contraction, prone on elbows, all fours, standing, walking. **Levels of control:** mobility, stability, controlled mobility (mobility on stability), skill.

Techniques

Facilitatory (to raise tone or start movement, as in hypotonia):

- Fast brushing over the skin of the muscle
- Quick icing (brief strokes of ice)
- Quick stretch; tapping over the muscle belly or tendon
- High-frequency vibration over the muscle
- Heavy joint compression (approximation) for co-contraction; resistance
- Fast vestibular stimulation (fast rocking, spinning)

Inhibitory (to lower tone, as in spasticity):

- Slow stroking down the back over the spinal muscles for about 3–5 minutes
- Neutral warmth: wrap the part or body in a blanket for about 10–20 minutes
- Prolonged (maintained) stretch
- Maintained pressure over the tendon insertion
- Light joint compression; slow rhythmic rocking or rolling
- Slow vestibular stimulation; prolonged cold

Three techniques for spasticity (S16): slow stroking, neutral warmth, prolonged stretch (or maintained tendon pressure, slow rocking, prolonged cold).

> Rood's techniques are classic exam content; their effects are short-lived and research support is limited. Use them to prepare for active practice, not instead of it.

3 Bobath concept (neurodevelopmental treatment, NDT)

Asked for 3 marks in W16 (three principles) and W19 (reflex-inhibiting patterns with an example).

Classic principles

- Abnormal tone and abnormal postural reflexes block normal movement, so first **inhibit** them.
- **Facilitate** normal automatic postural reactions (righting, equilibrium, protective reactions) and normal movement patterns.
- Use **handling** through **key points of control** (head, shoulder girdle, pelvis proximally; hands and feet distally).
- Treat the person as a whole, and include the affected side in all activity (bilateral, symmetrical).

Reflex-inhibiting patterns (RIPs)

Positions or movements opposite to the abnormal spastic pattern, used to reduce tone before facilitating movement.

- **Example, hemiplegic arm (flexor pattern):** shoulder forward and externally rotated, elbow extended, forearm supinated, wrist and fingers extended, thumb abducted; for example, sitting with weight borne on the extended arm placed beside or behind the body.
- **Example, hemiplegic leg (extensor pattern):** hip and knee flexed, hip abducted and externally rotated, ankle dorsiflexed.
- **Example, child with extensor thrust:** a flexed position (head, trunk, hips and knees bent) in side lying or carried in flexion.

Contemporary Bobath and evidence

- The modern Bobath concept describes itself as a problem-solving approach focused on postural control, selective movement and function, rather than inhibition followed by facilitation.
- After stroke, it is not superior to other approaches (Pollock, 2014). In cerebral palsy, NDT in its original passive form is rated ineffective for motor goals (Novak et al., 2020).

4 Brunnstrom's movement therapy

- Built on the stages of motor recovery after stroke (flaccidity, synergies with spasticity, voluntary synergies, movements out of synergy, isolated movement); see the stages and synergy patterns on the stroke page (p. 50).
- **Principle:** in early stages, use associated reactions, tonic reflexes and synergies to bring out movement; then train the person to break out of synergy into isolated movement.
- **Phenomena used:** associated reactions (effort in the strong limb raises tone in the weak one); **Souques' phenomenon** (lifting the arm above about 150° can release finger extension); **Raimiste's phenomenon** (resisted hip abduction or adduction of the strong leg causes the same movement in the weak leg); homolateral limb synkinesis.
- **Today:** the stages remain useful for assessment, but deliberately training synergies is not recommended; aim for selective, task-specific movement.

5 Motor Relearning Programme (Carr and Shepherd)

Asked for 3 marks in S18 (three principles) and 7 marks in W18 (MRP for the hemiplegic upper limb).

Principles

- Recovery is relearning of real-life tasks, so practise the **tasks themselves** (task-specific) in the right context.
- The patient is an **active learner**; the therapist teaches, using instructions, demonstration and feedback.
- **Eliminate unnecessary muscle activity** and compensation; practise the missing components.
- **Practice and carry-over:** many repetitions, with practice continued on the ward and at home.
- Based on the biomechanics of normal movement and motor learning science.

Four steps for each task

1. Analyse the task: compare with normal and find the missing components.
2. Practise the missing components.
3. Practise the whole task.
4. Transfer the training into daily life.

Seven sections: upper limb function; oro-facial function; sitting up over the side of the bed; balanced sitting; standing up and sitting down; balanced standing; walking.

MRP for the hemiplegic upper limb (W18)

- **Analyse:** reaching (shoulder flexion and abduction with the shoulder girdle moving correctly, elbow control), grasping and releasing (wrist extension with the fingers flexing, opposition), manipulating.
- **Missing components:** in lying, reach up to the ceiling and control the arm at different points of range; in sitting, reach forward and sideways to objects; wrist extension while grasping a cup; release; thumb opposition to each finger; turning the hand over.
- **Whole task:** drink from a cup, pick up and place objects, comb hair, write.
- **Avoid:** shoulder hitching, trunk leaning; protect the shoulder; stop soft-tissue shortening (arm positioned and stretched).
- **Transfer:** practice sheets for the ward and home, with family help.

6 Proprioceptive neuromuscular facilitation (PNF)

Asked for 7 marks in W25 (principles of PNF).

Philosophy

A positive approach that builds on what the person can do, works on function, uses the strong parts to help the weak, treats the whole person, and uses motor learning.

Basic principles (procedures)

1. **Resistance:** the right amount to facilitate contraction and motor learning.
2. **Irradiation and reinforcement:** strong muscles spread activity to weaker ones in the pattern.
3. **Manual contact:** a lumbrical grip over the muscles that should work guides direction.
4. **Body position and body mechanics** of the therapist in line with the movement.
5. **Verbal commands:** short, clear, well-timed.
6. **Vision:** the patient watches the movement.
7. **Traction and approximation:** traction facilitates movement; approximation facilitates stability and weight-bearing.
8. **Stretch:** the stretch stimulus and the stretch reflex facilitate contraction.
9. **Timing:** normal timing from distal to proximal; timing for emphasis to strengthen a weak part.
10. **Patterns:** diagonal, spiral mass movements.

Neurophysiological principles behind them (Sherrington): after-discharge, temporal and spatial summation, irradiation, successive induction, reciprocal innervation.

Patterns of the limbs

Pattern	Upper limb (shoulder)	Lower limb (hip)
D1 flexion	Flexion, adduction, external rotation; wrist and fingers flex (hand to opposite ear)	Flexion, adduction, external rotation; ankle dorsiflexion with inversion
D1 extension	Extension, abduction, internal rotation; wrist and fingers extend	Extension, abduction, internal rotation; plantarflexion with eversion
D2 flexion	Flexion, abduction, external rotation; wrist and fingers extend (drawing a sword upward)	Flexion, abduction, internal rotation; dorsiflexion with eversion
D2 extension	Extension, adduction, internal rotation; wrist and fingers flex (sheathing the sword)	Extension, adduction, external rotation; plantarflexion with inversion

Techniques

Group	Technique and use
Agonist techniques	**Rhythmic initiation** (passive, then assisted, then active and resisted: for difficulty starting, rigidity); **combination of isotonics** (concentric, eccentric and holding of one group: control); **repeated stretch** (to start or strengthen movement)
Reversal of antagonists	**Dynamic reversals** (movement in one direction then the other: coordination, endurance); **stabilising reversals** and **rhythmic stabilisation** (holding against changing resistance: stability, balance)
Relaxation	**Contract-relax** and **hold-relax** (to gain range; hold-relax also for pain)

7 Mental imagery, CIMT and transfer of learning

- **Mental (motor) imagery:** rehearsing a movement in the mind without moving. Added to physical practice, it improves upper limb function after stroke (low-quality evidence, Cochrane 2020). Use short, guided sessions, from the person's own view, of tasks they are practising.

- **Constraint-induced movement therapy:** restraint of the strong hand plus intensive, shaped practice with the weak hand; for people with some active wrist and finger extension. Criteria and doses are on the stroke page (p. 50); in children it is rated effective (Novak et al., 2020).
- **Transfer of learning:** practise in the setting where the task will be used (context-specific); move from part practice to whole-task practice; bilateral transfer (practice with one limb improves the other); vary practice so skills generalise.

8 Exam corner

Past questions (MUHS, 2016–2026)

W25 Q2(c)
Explain the principles of proprioceptive neuromuscular facilitation.

W18 Q2(c)
Motor Relearning Programme for the hemiplegic upper limb.

S17 Q2(d)
Principles and techniques of Rood's approach.

3-mark questions
Three Rood techniques to manage spasticity (S16); three principles of Bobath NDT (W16); three principles of the Motor Relearning Programme (S18); reflex-inhibiting patterns with an example (W19); three principles of Rood's approach (S22).

Pages for this topic: tone management (p. 216), motor control (p. 11), plasticity and motor learning (p. 15), stroke (p. 50).

Keywords to underline

sensory stimulation ontogenetic sequence key points of control reflex-inhibiting pattern synergy task-specific practice missing components irradiation diagonal patterns rhythmic initiation no approach superior

Answer skeleton: principles and techniques of an approach (7 marks)

1. **[Background, 1]** Who developed it and the idea behind it.
2. **[Principles, 3]** Four to six principles, each one line.
3. **[Techniques, 2–3]** Three or four techniques with their use, and one example on a patient.
4. **[Evidence, 0–1]** One line: used within task-oriented practice; no approach superior.

Draw these

- The D1 and D2 diagonals of the arm as arrows across the body
- A reflex-inhibiting pattern for the hemiplegic arm

Viva questions

1. Name three facilitatory and three inhibitory Rood techniques.
2. What are key points of control?
3. What are the four steps of the Motor Relearning Programme?
4. What is irradiation in PNF?
5. Describe the D2 flexion pattern of the upper limb.

6. What is Souques' phenomenon?

Linked practical spots

- Demonstrate a PNF pattern with rhythmic initiation; quick icing and brushing; a reflex-inhibiting position.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write three principles of the Motor Relearning Programme.**

Answer
Task-specific practice of real-life tasks in context; the patient as an active learner with instruction and feedback; eliminate unnecessary muscle activity and practise missing components; repetition with carry-over into daily life.

3 m **What is a reflex-inhibiting pattern? Give an example.**

Answer
A position or movement opposite to the abnormal spastic pattern, used in the Bobath approach to reduce tone. Example: for a flexed hemiplegic arm, sitting with weight borne on the arm with the shoulder externally rotated, elbow straight, forearm supinated and wrist and fingers extended.

7 m **Explain the principles of PNF.**

Answer
Philosophy: positive, functional, uses strong parts, whole person, motor learning. Basic principles: optimal resistance; irradiation and reinforcement; manual contact (lumbrical grip); body position and mechanics; clear verbal commands; vision; traction and approximation; stretch stimulus and reflex; normal timing and timing for emphasis; diagonal spiral patterns (D1, D2). Neurophysiological basis: after-discharge, summation, irradiation, successive induction, reciprocal innervation. Name a few techniques: rhythmic initiation, combination of isotonics, rhythmic stabilisation, contract-relax.

LAQ **Plan upper limb retraining for a man 3 months after stroke with some shoulder and elbow movement but poor hand control, using components from different approaches. (Practice case)**

Answer
Assessment of components (reach, grasp, release, manipulation); MRP: analyse, practise missing components (reaching to targets, wrist extension with grasp, release, opposition), then whole tasks (cup, comb, writing) with feedback and many repetitions; PNF patterns with rhythmic initiation for starting movement and combination of isotonics for control; mental imagery of the same tasks; modified CIMT if he has enough wrist and finger extension; shoulder protection and positioning; home practice programme.

MCQ **Which is an inhibitory Rood technique?**

A. Fast brushing

B. Quick icing

C. Slow stroking down the back

D. Tapping

Answer
C. The others facilitate.

MCQ **In the D1 flexion pattern of the upper limb, the shoulder moves into:**

A. Flexion, abduction, internal rotation

B. Flexion, adduction, external rotation

C. Extension, abduction, internal rotation

D. Extension, adduction, external rotation

Answer

B. The hand moves towards the opposite ear.

MCQ **The 2014 Cochrane review of stroke rehabilitation approaches concluded that:**

A. Bobath is superior

B. Brunnstrom is superior

C. No single approach is more effective; combining components works

D. Rehabilitation is not effective

Answer

C. Use evidence-based components, not one named approach.

Last recall check. Write the principles of Rood, Bobath, MRP and PNF in one line each, and two techniques of each.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Pollock A, et al. Physical rehabilitation approaches for the recovery of function and mobility following stroke. *Cochrane Database Syst Rev* 2014;(4):CD001920.
3. Barclay RE, et al. Mental practice for treating upper extremity deficits in individuals with hemiparesis after stroke. *Cochrane Database Syst Rev* 2020;(5):CD005950.
4. Novak I, et al. State of the evidence traffic lights 2019. *Curr Neurol Neurosci Rep* 2020;20:3.
5. Structure and classical content: Adler, Beckers and Buck, *PNF in Practice*; Carr and Shepherd, *A Motor Relearning Programme for Stroke*; Bobath, *Adult Hemiplegia: Evaluation and Treatment*; Umphred, *Neurological Rehabilitation* (Rood and Brunnstrom); O'Sullivan and Schmitz, *Physical Rehabilitation*.

TP2 Managing tone: spasticity, rigidity and hypotonia

> Tone came up in 9 papers: tone-reducing and tone-facilitating techniques for 7 marks, and the Modified Ashworth Scale, slow icing, methods to reduce tone and the classification of tone for 3 marks. Answer in two steps: first measure and explain the abnormal tone, then choose techniques that change it for long enough to practise a real task, because tone techniques alone do not build function.

Syllabus	Treatment programme 2: techniques based on neurophysiological principles (tone); rigidity is also on the Parkinson's page
Priority	**High yield.** In 9 of 20 papers (2016–2026): 4 seven-mark and 6 three-mark questions.
Asked as	7m W17 · S19 · W20 · W24 · 3m W16 · W18 · S19 · S20 · W21 · S23
Usual split	7 (define tone 3 and physiotherapy for hypotonia 4; or a list of techniques with how and why); 3 (a scale, a technique, or three methods)
Checked against	Modified Ashworth Scale (Bohannon and Smith, 1987); Lance's definition of spasticity (1980); Cochrane review of stretch for contractures (Harvey et al., 2017); AHA/ASA stroke rehabilitation guideline (2026) on botulinum toxin. Full list under Sources (p. 221). Content checked 20 October 2026.

1 Muscle tone: definition and classification

Asked for 3 marks in S20 (define and classify) and with hypotonia for 7 marks in W17.

- **Muscle tone** is the resistance of a relaxed muscle to passive stretch. It has a **neural** part (the stretch reflex and descending control) and a **non-neural** part (the stiffness of muscle and connective tissue).
- Normal tone is high enough to hold posture against gravity and low enough to allow easy, selective movement.

Type	Features	Typical lesion
Spasticity	Velocity-dependent: more resistance with faster stretch; clasp-knife; brisk reflexes, clonus	Upper motor neuron: stroke, SCI, MS, CP
Rigidity	Same resistance at any speed, in flexors and extensors (lead-pipe; cogwheel with tremor)	Basal ganglia: Parkinson's disease
Dystonia	Sustained or intermittent contractions causing twisting postures	Basal ganglia: dyskinetic CP, focal dystonias
Hypotonia (flaccidity)	Low resistance, floppy, hypermobile joints, reduced reflexes	LMN, cerebellum, early stroke or spinal shock, Down's syndrome
Fluctuating tone	Tone swings between low and high	Dyskinetic (athetoid) CP

Spasticity (Lance, 1980): a motor disorder with a velocity-dependent increase in tonic stretch reflexes and exaggerated tendon jerks, from hyperexcitability of the stretch reflex, as one part of the upper motor neuron syndrome. The syndrome also has **negative** features (weakness, loss of dexterity, fatigue) that limit function more than the spasticity itself.

2 Measuring tone

Modified Ashworth Scale (3 marks in S19 and W21)

Grade	Meaning
0	No increase in tone
1	Slight increase: a catch and release, or minimal resistance at the end of the range
1+	Slight increase: a catch followed by minimal resistance through less than half of the remaining range

Grade	Meaning
2	More marked increase through most of the range, but the part moves easily
3	Considerable increase; passive movement is difficult
4	The part is rigid in flexion or extension

How to test: the patient lies relaxed; move the part through its full range in about one second, at most three times; grade the first movement.

- **Modified Tardieu Scale:** move the joint slowly (R2, full passive range) and as fast as possible (R1, the angle of the "catch"). A large gap between R2 and R1 means a strong dynamic (reflex) component that may respond to stretch, casting or botulinum toxin; a small gap means fixed contracture.
- Also: spasm frequency scores, the pendulum test, clonus count; and always the effect of tone on function.

3 Spasticity: when and why to treat

- **Treat when it causes a problem:** pain, spasms, poor hygiene or dressing, contracture, poor seating, or interference with movement.
- **Sometimes it helps:** leg extensor tone can help standing and transfers; do not remove it without checking function.
- **Look for triggers first:** pain, full bladder or urine infection, constipation, pressure sores, ingrown toenails, tight clothing, anxiety, fatigue.
- **Team options:** oral drugs (baclofen, tizanidine, dantrolene); **botulinum toxin** for focal spasticity followed by stretching, splinting or casting and task practice (a strong recommendation after stroke, AHA/ASA 2026); intrathecal baclofen for severe generalised spasticity; nerve blocks; surgery.

4 Techniques to reduce tone

Asked for 7 marks in S19 (tone-inhibiting techniques) and W20 (physiotherapy of spasticity), and for 3 marks in S23 (three methods to reduce tone) and W18 (slow icing).

Technique	How	Why it works
Positioning	Positions opposite to the spastic pattern in lying, sitting and standing; avoid long periods in the spastic pattern	Keeps muscles at length and reduces reflex activity
Prolonged stretch	Slow, sustained stretch held for minutes, by positioning, a splint, a standing frame or a cast	Reduces stretch reflex excitability for a short time and keeps length
Weight-bearing	Through the extended arm in sitting; standing on a plantigrade foot	Prolonged stretch plus joint compression
Slow icing (prolonged cold)	An ice pack wrapped in a damp towel, or cold immersion, over the spastic muscle for about 15–20 minutes	Cooling slows nerve conduction and calms the muscle spindle, so tone drops for a short time: use the window for stretching and active practice
Neutral warmth and slow stroking	Wrap the limb in a blanket; slow strokes down the back	Calming sensory input (Rood)
Rhythmic rotation	Slow, rhythmic rotation of the trunk or limb through a gradually larger range	Relaxes the muscle before movement
Splints and serial casting	Resting hand splint; AFO; casts changed every few days to gain range	Low-load, long-duration stretch
Electrical stimulation	Stimulate the antagonist (for example wrist extensors in a flexed hand) during task practice	Reciprocal inhibition and active movement

Technique	How	Why it works
Active task practice	Strengthen the weak antagonists; practise reaching and walking	Function improves when weakness and control are treated, not tone alone

Three methods to reduce tone (S23): prolonged stretch with positioning or a splint; prolonged cold (slow icing); weight-bearing. Others: neutral warmth, rhythmic rotation, reflex-inhibiting patterns, electrical stimulation of antagonists.

Classic answer vs current evidence: stretching for spasticity

Classic	Current	Write in the exam
Daily passive stretching reduces spasticity and prevents contractures.	Regular stretch for under 7 months has no clinically important effect on joint mobility, and little or no effect on spasticity, pain or activity (Cochrane). Combine positioning, splints or casts with active task practice, and use botulinum toxin with therapy for problem focal spasticity.	"Treat triggers; positioning, prolonged stretch with splints or casts, weight-bearing and cold to reduce tone for practice; active task training; botulinum toxin with therapy for focal spasticity."

Recall check. Name three ways to reduce tone and say why each works.

5 Hypotonia and facilitation techniques

Asked for 3 marks in W16 (three facilitatory techniques) and 7 marks in W17 (physiotherapy of hypotonia) and W24 (facilitatory techniques to improve tone).

Technique	How	Why it works
Quick stretch and tapping	A brisk stretch or tap over the muscle belly or tendon just before asking for movement	Excites the muscle spindle and stretch reflex
Approximation (joint compression)	Press through the long axis of the limb or trunk in weight-bearing positions	Joint receptors facilitate co-contraction around the joint
Resistance	Light, graded resistance to movement or holding	Recruits more motor units; irradiation (PNF)
Fast brushing and quick icing	Brief stimulation over the skin of the target muscle	Facilitatory sensory input (Rood)
Vibration	High-frequency vibration over the muscle belly	Tonic vibration reflex
Weight-bearing and bouncing	Prone on elbows, four-point kneeling, sitting on a ball with bouncing, standing	Compression and quick changes in load activate postural muscles
Fast vestibular input	Brisk rocking or swinging, used carefully	Increases postural tone
PNF techniques	Repeated stretch, rhythmic stabilisation, stabilising reversals	Build strength and stability
Electrical stimulation	Stimulate weak muscles (for example shoulder muscles to prevent subluxation) during functional activity	Produces contraction while voluntary control returns

Physiotherapy of hypotonia (W17)

- Support and protect hypermobile joints (shoulder support in early stroke; avoid overstretching).
- Positioning against gravity with support: prone over a wedge, supported sitting, standing frame.
- Facilitation (table above) just before asking for active movement.
- Co-contraction and holding: weight-bearing in kneeling and standing, rhythmic stabilisation.

- Strengthening and functional training: head and trunk control, sit-to-stand, walking.
- Teach carers handling and safe positions.

6 Precautions

Take care

- Cold: check skin sensation and circulation; avoid in cold sensitivity, Raynaud's phenomenon and over numb skin without close checks.
- Never force a spastic joint; it causes pain and more spasticity.
- Facilitation: quick, intense stimuli can increase spasticity in another part of the body; stop if tone rises unwanted.
- Fast vestibular input: avoid in seizures, raised intracranial pressure and with nausea.
- Splints and casts: check the skin daily for red marks.

7 Exam corner

Past questions (MUHS, 2016–2026)

W24 Q2(d)
Describe the facilitatory techniques to improve tone.

W20 Q2(a)
Describe the physiotherapy management of spasticity.

S19 Q2(b)
Explain tone-inhibiting techniques.

W17 Q2(d), 3 + 4 marks
What is muscle tone? Describe the physiotherapy management of hypotonia.

3-mark questions
Three facilitatory techniques for hypotonia (W16); slow icing (W18); Modified Ashworth Scale (S19, W21); define and classify muscle tone (S20); three methods to reduce tone (S23).

Pages for this topic: neurotherapeutic approaches (p. 209), Parkinson's disease (rigidity) (p. 164), stroke (spastic hand and shoulder) (p. 50), orthoses and splints (p. 249).

Keywords to underline

resistance to passive stretch velocity-dependent upper motor neuron syndrome Modified Ashworth Tardieu triggers prolonged stretch slow icing weight-bearing quick stretch approximation task practice

Answer skeleton: physiotherapy of spasticity (7 marks)

1. **[Define and assess, 1–2]** Lance's definition; Modified Ashworth and Tardieu; effect on function; triggers.
2. **[Techniques, 4]** Positioning; prolonged stretch with splints or casts; weight-bearing; slow icing; neutral warmth and rhythmic rotation; electrical stimulation of antagonists; active task practice and strengthening.
3. **[Team, 1]** Botulinum toxin with therapy for focal spasticity; oral drugs; intrathecal baclofen.

Draw these

- The stretch reflex arc (spindle, Ia afferent, alpha motor neuron) with the descending inhibition lost in UMN lesions

Viva questions

1. Define spasticity.
2. How do you grade the Modified Ashworth Scale?
3. What does the gap between R1 and R2 in the Tardieu scale tell you?
4. What is slow icing, and how does it reduce tone?
5. Name three triggers that worsen spasticity.
6. Name three facilitatory techniques.

Linked practical spots

- Grade tone with the Modified Ashworth Scale; resting hand splint; ice pack application; demonstrate approximation.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Describe the Modified Ashworth Scale.**

Answer

0 no increase; 1 slight increase with a catch and release or minimal resistance at the end of range; 1+ catch then minimal resistance through less than half the range; 2 marked increase through most of the range but easily moved; 3 considerable increase, passive movement difficult; 4 rigid in flexion or extension.

3 m **What is slow icing?**

Answer

Prolonged cold over a spastic muscle (an ice pack or cold immersion for about 15–20 minutes). It slows nerve conduction and calms the muscle spindle, so tone falls for a short time; the window is used for stretching and active practice. Quick icing, by contrast, facilitates.

7 m **What is muscle tone? Describe physiotherapy for hypotonia. (3 + 4)**

Answer

Tone (3): resistance of a relaxed muscle to passive stretch, with neural and non-neural parts; types: normal, hypertonia (spasticity, rigidity, dystonia), hypotonia, fluctuating.

Hypotonia (4): protect joints; positioning against gravity with support; facilitation (quick stretch, tapping, approximation, resistance, fast brushing, quick icing, vibration, bouncing); co-contraction in weight-bearing; rhythmic stabilisation; electrical stimulation; strengthening and functional training; carer education.

LAQ **A man 4 months after stroke has Modified Ashworth grade 2 elbow and wrist flexors that make dressing difficult. Plan the management of his spasticity. (Practice case)**

Answer

Assess: MAS, Tardieu (R1 and R2), range, pain, triggers, function (dressing). Physiotherapy: treat triggers; positioning of the arm out of the flexor pattern; weight-bearing through the extended arm; prolonged stretch with a resting hand splint at night; slow icing or rhythmic rotation before practice; electrical stimulation of wrist and finger extensors during reaching and grasping; task practice for dressing; strengthening of extensors. Refer for botulinum toxin to the elbow and wrist flexors, followed by stretching, splinting and task practice; review range and function.

MCQ **Resistance that increases with the speed of passive movement is typical of:**

A. Rigidity

B. Spasticity

C. Hypotonia

D. Dystonia

Answer

B. Rigidity is the same at any speed.

MCQ **Which technique facilitates tone?**

A. Prolonged stretch

B. Neutral warmth

C. Approximation in weight-bearing

D. Slow stroking

Answer

C. Joint compression facilitates co-contraction; the others inhibit.

MCQ **A Modified Ashworth grade of 1+ means:**

A. No increase in tone

B. A catch followed by minimal resistance through less than half the range

C. Passive movement is difficult

D. The part is rigid

Answer

B.

Last recall check. Write the Modified Ashworth grades, three tone-reducing and three facilitating techniques with reasons.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Bohannon RW, Smith MB. Interrater reliability of a modified Ashworth scale of muscle spasticity. *Phys Ther* 1987;67:206–7.
3. Lance JW. Symposium synopsis. In: Feldman RG, Young RR, Koella WP, eds. *Spasticity: Disordered Motor Control*. Year Book Medical; 1980.

4. Harvey LA, et al. Stretch for the treatment and prevention of contractures. *Cochrane Database Syst Rev* 2017;(1):CD007455.
5. American Heart Association/American Stroke Association. Guideline for adult stroke rehabilitation and recovery (2026): botulinum toxin for focal spasticity.
6. Structure and classical content: Umphred, *Neurological Rehabilitation*; O'Sullivan and Schmitz, *Physical Rehabilitation*; Stokes, *Physical Management for Neurological Conditions*.

TP2 Balance and coordination: systems, strategies, assessment and training

> Balance came up in 4 papers: training static and dynamic standing balance and evaluating sensory and motor strategies for 7 marks, and the motor strategies and ways to improve dynamic balance for 3 marks. Balance training is also part of almost every long case, so learn one progression you can adapt to any patient.

Syllabus	Treatment programme 2: coordination and balancing exercises using techniques based on neurophysiological principles
Priority	**Medium yield.** In 4 of 20 papers (2016–2026): 2 seven-mark and 2 three-mark questions, and part of most long cases.
Asked as	7m S16 · W18 · 3m S17 · S18
Usual split	7 (systems or strategies 2, assessment or progression 4–5); 3 (three strategies with one line each)
Checked against	Horak and Nashner on postural strategies; BESTest (Horak et al., 2009); Sherrington et al., exercise to prevent falls (2017). Full list under Sources (p. 227). Content checked 20 October 2026.

1 What balance needs

- **Balance** (postural control) is keeping the centre of mass over the base of support, in stillness (static), during movement (dynamic) and when disturbed (reactive).
- **Sensory systems:** vision, the vestibular system and somatosensation (pressure under the feet and joint position). The brain reweights them by situation: for example, more vestibular and vision on foam.
- **Central processing:** cerebellum, basal ganglia, brainstem and cortex integrate the information and choose a response.
- **Motor systems:** strength, range of motion (especially ankle), and coordinated postural strategies; **anticipatory** adjustments before a voluntary movement and **reactive** ones after a disturbance.
- **Cognition** and attention: balance worsens when attention is divided (dual tasks).

2 Motor strategies of balance

Asked for 3 marks in S18 (three motor strategies).

Strategy	When it is used	How it works
Ankle strategy	Small, slow disturbances on a firm, wide surface	The body sways as one piece about the ankles; muscles fire from distal to proximal
Hip strategy	Larger, faster disturbances, or a narrow or soft surface	Quick flexion or extension at the hips moves the centre of mass; muscles fire from proximal to distal
Stepping (or reaching) strategy	The centre of mass moves outside the base of support	A rapid step or a reach to grab support creates a new base

A **suspensory strategy** (lowering the centre of mass by bending the knees) is sometimes added.

3 Evaluating sensory and motor strategies

Asked for 7 marks in W18.

Sensory strategies

- **Modified Clinical Test of Sensory Interaction in Balance (mCTSIB):** stand 30 seconds in four conditions: firm surface eyes open, firm eyes closed, foam eyes open, foam eyes closed. Falls with eyes closed on a firm surface

suggest somatosensory loss; falls on foam with eyes closed suggest vestibular loss or poor sensory reweighting.

- **Romberg and sharpened Romberg:** loss of balance when the eyes close points to sensory (proprioceptive) ataxia.
- **Sensory Organisation Test** on computerised posturography (six conditions) where available.

Motor strategies

- **Reactive responses:** push and release test or a nudge in each direction; observe the ankle, hip or stepping response and whether it is quick and effective. The pull (retropulsion) test in Parkinson's disease.
- **Limits of stability and anticipatory control:** functional reach test (reach forward along a ruler at shoulder height; less than about 15 cm suggests a high fall risk); multidirectional reach; rising on toes.
- **Systems tools:** BESTest or Mini-BESTest (biomechanical constraints, stability limits, anticipatory adjustments, reactive responses, sensory orientation, dynamic gait).
- **Function and fall risk:** Berg Balance Scale (0–56), Timed Up and Go, Dynamic Gait Index or Functional Gait Assessment, dual-task walking.

4 Training static and dynamic standing balance

Asked for 7 marks in S16 and for 3 marks in S17 (three strategies to improve dynamic balance).

Ways to make standing balance harder (progression)

Change	Easier	Harder
Base of support	Feet apart	Feet together, semi-tandem, tandem, single leg
Surface	Firm floor	Foam, wobble board, uneven ground
Vision	Eyes open	Eyes closed, moving visual scenes
Upper limb support	Hands on parallel bars	No hands, then holding an object
Head	Still	Turning and nodding
Attention	Single task	Talking, counting backwards, carrying a tray

Static balance

- Hold positions in the progression above, increasing the time (for example 30 seconds) before making the task harder.
- Weight evenly on both legs with a mirror or scale feedback (after stroke).

Dynamic balance (S17: three strategies)

- **Reaching** beyond arm's length in all directions, to the floor and overhead.
- **Weight shifts and stepping:** forward, backward, sideways, onto a step, over obstacles; turning.
- **Reactive (perturbation) training:** expected, then unexpected pushes or pulls to train ankle, hip and stepping responses.
- **Walking variations:** changing speed and direction, narrow paths, head turns, carrying objects, dual tasks.
- **Functional and group activities:** tai chi, dance, ball games (catching and throwing).

Dose

For preventing falls in older people, exercise programmes that challenge balance and total more than **3 hours a week** have the largest effect (a 39% fall reduction; Sherrington et al., 2017). Balance must be challenged near the limit of ability, safely.

Recall check. Make a five-step progression of standing balance for a patient who can stand only with feet apart and eyes open.

5 Coordination

- **Tests:** finger-nose, finger-finger, heel-shin, rapid alternating movements (dysdiadochokinesia), rebound test, drawing a circle with the foot, tapping; each with eyes open and closed to separate cerebellar from sensory problems.
- **Frenkel's exercises:** slow, precise, repeated movements of increasing difficulty, first in lying, then sitting, standing and walking, watching the limb and moving to counts or floor markings. Vision replaces the lost position sense, so they suit **sensory ataxia** best.
- **Other methods:** PNF rhythmic stabilisation and stabilising reversals for proximal stability; slow reversals for smooth control; approximation; light weights on the wrist or ankle for some patients; task practice of real activities.

6 Precautions

> **Take care**
>
> - Use a gait belt, parallel bars or a harness; stand on the weaker side; clear the area of obstacles.
> - Check blood pressure in people with orthostatic hypotension before standing work.
> - Progress one variable at a time; the task should be challenging but safe.

7 Exam corner

Past questions (MUHS, 2016–2026)

S16 Q2(b)
Describe techniques to train static and dynamic standing balance.

W18 Q2(b)
Evaluating the sensory and motor strategies of balance.

3-mark questions
Three strategies to improve dynamic balance (S17); three motor strategies of balance (S18).

Pages for this topic: cerebellar disease and ataxia (p. 155), vestibular disorders (p. 116), Parkinson's disease (p. 164), scales (p. 21).

Keywords to underline

centre of mass, base of support, sensory reweighting, ankle, hip, stepping strategies, anticipatory and reactive, mCTSIB, functional reach, perturbation training, dual task, more than 3 hours a week

Answer skeleton: training static and dynamic standing balance (7 marks)

1. **[Assessment, 1]** Systems and strategies; a scale (Berg, mCTSIB).
2. **[Static, 2]** Progression of base, surface, vision, support, head and attention.
3. **[Dynamic, 3]** Reaching, weight shifts and stepping, perturbations, walking variations, functional activities.
4. **[Dose and safety, 1]** Challenging, more than 3 hours a week for fall prevention; guarding.

Draw these

- Stick figures showing the ankle, hip and stepping strategies

Viva questions

1. Name the three motor strategies of balance.
2. What does falling on foam with the eyes closed suggest?
3. What is the functional reach test?
4. How do you progress static balance?
5. Why do Frenkel's exercises suit sensory ataxia?

Linked practical spots

- Foam pad, wobble board, ruler for functional reach; demonstrate the push and release test.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write three motor strategies of balance.**

Answer
Ankle strategy (small, slow sway about the ankles on a firm surface); hip strategy (larger, faster disturbances or narrow surface, movement at the hips); stepping strategy (a step or reach when the centre of mass leaves the base).

3 m **Give three strategies to improve dynamic balance.**

Answer
Reaching beyond the base in all directions; stepping and walking with changes of speed, direction and surface; perturbation (reactive) training; also dual-task walking and tai chi.

7 m **How do you evaluate the sensory and motor strategies of balance?**

Answer
Sensory: mCTSIB in four conditions (firm and foam, eyes open and closed); Romberg and sharpened Romberg; posturography if available. Motor: push and release or nudge tests in each direction (observe ankle, hip, stepping); functional and multidirectional reach for limits of stability; anticipatory control (rising on toes, reaching); BESTest or Mini-BESTest; function and fall risk (Berg, TUG, DGI).

LAQ **A 70-year-old man with peripheral neuropathy has fallen twice in a month, mostly at night. Plan the assessment and balance training. (Practice case)**

Answer
Assessment: history of falls; sensation; mCTSIB (expect falls with eyes closed); Romberg; Berg, TUG, functional reach; strength and ankle range; vision; home hazards and lighting. Training: static progression with vision first, then reduced vision; stepping and reactive training; walking on varied surfaces; ankle strength; more than 3 hours a week of challenging balance exercise; night lights, a stick, footwear; foot care.

MCQ **On a firm surface, a small slow push backwards is usually met with:**

A. Ankle strategy

B. Hip strategy

C. Stepping strategy

D. Reaching strategy

Answer
A.

MCQ **A patient stands well on foam with eyes open but falls on foam with eyes closed. The likely problem is:**

A. Visual loss

B. Vestibular loss

C. Weak hip abductors

D. Anxiety only

Answer
B. On foam with eyes closed, only vestibular input is reliable.

MCQ **Frenkel's exercises are most useful for:**

A. Sensory ataxia

B. Spasticity

C. Rigidity

D. Flaccid paralysis

Answer
A. Vision substitutes for lost position sense.

Last recall check. Write the three balance systems, three strategies, the mCTSIB conditions and a balance progression.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Horak FB, Nashner LM. Central programming of postural movements: adaptation to altered support-surface configurations. *J Neurophysiol* 1986;55:1369–81.
3. Horak FB, Wrisley DM, Frank J. The Balance Evaluation Systems Test (BESTest) to differentiate balance deficits. *Phys Ther* 2009;89:484–98.
4. Sherrington C, et al. Exercise to prevent falls in older adults: an updated systematic review and meta-analysis. *Br J Sports Med* 2017;51:1750–8.
5. Duncan PW, et al. Functional reach: a new clinical measure of balance. *J Gerontol* 1990;45:M192–7.

TP3 Tools and adaptive equipment for neuro-rehabilitation: balls, tilt boards, bolsters, wedges, benches and mats

> The syllabus names the equipment of a neuro gym: vestibular balls, tilt boards, bolsters, wedges, graded benches and therapeutic mats. It was not asked in theory in 2016–2026, but you will use and name these in the practical; for each piece know what it trains, one or two activities, how to progress, and how to keep the patient safe.

Syllabus	Treatment programme 3: tools and adaptive equipment used for neuro-rehabilitation, such as vestibular balls, tilt boards, bolsters, wedges, graded benches and therapeutic mats
Priority	**Syllabus.** Not asked on its own in 2016–2026; a practical-exam topic.
Asked as	Not asked on its own in 2016–2026
Usual split	Likely 3 marks (uses of one piece of equipment) or a practical demonstration
Checked against	Standard neuro-rehabilitation practice; safety points consistent with the balance, cerebral palsy and spinal cord pages. Full list under Sources (p. 231). Content checked 20 October 2026.

1 The named equipment

Equipment	Trains	Example activities and progression
Therapy (vestibular, Swiss) ball	Postural and equilibrium reactions, protective reactions, trunk control, vestibular input; relaxation by slow rocking	Child prone over the ball for head and arm weight bearing; sitting on the ball with weight shifts; tipping slowly to elicit protective arm extension. Progress: larger movements, less support, faster tilts; a peanut-shaped ball for more stability
Tilt and wobble boards	Equilibrium (tilting) reactions, ankle and hip strategies, standing and sitting balance	Sit, kneel or stand on the board while it tilts in one direction (rocker) or all directions (wobble). Progress: from sitting to standing, eyes open to closed, holding on to free hands, adding a reaching or catching task
Bolsters and rolls	Weight bearing through arms, hip abduction (straddle sitting), positioning, trunk control	Prone over a roll for elbow and hand weight bearing; straddle sitting to reduce adductor tone and train sitting balance; side-to-side rolling for reactions
Wedges	Head and upper-trunk control in prone, positioning	Prone on a wedge with forearm support and toys in front (head lifting); supported half-lying for feeding or chest care. Progress: lower wedge, then flat prone
Graded benches and steps	Sit to stand, step-ups, stair training, seated balance at different heights	Sit to stand from a higher bench, lowered stepwise; step-ups on graded steps; benches as stable seats for reaching
Therapeutic mats	Bed and floor mobility, rolling, crawling, kneeling, getting up from the floor	Mat programme: rolling, prone on elbows, four-point kneeling, kneeling, half-kneeling, floor to stand; push-ups in long sitting for spinal cord injury

2 Other common equipment

- **Parallel bars:** early standing and walking with safe hand support; weight shifting; move to a walker or stick as soon as possible.
- **Tilt table:** gradual uprighting for orthostatic hypotension; early weight bearing; stretch of the calves.
- **Standing frame and prone stander:** supported standing for children and adults who cannot stand alone.
- **Treadmill** (with or without body-weight support): repetitive walking practice and endurance.
- **Mirror:** visual feedback for posture and symmetry; mirror therapy for the arm after stroke.

- **Foam pads, balance beams, cones and obstacles:** balance on unstable surfaces, narrow-base and obstacle walking.
- **Weights, resistance bands, pulleys and static cycles:** strengthening and fitness.
- **Hand-function boards, pegboards and putty:** grasp, release and dexterity.

3 Safety and good practice

Take care

- Guard the patient on unstable equipment; use a gait belt; clear the space; mats under balls and boards.
- Check the ball is properly inflated and the right size (hips and knees at about 90° when sitting).
- Avoid fast spinning or rocking in people prone to seizures or autonomic problems; watch for dizziness and blood pressure changes.
- Children with atlantoaxial instability (Down's syndrome): no forced neck flexion or trampoline-type bouncing.
- Clean equipment between patients; use latex-free balls and bands where latex allergy is possible (spina bifida).
- Equipment supports the task; the goal is always a real-life skill without it.

Recall check. Name two uses each for a therapy ball, a tilt board and a bolster, and one way to progress each.

4 Exam corner

Past questions (MUHS, 2016–2026)

Not asked on its own

Expect a practical demonstration or a short note on the uses of one piece of equipment.

Pages for this topic: balance (p. 223), cerebral palsy (p. 275), development (p. 267), spinal cord (p. 77).

Keywords to underline

equilibrium reactions protective reactions weight bearing graded progression mat programme guarding task transfer

Answer skeleton: uses of a piece of equipment (3 marks)

1. Description.
2. What it trains, with two activities and a progression.
3. One precaution.

Viva questions

1. How do you use a therapy ball to elicit protective reactions?
2. What is the difference between a rocker board and a wobble board?
3. Why is straddle sitting on a bolster useful in spastic diplegia?
4. How do you progress a mat programme in paraplegia?

Linked practical spots

- Identify and demonstrate: therapy ball, peanut ball, rocker and wobble boards, bolster, wedge, graded steps, mat, standing frame, tilt table.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Uses of a therapy (vestibular) ball in neuro-rehabilitation.**

Answer

Trains postural, equilibrium and protective reactions and trunk control, gives vestibular input, and can relax by slow rocking. Activities: prone over the ball for head and arm weight bearing; sitting with weight shifts; tipping to elicit protective extension. Progress by larger, faster movements and less support. Guard the patient and use a mat.

3 m **Uses of a wedge.**

Answer

Prone on a wedge with forearm support to train head and upper-trunk control (toys in front); positioning for comfort, feeding or chest care in half-lying. Progress to a lower wedge and then flat prone.

7 m **Describe the equipment used to train balance in neurological patients.**

Answer

Therapy and peanut balls (sitting reactions); rocker and wobble boards (standing and sitting tilting reactions); foam pads and beams (unstable and narrow surfaces); parallel bars (early standing with support); steps and benches (step-ups, stairs); cones and obstacles (dynamic balance); mirror (feedback). Give progressions (support, base, eyes, surface, speed, dual task) and safety (guarding, gait belt, mats).

LAQ **Practice: plan a one-hour gym session using equipment for a 4-year-old with spastic diplegia who sits alone but cannot stand without support. (15)**

Answer

Warm-up: prone over a roll with forearm weight bearing and reaching (5 min). Straddle sitting on a bolster with weight shifts and reaching across the midline (10 min). Ball: sitting with tilts for equilibrium and protective reactions (10 min). Kneeling and half-kneeling on the mat with a bench in front (10 min). Sit to stand from a graded bench, lowering the height (10 min). Standing in a standing frame with play at a table, then supported standing on a rocker board with hands held (10 min). Close with calf and hamstring stretches and parent teaching (5 min). Guard throughout; record repetitions and the bench height reached.

MCQ **A board that tilts in all directions is a:**

A. Rocker board

B. Wobble board

C. Wedge

D. Bolster

Answer

B.

MCQ **Straddle sitting on a bolster mainly helps a child with:**

A. Adductor spasticity

B. Hypermobile knees

C. Foot drop

D. Torticollis

Answer

A. It holds the hips abducted while training sitting balance.

MCQ **When sitting on a correctly sized therapy ball, the hips and knees are at about:**

A. 45°

B. 90°

C. 120°

D. Full extension

Answer

B.

Last recall check. List the six named pieces of equipment with what each trains and one safety point.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Standard neuro-rehabilitation texts on therapeutic gym equipment and paediatric handling.

TP4 Transfers, functional re-education, posture and gait training

> Transfers and gait training are part of nearly every long-case management answer, and named gaits (hemiplegic, parkinsonian, ataxic, high-stepping) appear as 7-mark questions on the condition pages. This page gives the shared method: break the task into parts, practise the whole task many times at a challenging level, and measure progress with walking speed and distance.

Syllabus	Treatment programme 4: application of transfer and functional re-education exercise, postural exercise and gait training
Priority	**Syllabus.** Not asked as its own question in 2016–2026; gait and transfer training sit inside most long-case answers, and named gaits are counted on the condition pages.
Asked as	Not asked on its own in 2016–2026; part of most long-case management answers
Usual split	Within long cases: transfers and gait training with progression and aids (4–7 of the management marks)
Checked against	Locomotor clinical practice guideline for chronic stroke, incomplete SCI and brain injury (Hornby et al., 2020); walking speed categories after stroke (Perry et al., 1995); Rancho Los Amigos gait phase terminology. Full list under Sources (p. 236). Content checked 20 October 2026.

1 Bed mobility and transfers

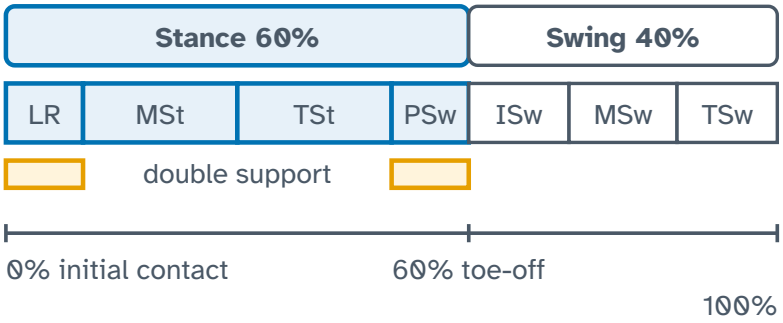
- **Rolling:** head and arm lead, knees bent; in hemiplegia practise rolling to both sides (towards the weak side is easier; towards the strong side trains the weak side).
- **Supine to sitting:** roll to side lying, legs over the edge, push up through the arm; reverse to lie down.
- **Sit to stand:** slide forward, feet back under the knees, lean the trunk forward ("nose over toes"), push up through both legs, stand tall; sit down by leaning forward and bending the hips and knees with control. Raise the seat height to make it easier, lower it to progress.
- **Transfers:** stand-pivot (in hemiplegia usually towards the stronger side, guarding the weak knee); sliding board or depression (lift) transfers in paraplegia; car, toilet and bath transfers; getting up from the floor after a fall.
- Teach families safe assistance and the equipment that replaces lifting (see lifting and wheelchairs (p. 257)).

2 Functional re-education and posture

- **Task practice:** sitting balance and reaching, standing up and sitting down, standing balance, walking, and reaching and manipulation, practised as whole tasks in real settings (see motor relearning (p. 209)).
- **Part practice** only for components the patient cannot yet do (for example weight shift onto the weak leg), then back to the whole task.
- **Repetition and intensity:** many repetitions, progressively harder (less support, faster, varied surfaces, dual tasks).
- **Posture:** sitting with the pelvis upright and weight even on both buttocks; standing with weight even on both feet; correct the habitual posture of the condition (flexed in Parkinson's disease, asymmetrical after stroke) with feedback (mirror, verbal cues) and strengthening of the extensors and trunk.

3 The normal gait cycle

Phases of the gait cycle (one limb)



LR loading response, MSt midstance, TSt terminal stance, PSw pre-swing, ISw initial swing, MSw mid-swing, TSw terminal swing (Rancho Los Amigos terms). Each double-support period is about 10% of the cycle; slower walking lengthens stance and double support.

Text version of the figure

- Stance (about 60%): loading response 0–10%, midstance 10–30%, terminal stance 30–50%, pre-swing 50–60%.
- Swing (about 40%): initial swing 60–73%, mid-swing 73–87%, terminal swing 87–100%.
- Double support at 0–10% and 50–60%.

- **Tasks of gait:** weight acceptance (initial contact, loading response), single-limb support (midstance, terminal stance), limb advancement (pre-swing and the swing phases).
- **Measures:** walking speed (10-metre walk test), endurance (6-minute walk test), step and stride length, cadence, symmetry.
- **Walking speed after stroke** predicts walking in the community: below about 0.4 m/s household walking, 0.4–0.8 m/s limited community walking, above about 0.8 m/s community walking (Perry et al., 1995).

4 Pathological neurological gaits

Gait	Features	Seen in
Hemiplegic	Slow, asymmetric; short stance on the weak side; stiff knee, foot drop, circumduction or hip hiking	Stroke (see stroke (p. 50))
Scissoring and crouch	Thighs cross (adductor spasticity); hips and knees bent, toe walking	Spastic diplegic CP
Festinating, shuffling	Flexed posture, short steps that speed up, little arm swing, freezing, turning en bloc	Parkinson's disease
Cerebellar ataxic	Wide-based, staggering, irregular steps; no better with eyes open	Cerebellar disease, MS
Sensory ataxic (stamping)	High steps, heels stamped down, eyes on the feet; worse in the dark; Romberg positive	Neuropathy, dorsal column disease
High-stepping (steppage)	Hip and knee lifted high to clear a dropped foot; foot slaps down	Peroneal nerve palsy, CMT, polio
Waddling	Trunk sways side to side; wide base; lordosis	Proximal muscle weakness: DMD, myopathies
Trendelenburg and hand-on-thigh	Pelvis drops on the swing side; hand pushes the thigh to keep the knee straight	Weak hip abductors; weak quadriceps (polio)

5 Principles of gait training

- **Walking practice improves walking:** in chronic stroke, incomplete spinal cord injury and brain injury, walking training at moderate to high intensity (or with added feedback such as virtual reality) improves walking speed and distance (Hornby et al., 2020).
- **Methods:** overground walking with variety (turns, slopes, obstacles, crowds), treadmill training with or without body-weight support, stepping practice, dual-task walking, cueing in Parkinson's disease.
- **Prerequisites to train alongside:** standing balance, weight shift onto each leg, strength (hip extensors, quadriceps, plantarflexors, dorsiflexors), range (hip extension, ankle dorsiflexion), endurance.
- **Orthoses and aids:** an AFO for foot drop (see orthoses (p. 249)); the least aid that keeps walking safe.
- **Progression:** parallel bars, then walker, then crutches or stick, then no aid; level floor, then slopes, stairs and outdoors; shorter to longer distances; supervised to independent.
- **Measure** walking speed and distance at the start and at reviews.

Walking aids and crutch gaits

- **Aid height:** handgrip at the wrist crease (or greater trochanter) with about 20–30° of elbow bend; stick in the hand opposite the weak leg.
- **Four-point:** one crutch, the opposite foot, the other crutch, the other foot; slow and most stable.
- **Two-point:** a crutch and the opposite foot together, then the other pair; faster, needs better balance.
- **Three-point:** both crutches with the affected leg, then the good leg (non- or partial weight bearing).
- **Swing-to and swing-through:** both crutches forward, then both legs swung up to or past them (paraplegia with KAFOs).
- **Stairs:** going up, the stronger leg leads; going down, the weaker leg and the stick or crutch lead ("up with the good, down with the bad").

Recall check. Describe the four-point and swing-through crutch gaits and say who uses each.

6 Exam corner

Where these skills are asked

Gait questions on condition pages

Hemiplegic gait (stroke), parkinsonian gait and gait training (Parkinson's disease), ambulation training after an L3 fracture (spinal cord), ataxic gait (ataxia), DMD gait deviations (DMD).

Long cases

"Short- and long-term management with emphasis on trunk control and ambulation", transfers and expected ambulation level.

Keywords to underline

task-specific practice, repetition and intensity, sit to stand, stand-pivot transfer, stance 60%, swing 40%, walking speed, treadmill training, progression of aids, crutch gaits, up with the good

Answer skeleton: gait training in a long case

1. Gait analysis: deviations in each phase and their causes.
2. Prerequisites: balance, weight shift, strength, range.
3. Walking practice with intensity and variety; treadmill if available; orthosis and aid.

4. Progression (aids, surfaces, distance, stairs) and outcome measures (speed, distance).

Draw these

- The gait cycle bar with its phases (section 3)

Viva questions

1. What percentage of the gait cycle is stance?
2. Name the phases of stance and swing.
3. What walking speed suggests a community walker after stroke?
4. How do you teach sit to stand?
5. Explain the three-point crutch gait.
6. Which leg leads going up stairs?

Linked practical spots

- Demonstrate sit to stand, a stand-pivot transfer, crutch and stick heights, crutch gaits and stair climbing; time a 10-metre walk.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the phases of the gait cycle.**

Answer
Stance (about 60%): initial contact and loading response, midstance, terminal stance, pre-swing. Swing (about 40%): initial swing, mid-swing, terminal swing. Two double-support periods of about 10% each.

3 m **Describe crutch gaits.**

Answer
Four-point (crutch, opposite foot, other crutch, other foot: most stable); two-point (crutch with opposite foot together: faster); three-point (both crutches with the affected leg, then the good leg: for non- or partial weight bearing); swing-to and swing-through (both crutches, then both legs: paraplegia with KAFOs).

7 m **Describe the principles of gait training in a neurological patient.**

Answer
Gait analysis to find the deviations and their causes; train prerequisites (balance, weight shift, strength, range); task-specific walking practice with many repetitions at moderate to high intensity, overground with variety and on a treadmill with or without body-weight support; feedback and cues; an orthosis for foot drop and the least aid needed; progression of aids, surfaces, distance, stairs and outdoor walking; measure walking speed and distance; home walking programme and falls safety.

LAQ **Practice: a 58-year-old woman 2 months after a right MCA stroke stands with support and walks 10 m in parallel bars with her left knee snapping back. Plan transfers and gait training. (5 + 10)**

Answer

Transfers (5): rolling both ways, supine to sit, sit to stand with feet back and trunk forward (raised seat, then lower), stand-pivot towards the right, toilet and car transfers; teach her daughter safe assistance. **Gait (10):** weight shift onto the left leg, step-ups and knee control in small ranges (0–15°), quadriceps and hip extensor strengthening, calf stretching; AFO if the ankle drops into plantarflexion and drives the knee back; walking practice in bars, then with a quad stick, increasing distance and speed; treadmill practice; turns and obstacles; stairs (up with the right, down with the left); measure 10-metre walk speed weekly.

MCQ **Stance phase occupies about:**

A. 40% of the gait cycle

B. 50%

C. 60%

D. 80%

Answer

C.

MCQ **A shuffling gait with steps that speed up suggests:**

A. Cerebellar disease

B. Parkinson's disease

C. DMD

D. Peroneal palsy

Answer

B.

MCQ **Going down stairs with a stick, which leads?**

A. The stronger leg

B. The weaker leg with the stick

C. Either

D. Both legs together always

Answer

B.

Last recall check. Draw the gait cycle, list the neurological gaits with one feature each, and write the progression of gait training.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Hornby TG, et al. Clinical practice guideline to improve locomotor function following chronic stroke, incomplete spinal cord injury, and brain injury. *J Neurol Phys Ther* 2020;44:49–100.
3. Perry J, et al. Classification of walking handicap in the stroke population. *Stroke* 1995;26:982–9.

4. Perry J, Burnfield JM. *Gait analysis: normal and pathological function*. 2nd ed. Thorofare: Slack; 2010 (Rancho Los Amigos phase terms).

TP5 Neurogenic bladder and bowel: types, assessment and training

> A 7-mark question on the types of bladder and their physiotherapy was in 5 papers, and a 3-mark list of types in 4 more. The answer always rests on one idea: where the lesion sits (brain, cord above the sacral segments, or the sacral cord and roots) decides how the bladder behaves, and that decides how it is emptied safely.

Syllabus	Treatment programme 5: bladder and bowel training (also Ch 5A(c), management including bladder training)
Priority	**High yield.** In 9 of 20 papers (2016–2026): 5 seven-mark and 4 three-mark questions, plus the type of bladder in the S26 spinal cord long case.
Asked as	LAQ part S26 · 7m W16 · W17 · S21 · W25 · S26 · 3m S17 · W20 · W22 · S23
Usual split	7 (types 3 + physiotherapy 4, or classify and manage one type in detail); 3 (list three types, or describe the reflex bladder)
Checked against	EAU Guidelines on Neuro-Urology (limited update, March 2026); Canadian SCI guidance on bladder management (adapted from the Canadian Urological Association, 2019); Consortium for Spinal Cord Medicine neurogenic bowel guideline (2020); NICE quality standard on supervised pelvic floor muscle training. Full list under Sources (p. 245). Content checked 19 October 2026.

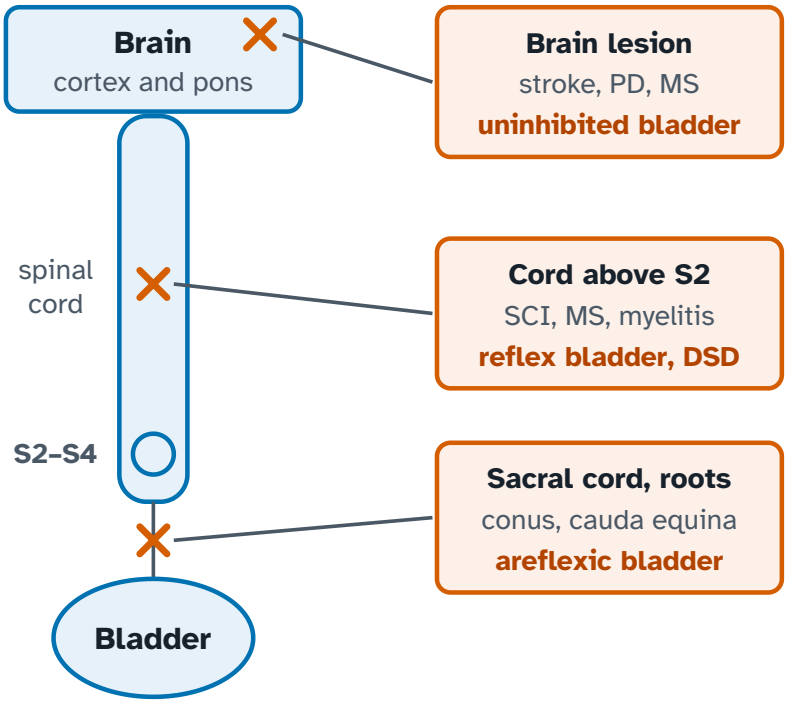
1 What it is

- **Neurogenic bladder** (neurogenic lower urinary tract dysfunction) is abnormal storage or emptying of urine caused by a lesion of the brain, spinal cord or peripheral nerves.
- **Neurogenic bowel** is loss of normal control of stool storage and emptying for the same reasons.
- Common causes: spinal cord injury, multiple sclerosis, stroke, Parkinson's disease, spina bifida, cauda equina lesions, diabetic neuropathy.
- **Why it matters:** a badly managed bladder causes infections, stones, autonomic dysreflexia, and in the long run kidney damage from high bladder pressure. Leaking urine also damages skin and stops people working and going out.

2 How the bladder is controlled

Nerve supply	From	Action
Sympathetic (hypogastric nerve)	T10–L2	Relaxes the detrusor and closes the bladder neck, so the bladder **stores**
Parasympathetic (pelvic nerve)	S2–S4	Contracts the detrusor, so the bladder **empties**
Somatic (pudendal nerve)	S2–S4	Keeps the external sphincter closed under voluntary control
Pontine micturition centre	Pons, under cortical control	Switches from storing to emptying and makes the sphincter relax as the detrusor contracts

Where the lesion is decides the type of bladder



Crosses mark the three lesion sites. DSD is detrusor-sphincter dyssynergia: the sphincter tightens when the bladder contracts.

3 Types of neurogenic bladder

Asked for 3 marks in S17, W20, W22 and S23, and for 7 marks with physiotherapy in W16, W17, S21, W25 and S26. Give the lesion-level types first, then the classic names.

By the level of the lesion (EAU 2026)

Lesion	Symptoms and residual urine	Urodynamics
Above the pons (brain)	Mainly storage symptoms: urgency, frequency, urge leaks; little urine left after voiding	Detrusor overactivity; the sphincter works normally
Spinal cord, between the pons and the sacral cord	Storage and emptying symptoms; residual urine usually raised	Detrusor overactivity with detrusor-sphincter dyssynergia
Sacral cord and below	Mainly emptying problems; residual urine raised	Weak or absent detrusor contraction

Classic types (Lapides)

Type	Lesion and examples	Features
Uninhibited	Brain: stroke, brain tumour, Parkinson's disease, MS, dementia	Sensation present; small capacity; urgency, frequency, urge incontinence; empties fully
Reflex (automatic)	Complete cord lesion above S2: SCI, MS, transverse myelitis	No sensation or voluntary control; empties by reflex when it fills; small capacity; sphincter dyssynergia, so emptying is incomplete
Autonomous (atonic)	Sacral cord or cauda equina; spinal shock	No sensation, no reflex contraction; large capacity; overflow dribbling; large residual
Sensory paralytic	Sensory nerves or posterior columns: diabetes, tabes dorsalis, vitamin B12 deficiency	No sense of fullness, so the bladder overstretches; large capacity and residual; late, weak voiding
Motor paralytic	Motor nerves to the detrusor: poliomyelitis, root injury	Sensation present but cannot empty; painful retention; large residual

Spastic vs flaccid bladder (7 marks in W16 and W17)

Feature	Spastic (UMN, reflex)	Flaccid (LMN, areflexic)
Lesion	Cord above S2–S4 (fracture above about T12)	Sacral cord or cauda equina (fracture at about T12–L1 or below); also spinal shock
Reflex arc	Intact but cut off from the brain	Broken
Detrusor and capacity	Overactive; small capacity	Underactive; large capacity
Sphincter	Tightens as the detrusor contracts (dyssynergia)	Weak, so stress leaks; or does not open
What the person notices	Sudden reflex leaks; spasms; incomplete emptying	Retention with overflow dribbling
Main danger	High pressure: reflux and kidney damage; autonomic dysreflexia at or above T6	Overstretching, infection, stones
Emptying	Intermittent catheterisation with drugs that calm the detrusor	Intermittent catheterisation

Reflex bladder in 3 marks (W22)

A reflex bladder follows a complete cord lesion above the sacral segments. Because the sacral reflex arc is intact but cut off from the brain, the bladder contracts by reflex when it fills, without sensation or voluntary control. The sphincter often tightens at the same time (dyssynergia), so pressure is high and emptying incomplete. Manage it with intermittent catheterisation and drugs that calm the detrusor.

4 Assessment

- **History:** how the person empties now (catheter, reflex leaks, straining), leaks, urgency, infections, autonomic dysreflexia, bowel pattern, fluids, medicines.
- **Bladder diary** for at least 3 days: fluids in, times and volumes of voiding or catheterisation, leaks.
- **Residual urine** after voiding: bladder scanner or catheter.
- **Urodynamics** (done by the urologist): bladder pressure, capacity, detrusor overactivity, dyssynergia.
- **Physio-specific:** hand function and sitting balance for self-catheterisation; transfers to the toilet or commode; dressing; cognition; pelvic floor contraction in incomplete lesions, MS and stroke; skin.
- **Home:** toilet type (an Indian squat toilet is not possible for most paraplegics), bathroom access, privacy, water for washing.

5 Bladder management and the physiotherapist's role

By type

Type	Management	Physio role
Uninhibited (stroke, PD, MS)	Bladder training; drugs for overactive bladder	Timed voiding by the clock; urge suppression; pelvic floor training; faster, safer walking and transfers to the toilet; a bedside commode; easy clothing
Reflex (cord above S2)	Intermittent catheterisation with antimuscarinic drugs; botulinum toxin into the detrusor if needed; a condom sheath only if urodynamics show safe pressure	Teach self-catheterisation skills; balance and hand function; autonomic dysreflexia signs and first steps
Areflexic (sacral cord, cauda equina)	Intermittent catheterisation; an indwelling or suprapubic catheter only if that is not possible	Teach self-catheterisation; pelvic floor training if there is any contraction; protect skin from leaks
Sensory paralytic	Timed voiding; catheterise if the residual stays high	Void by the clock even without an urge

Type	Management	Physio role
Motor paralytic	Intermittent catheterisation	As for areflexic

Clean intermittent catheterisation (CIC)

- The preferred method after SCI where possible (Canadian SCI guidance, adapted from the Canadian Urological Association, 2019).
- **How often:** on average **4–6 times a day**, keeping each volume **under 400–500 mL** (EAU 2026). Plan fluids with the team so the volumes stay in range.
- **Catheter:** usually **12–16 Fr**; hydrophilic catheters lower the infection rate.
- **Teach:** hand washing; position (long sitting on the bed, or in the wheelchair); balance with both hands free; clothing; cleaning; recording volumes. Women may need a mirror and a leg spreader; people with C6–C7 tetraplegia need adapted aids.

Bladder training for the uninhibited bladder

1. **Timed voiding:** go to the toilet at set times by the clock, before the bladder triggers; lengthen the interval as control improves.
2. **Urge suppression:** when an urge comes, stop and stand or sit still; squeeze the pelvic floor quickly a few times; relax and breathe; walk calmly to the toilet when the urge fades.
3. **Pelvic floor muscle training:** helps urinary symptoms and leaks in MS, stroke and incomplete SCI, though the evidence is limited (EAU 2026). Supervise it for **at least 3 months**, as for stress incontinence (NICE).
4. **Electrical stimulation:** tibial nerve stimulation and TENS may help an overactive bladder in MS and stroke; more trials are needed.
5. **Lifestyle:** steady fluids through the day, less tea and coffee late in the day, no constipation.

Classic answer vs current guidance: emptying by tapping or pressing

Classic	Current	Write in the exam
Spastic bladder: trigger reflex voiding by suprapubic tapping or stroking the thigh. Flaccid bladder: empty by pressing on the lower abdomen (Credé) or straining (Valsalva).	These raise bladder pressure, can damage the kidneys and can set off autonomic dysreflexia. Discourage them unless urodynamics show the pressure stays safe (EAU 2026). Clean intermittent catheterisation is preferred.	"Clean intermittent catheterisation 4–6 times a day, with volumes under 400–500 mL. Tapping, Credé and Valsalva only if urodynamics show safe pressures."

Recall check. List the differences between a spastic and a flaccid bladder, and how each is emptied today.

6 Bowel training

Type	Lesion and features	Bowel programme
Reflexic (UMN) bowel	Cord above the conus. The anal sphincter stays tight and the reflex works, so stool is held and constipation is common	**Daily**, or at least **3 times a week**; a suppository or mini enema; digital rectal stimulation; sitting upright on a commode where possible
Areflexic (LMN) bowel	Conus or cauda equina. The sphincter is slack and there is no reflex emptying, so stool leaks	**Once or more each day**; manual removal of stool; a firmer stool to prevent leaks

- Same time each day, often after a meal or a hot drink, to use the gastrocolic reflex.
- Aim for soft, formed stool (Bristol type 3–4), with the session finished in **under 30 minutes**, and never longer than 1 hour.

- Increase fibre slowly; a high-fibre diet is not right for everyone. Drink enough, balanced against the bladder plan.
- Physio role: sitting balance and transfers to the commode or shower chair; adapted aids (suppository inserter, digital stimulator) for weak hands; daily activity and exercise.

Classic answer vs current guidance: abdominal massage and straining

Classic	Current	Write in the exam
Clockwise abdominal massage and straining to empty the bowel.	The 2020 bowel guideline advises against abdominal massage and the Valsalva manoeuvre for emptying.	"A regular bowel programme: rectal stimulation for a reflexic bowel, manual removal for an areflexic bowel, upright on a commode, stool consistency controlled."

7 Precautions

Take care

- In SCI at or above T6, a full bladder or bowel can trigger autonomic dysreflexia: check drainage before exercise and know the first steps (see the spinal cord page (p. 81)).
- Watch for urine infection: fever, cloudy or smelly urine, more spasms or leaks, new autonomic dysreflexia.
- Keep skin dry and clean after leaks; check under condom sheaths.
- Never clamp a catheter for long or block drainage tubing on the tilt table or during transfers.

8 Exam corner

Past questions (MUHS, 2016–2026)

S26 Q2

Describe the types of bladder and their management. (The paper printed it as "Describe the Bladder and Management for the same".)

S26 Q4(a), part of a long case

A man with a T9–T10 fracture senses a full bladder but cannot void without stimulation: explain the type of bladder. Answer on the spinal cord page (p. 90).

W25 Q2(a)

Describe types of bladder and the physiotherapy management of the same.

S21 Q2(d)

Classify neurological bladder. Write assessment and physiotherapy management of any one in detail.

W17 Q2(f), 3 + 4 marks

Explain the difference between spastic and flaccid bladder. Write physiotherapy management for both.

W16 Q2(b)

Explain the flaccid and spastic bladder with their physiotherapy management.

3-mark questions

Enumerate three types of neurological bladder (S17); enlist types of pathological bladders (W20, S23); reflex bladder (W22).

Pages for this topic: spinal cord injury (p. 77), multiple sclerosis (p. 133), spina bifida (p. 298), stroke (p. 50).

Keywords to underline

level of lesion uninhibited reflex autonomous detrusor-sphincter dyssynergia residual urine clean intermittent catheterisation 400–500 mL timed voiding pelvic floor training autonomic dysreflexia

Answer skeleton: types of bladder and physiotherapy (7 marks)

1. **[Control, 1]** One line on the three nerve supplies and the pontine centre; draw the lesion-level figure.
2. **[Types, 2–3]** Lesion-level types with symptoms; the five classic names; spastic vs flaccid table.
3. **[Physiotherapy, 3–4]** Assessment (diary, residual urine); intermittent catheterisation teaching; bladder training and pelvic floor training for the uninhibited bladder; toilet transfers and access; autonomic dysreflexia; infection signs.

Draw these

- The lesion-level figure (section 2)
- The spastic vs flaccid table as two columns

Viva questions

1. Which nerves store urine and which empty it?
2. What is detrusor-sphincter dyssynergia, and why is it dangerous?
3. Which bladder follows a cauda equina lesion?
4. How often should a person self-catheterise, and what volume is safe?
5. Why are Credé and suprapubic tapping discouraged now?

Linked practical spots

- Catheters (Foley, intermittent, condom sheath); urine bag; commode chair.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Enumerate three types of neurological bladder.**

Answer
Uninhibited (brain lesion: stroke, MS); reflex or automatic (cord lesion above S2: SCI); autonomous or atonic (sacral cord or cauda equina). Also sensory paralytic and motor paralytic.

3 m **What is a reflex bladder?**

Answer
The bladder after a complete cord lesion above the sacral segments. The sacral reflex arc is intact but cut off from the brain, so the bladder empties by reflex when full, without sensation or control, often against a tight sphincter (dyssynergia). Managed with intermittent catheterisation and antimuscarinic drugs.

7 m **Explain the difference between spastic and flaccid bladder. Write physiotherapy management for both. (3 + 4)**

Answer

Difference (3): spastic: UMN lesion above S2, reflex arc intact, overactive small bladder, dyssynergia, reflex leaks, high pressure and autonomic dysreflexia risk. Flaccid: LMN lesion of the sacral cord or cauda equina, reflex arc broken, large underactive bladder, overflow dribbling, infection and stones.

Physiotherapy (4): assessment with a bladder diary and residual urine; teach clean intermittent catheterisation 4–6 times a day with volumes under 400–500 mL (balance, positioning, hand skills); fluid planning; pelvic floor training where any contraction exists; transfers to the toilet and an accessible toilet; autonomic dysreflexia and infection signs. Avoid tapping, Credé and Valsalva unless urodynamics show safe pressures.

LAQ **A 55-year-old man had a T9–T10 fracture fixed 2 weeks ago. He senses a full bladder but cannot void without stimulation. Explain the type of bladder. (Part of S26 Q4(a))**

Answer

A reflex (UMN, spastic) bladder: the lesion is above the sacral micturition centre, so the reflex arc is intact but cut off from the brain, and it empties by reflex when stimulated, often with dyssynergia. His sense of fullness suggests some spared sensory fibres. Manage with clean intermittent catheterisation and antimuscarinic drugs; avoid triggered voiding unless urodynamics show safe pressures; teach infection signs.

MCQ **A patient with a cauda equina lesion is most likely to have:**

A. Uninhibited bladder

B. Reflex bladder

C. Autonomous (areflexic) bladder

D. Normal bladder

Answer

C. The sacral reflex arc is broken, so the bladder is atonic with a large residual.

MCQ **During intermittent catheterisation, the bladder volume should ideally stay below:**

A. 150–200 mL

B. 400–500 mL

C. 800 mL

D. 1000 mL

Answer

B. EAU 2026: as a rule, not more than 400–500 mL, with catheterisation 4–6 times a day.

MCQ **Which is the first-choice method for a reflex bladder after SCI?**

A. Suprapubic tapping

B. Credé manoeuvre

C. Clean intermittent catheterisation

D. Permanent indwelling catheter

Answer

C. Tapping and Credé raise bladder pressure; indwelling catheters carry more complications.

Last recall check. Draw the lesion-level figure and write the spastic vs flaccid table from memory.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. European Association of Urology. EAU Guidelines on Neuro-Urology, limited update March 2026 (lesion-level patterns after Panicker et al., *Lancet Neurol* 2015).
3. Canadian Spinal Cord Injury Practice (CAN-SCIP) guideline: bladder management, adapted from the Canadian Urological Association guideline (2019).
4. Consortium for Spinal Cord Medicine. Management of neurogenic bowel dysfunction in adults after spinal cord injury. Paralyzed Veterans of America; 2020.
5. National Institute for Health and Care Excellence. Quality standard QS77: urinary incontinence in women (supervised pelvic floor muscle training for at least 3 months).
6. Lapides J. Neuromuscular vesical and ureteral dysfunction. In: Campbell's Urology, 1970 (classic types).

TP6 Developing a philosophy of caring: person- and family-centred neurological rehabilitation

> The syllabus asks you to develop a philosophy of caring; it was not asked on its own in 2016–2026, but it shows in every long case (goals agreed with the patient, family education) and is marked in the practical as the affective domain. The core idea: the person and family are partners, their goals lead, and how you treat them matters as much as what you do.

Syllabus	Treatment programme 6: developing a philosophy for caring
Priority	**Syllabus.** Not asked on its own in 2016–2026; marked as attitude (affective domain) in the practical.
Asked as	Not asked on its own in 2016–2026
Usual split	Likely 3 marks (principles of family-centred care) or part of a long-case answer
Checked against	Principles of person- and family-centred care and medical ethics as used in rehabilitation; consistent with the ICF page. Full list under Sources (p. 248). Content checked 20 October 2026.

1 Person-centred care

- Start from the person's life, values and goals (work, family roles, faith, hobbies), not only from the diagnosis.
- Set goals together and write them in the person's words (see ICF and goals (p. 4)).
- Respect autonomy: explain options, risks and benefits; accept informed choices, including saying no.
- Build self-management and confidence (self-efficacy) so progress continues after discharge.

2 Family-centred care in children

- Parents know their child best and are the main therapists: they decide priorities with you.
- Give clear, honest information in plain language; teach handling, positioning and a short home programme that fits the family's day.
- Build on the family's strengths; respect culture, beliefs and resources; support siblings and grandparents in joint families.
- Coordinate with school and other services; plan transitions (preschool, school, adulthood).

3 Communication and the therapeutic relationship

- Introduce yourself, explain what you will do and why, and ask permission before touching (consent every session).
- Listen actively; check understanding (ask the person to explain back); use the person's language and avoid jargon.
- Adapt to aphasia, hearing loss, neglect or cognitive problems: short sentences, gestures, pictures, demonstration, a quiet room.
- Show empathy and give realistic hope; praise effort and progress.
- Protect dignity and privacy: drape, close curtains, chaperone where appropriate, respect modesty and cultural norms.

4 Ethics in everyday practice

Principle	In practice
Autonomy	Informed consent; the person's choices; involve them in decisions even when capacity is reduced
Beneficence	Evidence-based treatment aimed at their goals

Principle	In practice
Non-maleficence	Safe handling, precautions, no unnecessary or ineffective treatment
Justice	Fair care regardless of income, caste, religion, gender or disability; help to access schemes
Confidentiality	Share information only with consent and the care team

5 Carers, teamwork and difficult situations

- **Carer burden:** ask how the carer is coping; teach safe handling; suggest respite, support groups and help with schemes.
- **Teamwork:** work with doctors, nurses, occupational and speech therapists, psychologists, social workers, orthotists and teachers; one shared plan.
- **Progressive and terminal disease** (MND, advanced Parkinson's, DMD): honest, compassionate conversations; plan ahead; focus on comfort, function and quality of life; respect advance wishes (see MND (p. 177)).
- **Reflective practice:** after difficult sessions, think about what went well and what to change; seek supervision.

Recall check. Give four principles of family-centred care and three ways to adapt communication for a patient with aphasia.

6 Exam corner

Past questions (MUHS, 2016–2026)

Not asked on its own

Shows up as patient and parent education in long cases and as attitude in the practical.

Keywords to underline

person-centred family-centred shared decision-making informed consent dignity empathy and realistic hope carer burden teamwork

Viva questions

1. What is family-centred care?
2. How do you take consent before treatment?
3. How would you talk to the family of a child newly diagnosed with DMD?
4. Name the four principles of medical ethics.

Linked practical spots

- In the practical, introduce yourself, take consent, drape and guard, and explain each step: this earns the affective-domain marks (see the practical exam (p. 358)).

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Principles of family-centred care.**

Answer

Parents are partners and decide priorities; honest, clear information; teaching handling and a home programme that fits the family's routine; building on family strengths and respecting culture and resources; coordinating with school and services.

7 m **Describe a philosophy of caring in neurological rehabilitation.**

Answer
Person- and family-centred goals; respect for autonomy with informed consent; communication adapted to the person (aphasia, hearing, cognition); empathy and realistic hope; dignity, privacy and cultural sensitivity; ethics (beneficence, non-maleficence, justice, confidentiality); support for carers; teamwork; honest planning in progressive disease; reflective practice.

MCQ **Respecting a patient's informed refusal of a treatment reflects:**

A. Beneficence

B. Autonomy

C. Justice

D. Non-maleficence

Answer
B.

MCQ **In family-centred care, therapy priorities are set by:**

A. The therapist alone

B. The doctor alone

C. The family together with the therapist

D. The school

Answer
C.

MCQ **The best way to check that a carer understood your instructions is to:**

A. Ask "Did you understand?"

B. Ask them to explain or show it back

C. Give a printed sheet only

D. Repeat it louder

Answer
B. The teach-back method.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Standard texts on person- and family-centred rehabilitation and medical ethics (autonomy, beneficence, non-maleficence, justice).

TP7 Orthoses and splints in neurological rehabilitation: prescription, AFOs, hand splints and making a temporary splint

> Orthoses came up in 5 papers: the AFO, the cock-up splint (twice) and the orthosis for radial nerve injury for 3 marks, and foot orthoses for children with cerebral palsy for 7 marks. Every orthosis answer names the problem it solves, the principle it works by (usually a three-point force system), and how you check that it fits and helps.

Syllabus	Treatment programme 7: prescription of appropriate orthotic devices and fabrication of temporary splints
Priority	**Medium yield.** In 5 of 20 papers (2016–2026): 1 seven-mark and 4 three-mark questions.
Asked as	7m S22 · 3m W17 · W21 · S25 · W25
Usual split	7 (types with what each does and who needs it); 3 (one orthosis: parts or design, principle, uses)
Checked against	Peroneal FES versus AFO trial (Bethoux et al., 2014); meta-analysis of AFO and FES on walking speed after stroke (Nascimento et al., 2020); Cochrane reviews of stretch for contracture (Harvey et al., 2017) and of assistive technology including orthoses for contracture after stroke (2024); manufacturer instructions for low-temperature thermoplastics. Full list under Sources (p. 255). Content checked 20 October 2026.

1 Definitions and aims

- An **orthosis** is an externally applied device that supports, aligns, protects or improves the function of a body part. **Splint** usually means a temporary or custom-made orthosis; **brace** is an older word for the same.
- **Static** orthoses hold a joint still; **dynamic** ones allow or assist movement (springs, elastics, hinges).
- **Aims:** improve function (walking, grasp); stabilise weak or unstable joints; prevent or reduce deformity; protect healing structures or overstretched muscles; position for comfort, skin and hygiene.
- **Named by the joints they cross:** FO (foot), AFO (ankle-foot), KAFO (knee-ankle-foot), HKAFO (hip-knee-ankle-foot), WHO (wrist-hand), CTLSO (cervico-thoraco-lumbo-sacral).

2 Biomechanical principles

- **Three-point force system:** two forces on one side and one opposite force in between control a joint. In a foot-drop AFO, the footplate lifts the forefoot, a strap holds the front of the ankle, and the calf section presses on the back of the upper calf.
- **Pressure = force ÷ area:** wide, well-moulded contact and long lever arms spread pressure and protect the skin, which matters most over wasted or numb limbs.
- **Ground reaction force:** where it passes relative to the knee decides whether the knee bends or straightens; a ground-reaction AFO uses this to push the knee back in stance.
- **Joint alignment:** the orthotic joint must line up with the anatomical joint axis, or the device rubs and pistons.
- **Materials:** thermoplastics (polypropylene for AFOs; low-temperature sheets for quick splints), metal and leather (adjustable, heavier), carbon fibre (light, springy).

3 Prescription and check-out

1. **Assess:** range of motion (fixed or flexible deformity), tone, strength, sensation and skin, balance and gait, cognition, home and goals.
2. **Set the aim** in ICF terms: for example "walk to the bus stop without tripping".
3. **Choose the design** (type, joints, stops, material) with the orthotist.
4. **Check the fit:** alignment, trim lines, pressure areas, ease of putting on; shoes that fit over it.

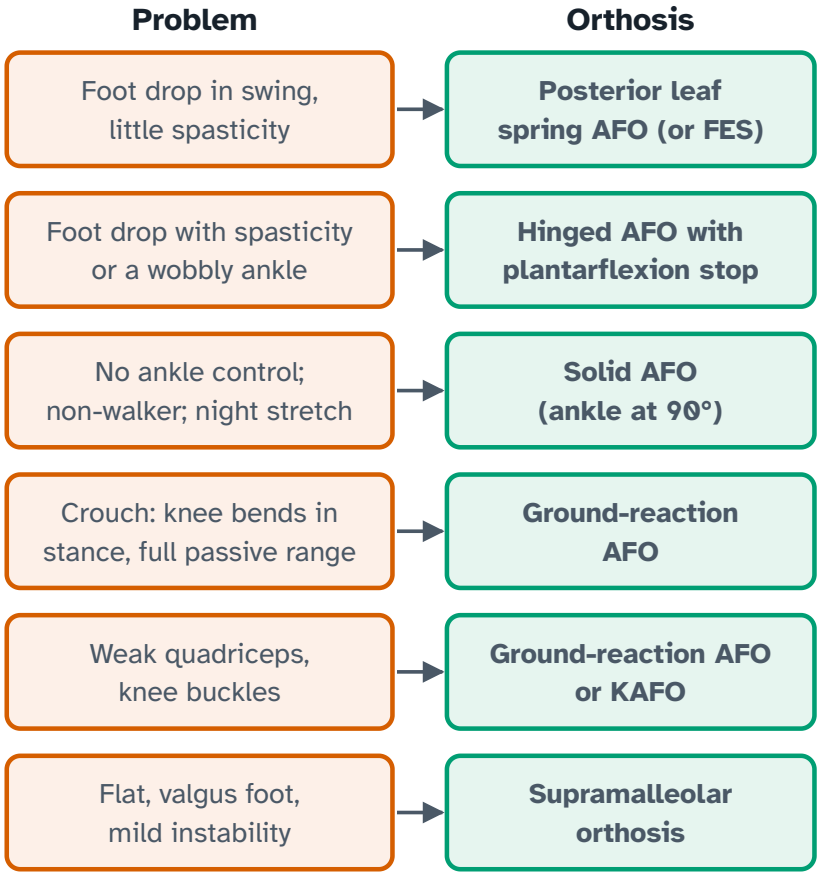
5. **Train:** donning and doffing, walking with it, a gradual wearing schedule.
6. **Review:** skin after each use, growth in children, wear and tear; measure the effect (gait speed, falls, Berg, task performance).

4 Lower limb orthoses

Ankle-foot orthosis (AFO; 3 marks in W21)

- **Parts:** calf section with a strap, ankle section (solid or hinged), footplate; worn inside a shoe.
- **Principle:** a three-point system that controls the ankle (and, through the ground reaction force, the knee).
- **Uses:** foot drop (stroke, peroneal nerve injury, MS, CMT), equinus and spasticity, ankle instability, crouch gait, knee hyperextension from a plantarflexed ankle.
- **Types:** posterior leaf spring, solid, hinged (articulated) with a plantarflexion stop, ground-reaction (anterior shell), and supramalleolar orthosis below it.
- **Evidence:** AFOs increase walking speed after stroke (Nascimento et al., 2020); peroneal nerve stimulation (FES) was as good as an AFO in a large trial (Bethoux et al., 2014).

Choosing an ankle-foot orthosis



A ground-reaction AFO needs a knee and hip that can straighten fully; if they cannot, or the hip is weak too, use a KAFO.

Text version of the chart

- Foot drop in swing with little spasticity: posterior leaf spring AFO (or peroneal FES).
- Foot drop with spasticity or a wobbly ankle: hinged AFO with a plantarflexion stop.
- No ankle control, a non-walker, or a night stretch: solid AFO with the ankle at 90°.
- Crouch with the knee bending in stance but full passive range: ground-reaction AFO.
- Weak quadriceps with the knee buckling: ground-reaction AFO if the knee straightens fully, otherwise a KAFO.
- Flat, valgus foot with mild instability: supramalleolar orthosis.

Above-knee and standing orthoses

- **KAFO:** for a weak quadriceps or an unstable knee (polio, spinal cord injury, spina bifida); knee joints with a drop lock (locked in standing), an offset joint (stable when weight is in front of it) or a Swiss lock.
- **HKAFO and reciprocating gait orthosis (RGO):** for paraplegia and high spina bifida; the RGO links the hips so that extending one hip swings the other leg forward.
- **Standing frames and parapodium:** supported standing for children and adults who cannot stand alone (bones, hips, bladder, development).

Foot orthoses for children with cerebral palsy (7 marks in S22)

Solid AFO, hinged AFO with plantarflexion stop, ground-reaction AFO, posterior leaf spring AFO, supramalleolar orthosis and dynamic (thin, moulded) AFO, each matched to the gait problem. The full answer is on the cerebral palsy page (p. 275).

5 Upper limb, shoulder and spinal orthoses

Orthosis	Used for	How it works
Static cock-up splint (W25)	Radial palsy at rest and night; carpal tunnel syndrome (in neutral); wrist support	Forearm-based splint holding the wrist in about 20–30° of extension, ending at the palmar crease so the fingers and thumb move freely
Dynamic cock-up (lively) splint (S25, W17)	Radial palsy, for daytime function	Wrist support plus an outrigger with elastic finger and thumb slings that re-extend the fingers after each active grasp
Knuckle-bender (anti-claw) splint	Claw hand (ulnar, Klumpke's)	Blocks knuckle hyperextension so the long extensors can straighten the fingers
Opponens or thumb web spacer	Median palsy	Holds the thumb in opposition and the web space open
Resting hand splint	Very weak or spastic hand after stroke or brain injury	Holds wrist and fingers in a functional or stretched position; little effect on contracture (section 7)
Shoulder supports	Flaccid hemiplegic shoulder, flail arm	Arm trough or lap tray in sitting; sling only for short periods when upright
Soft collar or head support	Head drop (MND, myasthenia)	Supports the head in a neutral position
Spinal braces (TLSO)	Spinal fracture stabilisation; selected neuromuscular scoliosis	Three-point control of the spine; limited role in paralytic scoliosis

More on splints for nerve injuries: peripheral nerve injuries (p. 94).

6 Making a temporary splint

1. **Decide the position** (for a cock-up splint, the wrist in about 20–30° of extension).
2. **Make a paper pattern** around the limb, marking landmarks (for a cock-up: two-thirds of the forearm to the distal palmar crease, clear of the thumb).
3. **Trace and cut** the pattern from low-temperature thermoplastic sheet.
4. **Heat** in a water bath at about 65–70 °C (usually 1–2 minutes, by sheet thickness), lift out and dry.
5. **Test the warmth on your own skin,** then mould it on the patient (over stockinette) in the chosen position and hold it for the working time (a few minutes) until it sets.
6. **Trim and smooth the edges,** flare them away from the skin, pad bony points, and fix straps (forearm, wrist and palm).
7. **Check the fit** after a short trial: red areas, numbness, function; spot-heat with a heat gun to adjust.
8. **Teach** the wearing schedule, skin checks and care: clean with cool water and keep away from heat (hot water, stoves, a closed car in the sun), which deforms it.

Serial casting: plaster or synthetic casts changed every few days to gain range step by step, for example for a fixed equinus.

7 Evidence, precautions and common mistakes

Classic answer vs current guidance: splinting to prevent contracture

Classic	Current	Write in the exam
Give every hemiplegic or spastic hand a resting splint to prevent contracture and reduce tone.	Sustained stretch from splints, positioning or casts has no clinically important effect on joint mobility in neurological conditions (Cochrane, 2017), and the 2024 Cochrane review of orthoses after stroke found only very low-certainty evidence. Use splints for a specific aim (function, protection, comfort, hygiene) and train active use.	"Splints are prescribed for a stated functional aim and reviewed; contracture is prevented mainly by active use, positioning and early movement."

Take care

- Check the skin after every use, especially over numb, wasted or swollen limbs; redness that does not fade soon after removal, blisters or numbness mean the orthosis must be adjusted.
- Fixed deformity must be corrected or allowed for before a rigid orthosis is fitted.
- Too heavy or too long an orthosis raises energy cost; use the lightest that does the job.
- Children grow: review fit every few months.
- Check that the patient can put it on and actually uses it.

8 Exam corner

Past questions (MUHS, 2016–2026)

S22 Q2(f), 7 marks
Describe various foot orthoses used for cerebral palsy children.

W21 Q1(b), 3 marks
Ankle foot orthosis.

W17 Q1(f), 3 marks
Describe the orthosis used for radial nerve injury.

S25 Q1(b), 3 marks
Explain uses and principle of dynamic cock-up splint.

W25 Q1(f), 3 marks
Describe cock-up splint.

Pages for this topic: cerebral palsy (p. 275), nerve injuries (p. 94), post-polio residual paralysis (p. 322), spina bifida (p. 298), wheelchairs (p. 257).

Keywords to underline

three-point force system, static and dynamic, prescription and check-out, posterior leaf spring, hinged AFO, ground-reaction AFO, KAFO drop lock, cock-up splint, outrigger, low-temperature thermoplastic, skin check

Answer skeleton: any orthosis (3 marks)

1. Definition and parts (a small labelled sketch).
2. Principle (three-point force system or the assist it gives).
3. Uses (conditions) and one precaution.

Draw these

- The AFO selection chart (section 4)
- A cock-up splint with the wrist in extension, and a dynamic splint with its outrigger

Viva questions

1. Explain the three-point force system with an example.
2. When would you choose a hinged AFO rather than a solid one?
3. What does a ground-reaction AFO need from the knee?
4. What is the difference between a static and a dynamic cock-up splint?
5. How do you check an orthosis before the patient goes home?
6. At what water temperature are low-temperature thermoplastics moulded?

Linked practical spots

- Identify: posterior leaf spring, solid, hinged and ground-reaction AFOs; KAFO with drop lock; static and dynamic cock-up splints; knuckle-bender; resting hand splint.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Ankle-foot orthosis. (W21)**

Answer
An orthosis crossing the ankle and foot: calf section and strap, ankle section, footplate, worn in a shoe. Works by a three-point force system controlling the ankle (and the knee through the ground reaction force). Types: posterior leaf spring, solid, hinged with plantarflexion stop, ground-reaction. Uses: foot drop (stroke, peroneal palsy, MS), spastic equinus, ankle instability, crouch gait. Check the skin and fit.

3 m **Describe the cock-up splint. (W25)**

Answer
A forearm-based wrist orthosis holding the wrist in about 20–30° of extension (neutral for carpal tunnel), ending at the distal palmar crease so the fingers and thumb are free. Static for rest and night; dynamic (lively) with an outrigger and elastic slings for daytime function in radial palsy. Prevents overstretching of the paralysed extensors and lets the flexors grip strongly.

7 m **Describe various foot orthoses used for children with cerebral palsy. (S22)**

Answer
Aims (stable base, control equinus, improve gait, protect range). Solid AFO (equinus with poor control, non-walkers, night stretch); hinged AFO with plantarflexion stop (walkers with enough calf length); ground-reaction AFO (crouch with good passive range); posterior leaf spring (mild foot drop); supramalleolar orthosis (flat valgus feet); dynamic thin AFO (mild tone problems). Prescription and check-out, skin checks, review with growth, and combine with stretching and strengthening.

LAQ **Practice: a 55-year-old man 4 months after stroke walks with his toes catching and his knee snapping back into hyperextension in stance; ankle dorsiflexion reaches neutral passively. Assess, choose an orthosis with reasons, and plan training. (5 + 5 + 5)**

Answer

Assess (5): ankle range (fixed or dynamic equinus), calf tone (Modified Ashworth), dorsiflexor and quadriceps strength, knee control, sensation and skin, balance, gait speed and TUG. **Choose (5):** an AFO set in slight dorsiflexion, hinged with a plantarflexion stop (or solid if the ankle is unstable): it clears the foot in swing, and because the ankle cannot drop into plantarflexion in stance, the shank tilts forward and the knee is no longer pushed back into hyperextension. **Train (5):** donning, a gradual wearing schedule, gait training with the AFO (heel strike, knee control drills, step practice), strengthening and calf stretching; review skin and gait speed in 2–4 weeks; consider peroneal FES as an equal alternative.

MCQ **A child with crouch gait and full passive knee extension is best helped by:**

A. Posterior leaf spring AFO

B. Ground-reaction AFO

C. Supramalleolar orthosis

D. No orthosis

Answer

B.

MCQ **The dynamic cock-up splint is mainly used for:**

A. Median nerve palsy

B. Ulnar claw hand

C. Radial nerve palsy

D. Erb's palsy

Answer

C.

MCQ **Low-temperature thermoplastic splints are usually softened in water at about:**

A. 30–35 °C

B. 45–50 °C

C. 65–70 °C

D. 100 °C

Answer

C.

Last recall check. Write the AFO chart, the two cock-up splints, and the eight steps of making a temporary splint.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Bethoux F, et al. The effects of peroneal nerve functional electrical stimulation versus ankle-foot orthosis in patients with chronic stroke: a randomized controlled trial. *Neurorehabil Neural Repair* 2014;28:688–97.

3. Nascimento LR, et al. Ankle-foot orthoses and continuous functional electrical stimulation improve walking speed after stroke: a systematic review and meta-analyses of randomized controlled trials. *Physiotherapy* 2020.
4. Harvey LA, et al. Stretch for the treatment and prevention of contractures. *Cochrane Database Syst Rev* 2017;(1):CD007455.
5. Mohammed Meeran RA, et al. Assistive technologies, including orthotic devices, for the management of contractures in adults after a stroke. *Cochrane Database Syst Rev* 2024;(9):CD010779.
6. Manufacturer instructions for low-temperature thermoplastic splinting materials (heating 60–72 °C, working time a few minutes).

TP8 Lifting and handling, wheelchairs and adaptive devices

> Wheelchair prescription was part of the S26 long case (a boy with Duchenne muscular dystrophy from a rural area), and lifting, wheelchairs and aids appear in the management of almost every severe case. Fit the device to the person, their body, their home and their roads, train them to use it, and review it as things change.

Syllabus	Treatment programme 8: lifting techniques, wheelchair modifications, adaptive devices
Priority	**Syllabus.** In 1 of 20 papers (2016–2026): wheelchair prescription inside the S26 long case; otherwise used within management answers.
Asked as	LAQ S26
Usual split	Part of a 15-mark long case (wheelchair prescription with short- and long-term goals)
Checked against	WHO guidelines on the provision of manual wheelchairs in less-resourced settings (2008); Cochrane review of manual handling training (Verbeek et al., 2011); weight-shift advice for wheelchair users with spinal cord injury; Indian accessibility standards (2021) and the ADIP scheme. Full list under Sources (p. 262). Content checked 20 October 2026.

1 Safe lifting and patient handling

- **Assess before you move anyone:** the task, the person (weight, ability, pain, lines and tubes, cognition), the environment (space, floor, height of surfaces) and the equipment available.
- **Plan and communicate:** explain, agree on commands ("ready, steady, stand"), and let the patient do as much as possible.
- **Use equipment** instead of lifting: slide sheets for moving in bed, transfer boards, stand-aids, and hoists for dependent patients.
- **Body mechanics:** wide base, bend the knees and hips, keep the load close, keep the back upright, turn with the feet not the trunk, work at a good height.
- **Never lift under the arms** (the "drag lift"): it injures the patient's shoulder, especially a hemiplegic shoulder, and the carer's back.

Classic answer vs current guidance: lifting technique

Classic	Current	Write in the exam
Teach correct lifting technique to prevent back injury in carers and staff.	Training in lifting technique alone did not prevent back pain in workers (Cochrane, Verbeek et al., 2011). Reduce manual lifting with equipment (hoists, slide sheets, boards) and plan each move.	"Risk assessment, use of handling equipment, patient participation, and good body mechanics; avoid manual lifting of dependent patients."

Transfers by level of help

- **Independent:** stand-pivot, or a sliding (transfer) board for paraplegia.
- **One helper:** stand-pivot towards the stronger side (hemiplegia), blocking the weak knee and guiding at the pelvis.
- **Two helpers or more support:** slide sheets and a board; a stand-aid if the person can bear some weight.
- **Dependent:** a mechanical hoist with the right sling. More on transfers: transfers and gait (p. 232).

2 The wheelchair

Parts

- Frame (folding or rigid); seat and backrest (sling or solid); cushion.
- Rear wheels with push rims; front castors; brakes; tyres (solid or pneumatic).
- Armrests (fixed, removable, desk-length); footrests or elevating legrests; footplates.
- Push handles; anti-tip bars; lap belt and postural supports.

Types

Type	Best for
Standard folding manual chair	Occasional or short-term use; easy transport
Lightweight rigid (active) chair	Active long-term users with good arms (paraplegia)
Attendant-propelled chair	Users who cannot self-propel; small rear wheels
Hemi-height or one-arm drive chair	Hemiplegia: low seat to propel with the good foot, or a double rim or lever on the good side
Tilt-in-space and reclining chairs	Poor trunk and head control, pressure relief, fatigue (MND, high tetraplegia, severe CP)
Powered chair	Weak arms or long distances (high tetraplegia, DMD, MND); joystick, chin or breath controls
Rough-terrain or hand-propelled tricycle	Rural and outdoor use in India

3 Wheelchair prescription (S26)

The WHO eight steps of wheelchair provision

1. Referral and appointment
2. Assessment
3. Prescription (selecting the chair)
4. Funding and ordering
5. Product preparation
6. Fitting and adjusting
7. User training
8. Follow-up, maintenance and repairs

Assessment

- **The person:** diagnosis and prognosis (stable or progressive), sitting balance, trunk and head control, arm strength, contractures and deformities (pelvic obliquity, scoliosis), skin and pressure risk, continence, breathing, cognition and vision, body size and growth.
- **Life and place:** home access and space, toilet, school or work, terrain (rough rural roads), transport, who pushes or helps, budget, local repair services.

Measurements (person seated, on the cushion to be used)

Dimension	Rule of thumb	If wrong
Seat width	Hip width plus about 5 cm	Too wide: leaning and hard to propel; too narrow: pressure on the hips
Seat depth	Buttock-to-back-of-knee length minus about 5 cm	Too deep: slides into a slumped posture; too short: high thigh pressure

Dimension	Rule of thumb	If wrong
Seat and footplate height	Thighs level; footplate clears the ground by about 5 cm	Footplate too high: pressure under the buttocks
Back height	Below the shoulder blades for active users; higher, with head support, for poor trunk control	Too low: poor trunk support; too high: blocks arm movement
Armrest height	Elbow height (arm by the side, elbow at 90°) plus about 2.5 cm	Too low: slumping; too high: shoulders hunched

Then choose the type, the seating (solid seat and back, lateral trunk supports, hip guides, pommel, headrest, lap belt at the hips) and a pressure-relieving cushion. The full DMD answer is on the DMD page (p. 334).

4 Wheelchair modifications by condition

Condition	Modifications
Hemiplegia	Hemi-height seat; propel with the good arm and foot; arm trough or lap tray for the weak arm; brake extension on the weak side
Paraplegia	Light rigid frame, low back, good cushion, removable armrests for board transfers
Tetraplegia (C5–C6)	Push-rim projections or coated rims, gloves, higher back, power chair for long distances; C4 and above: power chair with chin or breath control and tilt
Cerebral palsy	Custom seating: pelvic belt, hip guides, pommel, lateral supports, headrest, tray; tilt for poor head control
DMD and other progressive weakness	Manual chair for distance early; power chair with tilt, recline, elevating legrests and spinal supports later
MND	Tilt-in-space, head support, plan the power chair early because the need changes quickly

5 Pressure care and wheelchair skills

- **Cushion:** foam, gel or air cell, matched to the risk; check it is not bottomed out.
- **Weight shifts:** every 15–30 minutes, holding each for at least a minute: push-up, forward lean, side lean, or tilting the chair.
- **Skin checks** twice daily with a mirror; manage moisture and posture.
- **Skills to teach:** brakes and footrests, propulsion with long smooth strokes, turning, ramps, kerbs (back-wheel balance with help, then alone), transfers in and out, falls and getting back in, folding and loading, simple maintenance (tyres, brakes).

6 Adaptive devices and home changes

Walking aids

- **Sticks (single-point, tripod, quadripod), crutches (axillary, elbow), walkers (standard frame, rollator).**
- **Height:** the handgrip at the level of the wrist crease (or greater trochanter) with the arm by the side, giving about 20–30° of elbow bend; axillary crutch tops about 5 cm below the armpit, weight taken through the hands.
- A stick goes in the hand opposite the weak leg.

Daily-living aids

- **Eating:** built-up or angled cutlery, plate guard, non-slip mat, rocker knife.
- **Dressing:** button hook, sock aid, long-handled shoe horn, elastic laces, Velcro.
- **Reaching:** long-handled reacher.
- **Bathroom:** raised toilet seat or commode chair, grab rails, shower chair or bath bench, non-slip mats.

- **Communication and control:** communication boards or devices, call bells, environmental controls for severe disability.

Home modifications

- **Ramps:** no steeper than 1 in 12 for short rises (gentler for longer ramps), at least 1200 mm wide, with handrails on both sides at 760–900 mm (Indian accessibility standards, 2021).
- Wide doorways and turning space; rails by the toilet and bed; remove loose rugs; good lighting; a ground-floor room where possible.

Getting devices in India

The **ADIP scheme** (Ministry of Social Justice and Empowerment) supplies aids such as wheelchairs, tricycles and callipers free or at subsidised cost to eligible people with a disability certificate (40% or more), through ALIMCO, national institutes and NGOs; income limits apply.

7 Exam corner

Past questions (MUHS, 2016–2026)

S26 Q4(b), long case

An 8-year-old boy with Duchenne muscular dystrophy from a rural area walks only with support, with tight plantar flexors and weak girdle and breathing muscles. Explain the steps involved in wheelchair prescription; short- and long-term goals.

Keywords to underline

risk assessment handling equipment no drag lift WHO eight steps seat width plus 5 cm seat depth minus 5 cm tilt-in-space pressure-relieving cushion weight shifts user training ramp 1:12

Answer skeleton: wheelchair prescription (part of a long case)

1. Assess the person (stage, trunk, arms, deformity, skin, breathing, growth) and the setting (home, school, terrain, carer, budget).
2. Choose the type and its features, with reasons.
3. Measure (seat width, depth, height, back and armrest height) and specify seating and cushion.
4. Fit, train the user and family, and plan review and repairs.

Draw these

- A side view of a wheelchair with its parts labelled

Viva questions

1. Name the parts of a wheelchair.
2. How do you measure seat width and seat depth?
3. Which wheelchair suits a patient with hemiplegia?
4. How often should a wheelchair user shift weight?
5. Why is the drag lift dangerous?
6. How high should a walking stick be?

Linked practical spots

- Identify the parts of a wheelchair; measure a person for a chair; set a stick, crutch or walker height; demonstrate a sliding-board transfer; name daily-living aids.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **List the parts of a wheelchair.**

Answer

Frame; seat and backrest; cushion; rear wheels with push rims; front castors; brakes; armrests; footrests or legrests with footplates; push handles; anti-tip bars; lap belt.

3 m **Write the principles of safe patient handling.**

Answer

Assess the task, the person, the environment and the equipment; plan and communicate; let the patient help; use equipment (slide sheets, boards, hoists) instead of lifting; good body mechanics (wide base, load close, bend knees, no twisting); never lift under the arms.

7 m **Explain the steps involved in wheelchair prescription. (S26)**

Answer

WHO steps: referral, assessment, prescription, funding and ordering, preparation, fitting, user training, follow-up. **Assessment:** diagnosis and progression, trunk and head control, arm strength, deformity, skin, breathing, cognition, growth; home, school, terrain, carer, budget. **Prescription:** type (manual or power, folding or rigid, tilt), seating (solid seat and back, lateral supports, hip guides, pommel, headrest, belt), cushion. **Measurements:** seat width = hips + 5 cm; depth = buttock to knee − 5 cm; footplate 5 cm off the ground; armrest = elbow height + 2.5 cm; back height by trunk control. **Fitting and training** in use, transfers, pressure relief and care; **review** with growth or disease progression.

LAQ **Practice: a 30-year-old man with a C6 complete tetraplegia (AIS A) is ready for discharge to a first-floor flat. Plan his wheelchair, transfers and adaptive devices. (5 + 5 + 5)**

Answer

Wheelchair (5): light manual chair with push-rim projections or coated rims and gloves for indoors, power chair for distance; high back, good air or gel cushion; tilt or a recline if he cannot do weight shifts; measured to fit. **Transfers (5):** sliding-board transfers with assistance using a level transfer and removable armrests; a hoist for car or floor transfers; carer training in safe handling. **Devices (5):** universal cuff and built-up cutlery, tenodesis use, wrist splints, shower chair, raised commode, pressure-relieving mattress; a ramp at 1:12 with rails or a lift, wider doors, ground-level access; ADIP scheme for equipment.

MCQ **Seat width of a wheelchair should be about:**

A. Equal to hip width

B. Hip width + 5 cm

C. Hip width + 15 cm

D. Shoulder width

Answer

B.

MCQ **A patient with right hemiplegia holds a walking stick in the:**

A. Right hand

B. Left hand

C. Either hand

D. Both hands

Answer

B. The hand opposite the weak leg.

MCQ **The steepest ramp allowed for a short rise under Indian accessibility standards is:**

A. 1:6

B. 1:8

C. 1:12

D. 1:30

Answer

C.

Last recall check. Write the WHO eight steps, the five measurements and the handling principles.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. Guidelines on the provision of manual wheelchairs in less resourced settings. Geneva: WHO; 2008.
3. Verbeek JH, et al. Manual material handling advice and assistive devices for preventing and treating back pain in workers. *Cochrane Database Syst Rev* 2011;(6):CD005958.
4. Consortium for Spinal Cord Medicine. Pressure ulcer prevention and treatment following spinal cord injury (clinical practice guideline), as summarised in seating reviews.
5. Ministry of Housing and Urban Affairs (CPWD). Harmonised Guidelines and Standards for Universal Accessibility in India, 2021.
6. Ministry of Social Justice and Empowerment. Scheme of Assistance to Disabled Persons for Purchase/Fitting of Aids and Appliances (ADIP).

TP9 Ergonomic advice and caregiver education in neurological conditions

> Ergonomic and caregiver advice appears inside many long cases, for example home-care advice to the parents of a child with cerebral palsy or DMD, or a home programme for a stroke survivor; it was not asked on its own in 2016–2026. Good advice is specific to the person's home, work and carers, written down, demonstrated, and checked by asking them to show it back.

Syllabus	Treatment programme 9: ergonomic advice for prevention and rehabilitation for patients, and education of parents and caregivers about handling patients
Priority	**Syllabus.** Not asked on its own in 2016–2026; part of home-care advice in long cases (for example the W17 CP case).
Asked as	Not asked on its own in 2016–2026
Usual split	Within long cases: home-care or parent advice (for example 5 marks in W17)
Checked against	Consistent with the handling evidence on the wheelchairs page (Cochrane, 2011) and the condition pages. Full list under Sources (p. 265). Content checked 20 October 2026.

1 Ergonomic advice for patients

- **Posture:** sit with the pelvis upright, feet supported, back supported; change position often; avoid long sustained postures.
- **Workstation:** screen at eye level, elbows at about 90°, forearms supported, chair and desk heights adjusted; for wheelchair users, knee clearance and items within easy reach.
- **Energy conservation** (MS, MND, myasthenia, post-polio, Parkinson's): plan, prioritise, pace, sit to work, rest before fatigue, do heavy tasks when strongest.
- **Joint protection:** use large joints and both hands, avoid sustained grips, use aids (built-up handles, trolleys), protect the hemiplegic shoulder.
- **Home layout:** frequently used items between hip and shoulder height, clear walkways, good lighting, rails and non-slip surfaces (see adaptive devices (p. 257)).
- **Return to work:** graded hours, task changes, rest breaks, travel plan, a workplace visit when possible.

2 Educating parents and caregivers

- **Handling and carrying:** positions matched to the child's tone (spastic: break up the stiff pattern with hips bent and apart; athetoid: limbs together and bent, close to the carer); see cerebral palsy (p. 275).
- **Positioning through the day:** side lying, prone on a wedge, supported sitting; avoid W-sitting and long periods lying on the back.
- **Feeding:** upright, head slightly forward, trunk supported; refer for swallowing assessment when there is coughing or chest infection.
- **Dressing:** dress the stiffer or weaker limb first and undress it last.
- **Transfers and turning in adults:** slide sheets, transfer boards, hoists; never lift under the arms (see lifting (p. 257)).
- **Home programme:** short, written, illustrated, built into daily routines (bath time, play, meals); demonstrate it and ask the carer to show it back.
- **Warning signs** to report (for example shunt problems, autonomic dysreflexia, breathing trouble, skin redness).

3 Protecting the carer

- Use equipment rather than lifting; set the bed and chairs at working height; keep loads close; avoid twisting.
- Training in lifting technique alone does not prevent back pain (Cochrane, 2011), so equipment and planning matter more.
- Watch for carer strain: tiredness, low mood, back pain; involve other family members, respite and support groups.

Recall check. Give five points of home-care advice for the parents of a child with spastic cerebral palsy.

4 Exam corner

Past questions (MUHS, 2016–2026)

W17 Q4(b), long case

A 2-year-old with spastic diplegia: home-care advice (5 marks); goals and physiotherapy. Answer on the CP page (p. 275).

W16 Q4(b), long case

A 5-year-old with DMD: home-care advice to parents. Answer on the DMD page (p. 334).

Keywords to underline

posture workstation energy conservation joint protection handling and carrying positioning home programme teach-back carer strain

Answer skeleton: home-care advice (5 marks)

1. Positioning and handling for the day and night.
2. Daily activities: feeding, dressing, bathing, toileting.
3. Home programme built into routines; play.
4. Equipment and home changes.
5. Warning signs and follow-up; carer's own health.

Viva questions

1. What is energy conservation? Give four techniques.
2. How should a parent carry a child with spastic diplegia?
3. How do you make sure a carer can do the home programme?
4. How would you set up a desk for a wheelchair user?

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Energy conservation techniques.**

Answer

Plan the day; prioritise important tasks; pace with rests before fatigue; sit to work; organise items within reach; use aids and trolleys; do demanding tasks when strongest; avoid heat (MS).

7 m **Home-care advice to the parents of a child with cerebral palsy.**

Answer
Positioning (side lying, prone on a wedge, supported sitting; no W-sitting), carrying matched to tone, feeding upright with head slightly forward, dressing the stiffer limb first, daily play practising head control, sitting and reaching, standing frame time, splint and skin care, no baby walkers, a short written home programme shown back by the parents, and warning signs and follow-up.

MCQ **When dressing a hemiplegic patient, the weak arm goes into the sleeve:**

A. First

B. Last

C. Either

D. Never

Answer
A. And comes out last when undressing.

MCQ **Which best protects a carer's back?**

A. Lifting technique training alone

B. Using handling equipment and planning moves

C. A back belt

D. Lifting quickly

Answer
B.

MCQ **A computer screen should be placed:**

A. Below the desk

B. At eye level

C. Far to one side

D. Above head height

Answer
B.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Verbeek JH, et al. Manual material handling advice and assistive devices for preventing and treating back pain in workers. *Cochrane Database Syst Rev* 2011;(6):CD005958.

Part 5B

Physiotherapy management: paediatric conditions

17 hours theory, 40 practical

5B	Developmental neurology: milestones, reflexes and reactions	267
5B(1)	Cerebral palsy	275
5B(2)	Down's syndrome	289
5B(3)	Neural tube defects: spina bifida and hydrocephalus	298
5B(4)	Obstetric brachial plexus injury	309
5B(5)	Infections of the nervous system	317
5B(6)	Post-polio residual paralysis	322
5B(7)	Duchenne muscular dystrophy and other myopathies	334
5B(8)	Spinal muscular atrophy and hereditary neuropathies	345
5B(9)	Paediatric extrapyramidal disorders	350

5B Developmental neurology: motor milestones, primitive reflexes and postural reactions

> Development came up in 6 papers: gross and fine motor milestones for 7 marks three times, and the ATNR, a primitive reflex, cortical reflexes and the parachute reaction for 3 marks. You need the normal sequence and ages to spot delay in cerebral palsy, Down's syndrome and spina bifida, and to plan treatment in the order a child normally learns to move.

Syllabus	Ch 5B Developmental neurology: knowledge of normal development and plasticity in development (applies to every paediatric condition)
Priority	**High yield.** In 6 of 20 papers (2016–2026): 3 seven-mark and 5 three-mark questions.
Asked as	7m W19 · S21 · S23 · 3m S20 · W20 ×2 · S21 · W25
Usual split	7 (gross motor table 4 and fine motor 3, or gross motor with principles); 3 (one reflex or reaction: how to elicit, the response, when it appears and goes, and what abnormal means)
Checked against	WHO Motor Development Study (2006); CDC developmental milestone checklists (Zubler et al., 2022); University of Iowa review of infant primitive reflexes (D'Alessandro, 2024); developmental milestone teaching chart, Department of Pediatrics, GMCH Chandigarh. Full list under Sources (p. 273). Content checked 20 October 2026.

1 Principles of development

- Development is continuous, from conception to maturity.
- The **sequence is the same** in all children, but the **rate varies**.
- **Cephalocaudal:** control comes from head to foot (head control, then sitting, then standing).
- **Proximodistal:** from the trunk and shoulders out to the hands and fingers.
- **General to specific:** mass movements are replaced by specific, individual actions.
- Primitive reflexes fade as higher control develops, so that new milestones can appear.
- For babies born before 37 weeks, use the **corrected age** (age from birth minus the weeks born early) until about 2 years.

2 Gross motor milestones

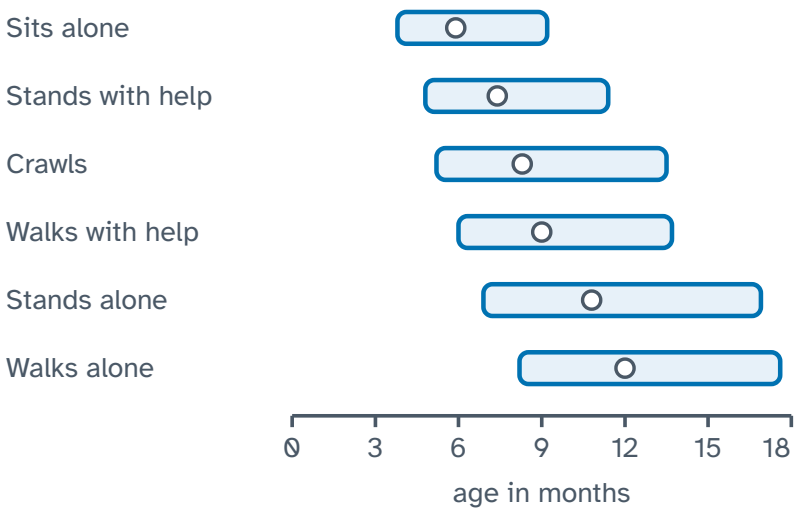
Asked for 7 marks in W19 (to 5 years), S21 (to 3 years) and S23 (to 5 years).

Age	Gross motor milestone
3 months	Holds the head steady (neck holding); lifts the chest on forearms in prone
5 months	Rolls over
6 months	Sits with support; takes weight on straight arms in prone
8 months	Sits without support
9 months	Stands holding on; crawls
12 months	Stands without support; walks with one hand held or alone but falls
15 months	Walks alone; creeps upstairs
18 months	Runs stiffly; climbs onto furniture
2 years	Runs well; walks upstairs two feet per step; jumps; kicks a ball
3 years	Walks upstairs with alternate feet; rides a tricycle

Age	Gross motor milestone
4 years	Hops on one foot; walks downstairs with alternate feet
5 years	Skips; hops well on either foot

These are the usual teaching ages (Indian paediatric charts). Real children vary widely, as the WHO study shows:

Normal age range for six gross motor milestones



Bars: the normal range (1st to 99th percentile) in 816 healthy children from Ghana, India, Norway, Oman and the USA; dots: the median age (WHO Motor Development Study, 2006). Walking alone anywhere between about 8 and 18 months is normal.

3 Fine motor milestones

Age	Fine motor milestone
3–4 months	Watches own hands; brings hands to the midline
4 months	Reaches with both hands (bidextrous reach)
6 months	Reaches with one hand (unidextrous reach); palmar grasp; transfers objects hand to hand by 6–7 months
9 months	Immature pincer grasp
12 months	Mature pincer grasp (thumb and index fingertips)
15 months	Imitates scribbling; tower of 2 blocks
18 months	Scribbles; tower of 3 blocks; feeds self with a spoon
2 years	Tower of 6 blocks; vertical and circular strokes
3 years	Tower of 9 blocks; copies a circle; dresses with help
4 years	Copies a cross; builds a bridge with blocks
5 years	Copies a triangle; ties shoelaces

4 Primitive reflexes

Automatic responses present at birth, controlled by the brainstem and spinal cord. They fade (are integrated) as the cortex matures, mostly by 4–6 months. A reflex that is **absent, asymmetrical, too strong (obligatory) or persists** beyond its usual age suggests brain damage, as in cerebral palsy.

Reflex	How to elicit and the response	Present
Rooting	Touch the cheek or corner of the mouth: the head turns towards it	Birth to 3–4 months

Reflex	How to elicit and the response	Present
Sucking	Finger or nipple in the mouth: rhythmic sucking	Birth to 3–4 months
Moro	Let the supported head drop back suddenly: arms spread and straighten with open hands, then come together, often with a cry	Birth to 3–4 months (normal up to 6)
Palmar grasp	Press the palm at the base of the fingers: the fingers grip	Birth to about 6 months
Plantar grasp	Press the sole below the toes: the toes curl	Birth to about 15 months
Stepping	Hold upright with the feet on a surface: alternate stepping movements	Birth to 2–3 months
Placing	Touch the top of the foot against a table edge: the foot lifts and is placed on the table	Birth to about 1 year
Galant	In prone suspension, stroke beside the spine: the trunk curves towards that side	Birth to 2–4 months (up to 6)
ATNR	Turn the head to one side: the arm and leg on the face side straighten, those on the skull side bend	Birth to about 6 months
STNR	In four-point or held prone: head bent, arms bend and legs straighten; head lifted, arms straighten and legs bend	Mid-infancy, gone by about 9–12 months
Babinski (extensor plantar)	Stroke the outer sole: the big toe goes up and the toes fan	Normal until 1–2 years

Asymmetrical tonic neck reflex (ATNR; 3 marks in S20 and S21)

- **Elicit:** child supine, head in midline; turn the head to one side (or watch when the child turns it).
- **Response:** the arm (and leg) on the face side extend and the arm (and leg) on the skull side flex: the "fencing" posture.
- **Normal:** present from birth, seen intermittently, and integrated by about 6 months, allowing the hands to come to the midline and the child to roll.
- **Abnormal:** an obligatory ATNR at any age (the child cannot get out of it), or persistence after about 6 months, suggests brain damage (cerebral palsy).
- **Effects of persistence:** cannot bring hands together or to the mouth, cannot roll or sit symmetrically, poor hand–eye coordination, asymmetrical posture with a risk of scoliosis and hip problems, and later difficulty crossing the midline in writing.
- **Physiotherapy:** midline positioning (side lying, symmetrical seating, toys in front), approach and play from the front.

Moro reflex in detail (example of "any one primitive reflex", W20)

Elicited by a sudden drop of the supported head; the arms abduct and extend with the hands opening, then adduct and flex. Present from birth, gone by 3–4 months (normal up to 6). **Absent:** severe brain depression or hypotonia. **Asymmetrical:** brachial plexus palsy, fractured clavicle or humerus, hemiplegia. **Persisting after 6 months:** cerebral palsy.

Recall check. Give the response and normal timing of the ATNR, and three effects if it persists.

5 Postural reactions

- **Righting reactions:** keep the head upright in space and the head, trunk and limbs in line (neck righting, body righting, labyrinthine and optical righting). The **Landau reaction** (held prone in the air, the head lifts and the trunk and legs extend) appears at about 3 months and is gone by 2 years.
- **Protective (parachute) reactions:** the arms shoot out to break a fall.

- **Equilibrium (tilting) reactions:** when the support surface tilts, the trunk curves towards the higher side and the limbs move out to keep balance; they persist for life.

Parachute reaction (3 marks in W25)

- **Elicit:** hold the child prone around the trunk and move the child suddenly head-first towards a surface.
- **Response:** both arms extend and abduct, with the hands open, to protect the head.
- **Timing:** the forward reaction appears at about 6–9 months and persists for life; sideways and then backward protective reactions follow over the next months.
- **Abnormal:** absent, delayed or asymmetrical (hemiplegic cerebral palsy, brachial plexus palsy, motor delay); without it, sitting and standing are unsafe.
- **Physiotherapy:** train protective reactions on a ball, roll or tilt board.

The reflex hierarchy and "cortical reflexes" (W20)

Classic levels (Fiorentino)

Level	Reflexes and reactions	When
Spinal	Flexor withdrawal, extensor thrust, crossed extension	First 2 months
Brainstem	ATNR, STNR, tonic labyrinthine reflex, positive and negative supporting reactions, associated reactions	Early infancy, integrated mostly by 4–6 months
Midbrain	Righting reactions (neck, body, labyrinthine, optical), Landau, protective reactions	From about 4–6 months; some stay for life
Cortical	Equilibrium (tilting) reactions in prone, supine, four-point, sitting, kneel-standing and standing; hopping and dorsiflexion reactions	From about 6 months (prone) to the second year (standing); for life

Cortical reflexes (3 marks): the equilibrium reactions in each position (prone, supine, four-point kneeling, sitting, kneel-standing, standing), plus the hopping and dorsiflexion reactions. Each appears after the child can get into that position (sitting reactions from about 7–8 months, standing reactions in the second year) and stays for life.

Classic answer vs current guidance: the reflex hierarchy

Classic	Current	Write in the exam
Development is the cortex progressively inhibiting lower reflexes; treatment must inhibit primitive reflexes before skills can appear.	Movement emerges from many interacting systems (strength, posture, perception, motivation, environment); reflexes and reactions are useful clinical signs, but treatment practises the functional skills themselves (see motor control (p. 11)).	Give the classic levels when asked, then add one line that the hierarchy is a teaching model and skills are trained directly.

6 Red flags and screening

When to worry (most children do these by the age shown; CDC, 2022)

- **6 months:** rolls from tummy to back; pushes up on straight arms in prone; leans on hands to sit.
- **9 months:** sits without support and gets into sitting alone; moves objects from hand to hand.
- **12 months:** pulls to stand and walks holding furniture; picks up small things with thumb and index finger.
- **18 months:** walks without holding on; scribbles; climbs on and off a chair.
- **2 years:** runs, kicks a ball, walks up a few stairs.
- **5 years:** hops on one foot.

Missing any of these needs a closer developmental check. Other red flags: loss of skills already gained; asymmetry of movement; a clear hand preference before about 12–18 months (weakness of the other hand); very low or high tone; primitive reflexes persisting beyond their age.

Screening and assessment tools

- **Trivandrum Developmental Screening Chart** (Indian, simple screening in the first years); Denver II.
- **Alberta Infant Motor Scale** (0–18 months), **Peabody Developmental Motor Scales**, Bayley scales.
- Hammersmith Infant Neurological Examination and General Movements assessment for early detection of cerebral palsy (see cerebral palsy (p. 275)).

7 How physiotherapy uses development

- Assess where the child is in the sequence, not just the age.
- Train the next skill in the sequence: head control, rolling, sitting, crawling, kneeling, pull to stand, cruising, walking.
- Use the key positions (prone, supine, side lying, sitting, four-point, kneeling, half-kneeling, standing) for strength, weight bearing and balance.
- Practise protective and balance reactions once the child can hold a position.
- Use play, toys and the family's daily routine; repetition matters more than any single technique.

8 Exam corner

Past questions (MUHS, 2016–2026)

W19 Q2(e), 7 marks
Describe the gross and fine motor development in a child up to the age of 5 years.

S21 Q2(c), 7 marks
Discuss gross motor and fine motor milestones up to age 3 years.

S23 Q2(b), 7 marks
Discuss the gross motor milestones till 5 years of age.

S20 Q1(a), S21 Q1(f), 3 marks
Asymmetrical tonic neck reflex.

W20 Q1(e), 3 marks
Describe in brief any one primitive reflex.

W20 Q1(f), 3 marks
Enumerate cortical reflexes.

W25 Q1(d), 3 marks
Write a note on parachute reactions.

Keywords to underline

cephalocaudal proximodistal sequence constant, rate varies corrected age neck holding 3 months sits alone 8 months walks alone 12–15 months pincer grasp integration obligatory ATNR parachute equilibrium reactions

Answer skeleton: milestones (7 marks)

1. **[Principles, 1]** Sequence constant, rate varies; head to toe; trunk to fingers.

2. **[Gross motor, 3]** The table in section 2 up to the age asked.
3. **[Fine motor, 3]** The table in section 3 up to the age asked (if asked); add one line on red flags.

Draw these

- The WHO windows chart (section 2)
- A stick figure of the ATNR "fencing" posture

Viva questions

1. At what age does a child hold the head, sit alone and walk alone?
2. When does the pincer grasp mature?
3. When is the ATNR integrated, and what does an obligatory ATNR mean?
4. What does an asymmetrical Moro reflex suggest?
5. Which reactions are called cortical?
6. Why do we use corrected age?

Linked practical spots

- Demonstrating the Moro, ATNR, parachute and equilibrium reactions on a doll or mannequin; blocks and crayons for fine motor testing.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Asymmetrical tonic neck reflex. (S20, S21)**

Answer
Turning the head of a supine infant to one side makes the face-side arm and leg extend and the skull-side limbs flex ("fencing" posture). Present from birth, integrated by about 6 months. Obligatory at any age, or persisting after 6 months, suggests cerebral palsy; persistence blocks hand-to-mouth and midline play and rolling, and causes asymmetry.

3 m **Write a note on parachute reactions. (W25)**

Answer
A protective reaction: the child, held prone and moved suddenly head-first towards a surface, extends and abducts both arms with open hands. Forward appears at about 6–9 months and persists for life, followed by sideways and backward reactions. Absent or asymmetrical in hemiplegia, brachial plexus palsy or delay; trained on a ball or roll.

7 m **Discuss gross motor and fine motor milestones up to age 3 years. (S21)**

Answer
Principles (one line). **Gross motor:** 3 months neck holding; 5 months rolls; 6 months sits with support; 8 months sits alone; 9 months stands holding, crawls; 12 months stands alone, walks with support; 15 months walks alone, creeps upstairs; 18 months runs; 2 years walks upstairs two feet per step, jumps, kicks a ball; 3 years alternate feet upstairs, tricycle. **Fine motor:** 4 months bidextrous reach; 6 months unidextrous reach, transfers by 6–7 months; 9 months immature pincer; 12 months mature pincer; 15 months tower of 2, imitates scribble; 18 months tower of 3, scribbles, spoon; 2 years tower of 6, strokes; 3 years tower of 9, copies a circle.

LAQ **Practice: a baby born at 32 weeks is now 9 months old and cannot sit alone. How would you assess development, what would worry you, and what would you advise? (5 + 5 + 5)**

Answer

Assess (5): corrected age is 7 months, so sitting alone is not yet overdue (WHO range up to about 9 months); check head control, rolling, prone on extended arms, supported sitting, reaching and transfer, tone, primitive reflexes (Moro, ATNR should be fading) and early reactions; use the Alberta Infant Motor Scale. **Worry (5):** no rolling, poor head control, persistent or obligatory ATNR, asymmetry, very high or low tone, early hand preference, loss of skills; refer for full assessment (Hammersmith examination). **Advise (5):** daily tummy time, supported sitting play with toys in front, reaching in side lying, rolling games, and review in 6–8 weeks.

MCQ **In the WHO study the median age of walking alone was:**

A. 9 months

B. 12 months

C. 15 months

D. 18 months

Answer

B. The normal range was about 8 to 18 months.

MCQ **Which reaction persists throughout life?**

A. Moro

B. ATNR

C. Parachute

D. Palmar grasp

Answer

C.

MCQ **A child usually copies a circle at:**

A. 18 months

B. 2 years

C. 3 years

D. 5 years

Answer

C. A cross at 4 years and a triangle at 5 years.

Last recall check. Write the gross and fine motor tables to 5 years, the ATNR, the Moro and the parachute reaction.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. WHO Multicentre Growth Reference Study Group. WHO Motor Development Study: windows of achievement for six gross motor development milestones. *Acta Paediatr Suppl* 2006;450:86–95.

3. Zubler JM, et al. Evidence-informed milestones for developmental surveillance tools. *Pediatrics* 2022;149(3):e2021052138; CDC "Learn the Signs. Act Early." milestone checklists.
4. D'Alessandro DM. What are examples of infantile primitive reflexes? PediatricEducation.org, University of Iowa, updated 2024.
5. Aulakh R. Developmental milestones (teaching chart). Department of Pediatrics, Government Medical College and Hospital, Chandigarh.
6. Fiorentino MR. *Reflex testing methods for evaluating C.N.S. development*. 2nd ed. Springfield: Thomas; 1973.

5B(1) Cerebral palsy: types, assessment, ICF and physiotherapy

> Cerebral palsy was a long case in 13 of the 20 papers (2016–2026), more than any other topic, and the same 5-year-old athetoid girl was asked four times. Short questions ask for causes, types, orthoses, carrying positions and the posture and movement problems of each type. Every answer rests on one idea: the brain lesion does not progress, but the child's muscles and bones keep growing around it, so active, goal-directed practice and prevention of deformity must start early.

Syllabus	Ch 5B(1) Cerebral palsy: aetiology and type, assessment, differential diagnosis, management
Priority	**Tier 1.** In 16 of 20 papers (2016–2026): 13 long cases, 6 seven-mark and 8 three-mark questions.
Asked as	LAQ S17 · W17 · S18 · W18 · W19 · S20 · S21 · S22 · W22 · W24 · S25 · W25 · S26 · 7m S18 · S19 · S22 · S24 · S26 ×2 · 3m W16 · W17 · S18 · W19 · W22 · W25 · S26 ×2
Usual split	15 (ICF with clinical reasoning 5, goals 3–5, management 7–10); 7-mark orthoses, carrying, posture and movement problems
Checked against	Rosenbaum consensus definition (2007); GMFCS expanded and revised (2007); Novak et al. early diagnosis (2017) and State of the Evidence Traffic Lights (2019, published 2020); AACPDM hip surveillance care pathway; Paleg et al. supported standing (2013); Indian prevalence meta-analysis (2019) and global prevalence analysis (2022); GMFCS motor development curves (2002). Full list under Sources (p. 287). Content checked 19 October 2026.

1 Definition

Cerebral palsy (CP) is a group of **permanent** disorders of the development of **movement and posture**, causing activity limitation, that are due to **non-progressive** disturbances in the developing fetal or infant brain. The motor disorder is often accompanied by disturbances of sensation, perception, cognition, communication and behaviour, by epilepsy, and by secondary musculoskeletal problems (international consensus, Rosenbaum et al., 2007).

- **Non-progressive lesion, changing picture:** the brain injury does not worsen, but the child's posture, tone and deformities change with growth.
- **How common:** about 3 per 1000 children in Indian studies (2.95 per 1000, 2019 meta-analysis). In high-income countries it is about 1.6 per 1000 live births and falling; the limited data from low- and middle-income countries show rates as high as 3.4 per 1000.

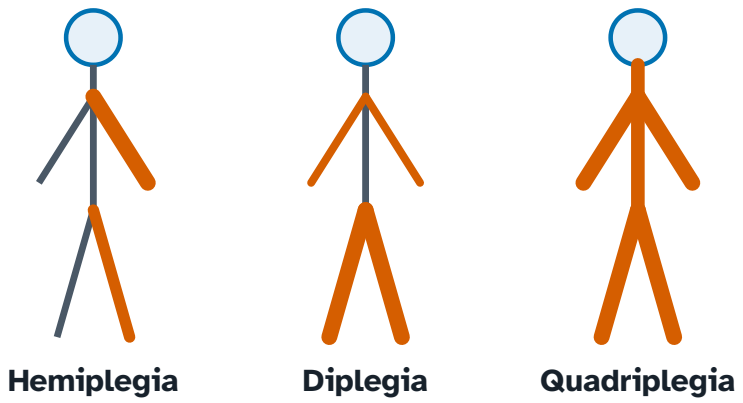
2 Classification and grading

By type of movement disorder and distribution (7 marks with causes in S24; 3 marks in S26)

Type	Main features	Usual brain lesion
Spastic (the commonest)	Increased tone that resists fast stretch, brisk reflexes, weakness, poor selective control. By distribution: **hemiplegia** (one side, arm more than leg), **diplegia** (legs much more than arms), **quadriplegia** (all four limbs and trunk)	Diplegia: periventricular leukomalacia in preterm babies. Hemiplegia: stroke or bleed on one side. Quadriplegia: widespread damage
Dyskinetic (athetoid, dystonic)	Involuntary movements (writhing athetosis, jerky chorea, twisting dystonic postures), fluctuating tone, poor control of head and trunk	Basal ganglia and thalamus: birth asphyxia at term, kernicterus
Ataxic	Low tone, poor balance and coordination, wide-based gait, intention tremor	Cerebellum
Hypotonic (3 marks in S26)	Floppy, poor head and trunk control, hypermobile joints, very delayed milestones; often changes to dyskinetic or ataxic later	Variable

Type	Main features	Usual brain lesion
Mixed	Usually spastic with dyskinetic features	Several areas

Distribution of spastic cerebral palsy



Thick orange lines show the affected parts; thicker means more affected. Hemiplegia: arm more than leg. Diplegia: legs much more than arms. Quadriplegia: all four limbs and the trunk.

By function: GMFCS (ages 6–12, expanded and revised 2007)

Level	Description
I	Walks without limitations; runs and jumps, with reduced speed, balance and coordination
II	Walks with limitations (long distances, uneven ground, crowds); needs a rail on stairs
III	Walks using a hand-held mobility device (walker, crutches); wheelchair for long distances
IV	Self-mobility with limitations; transported or uses powered mobility in most settings
V	Transported in a manual wheelchair; limited head and trunk control

Hand function is graded with the **Manual Ability Classification System** (MACS, levels I–V), and communication with the CFCS.

3 Causes and risk factors

Asked for 3 marks in W16, W17, S18, W19 and W22, and for 7 marks with the classification in S24.

Period	Causes and risk factors
Prenatal (before labour)	Maternal infections (rubella, cytomegalovirus, toxoplasmosis, Zika); multiple pregnancy; placental problems and poor fetal growth; maternal pre-eclampsia, thyroid disease, epilepsy; brain malformations and genetic disorders; alcohol and some drugs; Rh incompatibility
Perinatal (around birth)	**Prematurity** (especially before 32 weeks) and low birth weight; **birth asphyxia** (hypoxic-ischaemic encephalopathy, "delayed cry"); difficult or prolonged labour; severe neonatal jaundice (kernicterus); neonatal infection; neonatal stroke
Postnatal (first 2 years)	Meningitis and encephalitis; head injury; near drowning; severe hypoglycaemia; very high bilirubin

For a 3-mark answer give three from the right period, for example prenatal: maternal infection, multiple pregnancy, placental insufficiency.

4 Pathophysiology to impairments

1. A **non-progressive brain lesion** in the developing brain disturbs the control of movement and posture.

2. **Primary impairments:** abnormal tone (spastic, fluctuating or low), weakness, poor selective motor control, poor balance and postural control, sensory and perceptual problems.
3. **Secondary impairments develop with growth:** spastic, weak muscles do not lengthen as fast as the bones grow, so **contractures** appear (calf, hamstrings, hip adductors and flexors); bones twist (femoral anteversion, tibial torsion); the hips migrate out of the socket; the spine curves (scoliosis).
4. **Compensations** (tertiary): toe walking, crouching, W-sitting, using one hand only.
5. **Associated problems:** epilepsy, intellectual disability, vision and hearing loss, speech and swallowing difficulty, drooling, constipation, low bone density.

? Why do contractures get worse when the brain lesion does not?

> Muscle grows in length in response to stretch during normal daily movement. A spastic, weak muscle is rarely stretched through its full range, so it fails to keep up as the bone grows, especially in growth spurts. This is why stretching through active use, standing, casting and orthoses must continue through childhood.

Recall check. Name three primary and three secondary impairments in spastic diplegia.

5 Clinical features by type

Posture and movement problems: spastic diplegia (7 marks in S26)

- Legs much more affected than arms; hips flexed, **adducted** and internally rotated; knees flexed; ankles in **equinus**.
- **Sitting:** on the sacrum with posterior pelvic tilt and a rounded back (tight hamstrings), or W-sitting for stability.
- **Standing and walking: scissoring**, toe walking, crouch or jump knee pattern, in-toeing; arms held up for balance (high guard).
- Often born preterm with periventricular leukomalacia; intelligence frequently normal.

Posture and movement problems: athetoid (dyskinetic) CP (7 marks in S26)

- **Involuntary, writhing movements** that increase with effort, excitement and attempts to move; **fluctuating tone** from low to high.
- Poor head control and poor **midline** control; the asymmetrical tonic neck reflex persists, turning the head pulls the body into asymmetry.
- **Poor proximal stability** (little co-contraction), so the child cannot hold a position and grades movement poorly; sudden extensor thrusts push the child out of the chair.
- Drooling, unclear speech and feeding difficulty; hearing loss after kernicterus; intelligence often normal.
- Walking, if achieved, is late, unsteady and wide-based, with the arms held high.

Spastic hemiplegia

- One side, arm more than leg; fisted hand with the thumb in the palm; equinus or equinovarus foot.
- A clear hand preference before 12 months of age is an early warning sign.
- Learned non-use of the affected hand; seizures more common.

Spastic quadriplegia

- All four limbs and the trunk; poor head control; often intellectual disability, epilepsy and feeding problems.
- **Spinal deviations (3 marks in W25):** neuromuscular **scoliosis** (long C-shaped curve), **pelvic obliquity**, kyphosis, increased lumbar lordosis with hip flexion contracture, and windswept hips.

Impairments in hypotonic CP (3 marks in S26)

Low tone with weak antigravity muscles; poor head and trunk control; hypermobile joints; "frog-leg" lying posture; very delayed milestones; shallow breathing and a weak cough; feeding difficulty; low endurance.

6 Investigations and early diagnosis

- **Early diagnosis (Novak et al., 2017):** CP can be diagnosed or flagged as "high risk" **before 6 months' corrected age**. Before 5 months: MRI plus **Prechtl's General Movements Assessment** (absent "fidgety" movements). After 5 months: MRI plus the **Hammersmith Infant Neurological Examination** (HINE). Early diagnosis allows early intervention while the brain is most plastic.
- **MRI:** periventricular leukomalacia, infarcts, basal ganglia injury, malformations.
- **Hip X-ray** (anteroposterior pelvis) for hip surveillance: the **migration percentage** shows how much of the femoral head lies outside the socket.
- EEG for seizures; hearing and vision tests; genetic and metabolic tests when the history or course is unusual.

Differential diagnosis (part of the syllabus): progressive conditions must be excluded, such as muscular dystrophy, spinal muscular atrophy, metabolic and neurodegenerative disorders, brain tumours and hereditary spastic paraplegia. Loss of skills, family history or a worsening course points away from CP.

7 Medical and surgical management

- **Spasticity:** botulinum toxin injections into overactive muscles, followed by stretching, casting, orthoses and task practice; oral baclofen; intrathecal baclofen pump for severe generalised spasticity; selective dorsal rhizotomy for selected children with spastic diplegia.
- **Orthopaedic surgery:** tendon lengthening (Achilles, hamstrings, adductors); bone surgery for torsion; several procedures in one sitting (single-event multilevel surgery) in walking children; hip surgery for displacement; spinal fusion for severe scoliosis.
- **Other:** antiepileptic drugs; nutrition support, sometimes gastrostomy; drugs for drooling and dystonia.

8 Complications and prevention

Complication	What to look for	Physio role
Hip displacement	Reduced abduction, pain, leg-length difference; worst in GMFCS IV–V	Hip surveillance from about age 2 or diagnosis; refer if migration is above 30% or abduction below 30° (AACPDM); positioning in abduction; standing programme
Contractures	Calf, hamstrings, hip flexors and adductors, elbow and wrist flexors	Active use, standing, casting and orthoses; passive stretching alone is not enough
Scoliosis	Curve, pelvic obliquity, sitting imbalance	Supportive seating, positioning, trunk strengthening; refer for bracing or surgery
Low bone density and fractures	Non-walkers; pain after handling	Supported standing; gentle handling; calcium and vitamin D as advised
Chest infection and aspiration	Coughing with feeds, fever	Upright feeding positions, chest care
Pain	Irritability, hip or back pain	Positioning, find and treat the cause

9 PT assessment

- **History:** pregnancy, birth (gestation, weight, delayed cry, NICU stay, jaundice), milestones, seizures, feeding, vision, hearing, school, family's goals, home (floor living, carrying, toilet).
- **Observation:** posture in lying, sitting and standing; involuntary movements; how the child moves and plays.
- **Gross motor:** GMFCS level; **Gross Motor Function Measure** (GMFM-66 or GMFM-88); milestones; head and trunk control.
- **Tone:** Modified Ashworth Scale; Modified Tardieu Scale (catch angle at fast speed against full range at slow speed); primitive reflexes and postural reactions.

- **Range and contractures:** Thomas test (hip flexors), hip abduction, popliteal angle (hamstrings), ankle dorsiflexion with knee straight and bent (calf); spine.
- **Strength and selective motor control.**
- **Balance and gait:** sitting and standing balance; observational gait analysis (scissoring, equinus, crouch); walking aids.
- **Hand function:** MACS level; grasp, release, bimanual use.
- **Others:** cognition, communication, vision, hearing, behaviour; equipment; parents' understanding and goals.

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case. For a child, participation means play, school and family life.

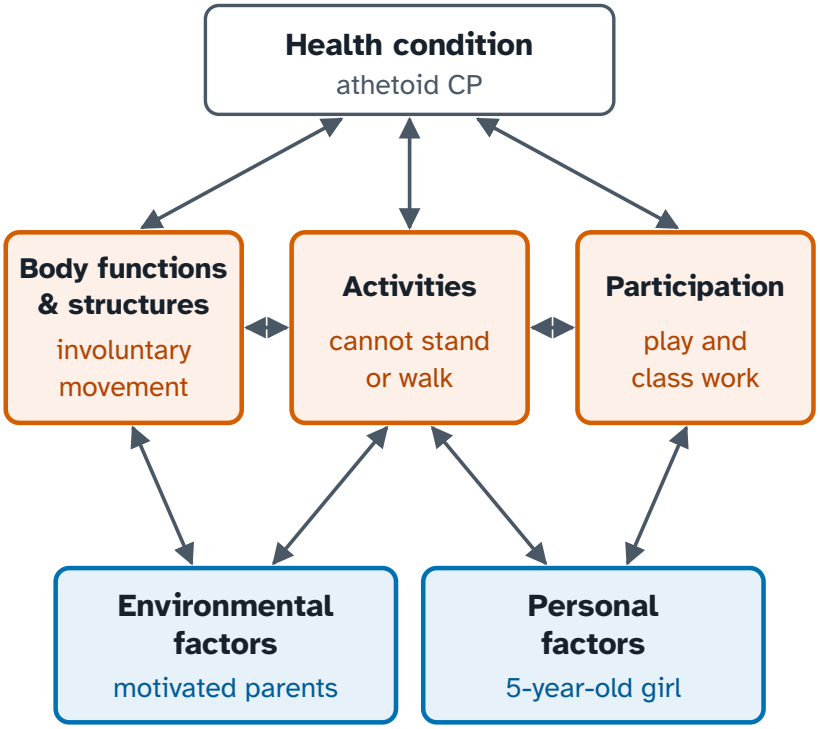
Worked example from a real paper (W22 wording, 5 + 7 + 3 marks; also asked in W19, S21 and W24)

> "Write ICF with clinical reasoning of a 5 year old athetoid cerebral palsy girl, she is able to maintain independent sitting but is unable to stand. She attends school regularly. Parents are financially sound and motivated for therapy. Formulate the short term goals with physiotherapy management."

Spot the ICF clues before you write

athetoid means involuntary movements, fluctuating tone and poor proximal stability. sits independently but cannot stand places her near GMFCS III–IV. attends school regularly is participation (and a facilitator). financially sound and motivated parents are strong facilitators: plan a home programme and equipment.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Dyskinetic (athetoid) cerebral palsy, 5 years	Basal ganglia lesion: involuntary movement, fluctuating tone

ICF component	From this case	Clinical reasoning
Body function and structure impairments	• Involuntary writhing movements of limbs and face • Fluctuating tone; poor co-contraction around hips, trunk and shoulders • Poor head and midline control; persistent asymmetrical tonic neck reflex • Poor standing balance and weight-bearing through the legs • Possible drooling, unclear speech, hearing loss • Risk of hip and spine deformity and of contractures	• Without proximal stability she cannot hold the trunk upright over the feet • Involuntary movement grows with effort, so movement must be graded • Persistent primitive reflexes block symmetrical postures
Activity limitations	Cannot stand or walk; difficulty with transfers, reaching and grasping, writing, feeding herself, toileting	Standing needs stable hips and trunk; hand use needs a stable base
Participation restrictions	Limited play with other children, classroom and playground activities, outings	Depends on adults to move around
Environmental factors	**Facilitators:** motivated, financially sound parents; regular school. **Barriers:** school access (stairs, toilets), lack of adapted seating; ask about transport.	Facilitators allow equipment, home programme and school adaptations
Personal factors	5-year-old girl; likely normal intelligence; motivated to play and learn	Play-based, goal-directed practice suits her

One-line functional diagnosis

A 5-year-old girl with athetoid cerebral palsy, with involuntary movements, fluctuating tone and poor proximal stability, who sits independently but cannot stand or walk, limiting mobility, self-care and play at school. Motivated parents and regular schooling are facilitators.

Practice cases from the papers: S18 4-year-old preterm boy with spastic diplegia and scissoring (gait training); S22 2-year-old hypotonic girl from a poor family (trunk control and ambulation); S25 7-year-old with spastic hemiplegia; W25 7-year-old with jerky movements and fluctuating tone (dyskinetic); S26 3-year-old with periventricular leukomalacia and scissoring.

Recall check. Fill the ICF for the S18 4-year-old with spastic diplegia who wants to go to school.

11 Problem list and goals

Goals for the 5-year-old athetoid girl (discuss them with the parents)

Problem	Short-term (6–8 weeks)	Long-term (6–12 months)
Head and trunk control	Holds head in midline while reaching in sitting	Sits on a school bench with both hands free for 15 minutes
Standing	Stands in a standing frame 60 minutes a day; bears weight evenly	Stands holding a support for 1 minute
Mobility	Moves from sitting to standing with help at a support	Walks short distances with a walker in the classroom
Hand use	Grasps and releases large objects at midline	Feeds herself with an adapted spoon
Deformity	Hip and ankle range maintained	No hip displacement on surveillance X-ray

12 PT management

Principles: **family-centred**, **goal-directed** (goals the child and family choose), **active, task-specific practice** through play, started **early**, with a **home programme** and prevention of deformity.

What the evidence supports (Novak et al., 2019 review)

Rated effective ("green"): constraint-induced movement therapy and bimanual training for the hand, goal-directed and task-specific training, treadmill training with or without body-weight support, strength training, fitness training, mobility training, weight-bearing, casting, hippotherapy, environmental enrichment, home programmes, botulinum toxin with therapy, and hip surveillance. Rated not effective ("red") for motor goals: neurodevelopmental therapy in its original passive form, and sensory integration.

Classic answer vs current evidence: NDT (Bobath)

Classic	Current	Write in the exam
Inhibit abnormal tone and reflexes with handling (reflex-inhibiting patterns, key points of control) before facilitating normal movement.	Passive NDT does not improve motor function. Use active, goal-directed task practice; handling only to set the child up for active movement.	"Goal-directed, task-specific active training through play, with handling and positioning to support it, and a home programme."

By stage of childhood

- **Infancy (early intervention):** parent coaching; positioning and carrying; head and trunk control in prone, sitting and supported standing; reaching and play; feeding position; environmental enrichment.
- **Pre-school:** sitting, standing and walking practice; standing frames; orthoses; walkers; play and transfers; casting or botulinum toxin when spasticity limits progress.
- **School age:** strength and fitness training, treadmill walking, CIMT and bimanual training for hemiplegia, gait training, school adaptations and sport.
- **Adolescence:** fitness and weight control, managing contractures through growth spurts, independence and transition to adult services.

Management for the athetoid (dyskinetic) child

- **Proximal stability:** weight-bearing through the arms (prone on elbows, four-point kneeling) and legs (kneeling, standing frame); approximation through the shoulders and hips; holding positions against gentle resistance (co-contraction).
- **Midline control:** head and hands in midline, symmetrical sitting; play with both hands together.
- **Graded movement:** slow, small-range movements with feedback; avoid excitement and effort that increase involuntary movement.
- **Seating:** stable chair with a pelvic belt, foot support and a table; hips, knees and ankles at about 90°.
- Standing frame, walker or gait trainer when ready; communication and feeding with the team.

Gait training in spastic diplegia (S18 long case)

- Reduce the barriers: stretch and cast tight calves and hamstrings; botulinum toxin to adductors or calves when indicated; strengthen hip extensors and abductors, quadriceps and calf.
- Correct the alignment: **AFO** for equinus; ground-reaction AFO for crouch; abduction during standing practice to stop scissoring.
- Practise walking: standing balance and weight shift, stepping with a wide base, walking in the parallel bars, then with a walker (posterior walkers encourage an upright posture), treadmill training with or without body-weight support, then outdoors and in school.
- Fitness and strength for endurance; regular review of range in growth spurts.

Trunk control and ambulation in hypotonic CP (S22 long case)

- Head and trunk extension in prone over a wedge or ball; supported sitting with gradually less support; reaching in sitting; weight shift.

- Joint compression (approximation) and weight-bearing to build co-contraction; quick-stretch and tapping over trunk extensors; bouncing on a ball.
- Supported standing in a frame; then cruising along furniture and walking with a walker.
- Parent education with low-cost equipment (rolled towels, a corner seat, a wooden standing box), suited to a poor family.

Carrying positions (7 marks in S18)

Child	How to carry	Rationale
Spastic, legs stiff and crossed (diplegia, quadriplegia)	Facing you, legs apart around your waist or astride your hip, arms forward over your shoulders	Hip flexion and abduction break the extensor and scissoring pattern
Child who arches backwards	Curled forward: head, shoulders, hips and knees bent, arms forward	Flexion reduces extensor thrust
Athetoid child	Arms and legs kept together and bent, close to your body; hands together in front; head in midline	Gives stability and midline control and limits involuntary movement
Floppy (hypotonic) child	Back against your chest, facing outwards, your arm under the hips and across the trunk; good head support	Supports the trunk and lets the child look around and lift the head
Hemiplegic child	Affected arm forward and the affected side lengthened over your arm	Counters the flexed, shortened affected side

Foot orthoses for children with CP (7 marks in S22)

Orthosis	What it does	Used for
Solid AFO	Holds the ankle at 90° in stance and swing	Equinus with poor control; standing in non-walkers; night stretch (resting AFO)
Hinged (articulated) AFO with plantarflexion stop	Allows dorsiflexion in stance, blocks plantarflexion	Walking children with enough calf length; more normal roll-over
Ground-reaction AFO	Anterior shell pushes the knee back in stance	Crouch gait (with good knee and hip range)
Posterior leaf spring AFO	Flexible; lifts the foot in swing	Mild foot drop with little spasticity
Supramalleolar orthosis (SMO)	Controls the subtalar joint and forefoot	Flat, valgus feet with mild instability
Dynamic AFO (thin, moulded)	Contoured footplate supports the arches and toes	Mild tone problems and balance in standing

Home-care advice to parents (W17 long case)

- Positioning through the day: side lying, prone over a wedge, supported sitting; avoid lying always on the back and avoid W-sitting.
- Carrying and handling as above; feeding upright with the head slightly forward; dressing the stiffer limb first and undressing it last.
- Daily play that practises head control, reaching, rolling and sitting; standing frame time.
- No baby walkers; regular review of hips, splints and skin; follow the home programme.

By type (summary)

Type	Key physiotherapy
Spastic diplegia	Strength, range through casting and AFOs, gait training, treadmill, fitness
Spastic hemiplegia	CIMT or bimanual training for the hand; AFO; symmetrical weight-bearing

Type	Key physiotherapy
Spastic quadriplegia	Positioning and seating, standing programme, hip and spine surveillance, chest care, communication of needs
Dyskinetic	Proximal stability, midline control, graded movement, stable seating
Ataxic	Balance training, wide-base activities, weighted walkers, slow controlled practice
Hypotonic	Head and trunk control, weight-bearing and co-contraction, supported standing

Supported standing (dose)

A review of standing programmes (Paleg et al., 2013) suggests standing **60–90 minutes a day, 5 days a week** to improve bone density; **60 minutes a day** with the hips in 30–60° of total abduction for hip stability; 45–60 minutes a day for hip, knee and ankle range; and 30–45 minutes a day for spasticity.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Weakness	Progressive strength training with body weight, free weights or functional tasks (sit-to-stand, step-ups): **2–4 times a week** on non-consecutive days, **1–3 sets** of **6–15 repetitions** at **50–85% of 1RM**, for at least **12–16 weeks**	Weak muscles limit walking, stairs and transfers, and strength grows only with progressive load
Poor fitness	Cycling, treadmill walking, swimming or active play: at least **20 minutes**, **2–3 times a week**, above **60% of peak heart rate**, for at least **8–16 weeks**	Children with CP are less fit than their peers, so they tire early at school and play
Short muscles and hip migration	Standing frame with the hips in abduction; casting; AFOs; active use through full range. Standing doses are in the box above	Muscles lengthen only under regular stretch during growth, and standing in abduction keeps the femoral head in the socket
Poor hand use (hemiplegia)	Constraint-induced movement therapy or bimanual training, given as play with home practice: about **40 hours** of practice in total to improve hand function	Intensive, repeated use drives motor learning and overcomes learned non-use
Poor walking	Treadmill training with or without body-weight support; overground walking practice on the child's own goals	Practice improves the task that is practised
Low daily activity	**60 minutes** of moderate-to-vigorous activity at least **5 days a week**; sitting **under 2 hours a day**, or a **2-minute** break every **30–60 minutes**	Daily activity builds fitness and bone and lowers long-term health risks
Poor head and trunk control (hypotonic, dyskinetic)	Weight-bearing and holding practice in prone on elbows, supported sitting, kneeling and standing, held for longer each week during play	Proximal stability is the base for hand use, sitting and standing

Strength, fitness and activity doses: Verschuren et al. (2016) recommendations for people with CP. Hand practice dose: Jackman et al. (2020) review.

Recall check. List five interventions rated effective for CP, and describe how to carry a spastic child and an athetoid child.

13 Precautions and contraindications

Take care

- Seizures: know the child's seizure plan; avoid triggers such as fatigue.
- Low bone density: gentle handling and stretching; no forceful range in non-walking children.
- Hips: do not force abduction; report pain or loss of abduction.
- Swallowing: feed upright; no feeding in lying.
- Skin under casts and orthoses: check daily for red marks.
- After botulinum toxin: increase stretching and practice in the following weeks; watch for unusual weakness.
- Fatigue: short, frequent, playful sessions.

14 Education and home programme

- Explain that CP does not get worse, but deformities can be prevented with daily practice and positioning.
- Teach handling, carrying, feeding, bathing and dressing positions; give a short written or picture home programme the family can fit into daily routines.
- Low-cost equipment: a corner seat, wedge, standing box or frame, adapted spoon and cup.
- Encourage inclusive schooling, play with other children and sport; advise the school on seating, toilets and access.
- Help the family get a disability certificate and benefits; support the parents' own wellbeing.

15 Prognosis and discharge criteria

- The GMFCS level predicts long-term gross motor function. Children reach about 90% of their gross motor potential by about 5 years of age, and those at more severe levels reach it earlier (Rosenbaum et al., 2002).
- Sitting independently by 2 years of age is a strong sign of later walking: in a classic study, every child who sat by 2 years walked eventually (Molnar and Gordon, 1976).
- Associated problems (epilepsy, intellectual disability, feeding difficulty) and severe quadriplegia shorten life expectancy; most children with milder CP live into adulthood.
- Therapy continues in blocks through childhood around goals; discharge from an episode when its goals are met and the family manages the home programme.

16 Exam corner

Past questions (MUHS, 2016–2026)

W19 Q3(b), S21 Q4(a), W22 Q3(a), W24 Q3(a)

The worked example in section 10 (asked four times).

S26 Q3(a)

3-year-old with difficulty standing and walking, scissoring gait, MRI showing periventricular leukomalacia, at a normal school. Diagnosis; short- and long-term goals with management.

W25 Q3(b), 5 + 10 marks

7-year-old boy with difficulty sitting and standing; he rolls to both sides but with many jerky movements, his tone swings between high and low, and he sits for a while and then falls into trunk flexion. Functional diagnosis with clinical reasoning; short- and long-term goals with management.

S25 Q4(a)

7-year-old with spastic hemiplegic CP, delayed milestones, difficulty walking and using the right hand. ICF and management.

S22 Q4(a), 7 + 8 marks

2-year-old hypotonic girl who rolls and sits with moderate support; uneducated, poor parents. Impairments; management for trunk control and ambulation.

S20 Q3(b), 5 + 10 marks

3-year-old girl born at 7 months with a delayed cry, increased leg tone and a crossing gait. ICF; management.

W18 Q4(b), 3 + 7 + 5 marks

2-year-old with spastic diplegia, leg spasticity grade 2, arms grade 1, difficulty standing and gripping. ICF; goals; management.

S18 Q4(b), 6 + 9 marks

4-year-old preterm boy with scissoring and severe adductor and calf spasticity who wants to go to school. Goals; management with gait training.

W17 Q4(b), 5 + 10 marks

2-year-old with spastic diplegia, no head control, breathless and vomiting. Home-care advice; goals and management.

S17 Q3(b)

5-year-old spastic diplegic from a well-educated family, at a normal school, who needs support to stand and walk. Assessment with ICF; treatment strategies.

7-mark questions

Carrying positions with rationale (S18); physiotherapy for spastic CP (S19); foot orthoses for CP (S22); classification and causes (S24); posture and movement problems in athetoid CP and in spastic diplegic CP (S26).

3-mark questions

Prenatal causes (W16, S18); causes (W17); aetiological factors (W19, W22); spinal deviations in quadriplegic CP (W25); impairments in hypotonic CP, classification (S26).

Pages for this topic: developmental neurology (p. 267), orthoses (p. 249), neurotherapeutic approaches (p. 209), wheelchairs and seating (p. 257).

Keywords to underline

non-progressive lesion · developing brain · periventricular leukomalacia · GMFCS · scissoring · equinus · crouch · fluctuating tone · proximal stability · goal-directed training · home programme · hip surveillance · standing frame

Answer skeleton: any CP long case

1. Restate: age, type and distribution, birth history clue, GMFCS level you estimate, school and family.
2. **[ICF, 5]** ICF diagram and table with clinical reasoning; one-line functional diagnosis.
3. **[Goals, 3–5]** Short-term (6–8 weeks) and long-term (6–12 months), agreed with the parents.
4. **[Management, 7–10]** Principles (family-centred, goal-directed, active play-based practice); management for the type (gait, trunk control, proximal stability, hand); orthoses and equipment; standing programme and hip surveillance; home programme and parent education; precautions.

Draw these

- The three distributions of spastic CP (section 2)
- The ICF model filled from the case
- A carrying position for a spastic child

Viva questions

1. Why is CP called non-progressive when deformities get worse?
2. Which type of CP follows periventricular leukomalacia?
3. What does GMFCS level III mean?
4. How would you carry an athetoid child, and why?
5. When do you refer a child for hip displacement?
6. Which orthosis would you choose for crouch gait?

Linked practical spots

- AFO types (solid, hinged, ground-reaction); GMFCS and Modified Ashworth as scale spots.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Enumerate three prenatal causes of cerebral palsy.**

Answer
Maternal infection (rubella, cytomegalovirus, toxoplasmosis); multiple pregnancy; placental insufficiency with poor fetal growth. (Also maternal pre-eclampsia or thyroid disease, brain malformations, Rh incompatibility.)

3 m **Enumerate three spinal deviations in a child with quadriplegic CP.**

Answer
Neuromuscular scoliosis (long C-curve) with pelvic obliquity; kyphosis; increased lumbar lordosis (with hip flexion contracture). Windswept hips often go with them.

7 m **Explain the posture and movement problems of a child with athetoid cerebral palsy.**

Answer
Involuntary writhing movements that increase with effort and excitement; fluctuating tone; poor head and midline control with persistent ATNR; poor proximal stability and co-contraction, so the child cannot hold positions or grade movements; sudden extensor thrusts; drooling, unclear speech, feeding difficulty, possible hearing loss; late, unsteady, wide-based walking with arms held high; intelligence often normal.

MCQ **A child walks in a crouch with good hip and knee range. The most suitable orthosis is:**

A. Posterior leaf spring AFO
B. Ground-reaction AFO
C. Supramalleolar orthosis
D. Hinged AFO without a stop

Answer
B. The anterior shell of a ground-reaction AFO pushes the knee back in stance.

MCQ **Which approach is rated not effective for improving motor function in CP?**

A. Goal-directed training

B. Constraint-induced movement therapy

C. Passive neurodevelopmental therapy

D. Treadmill training

Answer

C. Passive NDT was rated red in the 2019 evidence review; the others are green.

MCQ **Which tool classifies the gross motor function of a child with CP?**

A. MACS

B. GMFCS

C. CFCS

D. Modified Ashworth Scale

Answer

B. GMFCS grades gross motor function (levels I–V). MACS grades hand use and CFCS grades communication; the Ashworth scale measures tone.

LAQ **A 2-year-old girl with hypotonic CP rolls to either side and comes to sitting with moderate support. Her parents are uneducated and poor. Describe her impairments with clinical reasoning, and the physiotherapy for trunk control and ambulation. (7 + 8, S22)**

Answer

Impairments with reasoning (7): low tone and weak antigravity muscles, so poor head and trunk control; poor co-contraction around hips and shoulders, so she cannot hold sitting; hypermobile joints; delayed milestones; poor sitting balance and protective reactions; possible feeding and breathing difficulty; risk of hip and foot deformity from lying in a frog-leg posture.

Physiotherapy (8): prone over a wedge or roll for head and trunk extension; supported sitting with less support each week, reaching in sitting and weight shift; approximation through shoulders and hips, and bouncing on a ball, for co-contraction; kneeling and supported standing in a low-cost wooden standing box; then cruising along furniture and a walker. Teach the parents with pictures, using a rolled towel, a corner seat and daily play routines. Protect hypermobile joints and do not overstretch.

Last recall check. Draw the three distributions, write the GMFCS levels, and list the orthoses with their uses.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Rosenbaum P, et al. A report: the definition and classification of cerebral palsy April 2006. *Dev Med Child Neurol* 2007;49(Suppl 109):8–14.
3. Palisano RJ, et al. Content validity of the expanded and revised Gross Motor Function Classification System. *Dev Med Child Neurol* 2008;50:744–50.
4. Novak I, et al. Early, accurate diagnosis and early intervention in cerebral palsy. *JAMA Pediatr* 2017;171:897–907.
5. Novak I, et al. State of the evidence traffic lights 2019: systematic review of interventions for preventing and treating children with cerebral palsy. *Curr Neurol Neurosci Rep* 2020;20:3.
6. American Academy for Cerebral Palsy and Developmental Medicine. Hip surveillance in cerebral palsy, care pathway.
7. Paleg GS, Smith BA, Glickman LB. Systematic review and evidence-based clinical recommendations for dosing of pediatric supported standing programs. *Pediatr Phys Ther* 2013;25:232–47.
8. Chauhan A, et al. Prevalence of cerebral palsy in Indian children: a systematic review and meta-analysis. *Indian J Pediatr* 2019;86:1124–30.

9. McIntyre S, et al. Global prevalence of cerebral palsy: a systematic analysis. *Dev Med Child Neurol* 2022;64:1494–506.
10. Verschuren O, et al. Exercise and physical activity recommendations for people with cerebral palsy. *Dev Med Child Neurol* 2016;58:798–808.
11. Jackman M, et al. What is the threshold dose of upper limb training for children with cerebral palsy to improve function? A systematic review. *Aust Occup Ther J* 2020;67:269–80.
12. Rosenbaum PL, et al. Prognosis for gross motor function in cerebral palsy: creation of motor development curves. *JAMA* 2002;288:1357–63.
13. Molnar GE, Gordon SU. Cerebral palsy: predictive value of selected clinical signs for early prognostication of motor function. *Arch Phys Med Rehabil* 1976;57:153–8.
14. Structure and classical content: Levitt, *Treatment of Cerebral Palsy and Motor Delay*; Finnie, *Handling the Young Child with Cerebral Palsy at Home*; Shepherd, *Physiotherapy in Paediatrics*; Tecklin, *Pediatric Physical Therapy*.

5B(2) Down's syndrome: physical traits, neurological impairments and physiotherapy

> Down's syndrome was a 7-mark question three times: physiotherapy twice (S20, W25) and the physical traits and neurological impairments once (S23). Low tone, lax joints and slow learning delay every motor milestone, so physiotherapy practises the skills themselves (standing, stepping, walking, balance), builds strength and fitness, and keeps the neck and heart safe.

Syllabus	Ch 5B(2) Down's syndrome
Priority	**Tier 2.** In 3 of 20 papers (2016–2026): 3 seven-mark questions.
Asked as	7m S20 · S23 · W25
Usual split	7 (physical traits 3 and neurological impairments 4 in S23; or a short intro, assessment and physiotherapy by age)
Checked against	AAP health supervision guideline for children and adolescents with Down syndrome (Bull et al., 2022); treadmill training trial (Ulrich et al., 2001); progressive resistance training trial (Shields and Taylor, 2010); gross motor schedule for Down syndrome (Winders et al., 2019). Full list under Sources (p. 297). Content checked 20 October 2026.

1 Definition

- **Down's syndrome** is a genetic condition caused by an extra copy of chromosome 21 (trisomy 21). It causes typical physical features, intellectual disability, low muscle tone, delayed development and a higher risk of several medical problems.
- It is the commonest chromosomal cause of intellectual disability.

2 Classification and grading

Genetic types (AAP, 2022)

Type	Share	Point to know
Non-disjunction (free trisomy 21)	About 96%	Chromosomes fail to separate, usually in the egg; risk rises with maternal age
Translocation	3–4%	Extra 21 attached to another chromosome; a parent may carry it, so the recurrence risk is higher
Mosaicism	1–2%	Only some cells are affected; often milder

Intellectual disability is usually moderate (IQ 35–50), sometimes mild (50–70) and occasionally severe (20–35).

3 Causes and risk factors

- Older maternal age (risk rises steeply after 35); a previous child with Down's syndrome; a parent carrying a balanced translocation.
- **India:** about 1 in 1000 live births (estimates range from about 1 in 800 to 1 in 1150).
- **Screening:** first- and second-trimester blood tests with nuchal translucency on ultrasound, or cell-free DNA testing; diagnosis by chorionic villus sampling or amniocentesis.

4 Pathophysiology to impairments

1. The extra chromosome 21 changes brain development: the brain, and especially the cerebellum and brainstem, is smaller, with fewer neurons and connections. This gives low tone, poor postural control, slow reactions and slow

motor learning.

2. Collagen differences make ligaments lax: hypermobile joints, flat feet, unstable patellae and hips, and a risk of atlantoaxial instability.
3. Low tone, lax joints and weak muscles together delay every motor milestone, and children often widen their base and lock their joints for stability.
4. Heart defects, sleep apnoea and frequent chest infections reduce exercise tolerance; hearing and vision problems make learning and balance harder.
5. Low activity and a low metabolic rate lead to obesity, which further limits movement.

Recall check. Name three reasons why a child with Down's syndrome walks late.

5 Clinical features

Physical traits (S23)

- **Head and face:** small, flat-backed (brachycephalic) head; flat facial profile and nasal bridge; upward-slanting eyes with epicanthic folds; white spots on the iris (Brushfield spots); small mouth with a protruding tongue; small, low-set ears; excess skin at the back of the neck.
- **Hands:** short, broad hands; a single palmar crease; a short, curved little finger (clinodactyly).
- **Feet:** a wide gap between the first and second toes with a deep groove on the sole; flat feet.
- **Body:** short stature, low tone, hypermobile joints.

Neurological impairments (S23)

- **Hypotonia** (low tone) and reduced muscle strength.
- **Intellectual disability**, usually moderate; speech and language delay.
- **Delayed motor milestones:** on average sitting at about 10 months, crawling at about 14 months and walking at about 26 months; the slowest children walk at about 40 months (Winders et al., 2019).
- **Poor balance and postural control,** slow reaction and movement times, difficulty with motor planning and sequencing.
- **Gait:** wide base, out-turned feet, knees locked in extension, little heel strike.
- Hearing loss (75%) and vision problems (60–80%); seizures (1–13%); autism in some (7–19%); early Alzheimer's disease in adults.

Associated medical problems (AAP, 2022)

Problem	How common	Physio relevance
Congenital heart disease (often atrioventricular septal defect)	40–50%	Know the heart status before exercise; watch breathlessness and colour
Obstructive sleep apnoea	50–79%	Daytime tiredness reduces participation
Thyroid disease	Up to 50% by adulthood	Hypothyroidism adds to tiredness and weight gain
Hip problems	2–8%	Check hip stability
Symptomatic atlantoaxial instability	1–2%	Know the warning signs (section 13)
Gastrointestinal atresias	12%	Early surgery and feeding difficulties

6 Investigations

- **Karyotype** to confirm the diagnosis and type (needed for genetic counselling).
- Echocardiogram; hearing and vision tests; thyroid tests; blood count.

- Cervical spine X-rays in the neutral position only if there are symptoms of atlantoaxial instability; routine screening of children without symptoms is not recommended (AAP, 2022).
- **Developmental assessment:** Alberta Infant Motor Scale (infants), Gross Motor Function Measure, Peabody Developmental Motor Scales.

7 Medical and surgical management

- Heart surgery in infancy when needed; treatment of thyroid, ear, eye and sleep problems.
- **Early intervention** from infancy: physiotherapy, occupational therapy and speech therapy, with the parents as the main therapists at home.
- Inclusive education and support for families.
- Surgical fixation of the upper neck for symptomatic atlantoaxial instability.

8 Complications and prevention

Complication	Physio role
Atlantoaxial instability with cord compression	Know the signs; avoid trampolines; advise about contact sports and gymnastics; support the neck in handling
Flat feet, knee and patellar problems	Foot orthoses (supramalleolar or insoles) when needed; strengthen around the knees
Obesity and low fitness	Daily active play, sport, aerobic and strength training
Chest infections	Active play, airway clearance when unwell
Early ageing and dementia (adults)	Keep active; falls prevention

9 PT assessment

- History: heart condition and surgery, hearing and vision, breathing and sleep, neck symptoms, milestones, family and play.
- Gross motor milestones and quality of movement; standardised tests (Alberta Infant Motor Scale, Gross Motor Function Measure, Peabody).
- Tone (hypotonia), joint laxity (Beighton score in older children), range of motion, strength.
- Postural control in sitting and standing; balance; gait (base, foot position, knee locking).
- Foot posture, hip stability, spine.
- Endurance (breathlessness, colour, heart rate), attention, understanding and communication.

10 PT diagnosis by ICF

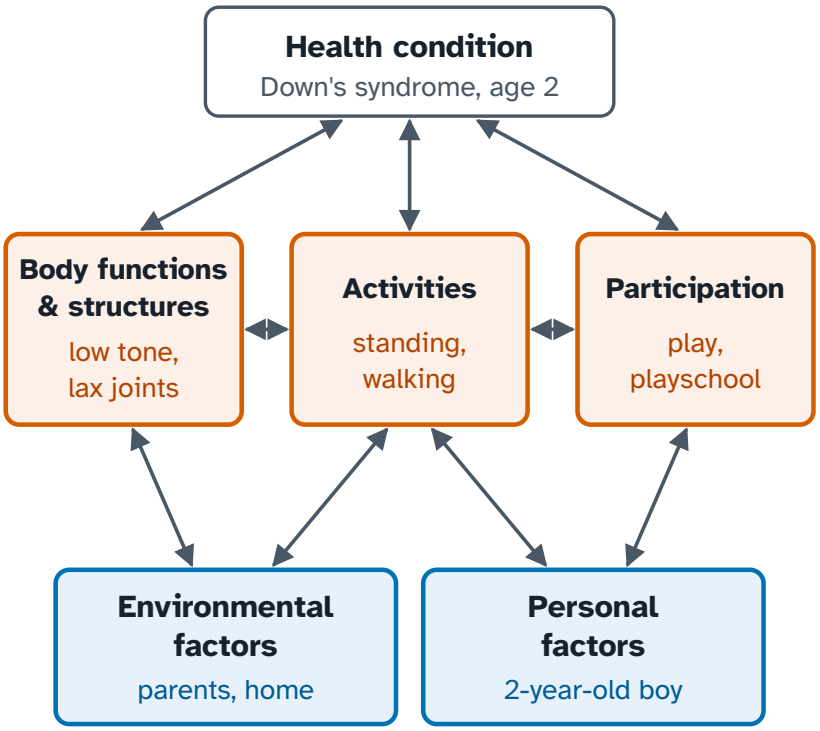
No Down's syndrome long case was set in 2016–2026, so this is a typical case for practice.

Practice case (typical presentation)

> A 2-year-old boy with Down's syndrome, whose heart defect was repaired at 6 months, sits alone and crawls on all fours but cannot pull to stand or cruise. He is floppy, with lax joints and flat feet, tires quickly, says a few words and has frequent colds. His parents want him to walk before he starts playschool.

> **Spot the ICF clues before you write**
>
> heart defect repaired: check his heart status and watch breathlessness. sits and crawls but cannot pull to stand: next skills are pulling to stand and cruising, then walking. floppy, lax joints, flat feet: strengthen and support, do not overstretch. tires quickly, frequent colds: short, frequent sessions. parents want him to walk before playschool: a home programme run by the parents.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Down's syndrome (trisomy 21); repaired heart defect	Delayed but steady motor progress expected
Body function and structure impairments	Hypotonia, lax joints, flat feet, weak trunk and leg muscles, poor balance, low endurance, speech delay	Low tone and laxity make holding positions hard; the heart history limits endurance
Activity limitations	Cannot pull to stand, cruise or walk	Needs leg strength and standing balance first
Participation restrictions	Play with other children; joining playschool	Walking widens play and learning
Environmental factors	**Facilitators:** motivated parents, home space, furniture to cruise along. **Barriers:** frequent illness; no therapy equipment at home	Parents can run daily practice with simple equipment
Personal factors	2-year-old boy who likes play	Train through play and songs

One-line functional diagnosis

A 2-year-old boy with Down's syndrome and a repaired heart defect, with low tone, lax joints and poor endurance, who sits and crawls but cannot yet pull to stand or walk, limiting his play and readiness for playschool.

Recall check. What is the next motor skill for this child, and how will you practise it at home?

11 Problem list and goals

Goals for the 2-year-old

Problem	Short-term (6–8 weeks)	Long-term (6–12 months)

Problem	Short-term (6–8 weeks)	Long-term (6–12 months)
Cannot stand or walk	Pulls to stand at furniture; stands holding on; cruises a few steps	Walks independently at home and outdoors
Poor balance	Stands with one hand held while playing	Squats to pick up toys and stands up without help
Low endurance	Plays actively for 10 minutes without breathlessness	Keeps up with playschool activities
Flat feet	Suitable footwear or orthoses assessed	Stable foot position in walking

12 PT management

By age

- **Infancy (0–1 year):** early intervention with the parents; handling in midline with trunk support; tummy time and play in prone for head and trunk control; weight bearing through the arms; rolling; supported then independent sitting; avoid long periods in the frog position (hips wide and turned out).
- **Toddler (1–3 years):** pull to stand, cruising, standing balance, stepping practice; **treadmill stepping** with a parent supporting the child, from when the child can sit alone; squat to stand; stairs with help; foot orthoses if the feet collapse.
- **Preschool and school age:** balance, running, jumping, ball skills and coordination through play; strength and endurance; swimming and other sport; motor planning with demonstration and repetition.
- **Adolescents and adults:** progressive resistance training, aerobic fitness and weight control; community sport; work-related tasks.

Answer for S20 and W25: physiotherapy for a child with Down's syndrome

1. **Assess** milestones, tone, laxity, balance, endurance, heart status and neck symptoms.
2. **Developmental training** in the order of milestones: head and trunk control, sitting, crawling, pull to stand, cruising, walking.
3. **Treadmill training:** supported stepping on a small treadmill at home, 8 minutes a day, 5 days a week, from sitting alone; infants trained this way walked about 101 days earlier (Ulrich et al., 2001).
4. **Strength and postural control:** weight-bearing play, squat to stand, step-ups, ball activities; balance practice on varied surfaces.
5. **Endurance and fitness:** active play, swimming, cycling, dance.
6. **Joint protection:** no overstretching of lax joints; foot orthoses for flat feet when needed.
7. **Teaching style and family:** short, frequent, play-based sessions with demonstration, simple words and praise; a daily home programme run by the parents; safety advice for the neck and heart.

Classic answer vs current guidance: improving tone or practising skills

Classic	Current	Write in the exam
Facilitate tone with tapping, joint approximation and vestibular stimulation, and wait until the child is "ready" to walk.	Practise the skills themselves: supported treadmill stepping helps infants walk earlier (Ulrich et al., 2001), and strength training builds leg strength in adolescents (Shields and Taylor, 2010). Facilitation is at most an add-on.	"Task-specific practice of standing, stepping and walking (including treadmill training), with strength, balance and fitness training."

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Delayed walking	Supported treadmill stepping, **8 min a day**, **5 days a week**, at home	Walked independently about 101 days earlier in a trial
Low tone and weak muscles (child)	Weight-bearing play, squat to stand, step-ups, climbing, daily	Strength and postural control for standing and walking
Weak legs (adolescent)	Progressive resistance training on machines, **twice a week** for 10 weeks	Increased leg strength in a trial (Shields and Taylor, 2010)
Poor balance	Standing and walking on varied surfaces, ball play, reaching games	Balance improves with specific practice
Lax joints, flat feet	Strengthen around the joints; supramalleolar orthoses or insoles if needed; no end-range stretching	Stability without stretching lax ligaments
Low fitness, weight gain	Active play, swimming, cycling, most days	Endurance and weight control

Recall check. Give the treadmill dose and the result of the Ulrich trial.

13 Precautions, contraindications and stop criteria

Take care

- **Atlantoaxial instability:** stop and refer for neck pain, head tilt, new weakness or stiffness, a change in walking, brisk reflexes, or a change in bladder or bowel control. Avoid trampolines; contact sports and gymnastics carry a risk of spinal cord injury (AAP, 2022). Support the head and neck during handling.
- **Heart:** know the cardiac history; stop for breathlessness, blue lips or unusual tiredness.
- **Lax joints:** no forceful or end-range stretching.
- **Hearing and vision:** face the child, use gestures and demonstration.

14 Education, lifestyle and home programme

- Parents are the main therapists: a short daily programme of play-based practice.
- Tummy time in infancy; standing and stepping practice; treadmill at home if available.
- Footwear and orthosis use; healthy diet and daily activity.
- Choosing safe sports; warning signs of neck problems.
- Regular health checks (heart, thyroid, hearing, vision) and inclusion at school.

15 Prognosis and discharge criteria

- Almost all children learn to walk, later than other children (on average at about 26 months).
- Function depends on the heart and other medical problems, the degree of intellectual disability and early intervention; many adults work and live with some support.
- **Discharge from an episode** when the walking or other goal is reached and the family runs the home programme; review at key stages (school, adolescence).

16 Exam corner

Past questions (MUHS, 2016–2026)

S20 Q2(d), 7 marks

Physiotherapy management of Down's syndrome.

S23 Q2(c), 7 marks

Write down the physical traits and neurological impairments seen in Down's syndrome.

W25 Q2(f), 7 marks

Discuss the physiotherapy management of a child with Down's syndrome.

Pages for this topic: developmental neurology (p. 267), managing hypotonia (p. 216), motor learning (p. 15).

Keywords to underline

trisomy 21 non-disjunction hypotonia ligamentous laxity delayed milestones intellectual disability atlantoaxial instability congenital heart disease treadmill training task-specific practice parent-led home programme

Answer skeleton: physiotherapy for Down's syndrome (7 marks)

1. **[Intro, 1]** Trisomy 21 with hypotonia, laxity and delay.
2. **[Assessment and aims, 1]** Milestones, tone, balance, heart and neck; aims.
3. **[Treatment, 4]** Developmental training in sequence; treadmill stepping; strength and balance; fitness; foot orthoses; teaching style.
4. **[Precautions and family, 1]** Neck and heart safety; parent-run home programme.

Draw these

- The ICF model filled for a child with Down's syndrome (section 10)

Viva questions

1. What are the three genetic types of Down's syndrome?
2. Name five physical traits.
3. Why do children with Down's syndrome walk late?
4. What is atlantoaxial instability, and what are its warning signs?
5. What did the Ulrich treadmill trial show?
6. Why should you not stretch the joints hard?

Linked practical spots

- Paediatric treadmill, supramalleolar orthosis, therapy ball, bench and wedge for developmental training.

Test yourself

Cover each answer and write yours on paper first, then check.

---

3 m **List the genetic types of Down's syndrome.**

Answer

Non-disjunction (free trisomy 21, about 96%), translocation (3–4%, may be inherited from a carrier parent), and mosaicism (1–2%, often milder).

3 m **What is atlantoaxial instability in Down's syndrome?**

Answer
Excess movement between the first and second cervical vertebrae due to lax ligaments; symptomatic in 1–2%. Signs: neck pain, head tilt, weakness, change in gait, brisk reflexes, bladder or bowel change. Routine X-rays are not advised without symptoms; avoid trampolines and take care with contact sports and gymnastics.

7 m **Write down the physical traits and neurological impairments seen in Down's syndrome. (S23)**

Answer
Physical traits: brachycephaly, flat face and nasal bridge, upslanting eyes, epicanthic folds, Brushfield spots, small mouth with protruding tongue, small ears, excess nuchal skin, short broad hands, single palmar crease, clinodactyly, sandal gap, flat feet, short stature, lax joints. **Neurological impairments:** hypotonia, intellectual disability (usually moderate), delayed milestones (sitting about 10 months, walking about 26 months on average), poor balance and postural control, slow reactions and motor planning, speech delay, hearing and vision problems, seizures in some, early dementia in adults.

LAQ **Practice: a 2-year-old with Down's syndrome and a repaired heart defect sits and crawls but cannot pull to stand. Write ICF with clinical reasoning, goals and physiotherapy. (5 + 3 + 7)**

Answer
ICF (5): the table in section 10 with the one-line diagnosis. **Goals (3):** short term: pulls to stand, stands holding, cruises; long term: walks independently, squats and recovers, keeps up at playschool. **Physiotherapy (7):** developmental training (pull to stand, cruising, stepping), treadmill stepping 8 minutes a day 5 days a week, strength and balance through play, endurance, foot orthoses if needed, heart and neck precautions, and a daily parent-run home programme.

MCQ **The commonest genetic type of Down's syndrome is:**

A. Translocation
B. Mosaicism
C. Non-disjunction
D. Deletion

Answer
C.

MCQ **Which activity should a child with Down's syndrome avoid?**

A. Swimming
B. Trampoline
C. Walking
D. Cycling

Answer
B. Because of the risk to the upper cervical spine.

MCQ **In the Ulrich trial, home treadmill training made infants with Down's syndrome walk independently about:**

A. 1 week earlier

B. 1 month earlier

C. 100 days earlier

D. 1 year earlier

Answer

C. 101 days.

Last recall check. Write the genetic types, ten physical traits, six neurological impairments and a seven-point physiotherapy plan.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Bull MJ, et al. Health supervision for children and adolescents with Down syndrome. *Pediatrics* 2022;149(5):e2022057010.
3. Ulrich DA, et al. Treadmill training of infants with Down syndrome: evidence-based developmental outcomes. *Pediatrics* 2001;108(5):e84.
4. Shields N, Taylor NF. A student-led progressive resistance training program increases lower limb muscle strength in adolescents with Down syndrome: a randomised controlled trial. *J Physiother* 2010;56:187–93.
5. Winders P, Wolter-Warmerdam K, Hickey F. A schedule of gross motor development for children with Down syndrome. *J Intellect Disabil Res* 2019.

5B(3) Neural tube defects: spina bifida (myelomeningocele) and hydrocephalus

Spina bifida and hydrocephalus came up in 9 papers: physiotherapy for meningomyelocele or spina bifida five times, physiotherapy for hydrocephalus once, and the types of spina bifida, its deformities and the features of hydrocephalus for 3 marks. Every answer turns on the neurological level, because the muscles that still work decide the deformities, the orthoses and how far the child will walk.

Syllabus	Ch 5B(3) Neural tube defects: spina bifida and hydrocephalus
Priority	**Tier 2.** In 9 of 20 papers (2016–2026): 6 seven-mark and 5 three-mark questions.
Asked as	7m W16 · S17 · W17 · W21 ×2 · W22 · 3m S17 · W19 · S20 · W20 · S25
Usual split	7 (deformities 2 and physiotherapy 5 in W17; otherwise physiotherapy by level and age); 3 (types of spina bifida, deformities, or types and features of hydrocephalus)
Checked against	Orthopaedic management of spina bifida, parts I and II (Swaroop and Dias, 2009 and 2011); the MOMS trial of prenatal repair (Adzick et al., 2011); Indian prevalence meta-analysis (Allagh et al., 2015); Spina Bifida Association information on tethered cord and shunt malfunction. Full list under Sources (p. 308). Content checked 20 October 2026.

1 Definition

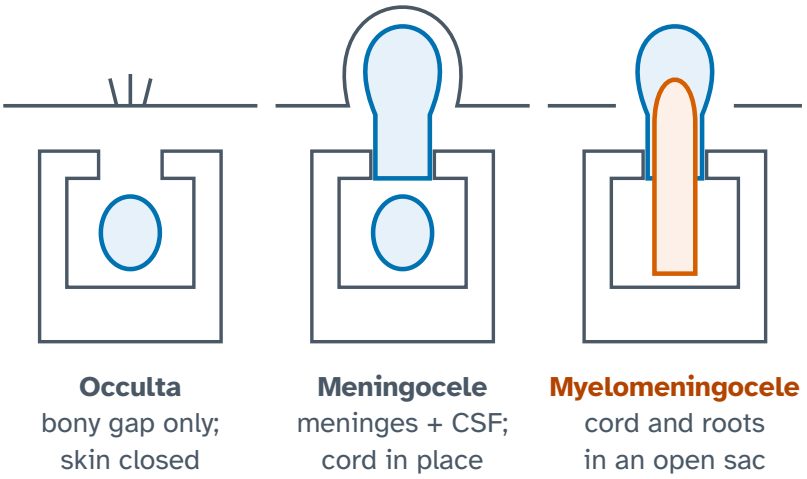
- **Neural tube defects** happen when the neural tube fails to close in the first 4 weeks of pregnancy: anencephaly and encephalocele at the head end, spina bifida along the spine.
- **Spina bifida** is incomplete closure of the vertebral arches, with or without a sac containing the meninges and the spinal cord.
- **Hydrocephalus** is too much cerebrospinal fluid (CSF) in the ventricles of the brain, usually under raised pressure, so the ventricles enlarge.

2 Classification and grading

Types of spina bifida (3 marks in S25)

- **Spina bifida occulta:** a gap in the bony arch only; the skin is closed, sometimes with a tuft of hair, dimple or fatty lump; usually no deficit.
- **Meningocele:** a skin-covered sac of meninges and CSF through the gap; the cord stays in the canal, so deficits are few.
- **Myelomeningocele** (meningomyelocele): the cord and nerve roots lie in the sac, which is often open; paralysis, sensory loss and bladder problems below the level. The commonest serious type.
- Others: lipomyelomeningocele (a fatty mass tethering the cord under closed skin) and myeloschisis (an open, flat cord plate).

Types of spina bifida



Draw this for the types question. Blue is the sac of meninges and CSF (and the normal cord); orange is the cord pulled into the sac, which is why the nerves below it fail.

Functional level in myelomeningocele

Level	Muscles working	Usual mobility
Thoracic and high lumbar	Trunk; hip flexors and adductors at most; no quadriceps	Stands and walks in childhood only with hip-spanning orthoses (reciprocating gait orthosis or HKAFO); most adults use a wheelchair
Low lumbar	Quadriceps and medial hamstrings; no gluteus medius or maximus	Walks with AFOs and crutches; about 80% stay community walkers as adults
Sacral	Quadriceps and gluteus medius; calf muscles weak in high sacral levels	High sacral: AFOs, gluteal lurch; low sacral: near-normal walking without orthoses

The level of the nerve damage, not the state of the hip, decides walking ability most (Swaroop and Dias, 2009).

Types of hydrocephalus (W19)

- **Non-communicating (obstructive):** CSF is blocked inside the ventricles, for example aqueduct stenosis, Chiari II malformation, tumours.
- **Communicating:** CSF flows out of the ventricles but is not absorbed, for example after bleeding in a premature baby or after meningitis.
- **Congenital** (with spina bifida, aqueduct stenosis, Dandy–Walker malformation) or **acquired**; **normal-pressure hydrocephalus** in older adults.

3 Causes and risk factors

- **Spina bifida** is multifactorial: low folate, a previous affected pregnancy, maternal diabetes and obesity, some antiepileptic drugs (valproate), high fever in early pregnancy, and genes.
- **India:** neural tube defects occur in about 4.5 per 1000 births in pooled studies, higher than in many countries (Allagh et al., 2015).
- **Prevention:** folic acid 400 micrograms daily from at least a month before conception until 12 weeks of pregnancy; a higher dose (4–5 mg) on medical advice after an affected pregnancy; food fortification.
- **Hydrocephalus:** Chiari II malformation with myelomeningocele, aqueduct stenosis, bleeding in premature babies, meningitis, tumours.

4 Pathophysiology to impairments

1. The exposed cord is damaged before and at birth, so muscles below the level are paralysed and flaccid and the skin is numb; some children also have spasticity.
2. Working muscles above the level pull without opposition, and posture adds to it, causing deformities: at L3–L4, hip flexors and adductors pull the hip into flexion and adduction and can dislocate it; at low lumbar levels, a working tibialis anterior with a paralysed calf pulls the foot into calcaneus.
3. The sacral roots (S2–S4) are affected in almost all, so the bladder and bowel are neurogenic.
4. The Chiari II malformation blocks CSF flow, causing hydrocephalus; about 8 in 10 babies repaired after birth need a shunt in the first year (MOMS trial).
5. Numb skin under pressure breaks down; unused bones become thin and break easily, especially after a cast.
6. As the child grows, the repaired cord can stay stuck (tethered) and be stretched, causing new loss.

Recall check. Why does a child with an L4 myelomeningocele tend to develop a calcaneus foot and dislocated hips?

5 Clinical features

- A sac on the back, usually lumbar or lumbosacral; paralysis and sensory loss below the level; neurogenic bladder and bowel; hydrocephalus.
- Learning, attention and hand-coordination difficulties are common when there is hydrocephalus.

Deformities in spina bifida (S17, W17)

Region	Deformities	Why
Spine	Scoliosis, kyphosis (often congenital at thoracolumbar levels), lordosis	Abnormal vertebrae, weak trunk muscles, tethered cord
Hip	Flexion contracture; flexion and adduction with subluxation or dislocation; flexion, abduction and external rotation in high lesions	Hip flexors and adductors work without the glutei; frog posture in high lesions
Knee	Flexion contracture (thoracic and high lumbar); valgus (low lumbar and sacral); extension contracture (rare, congenital)	Sitting posture and weak quadriceps; gait with weak glutei
Rotation	Persistent femoral anteversion; external tibial torsion (acquired) or internal torsion (with clubfoot)	Muscle imbalance and abnormal loading
Foot	Clubfoot and vertical talus (congenital); calcaneus (low lumbar); cavovarus and claw toes (sacral); valgus	Imbalance between working and paralysed muscles

Clinical features of hydrocephalus (W19, S20, W20)

- **Infants:** a head growing too fast (crossing the head-circumference centiles), a full or bulging anterior fontanelle, widened sutures, prominent scalp veins, "setting-sun" eyes (looking down), irritability, vomiting, poor feeding, drowsiness, a high-pitched cry, stiff legs, delayed milestones, seizures.
- **Older children and adults:** headache (worse in the morning), vomiting, drowsiness, swelling of the optic disc, double vision, poor balance.
- **Normal-pressure hydrocephalus** in older adults: walking difficulty, memory loss and urinary incontinence.

Tethered cord: signs of new deterioration

Back pain that worsens with activity, new weakness or loss of walking, changing foot shape, increasing scoliosis, and new bladder or bowel problems. Report these at once.

6 Investigations

- **Before birth:** maternal serum alpha-fetoprotein, the anomaly ultrasound scan, and sometimes amniotic fluid tests.
- **After birth:** MRI of the brain and spine (Chiari malformation, hydrocephalus, tethering), cranial ultrasound in infants, serial head circumference, X-rays of the spine and hips.
- **Bladder:** kidney ultrasound and urodynamic studies.
- **Physiotherapy:** repeated muscle charting and sensory testing to fix the level and pick up any change.

7 Medical and surgical management

- **Closure of the back** in the first days of life to prevent infection. In some centres the defect is repaired before birth: in the MOMS trial this cut the need for a shunt from 82% to 40% and improved walking at 30 months, at the cost of more premature births.
- **Hydrocephalus:** a ventriculoperitoneal (VP) shunt, or endoscopic third ventriculostomy. Shunts can block, become infected or drain too much.
- **Bladder and bowel:** clean intermittent catheterisation, medicines, a bowel routine (see bladder and bowel (p. 238)).
- **Orthopaedics:** clubfoot casting or surgery; contracture releases; for the hip, the aim is a level pelvis and free movement rather than putting a dislocated hip back; spinal fusion for severe curves; untethering surgery for a tethered cord.

8 Complications and prevention

Complication	Physio role
Shunt blockage or infection	Know the signs (vomiting, drowsiness, irritability, headache, bulging fontanelle, fever); stop and refer at once
Pressure sores	Daily skin checks of feet, buttocks and areas under orthoses; well-fitting orthoses and shoes; pressure relief
Fractures	Gentle handling and range of motion, especially after a cast; standing programme
Contractures and deformity	Positioning, prone lying, stretching, standing, orthoses
Tethered cord	Regular muscle charting; report any new weakness or change in gait
Latex allergy	Latex-free gloves, balls, bands and equipment from birth
Obesity, low fitness	Regular active play, sport and wheelchair fitness

9 PT assessment

- History: pregnancy and birth, closure, shunt and other surgery, milestones, bladder and bowel routine, school.
- **Muscle chart** of the legs and trunk, and sensory testing (touch and pin), to find the motor and sensory level; tone (any spasticity).
- Range of motion and deformity: hips (flexion contracture, adduction, stability), knees, feet, spine; limb lengths.
- Head circumference and shunt signs in infants; skin.
- Motor development and mobility: head control, sitting, rolling, crawling, standing, walking with orthoses, wheelchair skills.
- Upper limb strength and coordination; cognition and attention; family and home.

10 PT diagnosis by ICF

No spina bifida long case was set in 2016–2026, so this is a typical lumbar case (as in W22) for practice.

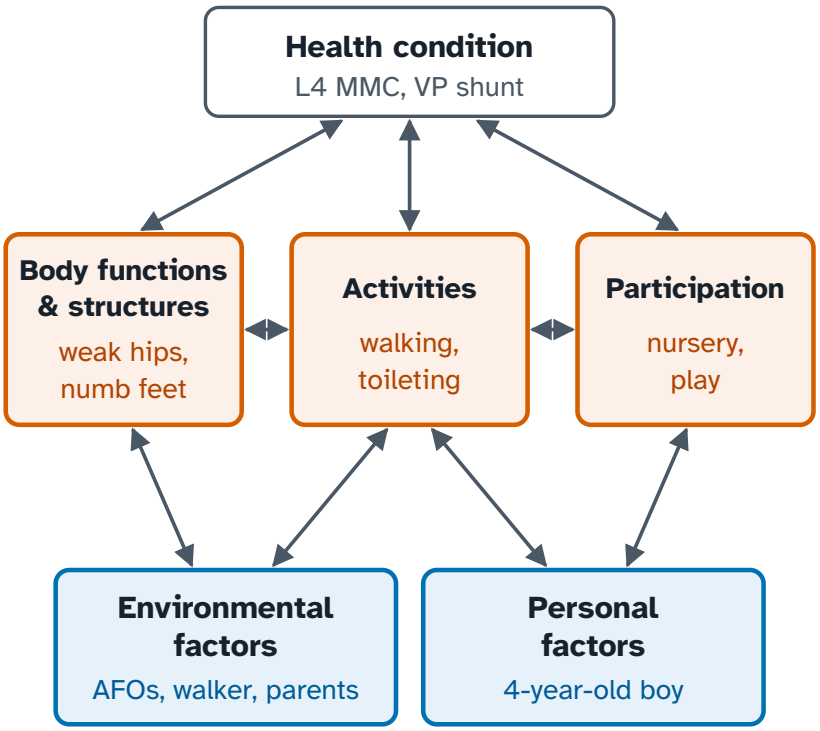
Practice case (typical presentation)

A 4-year-old boy has a myelomeningocele at L4, closed at birth, and a VP shunt. His hip flexors, adductors and quadriceps work; his glutei and calf muscles do not; sensation is absent below the knees. He walks short distances with AFOs and a walker, has a red mark on his left heel, is catheterised by his mother, and attends nursery school.

Spot the ICF clues before you write

L4, quadriceps working: low lumbar level, so AFOs and crutches are realistic and he should stay a walker. no glutei: hip flexion–adduction pull, so watch the hips. sensation absent below the knees and red mark on his heel: pressure-sore risk. VP shunt: know the warning signs. nursery school: plan mobility and toileting there.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Myelomeningocele at L4, closed at birth; VP shunt; neurogenic bladder	Low lumbar level: good walking prospects
Body function and structure impairments	Paralysis of glutei and calves; absent sensation below the knees; risk of hip flexion–adduction and calcaneus deformity; heel pressure mark; neurogenic bladder	Unopposed hip flexors and adductors and tibialis anterior; numb skin under load
Activity limitations	Walks only short distances with a walker; floor-to-stand, stairs and toileting need help	Weak hip extensors and abductors and no push-off
Participation restrictions	Keeping up with friends at nursery and play; toileting at school	Slow walking and catheter needs
Environmental factors	**Facilitators:** AFOs, walker, involved parents, nursery place. **Barriers:** distance, rough ground, latex items, no wheelchair for long distances	Crutches and a light wheelchair widen his world
Personal factors	4-year-old boy, keen to play	Use play to train

One-line functional diagnosis

A 4-year-old boy with an L4 myelomeningocele and a shunt, with paralysed glutei and calves and numb feet, who walks only short distances with AFOs and a walker, is at risk of hip, foot and skin problems, and needs help at nursery.

Recall check. Which muscles decide whether a child with spina bifida can walk with AFOs alone, and which orthosis is needed without them?

11 Problem list and goals

Goals for the 4-year-old

Problem	Short-term (6–8 weeks)	Long-term (6–12 months)
Limited walking	Walks around the house with AFOs and a walker; starts forearm crutches	Walks around nursery and home with AFOs and crutches; light wheelchair for long distances
Hip and foot deformity risk	Full hip extension in prone lying; parents do daily stretches	Level pelvis, no fixed contracture, plantigrade feet in AFOs
Numb skin	Heel mark healed; parents check skin twice daily	No pressure sores
Transfers and toileting	Floor to stand with support	Independent transfers; ready to learn self-catheterisation later

12 PT management

By age

- **Newborn, before closure:** prone or side lying to protect the sac; gentle handling; record which muscles move.
- **After closure (infancy):** positioning, gentle range of motion, teaching parents to handle and carry; development: head control, prone play, rolling, sitting, arm strength.
- **Toddler:** a standing frame from about 12 months (or once head control develops); a parapodium or reciprocating gait orthosis from about 2 years in high lesions; AFOs or KAFOs with a walker in lumbar lesions.
- **School age:** gait training with orthoses and crutches; wheelchair skills for distances; strength and fitness; independence in self-care and, later, self-catheterisation.
- **Adolescence and adulthood:** keep walking where possible, add a wheelchair when needed; weight control, skin care and sport.

Answer for W22: physiotherapy of a lumbar myelomeningocele

1. **Assess** the motor and sensory level by muscle charting; hips, knees, feet and spine; skin.
2. **Strengthen** working muscles (hip flexors and adductors, quadriceps, trunk) and the arms and shoulders for crutches and transfers.
3. **Prevent deformity:** prone lying, stretching of hip flexors and adductors, hamstrings and calf; positioning; night splints.
4. **Orthoses by the muscles working:** no quadriceps (thoracic, high lumbar): reciprocating gait orthosis or HKAFO with a walker, plus a wheelchair; weak quadriceps: KAFO; strong quadriceps (low lumbar, as at L4): AFOs with crutches; sacral: AFOs or none.
5. **Gait training:** standing balance, walker then crutches (swing-through or four-point), stairs, falls.
6. **Wheelchair** for long distances and wheelchair skills.
7. **Skin, bladder and latex care;** parent and school education; watch for shunt and tethered cord signs.

Physiotherapy for hydrocephalus (7 marks in W21)

- **Handling:** always support the heavy head and neck when lifting and carrying; teach parents.
- **Positioning:** change the head position every few hours to prevent scalp pressure sores and flattening; avoid long pressure over the shunt valve; supported side lying and prone on a wedge.
- **Development:** head control (prone on elbows, supported sitting), rolling, sitting balance, reaching, standing in a frame; follow the developmental sequence.
- **Tone and range:** stretching and positioning for stiff legs; strengthening of weak muscles.
- **Sensory, visual and cognitive stimulation** through play; adapted seating with head support.
- **Education:** shunt warning signs (vomiting, drowsiness, irritability, bulging fontanelle, fast head growth, fever) and when to seek help.

Classic answer vs current guidance: the dislocated hip

Classic	Current	Write in the exam
Reduce every dislocated hip surgically so the child can walk.	Aim for a level pelvis and free hip movement rather than an X-ray-perfect hip; walking depends mainly on the neurological level (Swaroop and Dias, 2009).	"Prevent hip flexion–adduction contracture and keep the pelvis level; plan orthoses and walking by the level."

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Weak working muscles and arms	Play-based resisted work, push-ups in sitting, crutch-walking drills, daily	Strength for walking, transfers and the wheelchair
Hip flexion–adduction pull	Prone lying daily, stretching, abduction positioning	Keeps the pelvis level and the hips mobile
Knee flexion contracture	Stretching, standing, serial casts or night splints	Over 20° makes orthoses and walking difficult
Foot deformity	Clubfoot casting (with the surgeon), AFOs for calcaneus or drop	A plantigrade foot that fits an orthosis
Cannot stand or walk	Standing frame from about 12 months; parapodium or reciprocating gait orthosis from about 2 years; KAFO or AFO with walker or crutches	Weight bearing for bones, hips and development
Numb skin	Skin checks twice daily (mirror for the child), padded well-fitting shoes, no heat on numb skin	Prevents pressure sores and burns

Recall check. Match each level (thoracic, high lumbar, low lumbar, sacral) with its orthosis and expected walking.

13 Precautions, contraindications and stop criteria

Take care

- Use only latex-free gloves and equipment.
- Stop and refer for shunt warning signs or new weakness, pain or bladder change (tethered cord).
- Thin bones: gentle range of motion, no forceful stretching; avoid crawling for 3–4 weeks after a period of immobilisation.
- No hot packs or heat on numb skin; check the skin after every orthosis session.
- Before closure, never put pressure on the sac.

14 Education, lifestyle and home programme

- Daily positioning, prone lying, stretches and a standing programme.
- Skin checks, orthosis wear and care, footwear.
- Bladder and bowel routine; latex avoidance; healthy weight.
- Shunt and tethered cord warning signs.
- Folic acid before and in any future pregnancy.
- School inclusion: accessible classroom, toilet access, a wheelchair for distances.

15 Prognosis and discharge criteria

- Walking depends mainly on the level: about 80% of children with low lumbar lesions stay community walkers as adults, while most with thoracic or high lumbar lesions use a wheelchair as adults (Swaroop and Dias, 2009).
- Hydrocephalus, shunt problems, a tethered cord and obesity can reduce function; most children now survive into adulthood.
- **Discharge from an episode** when goals are met, orthoses fit, the family runs the home programme and knows the warning signs; follow-up continues through growth.

16 Exam corner

Past questions (MUHS, 2016–2026)

W16 Q2(f), S17 Q2(b), 7 marks

Physiotherapy management of meningomyelocele.

W17 Q2(b), 7 marks (2 + 5)

Describe musculoskeletal deformities commonly seen in infantile spina bifida and its physiotherapy management.

W21 Q2(f), 7 marks

Physiotherapy management for spina bifida.

W21 Q2(b), 7 marks

Physiotherapy management for hydrocephalus.

W22 Q2(a), 7 marks

Physiotherapy management of myelomeningocele at lumbar level.

S17 Q1(e), 3 marks

Enumerate the deformities in spina bifida children.

S25 Q1(e), 3 marks

Enumerate types of spina bifida.

W19 Q1(c), 3 marks

What are the types and clinical features of hydrocephalus?

S20 Q1(f), W20 Q1(c), 3 marks

Clinical features of hydrocephalus.

Pages for this topic: bladder and bowel (p. 238), orthoses (p. 249), wheelchairs (p. 257), development (p. 267).

Keywords to underline

neural tube defect myelomeningocele neurological level muscle charting hip flexion–adduction calcaneus foot standing frame reciprocating gait orthosis KAFO, AFO skin care latex-free VP shunt tethered cord folic acid

Answer skeleton: physiotherapy of spina bifida (7 marks)

1. **[Intro, 1]** Define myelomeningocele and name the level.
2. **[Assessment and aims, 1]** Muscle chart and sensory level; aims by level.
3. **[Treatment by age, 4]** Positioning and handling; development; standing frame; orthoses and gait training by level; wheelchair; strengthening; deformity prevention.
4. **[Care and education, 1]** Skin, bladder, latex, shunt and tethered cord signs.

Draw these

- The three types of spina bifida (section 2)
- A table of levels with orthoses and expected walking

Viva questions

1. What is the difference between a meningocele and a myelomeningocele?
2. Which muscle decides whether a child needs a KAFO or only an AFO?
3. Why does an L4 lesion cause a calcaneus foot?
4. Name five signs of shunt malfunction.
5. What are the signs of a tethered cord?
6. Why must equipment be latex-free?

Linked practical spots

- Standing frame, parapodium, reciprocating gait orthosis, HKAFO, KAFO, AFO, forearm crutches, rollator walker.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Enumerate the types of spina bifida. (S25)**

Answer
Spina bifida occulta (bony gap, skin closed); meningocele (sac of meninges and CSF, cord in place); myelomeningocele (cord and roots in an open sac, the commonest serious type); also lipomyelomeningocele and myeloschisis.

3 m **Write the clinical features of hydrocephalus. (S20, W20)**

Answer
In infants: rapidly growing head, bulging fontanelle, widened sutures, prominent scalp veins, setting-sun eyes, irritability, vomiting, poor feeding, drowsiness, high-pitched cry, stiff legs, delayed milestones, seizures. In older children: morning headache, vomiting, drowsiness, papilloedema.

7 m **Describe the musculoskeletal deformities in infantile spina bifida and its physiotherapy management. (W17, 2 + 5)**

Answer

Deformities (2): spine: scoliosis, kyphosis; hip: flexion contracture, flexion–adduction with dislocation; knee: flexion contracture or valgus; feet: clubfoot, vertical talus, calcaneus, cavovarus. **Physiotherapy (5):** muscle charting for the level; positioning and handling; prone lying and stretching of hip flexors, adductors, hamstrings and calves; assisting clubfoot casting and splints; developmental training; standing frame from about 12 months; orthoses and gait training by level; strengthening of working muscles and arms; skin care, latex precautions and parent education on shunt and tethered cord signs.

LAQ **Practice: a 4-year-old boy with an L4 myelomeningocele and a VP shunt walks short distances with AFOs and a walker and has a heel pressure mark. Write ICF with clinical reasoning, goals and physiotherapy. (5 + 3 + 7)**

Answer

ICF (5): the table in section 10 with the one-line diagnosis. **Goals (3):** short term: walks at home with AFOs and walker, starts crutches, heel healed, full hip extension; long term: walks at nursery with AFOs and crutches, wheelchair for distance, no sores, level pelvis. **Physiotherapy (7):** the W22 answer in section 12 adapted to him, plus play-based training and education of parents and nursery staff.

MCQ **In a meningocele, the sac contains:**

A. Spinal cord and roots

B. Meninges and CSF only

C. Fat only

D. Nothing; the skin is closed

Answer

B.

MCQ **A child with spina bifida who has strong quadriceps but no glutei most likely walks with:**

A. No orthosis

B. AFOs and crutches

C. A reciprocating gait orthosis

D. Cannot walk

Answer

B. This is the low lumbar group.

MCQ **"Setting-sun" eyes in an infant suggest:**

A. Hydrocephalus

B. Facial palsy

C. Spina bifida occulta

D. Down's syndrome

Answer

A.

Last recall check. Write the types of spina bifida, the deformities by region, the levels with orthoses, and the shunt warning signs.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Swaroop VT, Dias L. Orthopedic management of spina bifida. Part I: hip, knee, and rotational deformities. *J Child Orthop* 2009;3:441–9.
3. Swaroop VT, Dias L. Orthopaedic management of spina bifida. Part II: foot and ankle deformities. *J Child Orthop* 2011;5:403–14.
4. Adzick NS, et al. A randomized trial of prenatal versus postnatal repair of myelomeningocele. *N Engl J Med* 2011;364:993–1004.
5. Allagh KP, et al. Birth prevalence of neural tube defects and orofacial clefts in India: a systematic review and meta-analysis. *PLoS ONE* 2015;10:e0118961.
6. Spina Bifida Association. Spinal cord tethering; signs of shunt malfunction (information sheets).

5B(4) Obstetric brachial plexus injury: types, differential diagnosis and physiotherapy

> Obstetric brachial plexus palsy came as a long case once (S19): a baby who has not moved one arm since a forceps delivery. The examiner wants three things: rule out the conditions that mimic it, refer early enough for surgery if the biceps does not recover, and teach the parents a home programme that keeps the shoulder from stiffening into internal rotation.

Syllabus	Ch 5B(4) Brachial plexus injuries (adult plexus injuries are on the nerve injuries page)
Priority	**Tier 2.** In 1 of 20 papers (2016–2026): 1 long case. The adult plexus questions are on the nerve injuries page.
Asked as	LAQ S19
Usual split	15 (differential diagnosis 5; short- and long-term goals with physiotherapy 10)
Checked against	Clinical review of neonatal brachial plexus palsy referral and management (J Hand Ther, 2020); reviews of surgical timing (biceps recovery, Toronto test, cookie test). Full list under Sources (p. 316). Content checked 20 October 2026.

1 Definition

Obstetric (neonatal) brachial plexus palsy is weakness of an infant's arm caused by stretching or tearing of the C5–T1 nerve roots during birth. Reported rates vary widely, from about **0.4 to 5 per 1000** births.

The anatomy, nerve degrees and adult injuries are on the nerve injuries page (p. 94).

2 Classification and grading

Narakas group	Roots	Picture
I	C5–C6 (Erb's)	No shoulder abduction or elbow flexion; hand works; the commonest and best outlook
II	C5–C7	As above, plus weak wrist and finger extension ("waiter's tip" with a dropped wrist)
III	C5–T1	Flail arm, no Horner's syndrome
IV	C5–T1 with Horner's	Flail arm with drooping eyelid and small pupil: T1 root avulsion; the worst outlook

- **Active Movement Scale** (0–7): grades each movement in infants, with and without gravity.
- **Mallet score** (in children who can follow commands, about 3 years and over): abduction, external rotation, hand to neck, hand to back and hand to mouth, each graded 1–5.

3 Causes and risk factors

- **Shoulder dystocia:** the shoulder gets stuck, and traction on the head stretches the upper roots.
- Large baby; maternal diabetes; prolonged or difficult labour; **forceps or vacuum delivery**; breech delivery.
- Some cases occur after a normal delivery or a caesarean section.

4 Pathophysiology to impairments

1. Traction stretches (neurapraxia), ruptures or avulses the roots; the upper roots are hit most often.
2. Because the deltoid, spinati and biceps are paralysed while the pectoralis major and subscapularis keep working, the shoulder is pulled into internal rotation.

3. Because the infant's shoulder joint is still forming, a persistent internal rotation pull deforms the glenoid and can subluxate the humeral head backwards (**glenohumeral dysplasia**).
4. Because the infant does not use the arm, the brain may "forget" it (developmental disregard), even when the nerves recover.
5. The arm may grow shorter and smaller; recovering nerves can cause muscles to fire together (co-contraction).

Recall check. Why is external rotation range the most important thing to protect in an infant with Erb's palsy?

5 Clinical features

- **Erb's posture:** arm by the side, internally rotated, elbow extended, forearm pronated ("waiter's tip"); the hand moves in group I.
- **Moro reflex** absent on the affected side; grasp reflex present in Erb's palsy but absent in a total palsy.
- **Total palsy:** flail, numb arm; Horner's syndrome if T1 is avulsed.
- Breathing difficulty if the phrenic nerve (C3–C5) is involved.
- **Later:** internal rotation contracture, posterior shoulder subluxation, elbow flexion contracture, winging of the scapula, a shorter arm, the "trumpet sign" (the elbow lifts when the child brings the hand to the mouth).

6 Investigations

- **X-ray** of the clavicle, humerus and shoulder to rule out a fracture.
- **Ultrasound** of the shoulder to check for dysplasia and subluxation.
- **MRI** (or CT myelography) for root avulsion when surgery is considered.
- **Nerve studies** help, but in infants they can look better than the clinical picture.
- Chest X-ray or ultrasound of the diaphragm if breathing is affected.

7 Medical and surgical management

- **Refer early:** every infant to a brachial plexus clinic by **1 month** of age, with management started before 3 months.
- **Nerve surgery** (grafts, nerve transfers) is considered when the biceps has not recovered by about **3 months** (or the Toronto test score is under 3.5), or when the child fails the "cookie test" (bringing a biscuit to the mouth with the arm held at the side) by about 9 months. A total palsy with Horner's syndrome is explored early. Infants who regain antigravity biceps by 4–5 months do about as well as those who have surgery.
- **Later surgery:** release of tight internal rotators, tendon transfers for external rotation, or a rotation osteotomy of the humerus.
- **Botulinum toxin** into tight internal rotators, with casting, in selected children.

8 Complications and prevention

Complication	What to look for	Prevention
Internal rotation contracture	Loss of passive external rotation with the arm at the side	Daily gentle external rotation with the scapula held; refer if range is lost
Glenohumeral dysplasia and posterior subluxation	Asymmetric skin folds, a clunk, loss of external rotation	Keep external rotation; early ultrasound and specialist review
Elbow flexion contracture	Cannot straighten the elbow fully	Elbow extension stretches; night splint or serial casting later
Developmental disregard	The child ignores the arm even with some power	Play that uses the arm; two-handed toys; constraint of the good arm in older infants

Complication	What to look for	Prevention
Skin injury	Bites or scratches on a numb hand	Watch the hand; keep nails short

9 PT assessment

- **History:** birth weight, delivery (forceps, shoulder dystocia), breathing problems, milestones, parents' concerns.
- **Observation:** resting posture, spontaneous movement, use in play, Horner's signs.
- **Movement:** Active Movement Scale for each joint; Moro and grasp reflexes.
- **Range:** passive external rotation with the arm at the side and in abduction (compare sides), abduction, elbow extension, supination.
- **Sensation:** reaction to touch and pinprick.
- **Development:** head control, rolling, sitting, weight-bearing through the arms, crawling.
- Mallet score once the child can follow commands.

10 PT diagnosis by ICF

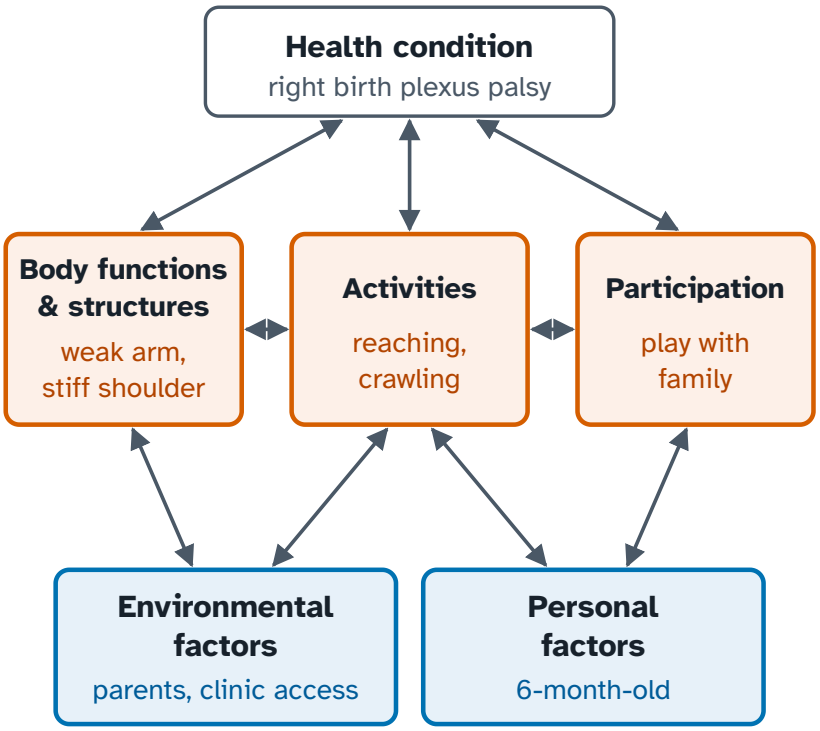
Draw the six-box ICF model, then fill it from the case. For an infant, activities are reaching, play and movement milestones.

Worked example from a real paper (S19, 5 + 10 marks)

> "A 6 months old baby is unable to move right upper limb since birth. Mother gives history of forceps delivery. All other developmental milestones are normal. Give differential diagnosis. Discuss short and long term goals and physiotherapy management." (One printed copy says 5 months.)

> **Spot the clues before you write**
>
> since birth and forceps delivery point to a birth injury of the plexus. other milestones normal argues against cerebral palsy. 6 months is past the usual 3-month point for surgery if the biceps has not recovered: refer to a plexus clinic now.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Right obstetric brachial plexus palsy after forceps delivery; group to be confirmed (Erb's or total)	Traction on the roots during delivery
Body function and structure impairments	• Weak or absent shoulder and elbow movement (and hand, if total) • Internal rotation posture; risk of contracture and shoulder dysplasia • Possible sensory loss and Horner's signs • Possible developmental disregard of the arm	• Upper roots supply deltoid, spinati and biceps • Working internal rotators pull the growing shoulder joint out of shape
Activity limitations	Cannot reach for toys, bring the hand to the mouth, prop on both arms or bear weight for crawling; asymmetric rolling and sitting balance	Six-month skills need both arms
Participation restrictions	Less two-handed play with parents and siblings	Play is the infant's participation
Environmental factors	**Facilitators:** parents who can do a home programme. **Barriers:** distance to a plexus clinic; worry and guilt	Parents deliver daily therapy
Personal factors	6-month-old with otherwise normal development	Good brain plasticity; normal milestones exclude a central cause

One-line functional diagnosis

A 6-month-old with right obstetric brachial plexus palsy after a forceps delivery, with weakness of the right arm and a risk of internal rotation contracture and shoulder dysplasia, who cannot reach, bring the hand to the mouth or bear weight on the arm, limiting two-handed play. Parents are the key facilitators.

Recall check. Name three conditions that can look like birth plexus palsy and how you tell them apart.

11 Problem list and goals

Goals for the 6-month-old (agree them with the parents)

Problem	Short-term (4–6 weeks)	Long-term (6–12 months)
Shoulder range	Full passive external rotation and abduction, matching the other side	No contracture; shoulder joint well formed on review
Active movement	Brings the hand towards the mouth with gravity eliminated	Hand to mouth and reaching overhead against gravity
Use in play	Uses the arm to hold toys at midline	Uses both hands in play; crawls on both arms
Parents	Do the home programme correctly several times a day	Attend plexus clinic reviews; surgical decision made on time

12 PT management

Principles: refer early; protect the shoulder; keep full range, especially external rotation; get the infant to use the arm through play; train the parents.

By phase

- **First 1–2 weeks:** gentle handling; support the arm; do not pull on it; dress the affected arm first and undress it last; position the arm supported, not trapped under the body. Rigid splinting in the newborn period is discouraged.
- **Early home programme:** gentle passive range several times a day (for example at each nappy change): shoulder external rotation with the arm at the side and the scapula held, abduction and flexion, elbow extension, forearm supination, wrist and finger extension. Sensory input with touch, massage and textures.
- **Active use through play:** toys placed to make the infant reach with the affected arm; hand-to-mouth play; prone on elbows and four-point weight-bearing; sitting with arm support; crawling.
- **Older infants and children:** constraint of the good arm for short play periods if the child ignores the weak arm; splints (resting wrist splint, elbow extension splint) for specific problems; serial casting or botulinum toxin for contractures under specialist care; rehabilitation after surgery as the surgeon advises.

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Loss of external rotation	Gentle passive external rotation with the scapula stabilised, **several times a day**, never forced	Protects the growing shoulder joint from dysplasia
Weak shoulder and elbow	Play in gravity-eliminated positions (side lying, supported sitting, water play), then against gravity	Early active use helps motor recovery and the brain's map of the arm
Developmental disregard	Toys on the affected side; two-handed play; short periods of constraint of the good arm when older	The child learns to use the arm as nerves recover
Wrist and finger drop (group II)	Resting wrist splint at night once past the newborn period; extension play	Keeps the hand in a functional position
Elbow flexion contracture	Elbow extension stretches; night extension splint or serial casting	Maintains reach

Recall check. Write a home programme for the parents in five lines.

13 Precautions, contraindications and stop criteria

Take care

- Never force a movement, especially external rotation and abduction; hold the scapula still while moving the shoulder.
- Do not lift the baby by the arms or pull on the affected arm.
- Stop and refer if the shoulder clunks, range is lost, or the arm swells or seems painful (fracture, dislocation).
- Protect a numb hand from biting and scratching.

14 Parent education and home programme

- Explain the injury without blame; many infants recover well, and early referral matters.
- Teach handling, dressing, positioning and the daily range-of-motion routine, with pictures.
- Play ideas that use the arm; praise all attempts.
- Keep all clinic appointments; record when new movements appear (for example the first time the hand reaches the mouth).

15 Prognosis and discharge criteria

- Many infants recover well, especially with upper plexus (C5–C6) injuries; reported rates vary between studies.
- **Good signs:** Erb's palsy only; antigravity biceps by about 3 months.
- **Poor signs:** total palsy; Horner's syndrome (T1 avulsion); no biceps recovery by 3–6 months.
- Even after good recovery, external rotation and abduction often stay limited, so the shoulder is reviewed until the child stops growing.

16 Exam corner

Past question (MUHS, 2016–2026)

S19 Q3(b), 5 + 10 marks

The worked example in section 10.

Pages for this topic: nerve injuries and the brachial plexus (p. 94), developmental neurology (p. 267), orthoses and splints (p. 249).

Keywords to underline

traction at birth shoulder dystocia Narakas group waiter's tip asymmetric Moro refer by 1 month biceps by 3 months external rotation glenohumeral dysplasia home programme

Answer: differential diagnosis in the S19 case (5 marks)

Condition	How it differs
Fracture of the clavicle or humerus (pseudoparalysis)	Pain on handling, swelling, crepitus; X-ray shows the fracture; movement returns as the fracture heals in 2–3 weeks
Septic arthritis or osteomyelitis of the shoulder	Fever, irritability, swelling and pain on movement; raised blood markers; an emergency
Hemiplegic cerebral palsy	UMN signs: increased tone, brisk reflexes, fisting; leg also involved; often other delays (here milestones are normal)
Birth injury of the cervical cord	Both arms or the legs affected; breathing and bladder problems

Condition	How it differs
Arthrogryposis	Stiff joints in several limbs from birth

Answer skeleton

1. **[Differential diagnosis, 5]** Obstetric plexus palsy (group by clinical picture) vs fracture, septic shoulder, hemiplegic CP, cord injury; tests that separate them.
2. **[Goals, 3–4]** Short-term: full external rotation, active use in play, parents trained. Long-term: hand to mouth and reaching, two-handed play, no shoulder deformity.
3. **[Management, 6–7]** Refer to a plexus clinic (surgery if no biceps by 3 months); handling and positioning; home range-of-motion programme; play-based active use and weight-bearing; constraint and splints later; precautions; parent education.

Draw these

- The brachial plexus with the upper roots marked (on the nerve injuries page (p. 94))
- The ICF model filled from the case

Viva questions

1. Which delivery problem most often causes Erb's palsy?
2. Why is the Moro reflex asymmetric?
3. When should a baby with no biceps recovery be referred for surgery?
4. What is glenohumeral dysplasia?
5. What is the cookie test?

Linked practical spots

- Resting wrist and hand splint; Active Movement Scale and Mallet score as scale spots.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the Narakas classification of obstetric brachial plexus palsy.**

Answer
I: C5–C6 (Erb's). II: C5–C7 (with wrist and finger drop). III: C5–T1, flail arm without Horner's. IV: C5–T1 with Horner's syndrome (T1 avulsion), the worst outlook.

7 m **Physiotherapy management of a baby with Erb's palsy.**

Answer
Refer to a plexus clinic by 1 month. Gentle handling and positioning; no pulling on the arm. Home programme several times a day: external rotation with the scapula held, abduction, flexion, elbow extension, supination, wrist and finger extension. Play-based active use, hand-to-mouth play, weight-bearing in prone and sitting. Sensory input. Constraint of the good arm and splints later if needed. Surgery if the biceps has not recovered by about 3 months. Educate the parents and review the shoulder regularly.

LAQ **A 6-month-old cannot move the right arm since a forceps delivery; other milestones are normal. Give the differential diagnosis, goals and physiotherapy. (5 + 10, S19)**

Answer
Use the differential diagnosis table, the goals in section 11 and the management in section 12. Stress that at 6 months a baby without biceps recovery needs a surgical opinion now, and that the home programme protects external rotation.

MCQ **Horner's syndrome in a baby with a total plexus palsy suggests:**

A. A clavicle fracture

B. Avulsion of the T1 root

C. Upper trunk stretch only

D. Cerebral palsy

Answer
B. The sympathetic fibres to the eye leave with T1; avulsion carries the worst outlook.

MCQ **The most important range to maintain in an infant with Erb's palsy is:**

A. Shoulder internal rotation

B. Shoulder external rotation

C. Elbow flexion

D. Wrist flexion

Answer
B. Loss of external rotation leads to glenohumeral dysplasia.

MCQ **By what age should an infant with birth plexus palsy be referred to a specialist clinic?**

A. 1 month

B. 6 months

C. 1 year

D. When the child starts school

Answer
A. By 1 month, so that management starts before 3 months.

Last recall check. Write the Narakas groups, the surgical trigger at 3 months, and five lines of home programme.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Clinical review of neonatal brachial plexus palsy: referral by 1 month and management before 3 months. *J Hand Ther* 2020 (doi 10.1016/j.jht.2019.12.004).
3. Reviews of surgical timing in birth brachial plexus palsy: absent biceps at 3 months, Toronto test score under 3.5, cookie test by 9 months; Al-Qattan and Waters on antigravity biceps by 4–5 months.
4. Narakas AO. Obstetrical brachial plexus injuries. In: Lamb DW, ed. *The Paralysed Hand*. Churchill Livingstone; 1987 (Narakas groups).
5. Structure and classical content: Tecklin, *Pediatric Physical Therapy*; Campbell, *Physical Therapy for Children*.

5B(5) Infections of the nervous system: meningitis, encephalitis and their after-effects

> Kernig's and Brudzinski's signs were a 3-mark question twice (W19, S21). Brain infections matter to physiotherapists for their after-effects: weakness, spasticity, hydrocephalus, hearing loss, seizures and learning problems that need rehabilitation, much as after brain injury or in cerebral palsy.

Syllabus	Ch 5B(5) Infectious disorders (adult infections and their sequelae are covered here too)
Priority	**Tier 3.** In 2 of 20 papers (2016–2026): Kernig's and Brudzinski's signs as 3 marks twice.
Asked as	3m W19 S21
Usual split	3 (each sign: how to elicit, positive response, meaning; one line on limits)
Checked against	WHO fact sheets on meningitis (2026) and Japanese encephalitis (2024); diagnostic accuracy of meningeal signs (Thomas et al., 2002). Full list under Sources (p. 321). Content checked 20 October 2026.

1 Meningitis

Definition and types

- **Meningitis** is inflammation of the meninges, usually from infection.
- **Bacterial** (the most serious): group B streptococcus in newborns; meningococcus, pneumococcus and Haemophilus influenzae in children and adults.
- **Tuberculous meningitis (TBM):** slower onset over weeks; common in India; often leaves hydrocephalus, strokes and cranial nerve palsies.
- **Viral** (usually milder), fungal and parasitic forms.

Causes and key features

- Fever, headache, neck stiffness, light sensitivity, vomiting, confusion; seizures, coma or limb weakness in severe cases; a non-blanching rash in meningococcal sepsis.
- **Infants:** irritability, poor feeding, drowsiness, a bulging fontanelle.
- **TBM stages** (modified British Medical Research Council): I, alert with no focal signs; II, drowsy or with focal signs; III, coma.

Kernig's and Brudzinski's signs (3 marks in W19 and S21)

Sign	How to elicit	Positive when
Kernig's sign	Patient supine; flex the hip and knee to 90°, then try to straighten the knee	Pain and resistance in the back of the thigh stop knee extension (stretch on inflamed nerve roots)
Brudzinski's neck sign	Patient supine; flex the neck passively, chin towards chest	The hips and knees flex involuntarily
Brudzinski's contralateral leg sign	Flex one hip and knee passively	The other leg flexes too

Meaning: signs of meningeal irritation (meningitis, subarachnoid haemorrhage). **Limit:** in adults with suspected meningitis each sign was positive in only about 5% of those with meningitis (Thomas et al., 2002), so a negative sign never rules meningitis out; neck stiffness is also often absent.

Medical point that matters for physiotherapy

- Diagnosis by lumbar puncture (CSF cells, protein, sugar, organisms), blood cultures, and CT or MRI before the puncture when needed.
- Urgent antibiotics (and steroids in some types); anti-tuberculous treatment for months in TBM; shunt for hydrocephalus.
- About 1 in 6 people with bacterial meningitis die, and about 1 in 5 survivors have lasting effects: hearing loss, seizures, limb weakness, vision, speech and memory problems, and amputations after sepsis (WHO).
- Every child should have a hearing test after bacterial meningitis.

ICF summary for a typical child after meningitis

ICF component	Typical findings	Clinical reasoning
Body functions and structures	Hemiparesis or spasticity, ataxia, hydrocephalus, hearing loss, seizures, attention and memory problems	Brain inflammation, infarcts and raised pressure damage motor and sensory systems
Activities	Lost or delayed milestones, walking, hand use, communication	Motor and sensory loss plus slower learning
Participation	School, play	Hearing and learning needs
Environmental factors	Family support, access to hearing aids and therapy	Early hearing aids and therapy improve outcome
Personal factors	Age, previous development	Younger brains reorganise more

Physiotherapy by key problem

Problem	Technique	Why
Acute stage, drowsy or in the ICU	Positioning with the head up about 30°, turning, chest care, gentle range of motion, splints for the feet and hands	Prevents chest infection, sores and contracture without raising brain pressure
Weakness and spasticity	Task practice, strengthening, stretching and positioning, orthoses	As for stroke or cerebral palsy
Lost milestones (children)	Developmental training in sequence; play-based practice	Regains head control, sitting, walking
Ataxia and balance	Balance and coordination training	Cerebellar or vestibular damage
Hearing loss, attention problems	Face the child, use gestures and demonstration, short sessions; refer for hearing aids	Learning depends on understanding

Precautions

- Infection control in the acute stage (masks, hand hygiene) as the team directs.
- Avoid neck flexion and head-down positions while brain pressure may be high; watch for seizures.
- Watch for shunt problems and new weakness after TBM (stroke, hydrocephalus).

Exam points

- For 3 marks: both signs with method, response and meaning, plus the low-sensitivity point.

2 Encephalitis

- **Encephalitis** is inflammation of the brain tissue, usually viral: fever, headache, confusion, behaviour change, seizures, coma, focal weakness.
- **Japanese encephalitis** is spread by Culex mosquitoes and is endemic in parts of India; about 1 in 250 infections causes severe illness, up to 30% of those with symptoms die, and 20–30% of survivors keep cognitive, behavioural or neurological problems (WHO, 2024). JE vaccine is in India's immunisation programme in endemic districts.

- **After-effects:** movement disorders (parkinsonism, dystonia), spasticity, cognitive and behavioural problems, epilepsy.
- **Physiotherapy:** as for brain injury: positioning and chest care in the acute stage, then graded task practice, tone management, balance and gait training, and behavioural strategies (see brain injury (p. 66)).

3 Other infections that need rehabilitation

- **Neurocysticercosis** (pork tapeworm cysts in the brain): a common cause of seizures in India; occasionally hydrocephalus.
- **Brain abscess:** focal weakness or seizures depending on the site.
- **Tuberculosis of the spine** (Pott's disease) and transverse myelitis: paraplegia (see spinal cord (p. 77)).
- **Post-infectious and infection-related conditions:** GBS (GBS page (p. 144)), poliomyelitis (PPRP (p. 322)), leprosy (neuropathies (p. 107)), HIV-related neuropathy and myelopathy.

Recall check. Name four lasting effects of bacterial meningitis and the physiotherapy for each.

4 Exam corner

Past questions (MUHS, 2016–2026)

W19 Q1(d), S21 Q1(a), 3 marks

What is Kernig's and Brudzinski's sign?

Keywords to underline

meningeal irritation Kernig's sign Brudzinski's sign low sensitivity tuberculous meningitis hydrocephalus hearing test Japanese encephalitis sequelae

Answer skeleton: Kernig's and Brudzinski's signs (3 marks)

1. Kernig's: method and positive response.
2. Brudzinski's: method and positive response.
3. Meaning (meningeal irritation) and the limitation (often negative even with meningitis).

Draw these

- Stick figures showing Kernig's test (hip at 90°, knee being straightened) and Brudzinski's neck sign (neck flexed, knees and hips bending)

Viva questions

1. How do you elicit Kernig's sign?
2. What is Brudzinski's neck sign?
3. Why can a negative Kernig's sign not rule out meningitis?
4. Name three complications of tuberculous meningitis.
5. Which test must every child have after bacterial meningitis?

Linked practical spots

- Demonstrate Kernig's and Brudzinski's signs on a model patient.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **What is Kernig's and Brudzinski's sign? (W19, S21)**

Answer

Kernig's: supine, hip and knee flexed to 90°; attempting to straighten the knee causes pain and resistance in the hamstrings. **Brudzinski's neck sign:** passive neck flexion causes involuntary flexion of the hips and knees. Both indicate meningeal irritation (meningitis, subarachnoid haemorrhage). They are often negative even when meningitis is present (sensitivity about 5% in adults), so they do not rule it out.

3 m **List the complications of tuberculous meningitis.**

Answer

Hydrocephalus, strokes from inflamed arteries (hemiparesis), cranial nerve palsies, vision loss, seizures, spinal arachnoiditis with paraparesis, cognitive and behavioural problems.

7 m **Physiotherapy for a child recovering from bacterial meningitis.**

Answer

Acute: positioning with head up, turning, chest physiotherapy, gentle range of motion, splints; avoid raising brain pressure. **Assessment:** milestones, tone, power, balance, hearing and vision, cognition. **Rehabilitation:** developmental training in sequence; task practice for sitting, standing and walking; tone management and orthoses; balance and coordination; play-based sessions with gestures for hearing loss; parent education and home programme; refer for hearing aids, seizure care and school support.

LAQ **Practice: an 8-year-old girl treated for tuberculous meningitis 2 months ago has a VP shunt, right hemiparesis and poor balance, and is back home. Write ICF with clinical reasoning, goals and physiotherapy. (5 + 3 + 7)**

Answer

ICF (5): body functions: right hemiparesis, spasticity, poor balance, fatigue, possible hearing or vision loss; activities: walking, stairs, writing, self-care; participation: school, play; environment: parents, school access, therapy distance; personal: young, keen to return to school. **Goals (3):** short term: walks indoors safely, uses the right hand as a helper; long term: walks to school, climbs stairs, writes with adaptations. **Physiotherapy (7):** task-specific gait and balance training, strengthening, stretching and an AFO if needed, constraint-based or bimanual hand practice, endurance, shunt warning signs, home programme and school liaison.

MCQ **Passive neck flexion producing hip and knee flexion is:**

A. Kernig's sign

B. Brudzinski's sign

C. Lhermitte's sign

D. Babinski's sign

Answer

B.

MCQ **Japanese encephalitis is spread by:**

A. Aedes mosquitoes

B. Culex mosquitoes

C. Contaminated water

D. Droplets

Answer
B.

MCQ **After bacterial meningitis, every child should have:**

A. An EMG

B. A hearing test

C. A muscle biopsy

D. A nerve conduction study

Answer
B.

Last recall check. Write both meningeal signs with their limits, and five lasting effects of brain infections with the physiotherapy for each.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. Meningitis fact sheet, 2026.
3. World Health Organization. Japanese encephalitis fact sheet, 6 August 2024.
4. Thomas KE, Hasbun R, Jekel J, Quagliarello VJ. The diagnostic accuracy of Kernig's sign, Brudzinski's sign, and nuchal rigidity in adults with suspected meningitis. *Clin Infect Dis* 2002;35:46–52.

5B(6) Post-polio residual paralysis (PPRP): stages, deformities, complications and physiotherapy

> Post-polio residual paralysis came up in 4 papers, three times for 7 marks: physiotherapy, three deformities, and the complications. Every answer starts from one idea: polio destroys anterior horn cells patchily, the leftover muscle imbalance, posture and growth produce the deformities, and physiotherapy strengthens what is left, stretches what is tight and braces what cannot be fixed.

Syllabus	Ch 5B(6) Post poliomyelitis residual paralysis (acute polio is also on the anterior horn cell page; reconstructive surgery belongs to the musculoskeletal syllabus)
Priority	**Tier 2.** In 4 of 20 papers (2016–2026): 3 seven-mark and 1 three-mark question.
Asked as	7m W18 · W22 · S25 · 3m S24
Usual split	7 (definition 1–2; clinical features, deformities or complications 2–3; physiotherapy 3–4); 3 (definition with the stages)
Checked against	Review of the orthopaedic treatment and bracing of polio (Joseph and Watts, 2015); EFNS guideline on post-polio syndrome (Farbu et al., 2006); WHO poliomyelitis fact sheet (2025); polio-free certification of the WHO South-East Asia Region (CDC, 2014). Full list under Sources (p. 332). Content checked 20 October 2026.

1 Definition

- **Poliomyelitis** is an infection by the poliovirus that destroys anterior horn cells (lower motor neurons) in the spinal cord and sometimes the brainstem, causing sudden, asymmetrical, flaccid paralysis with normal sensation.
- **Post-polio residual paralysis (PPRP)** is the permanent weakness, wasting and deformity that remain once recovery from paralytic polio has stopped, about 2 years after the illness (the residual stage).
- It is different from **post-polio syndrome**: new weakness, fatigue and pain that start many years (at least 15) after a long stable period.

2 Classification and grading

Stages of paralytic polio

Stage	What happens	Physio focus
1. Acute (paralysis) stage: the first few weeks	Fever, painful muscle spasm, paralysis reaching its maximum within days	Rest, positioning and splints in the functional position to prevent deformity; gentle passive movement as pain allows; chest care if breathing is weak
2. Recovery (convalescent) stage: up to about 2 years	Surviving neurons recover and sprout; most recovery comes in the first 6 months	Muscle charting, graded re-education and strengthening, standing and walking in orthoses, stretching
3. Residual stage (PPRP): after about 2 years	No further recovery; imbalance, posture and growth cause deformity and shortening	Strengthen what is left, stretch, orthoses, gait training, physiotherapy around surgery

Some people develop **post-polio syndrome** decades later (section 8).

Types of polio infection

- Most infections cause no symptoms or a mild illness; about **1 in 200** causes irreversible paralysis, usually of the legs (WHO).
- **Spinal paralytic** (the commonest paralytic form): limb and trunk paralysis.

- **Bulbar**: brainstem involvement with swallowing, speech and breathing problems; **bulbospinal** is both.

Grading: each muscle is graded on the MRC scale (0–5) and recorded on a **muscle chart**, which guides orthoses and surgery.

3 Causes and risk factors

- **Poliovirus** (an enterovirus, types 1, 2 and 3), spread mainly by the faecal–oral route; it mainly affects children under 5 (WHO).
- **Unvaccinated** children are at risk; oral and injectable polio vaccines prevent it.
- **India:** the last wild polio case was on 13 January 2011 (Howrah, West Bengal), and the WHO South-East Asia Region, including India, was certified polio-free on 27 March 2014. Most patients with PPRP in India today are adults who had polio as children.
- **What makes deformity worse:** marked muscle imbalance, lying in the frog position, growth of the bones while weak muscles and contracted soft tissues lag behind, and no splinting, orthosis or physiotherapy.

4 Pathophysiology to impairments

1. The virus multiplies in the gut, enters the blood and destroys anterior horn cells, so the muscles they supply become flaccid and waste and reflexes are lost. Sensory and autonomic neurons are spared, so sensation and bladder control are normal.
2. The damage is patchy and asymmetrical. Muscles whose anterior horn cells lie in a small area of the cord are hit hardest: tibialis anterior, quadriceps, deltoid and opponens pollicis.
3. Surviving neurons sprout to take over orphaned muscle fibres (very large motor units), which explains recovery up to about 2 years. These overworked units may fail decades later (post-polio syndrome).
4. Because strong muscles pull against weak ones, and gravity and posture add to it, contractures and deformities develop; a joint with no muscle control becomes unstable (flail).
5. The paralysed limb grows less (shortening), carries less weight (thin bones) and loses its muscle pump (a cold, bluish limb).

Recall check. Why does polio cause deformity years after the paralysis itself has stopped changing?

5 Clinical features

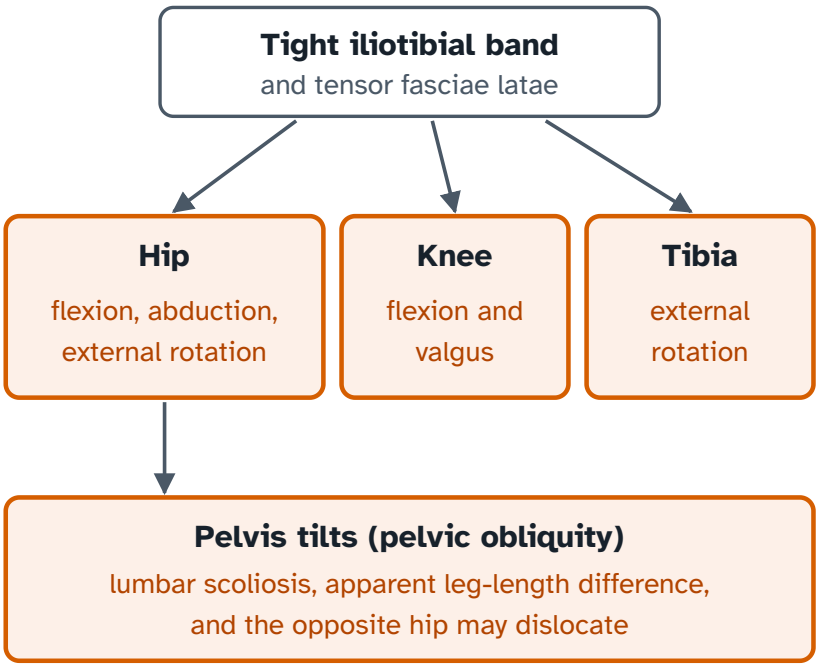
- Asymmetrical flaccid weakness and wasting, legs more than arms; reduced or absent reflexes; normal sensation, bladder and bowel.
- The affected limb is thin, shorter, and often cold and bluish.
- **Gait:** hand-on-thigh gait (weak quadriceps), Trendelenburg gait (weak hip abductors), backward lurch (weak gluteus maximus), high-stepping gait (foot drop), short-limb gait.

Deformities (W22: any three in detail)

Deformity	Why it develops	Signs and management
Hip flexion, abduction and external rotation	Tight iliotibial band and tensor fasciae latae (with sartorius and hip flexors), often from lying in the frog position	Positive Thomas and Ober tests; pelvic obliquity and lumbar scoliosis. Prone lying, stretching; soft-tissue release if fixed
Knee flexion	Tight hamstrings and iliotibial band; sitting or crawling for long periods	Crouched gait, knee unstable. Mild: serial casts; moderate: hamstring release
Genu recurvatum	Weak quadriceps: the person locks the knee backwards to stay stable; equinus adds to it	Knee bends backwards in stance. Leave mild recurvatum alone; over about 15° needs a brace or osteotomy

Deformity	Why it develops	Signs and management
Knee valgus and external rotation of the tibia	Tight iliotibial band pulling on the outer knee	Knock knee, foot turned out. Stretching; release of the band above the knee (Yount's release)
Equinus (often equinovarus or equinocavovarus)	Weak dorsiflexors (tibialis anterior) with a working calf, plus gravity	Foot drop, high-stepping gait, tripping. Calf stretching, AFO; tendon transfer later
Calcaneus (calcaneovalgus or calcaneocavus)	Weak calf with working dorsiflexors	Walks on the heel with no push-off. AFO; tendon transfer to the heel or fusion later
Paralytic scoliosis	Weak trunk muscles, or pelvic obliquity from a hip contracture	Early thoracic curves can impair breathing. Trunk strengthening, seating, breathing exercises; spinal surgery for severe curves
Upper limb	Deltoid (most often), elbow flexors, opponens pollicis	Flail or subluxed shoulder, weak elbow flexion, no thumb opposition. Support, strengthening; arthrodesis or tendon transfer

What a tight iliotibial band does



Draw this for the deformity questions. The band shortens when the child lies with hips bent, apart and turned out (the frog position) during the acute illness, then fails to grow with the bone.

Text version of the flowchart

1. A tight iliotibial band and tensor fasciae latae pull the hip into flexion, abduction and external rotation.
2. At the knee the band causes flexion and valgus.
3. It turns the tibia outwards (external rotation).
4. The hip abduction contracture tilts the pelvis (pelvic obliquity), giving a compensatory lumbar scoliosis and an apparent leg-length difference; the opposite hip can dislocate.

6 Investigations

- **Acute flaccid paralysis:** two stool samples, at least 24 hours apart, within 14 days of onset, tested for poliovirus (national surveillance).
- **PPRP** is diagnosed from the history and examination: a detailed muscle chart, range of motion, deformities and limb lengths.

- **X-rays:** deformities, hip subluxation, scoliosis; a scanogram or block test for limb-length difference.
- **EMG and nerve conduction** if the diagnosis is unclear: chronic denervation with large motor units, normal sensory conduction.
- Breathing tests if the trunk is weak or the scoliosis is severe; bone density in adults with fractures.

7 Medical and surgical management

- **Prevention:** vaccination (oral and injectable polio vaccines; the Pulse Polio programme in India). There is no antiviral cure; acute care is supportive.
- **Surgery** is planned in the residual stage, once recovery has stopped (usually after 2 years), using the muscle chart:

Problem	Surgery
Contracture	Soft-tissue release (iliotibial band, hip flexors, hamstrings, tendo Achillis lengthening)
Muscle imbalance or weakness	Tendon transfer from the strong side of the joint, only if the donor muscle is grade 4–5 (it usually loses about one grade); for example hamstrings to the quadriceps, tibialis posterior to the dorsum of the foot, or a superficialis tendon to restore thumb opposition
Unstable or deformed joint	Osteotomy (for example supracondylar extension osteotomy of the femur near skeletal maturity) or arthrodesis (foot, shoulder, wrist, spine)
Shortening	Shoe raise; lengthening of the short limb or growth arrest of the long one. A difference under 2 cm can be ignored
Scoliosis	Spinal fusion for severe or progressive curves

8 Complications and prevention

Asked for 7 marks in S25 (clinical features and physiotherapy of PPRP complications).

Complication	What you see	Physiotherapy
Contractures and deformities	Hip, knee and foot deformities (section 5)	Stretching, prone lying, serial casts, night splints, orthoses; rehabilitation after release surgery
Joint instability	Knee buckling, flail joints, hip subluxation	Strengthen the muscles left; floor-reaction orthosis or KAFO; do not correct a helpful mild recurvatum
Limb-length difference	Short-limb gait, pelvic tilt, back pain	Shoe raise, gait training, back and hip strengthening
Scoliosis and pelvic obliquity	Uneven shoulders and pelvis, poor sitting balance, breathlessness	Trunk strengthening, seating support, breathing exercises; release of hip contracture
Osteoporosis and fractures	Fractures (often the femur) after minor falls	Weight bearing in orthoses, falls prevention, gentle force when stretching
Overuse pain and early arthritis	Pain in the stronger leg, the knee pushed with the hand, and shoulders and wrists from crutches	Joint protection, correct crutch use, weight control, pacing
Cold, bluish limb and chilblains	Cool skin, poor circulation	Warm clothing, skin care, active movement
Post-polio syndrome	New weakness, fatigue, pain and cold intolerance after many stable years	Pacing, energy conservation, moderate non-fatiguing exercise, orthosis review, weight loss

Post-polio syndrome (Halstead criteria, recommended by the EFNS)

1. A confirmed history of paralytic polio.
2. Partial or fairly complete recovery after the acute illness.
3. At least 15 years of neurological and functional stability.

4. Two or more new problems after the stable period: marked fatigue, muscle or joint pain, new weakness, new wasting, loss of function, cold intolerance (new weakness is the key one).
5. No other medical explanation.

The EFNS guideline rates supervised aerobic and muscle training (isometric and isokinetic) as a safe, effective way to prevent further decline in people with moderate weakness (Level B); training in warm water without swimming is particularly useful.

9 PT assessment

- **History:** age at polio, recovery, previous surgery and orthoses, falls, pain, fatigue, any new weakness, work and home.
- **Muscle chart:** manual muscle testing (MRC 0–5) of every muscle on both sides, dated, so change and transfer options can be judged.
- **Range and deformity:** Thomas test (hip flexion), Ober's test (iliotibial band), popliteal angle (hamstrings), ankle dorsiflexion, recurvatum angle.
- **Limb length:** true and apparent; girth.
- **Spine and posture:** scoliosis, pelvic obliquity, sitting balance.
- **Gait and function:** pattern, aids, orthoses, stairs, transfers; endurance (for example 6-minute walk).
- Skin under orthoses, circulation, bone health and breathing if the trunk is weak.

10 PT diagnosis by ICF

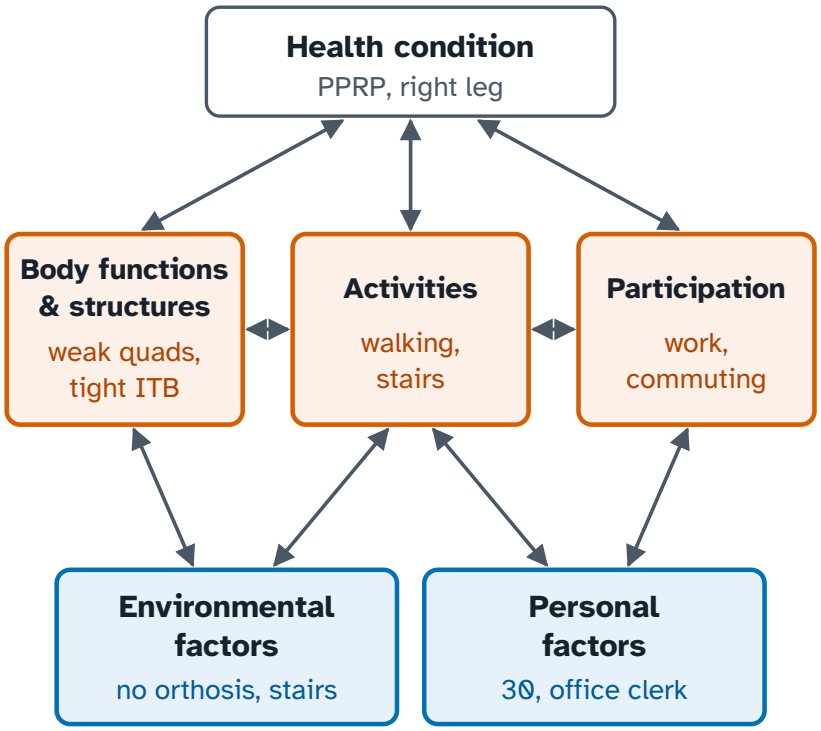
No PPRP long case was set in 2016–2026, so this is a typical case for practice. Draw the six-box ICF model and fill it the same way.

Practice case (typical presentation)

> A 30-year-old woman who had polio at the age of 2 works as an office clerk. Her right quadriceps is grade 2 and her right ankle dorsiflexors grade 1. She walks pushing her right thigh with her hand, has a tight right iliotibial band (hip flexion–abduction contracture and knee valgus), a foot drop and a right leg 3 cm shorter. For a year she has had right knee pain and tires by evening.

> **Spot the ICF clues before you write**
>
> polio at the age of 2: residual stage for decades, so no more nerve recovery; keep post-polio syndrome in mind. quadriceps grade 2: the knee buckles in stance, hence the hand-on-thigh gait; an orthosis can stabilise it. dorsiflexors grade 1: foot drop needs an AFO. tight iliotibial band: stretch it and watch the pelvis and spine. 3 cm shorter: a shoe raise. knee pain, tires by evening: overloaded joints and a costly gait, so pace her day.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	PPRP of the right leg, residual stage for 28 years	No further nerve recovery; problems come from weakness, imbalance and deformity
Body function and structure impairments	Right quadriceps grade 2, dorsiflexors grade 1 (foot drop); tight iliotibial band with hip flexion–abduction contracture and knee valgus; 3 cm shortening; knee pain; low endurance	The weak quadriceps cannot hold the knee in stance; imbalance and growth caused the contracture and shortening
Activity limitations	Walks slowly with a hand on the thigh; difficulty with stairs, slopes and uneven ground; trips; tires by evening	Without a hand or orthosis the knee buckles; a limping gait costs more energy
Participation restrictions	Commuting, a full working day, social outings	Fatigue and pain shorten the day
Environmental factors	**Facilitators:** a desk job; family. **Barriers:** no orthosis or shoe raise; stairs at work; crowded public transport	An orthosis and a shoe raise are quick wins
Personal factors	30-year-old woman who has coped for years; may dislike a visible orthosis	Involve her in choosing a light, discreet orthosis

One-line functional diagnosis

A 30-year-old clerk with PPRP of the right leg (quadriceps grade 2, foot drop, tight iliotibial band, 3 cm shortening) who walks with a hand on her thigh, has knee pain and tires by evening, limiting her commute and working day.

Recall check. Which finding in this case explains the hand-on-thigh gait, and what orthosis would remove the need for it?

11 Problem list and goals

Goals for the 30-year-old clerk

Problem	Short-term (4–6 weeks)	Long-term (3–6 months)
Knee buckling and foot drop	Fitted with a floor-reaction orthosis; walks indoors without pushing on her thigh	Walks to work and on stairs safely with the orthosis
Tight iliotibial band	Gains hip extension and adduction range; does daily prone lying and stretches	No progression of the contracture, pelvic tilt or scoliosis
Shortening	Shoe raise fitted	Level pelvis while walking
Knee pain and fatigue	Less pain; uses pacing at work	Completes a working day without exhaustion

12 PT management

By stage

- **Acute stage:** rest; keep the limbs in neutral (no frog position) with pillows, sandbags and splints in the functional position; warmth for painful spasm; gentle passive movement once pain allows; chest care if breathing is weak.
- **Recovery stage (up to 2 years):** muscle chart every few months; graded re-education (grades 0–1: passive movement and facilitation; grade 2: assisted and gravity-eliminated work, hydrotherapy; grade 3 and above: active and resisted work, short of fatigue); stand and walk early in orthoses; stretching and night splints.
- **Residual stage (PPRP):** strengthen partially weak muscles; stretch contractures; prescribe orthoses and a shoe raise; gait and functional training; physiotherapy before and after surgery; in adults, pacing and joint protection.

Answer for W18: physiotherapy for PPRP (7 marks)

1. **Assess:** muscle chart, contractures, limb length, gait, function.
2. **Strengthen** muscles of grade 3 or more against resistance, and grade 1–2 muscles with assisted and gravity-eliminated work, avoiding fatigue.
3. **Stretch** the iliotibial band, hip flexors, hamstrings and calf; prone lying; serial casts or night splints.
4. **Orthoses:** AFO for foot drop; floor-reaction orthosis or KAFO for a weak quadriceps; shoe raise; crutches or a stick.
5. **Gait and function:** walking with the orthosis, stairs, transfers, daily activities.
6. **Around surgery:** prepare and re-educate after tendon transfers and releases.
7. **Education:** home programme, skin checks, pacing, warning signs of post-polio syndrome.

Rehabilitation after a tendon transfer

- **Before:** strengthen the donor muscle (it must be grade 4–5), teach its action, and stretch any contracture.
- **After the immobilisation the surgeon sets:** start by asking for the donor's old action (for example "bend the knee" after a hamstring-to-quadriceps transfer) while the tendon now produces the new one; use touch, mirror or EMG biofeedback; progress to the new action in function.
- Protect the transfer from stretching; after foot transfers an AFO is usually worn for about 6 months.

Classic answer vs current guidance: correcting deformities

Classic	Current	Write in the exam
Stretch and correct every contracture and deformity.	With a weak quadriceps, mild genu recurvatum or mild equinus keeps the knee straight in stance, so leave them alone; correct recurvatum beyond about 15° (Joseph and Watts, 2015).	"Mild recurvatum or equinus that stabilises a knee with weak quadriceps is not corrected."

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Weak muscles, grade 3 or more	Progressive resisted exercise, **8–12 repetitions**, **2–3 days a week**, with rests; stop short of fatigue	Builds what remains without overworking the enlarged motor units
Weak muscles, grade 1–2	Assisted active and gravity-eliminated work; hydrotherapy	Lets weak muscles move through range
Contractures	Daily prone lying and sustained stretching; serial casts for a mild knee flexion contracture	Slows progression; fixed contractures need release
Knee instability (quadriceps grade 3 or less)	Floor-reaction orthosis if the knee has no deformity; KAFO with a drop-lock knee if the hip is also weak	Stops buckling in stance and frees the hand
Foot drop	AFO (posterior leaf spring)	Clears the foot in swing; prevents fixed equinus
Shortening over 2 cm	Shoe raise	Levels the pelvis and saves energy
Fatigue and overuse (adults)	Pacing, energy conservation, light orthoses, weight control	Protects joints and ageing motor units

Recall check. List the orthosis you would choose for foot drop, for a quadriceps of grade 2 with a normal hip, and for a weak quadriceps with a weak hip.

13 Precautions, contraindications and stop criteria

Take care

- Do not stretch away a mild recurvatum or equinus that keeps a weak knee stable.
- Avoid exercise to exhaustion; reduce it if pain, weakness or fatigue lasts into the next day.
- Thin bones: use gentle, sustained force when stretching, and protect against falls.
- Check the skin under orthoses: wasted limbs have little padding over bony points.
- After a tendon transfer, do not stretch the transferred tendon.
- New weakness, fatigue, breathing or swallowing problems in an adult: refer (possible post-polio syndrome).

14 Education, lifestyle and home programme

- Daily stretches and prone lying; strengthening a few times a week.
- Wear and care of orthoses and the shoe raise; check the skin daily.
- Pacing, rest breaks, a healthy weight and joint protection (correct crutch height and grip).
- Report new weakness, unusual fatigue, pain or breathing problems.
- Make sure children in the family complete their polio vaccination.

15 Prognosis and discharge criteria

- After about 2 years no further nerve recovery occurs; function then depends on the pattern of weakness, the control of deformity, orthoses and surgery.
- Many people walk and work independently with orthoses; some develop post-polio syndrome after at least 15 stable years.
- **Discharge when:** orthoses fit and are used, walking and transfers are safe, the home programme is independent, and the warning signs are known.

16 Exam corner

Past questions (MUHS, 2016–2026)

W18 Q2(f), 7 marks
Physiotherapy management for post polio residual paralysis.

W22 Q2(c), 7 marks
Define PPRP. Describe any 3 deformities seen in PPRP in detail.

S25 Q2(e), 7 marks
What is post polio residual paralysis? Describe the clinical features and physiotherapy management of PPRP complications.

S24 Q1(c), 3 marks
What is PPRP?

Pages for this topic: anterior horn cell diseases (p. 177), orthoses and splints (p. 249), neurological assessment (p. 29).

Keywords to underline

anterior horn cells asymmetrical flaccid paralysis sensation normal residual stage after 2 years muscle imbalance muscle chart iliotibial band contracture hand-on-thigh gait floor-reaction orthosis KAFO shoe raise tendon transfer post-polio syndrome

Answer skeleton: a 7-mark PPRP question

1. **[Definition, 1–2]** PPRP and the three stages of polio, with timing.
2. **[Features or deformities, 2–3]** Asymmetrical flaccid weakness with normal sensation (or the complications for S25); for each deformity give the cause, the signs and the treatment; draw the iliotibial band chain.
3. **[Physiotherapy, 3–4]** Muscle chart; strengthening by grade; stretching and positioning; orthoses and shoe raise; gait training; rehabilitation around surgery; education and post-polio syndrome.

Draw these

- The effects of a tight iliotibial band (section 5)
- A timeline of the three stages: acute (weeks), recovery (up to 2 years), residual (after 2 years)
- A muscle chart: muscles down the side, right and left grades with dates across

Viva questions

1. Which cells does the poliovirus destroy, and why is sensation normal?
2. When does the residual stage begin?
3. What deformities does a tight iliotibial band cause?
4. Why is mild genu recurvatum left alone when the quadriceps is weak?
5. How strong must a muscle be before it is used for a tendon transfer?
6. What is post-polio syndrome?

Linked practical spots

- Floor-reaction (ground-reaction) AFO, KAFO with a drop-lock knee, posterior leaf spring AFO, shoe raise, axillary and elbow crutches.

- Thomas test, Ober's test, limb-length measurement, a muscle chart.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **What is PPRP? (S24)**

Answer

Post-polio residual paralysis is the permanent asymmetrical flaccid weakness, wasting and deformity left after paralytic poliomyelitis once recovery has stopped, about 2 years after the illness. Polio passes through the acute stage (weeks), the recovery stage (up to about 2 years) and the residual stage. Sensation is normal; deformities come from muscle imbalance, posture and growth.

7 m **Define PPRP. Describe any 3 deformities seen in PPRP in detail. (W22)**

Answer

Definition as above. **1. Hip flexion, abduction and external rotation:** a tight iliotibial band and tensor fasciae latae, often from the frog position in the acute stage; positive Thomas and Ober tests; leads to pelvic obliquity, lumbar scoliosis and knee valgus. Treat with prone lying, stretching, and release if fixed. **2. Genu recurvatum:** weak quadriceps, so the person locks the knee backwards to stand (with equinus adding to it); leave mild recurvatum, brace or osteotomy if over about 15°; floor-reaction orthosis. **3. Equinus (foot drop):** weak tibialis anterior with a working calf and gravity; high-stepping gait and tripping; calf stretching, AFO, and a tendon transfer (for example tibialis posterior) when a grade 4–5 donor is available.

7 m **Describe the clinical features and physiotherapy management of PPRP complications. (S25)**

Answer

Features: asymmetrical flaccid weakness, wasting, lost reflexes, normal sensation, a short cold limb, abnormal gait. **Complications with physiotherapy:** contractures and deformities (stretching, positioning, casts, orthoses); joint instability (strengthening, floor-reaction orthosis or KAFO); limb-length difference (shoe raise, gait training); scoliosis and pelvic obliquity (trunk strengthening, seating, breathing exercises); osteoporosis and fractures (weight bearing, falls prevention); overuse pain and arthritis (joint protection, crutch advice, pacing); post-polio syndrome (pacing, moderate non-fatiguing exercise, orthosis review).

LAQ **Practice: a 30-year-old clerk who had polio at 2 has right quadriceps grade 2, foot drop, a tight iliotibial band and 3 cm shortening, and walks with a hand on her thigh. Write ICF with clinical reasoning, goals and physiotherapy. (5 + 3 + 7)**

Answer

ICF (5): the table in section 10 with the one-line diagnosis. **Goals (3):** short term: orthosis fitted and walking indoors without hand support, more hip range, shoe raise; long term: safe community walking and stairs, no progression of deformity, a full working day without exhaustion. **Physiotherapy (7):** muscle chart; resisted exercise for grade 3+ muscles and assisted work for weaker ones, short of fatigue; iliotibial band and hip flexor stretching with prone lying; floor-reaction orthosis (or KAFO if the hip is weak) and shoe raise; gait training on level ground, slopes and stairs; pacing and joint protection; home programme; watch for post-polio syndrome.

MCQ **The poliovirus destroys:**

A. Dorsal root ganglia

B. Anterior horn cells

C. The neuromuscular junction

D. The corticospinal tract

Answer
B. Hence flaccid paralysis with normal sensation.

MCQ **The residual stage of polio begins about:**

A. 2 weeks after onset

B. 6 months after onset

C. 2 years after onset

D. 15 years after onset

Answer
C. Fifteen or more stable years is part of the definition of post-polio syndrome.

MCQ **A tight iliotibial band causes all of these except:**

A. Hip flexion

B. Hip abduction

C. Hip internal rotation

D. External rotation of the tibia

Answer
C. It causes external rotation of the hip.

MCQ **A muscle chosen for tendon transfer should be at least:**

A. Grade 1

B. Grade 2

C. Grade 3

D. Grade 4

Answer
D. A transferred muscle usually loses about one grade.

Last recall check. Write the three stages of polio, three deformities with their causes, and seven points of physiotherapy for PPRP.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Joseph B, Watts H. Polio revisited: reviving knowledge and skills to meet the challenge of resurgence. *J Child Orthop* 2015;9:325–38.
3. Farbu E, et al. EFNS guideline on diagnosis and management of post-polio syndrome. *Eur J Neurol* 2006;13:795–801.
4. World Health Organization. Poliomyelitis fact sheet, 2 April 2025.
5. Bahl S, et al. Polio-free certification and lessons learned: South-East Asia Region, March 2014. *MMWR* 2014;63(42):941–6.

6. Regmi A, Iyengar KP, et al. Lower limb fractures in patients with post-polio residual paralysis: a systematic review and pooled analysis. *Indian J Orthop* 2026.

5B(7) Duchenne muscular dystrophy: Gowers' sign, gait, complications and physiotherapy

> Duchenne muscular dystrophy was a long case in 6 papers and a 7-mark question in 7, usually a boy of 4–8 who falls often and cannot get up from the floor. Every answer follows the stage of the disease: keep the child walking, stretching and active without damaging fragile muscle, then protect posture, breathing and independence in a wheelchair.

Syllabus	Ch 5B(7) D.M.D. and other myopathies (adult myopathies have their own page)
Priority	**Tier 1.** In 11 of 20 papers (2016–2026): 6 long cases, 7 seven-mark and 2 three-mark questions.
Asked as	LAQ S16 · W16 · S20 · W20 · W21 · S26 · 7m S16 · S18 · W22 · S24 · W24 · S25 · S26 · 3m S16 · S24
Usual split	15 (ICF with clinical reasoning 5, goals and management 10; or investigations 3, goals 5, management 7; S26 asked wheelchair prescription and goals)
Checked against	DMD care considerations, parts 1 and 2 (Birnkrant et al., 2018). Full list under Sources (p. 344). Content checked 20 October 2026.

1 Definition

Duchenne muscular dystrophy (DMD) is an X-linked recessive disease in which the muscle protein **dystrophin** is absent, so muscle fibres are damaged by their own contractions and are slowly replaced by fat and fibrous tissue. It causes progressive, mainly proximal weakness that begins in early childhood and later affects breathing and the heart.

- It affects boys: about **16–20 per 100,000** live male births (about 1 in 5,000–6,000).
- Signs usually appear between 2 and 5 years of age.

2 Classification and grading

Common muscular dystrophies

Type	Key features
Duchenne	X-linked; boys; onset 2–5 years; dystrophin absent; loss of walking around 10–14 years
Becker	X-linked; dystrophin reduced or abnormal; milder and later; many walk into adult life
Limb-girdle	Both sexes; shoulder and pelvic girdle weakness; onset childhood to adulthood
Facioscapulohumeral	Face, shoulder blades (winging) and upper arms; often asymmetrical
Myotonic	Adults; slow release of grip (myotonia), distal weakness, cataracts, heart conduction problems

The five stages of DMD (2018 care considerations)

1. **At diagnosis:** mild delay; falls; family learns about the disease.
2. **Early ambulatory:** Gowers' sign, waddling gait, toe walking; can still climb stairs.
3. **Late ambulatory:** walking is laboured; loses the ability to climb stairs and rise from the floor.
4. **Early non-ambulatory:** can self-propel a wheelchair for a time; posture is maintained; scoliosis may start.
5. **Late non-ambulatory:** arm function and posture are hard to maintain; breathing and heart problems grow.

3 Causes and risk factors

- A mutation of the **DMD gene** on the X chromosome, most often a deletion of one or more exons, so no working dystrophin is made.
- **Inheritance:** a carrier mother passes the gene to half of her sons (affected) and half of her daughters (carriers). Some cases arise from new mutations.
- A family history on the mother's side, such as an affected maternal cousin or uncle (W20 case), is a strong clue.

4 Pathophysiology to impairments

1. Dystrophin anchors the inside of the muscle fibre to its membrane. Without it, the membrane tears during contraction, especially **eccentric** (lengthening) contraction.
2. Calcium floods in, the fibre dies, and repair cannot keep up, so muscle is replaced by fat and fibrous tissue.
3. Because fat and fibrous tissue enlarge some muscles, the calves look big but are weak (**pseudohypertrophy**).
4. Because proximal muscles (hip extensors, quadriceps, shoulder girdle) fail first, the child struggles to rise from the floor, climb stairs and run.
5. Because weak muscles and fibrosis shorten, **contractures** develop (ankle plantarflexors, iliotibial band, hip flexors, later knees, elbows and wrists).
6. Because trunk and breathing muscles weaken, **scoliosis** and restrictive lung disease follow after walking is lost; the heart muscle is also affected (cardiomyopathy).

? Why avoid eccentric and heavy resistance exercise?

> Lengthening contractions and high loads put the most stress on the fragile membrane, so they cause more fibre damage than the muscle can repair. Submaximal activity such as swimming and cycling keeps the child fit without that damage.

Recall check. Why are the calves big but weak in DMD, and which muscles weaken first?

5 Clinical features

- **Early signs:** late walking; frequent falls; difficulty running, jumping and climbing stairs; toe walking; enlarged calves; Gowers' sign; learning or speech delay in some.
- **Later:** loss of walking; contractures; scoliosis; weak cough and breathing; cardiomyopathy; weight gain on steroids, weight loss later.

Gowers' sign (3 marks in S16 and S24)

Gowers' sign: rising from the floor in four steps



The child uses the arms to "climb up" his own legs because the hip and knee extensors are too weak to lift the trunk.

Gowers' sign is the way a child with weak hip and knee extensors gets up from the floor: he turns face down, rises onto hands and knees, then hands and feet with the hips high, walks his hands up onto his knees and thighs, and pushes the trunk upright. It shows proximal lower-limb weakness, typical of DMD but also seen in other myopathies and spinal muscular atrophy.

Gait deviations (7 marks with Gowers' sign in S26)

- **Waddling (Trendelenburg) gait:** the pelvis drops and the trunk sways over the stance leg because the hip abductors are weak.
- **Increased lumbar lordosis** with the shoulders held back: weak hip extensors, so the child keeps the line of gravity behind the hips.
- **Wide base** for balance.
- **Toe walking** (equinus) from tight calves and weak dorsiflexors; no heel strike.
- **Knee locked back** in stance as the quadriceps weaken.
- Arms held behind the trunk; slow, short steps; frequent falls; cannot run or jump.

6 Investigations

S16 asked to plan investigations (3 marks).

- **Creatine kinase (CK):** very high, many times normal, from early life.
- **Genetic testing:** deletion and duplication testing of the DMD gene first, then sequencing if needed; it confirms the diagnosis and the mutation type.
- **Muscle biopsy:** absent dystrophin on staining, only if genetic tests are unclear.
- **EMG:** myopathic pattern (small, brief, polyphasic potentials).
- **Heart:** ECG, echocardiogram or cardiac MRI.
- **Lungs:** FVC, peak cough flow, maximum inspiratory and expiratory pressures.
- **Bones:** spine X-ray for vertebral fractures; bone density.

7 Medical and surgical management

- **Glucocorticoids** started before substantial physical decline: prednisone or prednisolone **0.75 mg/kg a day** or deflazacort **0.9 mg/kg a day**. They prolong walking: in large studies the mean age at losing walking rose from 10.0 years (little or no steroid) to 11.2 years on daily prednisone and 13.9 years on daily deflazacort.
- Steroids must never be stopped suddenly (adrenal crisis); watch weight, bones, behaviour and growth.
- **Newer treatments** for some mutations: exon-skipping drugs and gene therapy, available in a few countries.
- **Heart:** regular checks and heart-failure medicines.
- **Breathing support:** lung volume recruitment, assisted cough and non-invasive ventilation at set thresholds (section 8).
- **Surgery:** Achilles tendon lengthening in selected walkers with a significant ankle contracture and good hip and knee extensor strength; posterior spinal fusion for a progressive curve after walking is lost.

8 Complications and prevention

Asked for 7 marks in S18 and W24.

Complication	What to know	Prevention and physio role
Contractures	Ankle plantarflexors, iliotibial band, hip flexors first; later knees, elbows, wrists, fingers	Home stretching **4–6 times a week**; night AFOs; standing; serial casting
Scoliosis	Rare while walking; common after walking is lost; worsens sitting and breathing	Check the spine at every visit; upright symmetrical seating with lateral supports; refer early; posterior spinal fusion for a progressive curve
Falls and fractures	20–60% have low-trauma limb fractures and up to 30% symptomatic vertebral fractures; a fracture can end walking	Fall-proof home and school; safe transfers; vitamin D and calcium; get the child back on his feet quickly after a fracture
Respiratory failure	Weak cough, chest infections, night-time hypoventilation (morning headache, poor sleep)	Lung volume recruitment when FVC is **60% or less**; assisted cough when FVC is under **50%**, peak cough flow under **270 L/min** or MEP under **60 cmH2O**; non-invasive ventilation as advised
Cardiomyopathy	Often silent until late	Regular cardiac review; avoid overexertion
Weight problems	Obesity on steroids; undernutrition and swallowing difficulty later	Activity, diet advice, dietitian
Pain, mood and learning	Back and joint pain; anxiety, depression; learning difficulty in some	Positioning, seating; school support; counselling

9 PT assessment

- **History:** milestones, falls, family history on the mother's side, steroid use, school, home (floor living, stairs, toilet), parents' concerns.
- **Timed tests:** time to rise from the floor (Gowers' time), 10-metre walk or run, four-stair climb; 6-minute walk.
- **Function:** North Star Ambulatory Assessment (walkers); Performance of the Upper Limb scale.
- **Strength:** manual muscle testing of proximal and distal groups.
- **Range of motion:** ankle dorsiflexion with the knee straight, iliotibial band (Ober test), hip flexors (Thomas test), hamstrings, elbows, wrists.
- **Posture and spine:** lordosis, scoliosis in sitting and standing.
- **Breathing:** FVC, peak cough flow, chest expansion, cough.
- Repeat at least every 6 months to define the stage.

10 PT diagnosis by ICF

Draw the six-box ICF model, then fill it from the case. In a child, participation means play, school and family life.

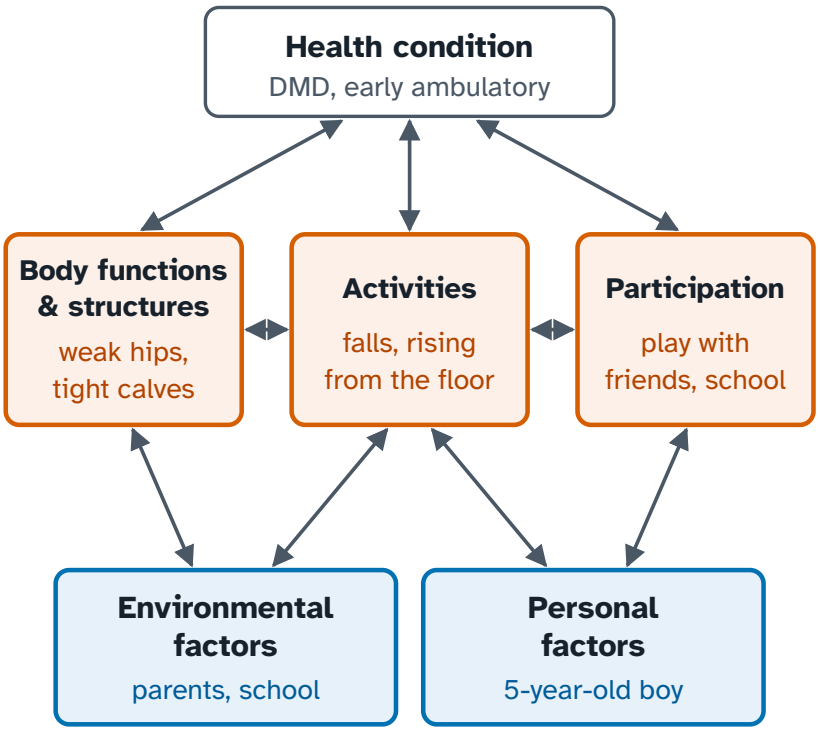
Worked example from a real paper (S20, 5 + 10 marks)

> "A 5 year old boy was brought to the PT OPD by his parents with complaints of frequent falls while playing. He has great difficulty in getting up from sitting position. On observation, size of calf muscles is noted to be increased. ICF with clinical reasoning. Detailed PT treatment with more emphasis on parent education and counselling."

Spot the ICF clues before you write

5-year-old boy with frequent falls, difficulty getting up and enlarged calves means DMD in the early ambulatory stage with proximal weakness and Gowers' sign. emphasis on parent education and counselling means the parents are the key facilitators: give a home programme and explain the future honestly.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box. The table below gives the full content.

ICF component	From this case	Clinical reasoning
Health condition	Duchenne muscular dystrophy, early ambulatory stage (confirm with CK and genetic testing)	Absent dystrophin; progressive muscle fibre loss
Body function and structure impairments	• Proximal weakness: hip extensors and abductors, quadriceps, shoulder girdle • Calf pseudohypertrophy; tightening of calves and iliotibial bands • Lumbar lordosis; waddling, toe-walking gait • Poor balance and low endurance	• Proximal muscles are lost first • Fat and fibrous tissue enlarge and stiffen the calves • Lordosis keeps the line of gravity behind weak hip extensors

ICF component	From this case	Clinical reasoning
Activity limitations	Frequent falls; difficulty rising from sitting or the floor (Gowers' sign); cannot run, jump or climb stairs like peers	Weak hip and knee extensors cannot lift the body against gravity
Participation restrictions	Left out of active play and games; slower at school activities	Fatigue and falls limit play with peers
Environmental factors	**Facilitators:** concerned parents who brought him early; school. **Barriers:** stairs, floor sitting at home and school, lack of knowledge about the disease	Parents will deliver most of the daily programme
Personal factors	5-year-old boy who wants to play	Exercise must be play-based and fun

One-line functional diagnosis

A 5-year-old boy with Duchenne muscular dystrophy in the early ambulatory stage, with proximal weakness, calf pseudohypertrophy and early tightness, who falls often and has great difficulty getting up, limiting play with friends. His parents are the main facilitators.

Practice cases from the papers: W20 5-year-old with falls for 2 years who cannot keep pace with peers and has an affected maternal cousin; W21 4-year-old who cannot rise from the ground, with large calves; S16 7-year-old with weight gain and calf hypertrophy (plan investigations); W16 5-year-old only son (home-care advice); S26 8-year-old from a rural area who walks only with support (wheelchair prescription).

Recall check. Fill the ICF for the S26 8-year-old. Which stage is he in, and how do the goals change?

11 Problem list and goals

Goals for the 5-year-old boy (agree them with the parents)

Problem	Short-term (4–6 weeks)	Long-term (6–12 months)
Tightness	Parents stretch ankles, knees and hips 4–6 times a week; night AFOs worn	Ankle dorsiflexion kept at least to neutral
Endurance and play	Swims or cycles for fun without next-day exhaustion	Keeps up with classmates in short play sessions
Falls	Home and school made safer	Fewer falls; no fractures
Walking	Walks around school with rests	Walking kept as long as possible
Family	Parents understand the disease and the home programme	Genetic counselling done; family copes and plans ahead

12 PT management

Principles (7 marks in S16, W22, S24 and S25): preserve walking and range for as long as possible; keep the child active without overworking fragile muscle; prevent deformity and falls; support breathing; keep him independent and included at every stage.

By stage

- **Early ambulatory:** home stretching and night AFOs; submaximal aerobic play (swimming, cycling); energy conservation (a buggy or wheelchair for long distances so he can still play); fall prevention; parent education.
- **Late ambulatory:** continue the above; serial casting for ankle tightness; KAFOs with locked knees to prolong standing and walking; a manual wheelchair for longer distances; standing frame.
- **Early non-ambulatory:** power wheelchair with custom seating; supported standing; daytime AFOs for foot position; arm and hand stretching; breathing exercises; spine checks.

- **Late non-ambulatory:** seating and pressure care; lung volume recruitment and assisted cough; arm supports and assistive technology; transfers with a hoist; pain management.

Classic answer vs current guidance: strengthening

Classic	Current	Write in the exam
Progressive resistance exercise to strengthen weak muscles.	Avoid eccentric and high-resistance exercise; give regular submaximal aerobic activity such as swimming and cycling, with rests and energy conservation (2018 care considerations).	"Submaximal aerobic activity (swimming, cycling) and functional play; no eccentric or heavy resistance work; stop before fatigue."

By impairment: technique, dose and why

Impairment	Technique and dose	Why
Contracture risk	Home stretching of ankles, knees and hips **4–6 times a week**; later wrists, hands and neck	Weak, fibrosing muscles shorten unless stretched regularly
Ankle tightness	Night AFOs from an early age; serial casting when needed	A prolonged low-load stretch at night works better than stretching alone
Low endurance	Swimming (from the early ambulatory stage) and cycling, submaximal, with rests	Keeps fitness and function without damaging muscle
Losing the ability to stand	KAFOs with locked knees; supported standing devices	Standing protects bones, range and the spine
Scoliosis risk	Symmetrical seating with a solid seat and back, lateral trunk supports, hip guides and a headrest	Supports a weak trunk after walking is lost
Weak cough	Breathing exercises; lung volume recruitment (FVC 60% or less); assisted cough (FVC under 50% or peak cough flow under 270 L/min)	Clears secretions and prevents chest infections
Fatigue	Energy conservation; rests between activities; a wheelchair for long distances	Saves energy for play and school

Wheelchair prescription (S26 long case)

1. **Assess:** stage, walking ability, arm strength, trunk control, contractures, spine, breathing, cognition; home, school and terrain (a rural area: rough roads, no ramps); who will push; budget and repair services.
2. **Choose:** first a light manual chair for long distances (pushed by family) while he still walks indoors; then a power chair when he can no longer walk, ideally with tilt, recline, elevating leg rests, standing support and adjustable seat height. In a rural setting, a sturdy frame with large pneumatic tyres that local mechanics can repair.
3. **Seating:** solid seat and back, lateral trunk supports, hip guides, adductor pads (pommel), headrest, pressure-relieving cushion, lap belt.
4. **Measure:** seat width about 5 cm more than hip width; seat depth about 5 cm less than buttock-to-knee length; seat height so the feet clear the ground with the footplate; armrest height about 2.5 cm above the elbow with the arm by the side; backrest high enough to support the trunk.
5. **Train and review:** teach the child and family to use, fold and maintain it; check fit and posture every few months as he grows.

Recall check. List the steps of wheelchair prescription and four seating features that protect the spine.

13 Precautions, contraindications and stop criteria

Take care

- Avoid eccentric work (jumping down, running downhill, lowering heavy loads) and heavy resistance.
- Stop before exhaustion; muscle pain or weakness lasting into the next day, or dark urine, means the activity was too hard.
- Fragile bones: no forceful stretching; safe transfers; prevent falls.
- Never stop steroids suddenly; in illness or injury the doctor may need to give extra steroid.
- Heart and breathing involvement: watch breathlessness and colour during exercise.

14 Parent education and home care

W16 asked for home-care advice to parents (3 marks); S20 for parent education and counselling.

- Explain the disease honestly and simply; the child did nothing wrong, and active play is good when it is not exhausting.
- Daily stretching routine and night splints; check the skin under splints.
- Fun activity: swimming, cycling, play; avoid jumping games and long walks to exhaustion; use a buggy or wheelchair for outings.
- Home safety: clear floors, rails on stairs, a western toilet or commode; chairs of the right height so he can rise easily.
- School: ground-floor classroom or lift, extra time, rest breaks, adapted sports, help on stairs.
- Healthy diet and weight control, especially on steroids; vitamin D and calcium as advised.
- Genetic counselling for the mother and sisters; disability certificate and benefits; parent support groups; emotional support for the child and siblings.

15 Prognosis and discharge criteria

- DMD is progressive. Without steroids, walking is lost at about 10 years on average; steroids delay this by one to several years.
- Scoliosis, breathing failure and cardiomyopathy follow in the teens; with steroids, breathing support and heart care, many now live into adulthood.
- Physiotherapy continues for life, with reviews at least every 6 months to stage the disease and update the programme.

16 Exam corner

Past questions (MUHS, 2016–2026)

S26 Q4(b)

8-year-old boy with DMD from a rural area who walks only with support when made to stand; tight plantarflexors; weak pelvic and shoulder girdle and respiratory muscles. Steps of wheelchair prescription; short- and long-term goals.

W21 Q3(b), 5 + 10 marks

4-year-old boy with recurrent falls, difficulty standing from the ground and hypertrophied calves. ICF with clinical reasoning; physiotherapy in detail.

W20 Q4(b), 5 + 10 marks

5-year-old boy with repeated falls for 2 years who cannot keep pace with other children and has difficulty getting up from sitting; his maternal cousin has the same problem. ICF with clinical reasoning; goals in detail.

S20 Q4(b), 5 + 10 marks

The worked example in section 10.

W16 Q4(b), 3 + 5 + 7 marks

5-year-old school-going boy, only son, with frequent falls, difficulty standing up from the floor and enlarged calves, diagnosed with DMD. Parental advice for home care; functional impairments with reasoning; short-term plan.

S16 Q3(b), 3 + 5 + 7 marks

7-year-old boy with frequent falls, difficulty getting up from the floor, weight gain over 6 months and calf hypertrophy. Investigations; goals; physiotherapy.

7-mark questions

Physiotherapy for DMD (S16, S24); complications of DMD (S18); managing the complications (W24); DMD and the principles of physiotherapy (W22); features and physiotherapy (S25); Gowers' sign and gait deviations (S26).

3-mark questions

Gowers' sign (S16, S24).

Pages for this topic: SMA and HSMN (p. 345), myopathies in adults (p. 186), wheelchairs (p. 257), orthoses (p. 249).

Keywords to underline

dystrophin X-linked proximal weakness pseudohypertrophy Gowers' sign waddling gait lordosis stretching 4–6 times a week night AFOs submaximal aerobic avoid eccentric scoliosis stage of disease

Answer skeleton: any DMD long case

1. Restate: age, falls, rising, calves, walking status, family history, home and school; name the stage.
2. **[ICF, 5]** ICF diagram and table with clinical reasoning; one-line functional diagnosis.
3. **[Goals, 3–5]** Short-term goals for range, safe activity and parents' skills; long-term goals to keep walking, prevent deformity and keep him at school and play.
4. **[Management, 7–10]** Stretching and night AFOs; submaximal aerobic activity; no eccentric or heavy resistance; energy conservation; falls and fracture prevention; standing, KAFOs, wheelchair and seating by stage; breathing; parent education and counselling; precautions.

Draw these

- Gowers' sign in four steps (section 5)
- The ICF model filled from the case

Viva questions

1. What is Gowers' sign, and which muscles are weak?
2. Why are the calves enlarged?
3. Why should a boy with DMD avoid eccentric exercise?
4. How often should home stretching be done?
5. When do you start assisted cough?
6. What seating features prevent scoliosis?

Linked practical spots

- Night AFO, KAFO with locked knee, wheelchair measurement, Gowers' sign demonstration.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Explain Gowers' sign.**

Answer
A child with weak hip and knee extensors rises from the floor by turning face down, getting onto hands and knees, then hands and feet with the hips high, walking the hands up onto the knees and thighs, and pushing the trunk upright. It shows proximal lower-limb weakness, classically in DMD.

3 m **Write three investigations for a boy suspected of DMD.**

Answer
Serum creatine kinase (very high); genetic testing of the DMD gene (deletion and duplication testing, then sequencing); muscle biopsy for dystrophin if genetics are unclear. Also ECG and echocardiogram, lung function tests.

7 m **Explain the complications of DMD and their management.**

Answer
Contractures (stretching 4–6 times a week, night AFOs, casting, standing); scoliosis after walking is lost (seating with lateral supports, spine checks, spinal fusion); falls and fractures (home safety, vitamin D, quick remobilisation); respiratory failure (lung volume recruitment at FVC 60% or less, assisted cough at FVC under 50% or peak cough flow under 270 L/min, non-invasive ventilation); cardiomyopathy (regular cardiac review); obesity on steroids (activity and diet); pain and low mood (positioning, support).

LAQ **A 5-year-old only son has frequent falls, difficulty standing up from the floor and enlarged calves (DMD). Give parental advice for home care, his functional impairments with reasoning, and a short-term plan. (3 + 5 + 7, W16)**

Answer
Home care (3): daily stretching and night splints; fun swimming and cycling but no exhausting play or jumping; safe floors and rails; chairs of the right height; diet; genetic counselling and support.

Impairments with reasoning (5): proximal weakness (fibre loss from absent dystrophin), so Gowers' sign and falls; calf pseudohypertrophy and early tightness; lordosis and waddling gait; poor endurance; balance problems.

Short-term plan (7): parents taught stretching of ankles, knees and hips 4–6 times a week; night AFOs; submaximal play-based aerobic activity with rests; balance games; posture; energy conservation with a buggy for long distances; home and school safety; review every 6 months.

MCQ **Which exercise should be avoided in DMD?**

A. Swimming

B. Cycling

C. Eccentric and high-resistance exercise

D. Stretching

Answer
C. Lengthening and heavy contractions damage the fragile muscle membrane.

MCQ **The inheritance of DMD is:**

A. Autosomal dominant

B. Autosomal recessive

C. X-linked recessive

D. Mitochondrial

Answer
C. Carrier mothers pass it to half their sons.

MCQ **The 2018 care considerations recommend home stretching:**

A. Once a month

B. Once a week

C. 4–6 times a week

D. Only after walking is lost

Answer
C. Daily preventive home stretching 4–6 times a week at ankles, knees and hips.

Last recall check. Draw Gowers' sign, list the gait deviations, and write the stretching, exercise and breathing thresholds.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Birnkrant DJ, et al. Diagnosis and management of Duchenne muscular dystrophy, part 1: diagnosis, and neuromuscular, rehabilitation, endocrine, and gastrointestinal and nutritional management. *Lancet Neurol* 2018;17:251–67.
3. Birnkrant DJ, et al. Diagnosis and management of Duchenne muscular dystrophy, part 2: respiratory, cardiac, bone health, and orthopaedic management. *Lancet Neurol* 2018;17:347–61.
4. Structure and classical content: Tecklin, *Pediatric Physical Therapy*; Campbell, *Physical Therapy for Children*; O'Sullivan and Schmitz, *Physical Rehabilitation* (wheelchair measurement).

5B(8) Spinal muscular atrophy and hereditary motor and sensory neuropathies (Charcot–Marie–Tooth disease)

> These two inherited lower motor neuron conditions are in the syllabus but were not asked on their own in 2016–2026. Learn them for short notes and vivas: SMA destroys anterior horn cells symmetrically from infancy, while Charcot–Marie–Tooth disease damages the long peripheral nerves from the feet upwards, so the physiotherapy is very different.

Syllabus	Ch 5B(8) S.M.A. and H.S.M.N.
Priority	**Syllabus.** Not asked on its own in 2016–2026.
Asked as	Not asked on its own in 2016–2026
Usual split	Likely 3 marks (types of SMA; features of CMT) or 7 marks (physiotherapy for either)
Checked against	SMA standards of care, parts 1 and 2 (Mercuri, Finkel et al., 2018); SMA genetics and incidence reviews; Charcot–Marie–Tooth reviews. Full list under Sources (p. 349). Content checked 20 October 2026.

1 Spinal muscular atrophy (SMA)

Definition and types

- **SMA** is an inherited (autosomal recessive) loss of anterior horn cells, causing symmetrical, proximal, lower motor neuron weakness. About 94% have deletion of both copies of the **SMN1** gene (exon 7) on chromosome 5; the number of copies of the back-up gene **SMN2** modifies severity.
- About 1 in 10,000 live births; about 1 in 40–50 people carry it.

Type	Onset	Best motor function without treatment
0	Before birth	Severe weakness at birth; breathing failure
1 (Werdnig–Hoffmann)	Before 6 months	Never sits; most died before 2 years without treatment
2	6–18 months	Sits but never walks
3 (Kugelberg–Welander)	After 18 months	Walks, but many lose walking later
4	Adult	Mild proximal weakness

Key features

- Symmetrical weakness and low tone, legs more than arms and proximal more than distal; absent reflexes; tongue fasciculations; tremor of the hands (minipolymyoclonus).
- Normal sensation and intelligence; the face is usually spared.
- Weak intercostals with a working diaphragm: bell-shaped chest and belly breathing; weak cough, chest infections.
- Scoliosis, contractures, hip subluxation, feeding difficulty (type 1).

Medical point that matters for physiotherapy

- Diagnosis by genetic testing (SMN1 deletion), increasingly through newborn screening.
- **Disease-modifying drugs** have transformed SMA: nusinersen (intrathecal, 2016), onasemnogene abeparvovec (one-time gene therapy, 2019) and risdiplam (oral, 2020). Treated babies, especially before symptoms, can reach milestones never seen before, so physiotherapy now aims at gaining skills, not only preventing complications.
- Cost and access remain major barriers in India.

Physiotherapy by functional level (SMA standards of care, 2018)

Level	Focus
Non-sitters	Positioning and supported seating, gentle range of motion, splints, chest physiotherapy and airway clearance, cough assist and NIV as advised, developmental play
Sitters	Seating and a wheelchair (often powered), spinal support, standing frame, orthoses, active play and arm function, contracture and scoliosis prevention, respiratory care
Walkers	Strength, endurance and balance training without exhaustion, aquatic exercise, falls prevention, orthoses, a wheelchair for long distances

Precautions

- Avoid exhausting exercise; watch breathing (night-time breathing problems, weak cough).
- Fragile bones: gentle handling.

2 Hereditary motor and sensory neuropathy (Charcot–Marie–Tooth disease)

Definition and types

- The commonest inherited neuropathy (about 1 in 2500 people): slowly progressive damage to the longest motor and sensory nerves.
- **CMT1** (HSMN I): demyelinating, with very slow nerve conduction; CMT1A (PMP22 gene duplication) is the commonest type.
- **CMT2** (HSMN II): axonal, with near-normal conduction velocity but small responses.
- Usually autosomal dominant, starting in childhood or adolescence.

Key features

- Distal weakness and wasting of the legs ("inverted champagne bottle" calves), then the hands.
- High-arched feet (pes cavus) with clawed toes; foot drop with a high-stepping gait; ankle sprains and falls.
- Reduced or absent ankle reflexes; distal sensory loss (glove and stocking); thickened nerves in some.

Physiotherapy by key problem

Problem	Physiotherapy
Foot drop, ankle instability	AFO (posterior leaf spring or carbon fibre), supportive high-top footwear
Calf and foot tightness, cavus	Daily calf and plantar fascia stretching, night splints; surgery for severe cavus
Weakness	Moderate strengthening of hip, knee and proximal muscles, avoiding overwork of very weak distal muscles
Balance and falls	Balance training with vision and a stick; home safety
Hand weakness	Hand exercises, splints, built-up handles, button hooks
Numb feet	Daily foot inspection, good footwear, callus care

Recall check. Compare SMA and CMT: which neurons, which distribution of weakness, and the main physiotherapy aim of each.

3 Exam corner

Past questions (MUHS, 2016–2026)

Not asked on its own

Expect short questions such as types of SMA, features of CMT, or physiotherapy for either.

Pages for this topic: anterior horn cell diseases (p. 177), neuropathies (p. 107), orthoses (p. 249), wheelchairs (p. 257).

Keywords to underline

SMN1 deletion SMN2 copies symmetrical proximal weakness tongue fasciculations non-sitter, sitter, walker nusinersen, risdiplam, gene therapy CMT1 demyelinating pes cavus inverted champagne bottle AFO

Draw these

- A table comparing SMA types 1–3 by onset and best motor function

Viva questions

1. Which gene is deleted in SMA, and what modifies severity?
2. How do the SMA types differ?
3. Why is the chest bell-shaped in SMA type 1?
4. What is the commonest type of CMT?
5. Why does a person with CMT have a high-stepping gait?

Linked practical spots

- AFO, cough assist device, standing frame, adapted seating.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Classify spinal muscular atrophy.**

Answer

Type 0 (prenatal, severe); type 1 Werdnig–Hoffmann (onset before 6 months, never sits); type 2 (6–18 months, sits but never walks); type 3 Kugelberg–Welander (after 18 months, walks, may lose walking); type 4 (adult, mild). Caused by SMN1 deletion; SMN2 copies modify severity.

3 m **Clinical features of Charcot–Marie–Tooth disease.**

Answer

Slowly progressive distal weakness and wasting of the legs (inverted champagne bottle), pes cavus and clawed toes, foot drop with high-stepping gait, absent ankle reflexes, distal sensory loss, later hand weakness; usually dominant inheritance, onset in childhood or adolescence.

7 m **Physiotherapy for a child with SMA type 2.**

Answer
A sitter: supported seating and a powered wheelchair; spinal support and scoliosis monitoring; standing frame for bones and hips; orthoses; range of motion and stretching for hip, knee and ankle contractures; active play and arm function; airway clearance, cough assist and NIV as advised; aquatic play; family education; review gains if on disease-modifying drugs.

LAQ **Practice: a 15-year-old boy with CMT1A trips often, has high-arched feet and weak ankles, and wants to play cricket. Write impairments, goals and physiotherapy. (5 + 3 + 7)**

Answer
Impairments (5): distal weakness (dorsiflexors, evertors, intrinsics), pes cavus and calf tightness, reduced ankle reflexes and distal sensation, poor balance, mild hand weakness. **Goals (3):** no trips at school, safe participation in cricket with an orthosis, maintained ankle range. **Physiotherapy (7):** carbon or leaf-spring AFOs and supportive footwear, calf stretching and night splints, proximal strengthening and moderate distal work, balance training, sport-specific agility with ankle support, foot care, and orthopaedic review for the cavus foot.

MCQ **A child who sits but has never walked most likely has SMA type:**

A. 1

B. 2

C. 3

D. 4

Answer
B.

MCQ **The commonest Charcot–Marie–Tooth type is due to:**

A. SMN1 deletion

B. PMP22 duplication

C. Dystrophin mutation

D. CTG repeat

Answer
B. CMT1A.

MCQ **Sensation in SMA is:**

A. Lost distally

B. Normal

C. Lost below a level

D. Lost on one side

Answer
B. Only anterior horn cells are affected.

Last recall check. Write the SMA types, the CMT features and the physiotherapy for each.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Mercuri E, Finkel RS, et al. Diagnosis and management of spinal muscular atrophy: part 1 (diagnosis, rehabilitation, orthopaedic and nutritional care). *Neuromuscul Disord* 2018;28:103–15.
3. Finkel RS, Mercuri E, et al. Diagnosis and management of spinal muscular atrophy: part 2 (pulmonary and acute care, medications, other systems, ethics). *Neuromuscul Disord* 2018;28:197–207.
4. Reviews of SMA genetics and treatment (SMN1 deletion, SMN2 copy number, nusinersen, onasemnogene abeparvovec, risdiplam) and of Charcot–Marie–Tooth disease.

5B(9) Paediatric extrapyramidal disorders: dyskinetic CP, kernicterus, Wilson's disease, chorea and dystonia

> Extrapyramidal problems in children came up as dyskinetic cerebral palsy: a long case in W25 and the posture and movement problems of athetoid CP for 7 marks in S26, both answered on the cerebral palsy page. This page adds the other childhood movement disorders and the shared rule for treating them: give the child a stable base and midline control, then practise useful tasks within it.

Syllabus	Ch 5B(9) Paediatric extrapyramidal disorders
Priority	**Tier 3.** In 2 of 20 papers (2016–2026) as dyskinetic cerebral palsy (W25 long case, S26 seven-mark question), answered on the cerebral palsy page.
Asked as	LAQ W25 · 7m S26
Usual split	15 (functional diagnosis 5, goals and physiotherapy 10, in W25); 7 (posture and movement problems of athetoid CP, S26)
Checked against	Standard paediatric neurology descriptions of involuntary movements, kernicterus, Wilson's disease, Sydenham's chorea and dopa-responsive dystonia. Full list under Sources (p. 354). Content checked 20 October 2026.

1 Types of involuntary movement

Movement	What it looks like
Dystonia	Sustained or intermittent muscle contractions causing twisting, repetitive movements or abnormal postures, often triggered or worsened by trying to move
Athetosis	Slow, continuous, writhing movements, mostly of the hands and feet
Chorea	Brief, irregular, flowing jerks that move from one body part to another
Ballismus	Large, flinging movements of a whole limb
Tremor	Rhythmic oscillation at rest, in posture or on action
Tics	Sudden, repetitive movements or sounds that can be held back for a while
Myoclonus	Sudden, shock-like jerks

All come from disordered basal ganglia circuits (and their links with the cortex and cerebellum); they usually disappear in sleep and worsen with effort, emotion and excitement.

2 Dyskinetic cerebral palsy

- The commonest extrapyramidal disorder in children: dystonic or choreo-athetoid movements, fluctuating tone, poor head and trunk control, drooling and unclear speech, often with normal intelligence.
- Linked to severe birth asphyxia (deep grey matter injury) and to kernicterus.
- The full answers (types, ICF of the 5-year-old athetoid girl, carrying, management, the W25 case and the S26 posture and movement problems) are on the cerebral palsy page (p. 275).

3 Kernicterus (bilirubin encephalopathy)

- Very high unconjugated bilirubin in a newborn crosses into the brain and damages the basal ganglia, brainstem nuclei and hearing pathways.
- **Classic picture:** athetosis (dyskinetic CP), limited upward gaze, sensorineural hearing loss, and enamel defects of the baby teeth; intelligence often spared.

- **Prevention:** early detection of jaundice, phototherapy and exchange transfusion.
- **Physiotherapy:** as for dyskinetic CP, with hearing checked early and communication through vision and gesture.

4 Wilson's disease (hepatolenticular degeneration)

- An inherited (autosomal recessive) defect of copper handling: copper builds up in the liver and brain (basal ganglia).
- **Neurological picture** in older children and young adults: tremor (including a coarse "wing-beating" tremor), dystonia, slow and slurred speech, drooling, parkinsonism, clumsy handwriting; behaviour and mood change.
- **Signs:** Kayser–Fleischer rings (brownish rings at the edge of the cornea); liver disease.
- **Treatment:** lifelong copper-removing drugs or zinc; liver transplant in liver failure. Treated early, much of the disability can improve.
- **Physiotherapy:** dystonia and tremor management (stable positions, weighted or built-up utensils, task practice), balance and gait training, speech and swallowing referral.

5 Sydenham's chorea

- A feature of rheumatic fever, weeks to months after a streptococcal throat infection, usually in school-age children.
- Chorea (often worse on one side), low tone and clumsiness, a "milkmaid's" grip that tightens and relaxes, emotional lability.
- Usually settles over weeks to months; the child needs long-term antibiotic prophylaxis because of the heart risk.
- **Physiotherapy:** safety during the active phase, rest and graded return to activity, handwriting and fine-motor support, and checking the heart before strenuous exercise.

6 Childhood dystonias and tics

- **Early-onset generalised dystonia** (genetic): often starts in a leg or foot in childhood and spreads; deep brain stimulation of the globus pallidus helps many.
- **Dopa-responsive dystonia:** childhood leg dystonia that is worse in the evening and better after sleep, with a dramatic, lasting response to low-dose levodopa; every child with unexplained dystonia should have a levodopa trial.
- **Tics and Tourette syndrome:** multiple motor tics and at least one vocal tic for more than a year, starting in childhood; physiotherapy has little role beyond advice.

7 Physiotherapy principles for involuntary movements

- **Stable base first:** seating with a pelvic belt, foot support and trunk supports; standing frames; work in positions where the child is steady.
- **Midline and symmetry:** hands together, head in midline, toys in front.
- **Graded, sustained holding** (weight bearing through arms and legs) to build proximal stability.
- **Task practice** of useful skills in stable positions; reduce effort and excitement that trigger overflow movements; calm surroundings.
- **Supports and devices:** wrist and thumb splints for function, weighted or built-up utensils, switches and communication aids.
- **Teamwork:** medicines (anticholinergics, baclofen, botulinum toxin for focal dystonia), speech and feeding therapy, and education.

Recall check. Name four childhood extrapyramidal disorders and one key physiotherapy principle for a child with athetosis.

8 Exam corner

Past questions (MUHS, 2016–2026)

W25 Q3(b), long case

A 7-year-old with jerky movements, fluctuating tone and truncal flexion in sitting (dyskinetic CP): functional diagnosis; goals and physiotherapy. Answer on the CP page (p. 275).

S26 Q2, 7 marks

Posture and movement problems in athetoid CP. Answer on the CP page (p. 275).

Keywords to underline

basal ganglia dystonia athetosis chorea fluctuating tone kernicterus Kayser–Fleischer ring dopa-responsive dystonia proximal stability midline control

Answer skeleton: a child with involuntary movements

1. Name the movement type and the likely cause.
2. Impairments: involuntary movement, fluctuating tone, poor proximal stability, speech and feeding.
3. Physiotherapy: stable seating and positions, midline control, sustained holding, task practice, devices; team and family.

Draw these

- A seated child with pelvic belt, foot support, lateral supports and a tray (stable base for an athetoid child)

Viva questions

1. How do athetosis, chorea and dystonia differ?
2. What is kernicterus, and which type of CP does it cause?
3. What is a Kayser–Fleischer ring?
4. Which childhood dystonia responds to levodopa?
5. Why does an athetoid child need a stable base?

Linked practical spots

- Adapted seating with pelvic belt and supports; built-up or weighted spoon; wrist splint.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Differentiate athetosis, chorea and dystonia.**

Answer

Athetosis: slow, continuous, writhing movements of the hands and feet. Chorea: brief, irregular jerks flowing between body parts. Dystonia: sustained contractions producing twisting movements or abnormal postures, worse with attempted movement.

3 m **What is kernicterus?**

Answer
Brain damage from very high newborn bilirubin, affecting the basal ganglia, brainstem and hearing pathways: athetoid (dyskinetic) CP, limited upward gaze, sensorineural hearing loss and dental enamel defects. Prevented by early treatment of jaundice (phototherapy, exchange transfusion).

7 m **Physiotherapy principles for a child with involuntary movements.**

Answer
Assess movement type, tone changes, postural control, function and seating. Provide a stable base (adapted seating with pelvic belt, foot and trunk supports; standing frame); midline and symmetrical positions; sustained holding and weight bearing for proximal stability; task practice in stable positions with low effort and calm surroundings; splints and adapted utensils; communication aids; work with medicines, speech and feeding therapy; family education and home programme.

LAQ **Practice: a 14-year-old girl with newly diagnosed Wilson's disease has dystonia of the right arm, tremor, slurred speech and falls. Write impairments, goals and physiotherapy. (5 + 3 + 7)**

Answer
Impairments (5): dystonia and tremor of the right arm, poor balance and gait, dysarthria, fatigue, possible mood change. **Goals (3):** short term: safe walking at home and school, writing with adaptations; long term: independent at school with improving control as treatment works. **Physiotherapy (7):** balance and gait training, falls prevention, task practice of writing and self-care in stable positions, weighted or built-up utensils, stretching for dystonic postures, endurance exercise, speech-therapy referral, education about lifelong medication.

MCQ **Kayser–Fleischer rings are seen in:**

A. Kernicterus
B. Wilson's disease
C. Sydenham's chorea
D. Tourette syndrome

Answer
B.

MCQ **Sydenham's chorea follows:**

A. Viral meningitis
B. Streptococcal throat infection
C. Poliomyelitis
D. Measles

Answer
B. As part of rheumatic fever.

MCQ **Childhood dystonia that is worse in the evening and responds dramatically to levodopa is:**

A. Dopa-responsive dystonia

B. Kernicterus

C. Spastic diplegia

D. Tourette syndrome

Answer

A.

Last recall check. Define each involuntary movement, list the childhood extrapyramidal disorders with one key feature each, and write the physiotherapy principles.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Standard paediatric neurology texts for involuntary movements, kernicterus, Wilson's disease, Sydenham's chorea and childhood dystonias.

Exam Answer writing for the Neuro Physiotherapy paper

> All 20 papers from 2016 to 2026 had the same shape: 80 marks in 3 hours, five of six 3-mark questions, five of six 7-mark questions, and two patient cases for 15 marks each. Marks follow structure as much as knowledge, so this page turns the notes into answers that score.

Syllabus	Exam skills for the whole Neuro Physiotherapy paper (51412)
Priority	**Essential.** Applies to every question in the paper.
Asked as	Every paper, 2016–2026
Usual split	Q1 5 × 3, Q2 5 × 7, Q3 and Q4 1 × 15 each (80 marks)
Checked against	MUHS IV BPTh Neuro Physiotherapy papers 2016–2026 (pattern and question counts from the session lists); MUHS syllabus (2012) printed scheme. Full list under Sources (p. 357). Content checked 20 October 2026.

1 The paper and a time plan

The syllabus prints a scheme with MCQs and 5-mark questions, but it was never used in 2016–2026. The real paper:

Question	Choice and marks	Time to spend
Read and choose	Tick your choices; plan the two cases	5 min
Q1 short answers	Any 5 of 6 × 3 = 15	About 6 min each (30 min)
Q2 short answers	Any 5 of 6 × 7 = 35	About 14 min each (70 min)
Q3 and Q4 long cases	Any 1 of 2 each × 15 = 30	About 30 min each (60 min)
Revise	Underline keywords; add missing doses and precautions	15 min

- About **2 minutes a mark**; never spend 20 minutes on a 3-mark answer.
- If you write slowly, do the two long cases first so 30 marks are safe.
- Choose the case whose **condition** you know best, not the shorter-looking one. The two options usually mix an upper and a lower motor neuron condition, or adult and paediatric.

2 What is asked most (2016–2026)

Topic	How often	Page
Cerebral palsy	Long case in 13 papers, plus short questions	Cerebral palsy (p. 275)
Stroke	In 18 papers: 11 long cases, 13 seven-mark questions	Stroke (p. 50)
Spinal cord injury, bladder, autonomic dysreflexia	In 18 papers: 9 long cases and a short question almost every paper	Spinal cord (p. 77), bladder (p. 238)
GBS	9 long cases, often ventilated or just out of the ICU	GBS (p. 144)
Peripheral nerves and plexus	In 14 papers; classification of nerve injury five times	Nerve injuries (p. 94)
Parkinson's disease, MS, DMD, brain injury	5–7 long cases each	PD (p. 164), MS (p. 133), DMD (p. 334), brain injury (p. 66)
Vestibular, approaches, tone, motor control and learning	Frequent 7-mark questions	Vestibular (p. 116), approaches (p. 209), tone (p. 216), motor control (p. 11)
Short-question favourites	GCS, UMN vs LMN, Modified Ashworth, ATNR, nerve injury classification, types of bladder, autonomic dysreflexia	Scales (p. 21), assessment (p. 29), development (p. 267)

Each page's priority line gives its exact count from the session lists.

3 Short answers: 3 and 7 marks

3 marks (about half a page)

1. One-line definition.
2. Three to five points, a small table or a labelled sketch.
3. One line of physiotherapy relevance.

> **Example: "Autonomic dysreflexia" (3 marks)**
>
> Definition: a sudden dangerous rise in blood pressure in spinal cord injury at or above T6, triggered by a stimulus below the lesion. Features: pounding headache, flushing and sweating above the lesion, slow pulse, blocked nose. Management: sit up and lower the legs, loosen clothing, check the bladder then the bowel, monitor blood pressure; the commonest cause is a full bladder.

7 marks (1.5–2 pages)

1. Definition or introduction (2 lines).
2. Classification, cause or the "why" (a flowchart or table is ideal).
3. The main body as 6–8 headed points: assessment points, aims, techniques with doses, by stage or by problem.
4. A labelled diagram where one exists.
5. Precautions.
6. One closing line: an outcome measure or the evidence.

Each condition page has an answer skeleton with marks for the questions actually asked.

4 Long answers: the 15-mark case

1. **Read the split** (for example 5 + 7 + 3 or 7 + 8) and give each part space in proportion.
2. **Underline the clues:** age, job and earning role, family, home (stairs, rural), time since onset, the main complaint, what the patient wants.
3. **Draw the ICF model,** then the ICF table with a "clinical reasoning" column, and a one-line functional diagnosis (see ICF (p. 4)).
4. **Problem list** in order of priority.
5. **Goals:** short- and long-term, SMART, with time frames, linked to the problems.
6. **Management:** by phase or stage, then by impairment with technique, dose and reason; precautions; education and home programme.
7. **Outcome measures** and discharge criteria in one or two lines.

Upper vs lower motor neuron cases

- **UMN** (stroke, SCI, MS, CP, brain injury): tone, motor control and task practice, balance and gait, cognition and perception, prevention of contracture.
- **LMN** (GBS, nerve injury, polio, neuropathy): protection of denervated muscle and numb skin, strengthening by grade, splints, breathing in GBS, re-education as reinnervation comes.

5 Diagrams that earn marks

- The ICF model (every long case).
- The motor pathway with UMN and LMN marked; the circle of Willis; incomplete cord syndromes.
- The autonomic dysreflexia flowchart; types of neurogenic bladder.

- Brachial plexus; nerve injury classification; the strength-duration curve.
- The gait cycle; the iliotibial band chain in polio; types of spina bifida; the neuromuscular junction.

Keep diagrams simple, labelled and large enough to read; a diagram never replaces the text.

6 Common mistakes and a final checklist

- Writing everything known about the condition instead of answering the parts asked.
- Goals without time frames; treatment without doses; no precautions.
- Ignoring the patient's job, family and home in the case.
- Old answers the guidelines have moved on from (for example passive stretching alone, bed rest, avoiding exercise in MS or myasthenia): check the "classic vs current" boxes.

Checklist before you hand in

- Question numbers and sub-letters correct; choices clear.
- Keywords underlined; headings visible.
- Every long case has the ICF model, functional diagnosis, goals with time frames, and doses.

Last week

- Day 1–3: the Tier 1 pages (stroke, CP, SCI, GBS, nerves, PD, MS, DMD, brain injury): ICF tables, goals, management skeletons.
- Day 4–5: 7-mark favourites (vestibular, approaches, tone, motor learning, Bell's palsy, spina bifida, MND).
- Day 6: 3-mark favourites (scales, UMN vs LMN, reflexes, autonomic dysreflexia, bladder types, nerve classification).
- Day 7: write two timed long cases and five short answers; read the recall checks.

Sources

1. MUHS BPTh syllabus (2012), printed theory scheme, and MUHS IV BPTh Neuro Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).

Exam The Neuro practical exam: long case, short case, spots and journal

> The Neuro practical is 80 marks: a long case (45), a short case with two skills (20), five spots (10) and the journal (5), marked on knowledge, hands-on skill and attitude. This pack follows that scheme and links each skill to the page that explains it.

Syllabus	Practical syllabus: clinical skills (sensory testing and re-education, muscle testing and re-education, electrical modalities, tone management, posture, transfers, functional re-education, gait, coordination, balance, fitness, outcome measures) and documentation of 8 cases with ICF
Priority	**Essential.** 80 practical marks plus 20 internal assessment.
Asked as	Long case 45 · Short case 20 · Spots 10 · Journal 5
Usual split	Long case 10 + 10 + 10 + 15; short case 2 × 10; spots 5 × 2; journal 5
Checked against	MUHS BPTh syllabus (2012), Neuro Physiotherapy practical examination scheme and clinical skills list. Full list under Sources (p. 360). Content checked 20 October 2026.

1 The marking scheme

Part	What is marked	Marks
Long case	(a) Subjective and physical examination 10; (b) evaluation and physiotherapy diagnosis (ICF) 10; (c) plan of care and goal setting 10; (d) demonstration of one important test and one treatment on the patient 15. Marked in cognitive, psychomotor and affective domains	45
Short case	Demonstration of two physiotherapy intervention skills for effective patient management, 10 each	20
Spots	5 spots, 2 marks each, 3 minutes per spot: EMG and nerve conduction studies, orthoses, prostheses, neurological assessment and scales	10
Journal	Documentation of cases with assessment, evaluation, diagnosis (ICF), prognosis, intervention and outcome	5

Internal assessment adds 20 marks. In the terminal and preliminary exams the spots carry 15 marks (5 × 3) and the journal is not marked.

2 Long case: assessment proforma

Subjective

- Demographics, dominance, occupation, family, home (stairs, toilet type), referral.
- Chief complaints in the patient's words; history of present illness (onset, progression, investigations, treatment); past, personal, family and social history; drugs; red flags.
- The patient's own goals.

Objective

- General: vital signs, consciousness (GCS), posture, deformity, skin, aids.
- Higher functions, cranial nerves, sensation, tone (Modified Ashworth), power (MRC or voluntary control), reflexes, coordination, balance (Berg, TUG), gait, function (Barthel or FIM) (see assessment (p. 29) and scales (p. 21)).
- Condition-specific: ISNCSCI in SCI, Hoehn and Yahr in PD, GMFCS in CP, muscle chart in LMN lesions.

Then

- Investigations seen; ICF table and functional diagnosis; problem list; goals; plan; outcome measures.

3 Presenting the long case (3–5 minutes)

1. One sentence: "Mr A, 54, bank manager, right MCA infarct 2 weeks ago, now..."
2. Key positive findings in ICF order: body functions and structures, activities, participation, environmental and personal factors.
3. The functional diagnosis in one line.
4. Two or three priority problems with short- and long-term goals.
5. The plan, then demonstrate the test and the treatment you chose (part d, 15 marks): explain, position, perform, check the response.

Choose a test and treatment that fit the main problem (for example the Modified Ashworth Scale and sit-to-stand training in stroke; ISNCSCI key muscles and a sliding-board transfer in paraplegia).

4 Short case: demonstrating skills (2 × 10 marks)

Skills from the syllabus

- Sensory testing and sensory re-education; muscle testing or voluntary control and muscle re-education.
- Electrical modalities for re-education or pain (see electrotherapy (p. 204)).
- Management of tone; facilitatory and inhibitory techniques (see tone (p. 216) and approaches (p. 209)).
- Postural assessment and correction; transfer training; functional re-education; gait assessment and training (see transfers and gait (p. 232)).
- Coordination testing and training; balance strategies (see balance (p. 223)).
- Fitness training; outcome measures and quality-of-life questionnaires.

Steps for any skill demonstration

1. Introduce yourself, explain, take consent; check precautions (blood pressure, skin, pain).
2. Position the patient and yourself; drape; guard.
3. Demonstrate with clear commands; give feedback; show progression.
4. Re-check the response and state the dose (sets, repetitions, frequency) and why.

5 Spots (5 × 2 marks, 3 minutes each)

Spot	What to say
EMG or nerve conduction report	Normal or abnormal; neurogenic or myopathic; demyelinating or axonal; the likely condition (for example slow conduction and absent F-waves in GBS) and what it means for physiotherapy (see assessment (p. 29))
Orthosis	Name (AFO type, KAFO, cock-up, knuckle-bender), parts, principle, indications, one precaution (see orthoses (p. 249))
Prosthesis	Level (below-knee, above-knee), parts (socket, suspension, pylon, foot), the usual indication (for example amputation for a diabetic foot), one gait problem
Assessment tool	Reflex hammer, tuning fork (128 Hz), two-point discriminator, goniometer: what it tests and how
Scale	Name, what it measures, scoring, interpretation (GCS, Modified Ashworth, ASIA, Berg, Hoehn and Yahr, Barthel) (see scales (p. 21))

6 The journal (5 marks)

- **Eight cases** using the ICF model: four upper motor neuron (stroke, SCI, brain injury, degenerative or demyelinating disease), two lower motor neuron (peripheral nerve or plexus injury, GBS), two paediatric (CP, myopathy, meningocele).
- For each: assessment, evaluation, diagnosis (ICF), prognosis, intervention and outcome, with measures at the start and end.
- Neat, complete, signed and up to date; the examiner may ask about any case in it.

7 The affective domain: what examiners watch

- Introduction, consent and explanation; respect, dignity and draping (see caring (p. 246)).
- Safety: guarding, gait belt, brakes on, checking blood pressure and skin, stopping for warning signs.
- Hand hygiene; tidy equipment; thanking the patient.
- Calm, clear answers; saying "I would check with the team" rather than guessing.

Last two weeks

- Practise the assessment proforma on classmates until it takes under 20 minutes.
- Rehearse presenting three typical cases aloud (stroke, SCI, CP) in ICF order.
- Practise each short-case skill with a partner, saying the commands and doses aloud.
- Revise spots with real reports, orthoses and scales; complete and check the journal.

Sources

1. MUHS BPTh syllabus (2012): Neuro Physiotherapy clinical skills, case documentation requirements and practical examination scheme.

Abbreviations

1RM: One-repetition maximum

ACA: Anterior cerebral artery

ADA: American Diabetes Association

ADIP: Assistance to Disabled Persons for purchase of aids and appliances (scheme)

ADL: Activities of daily living

AFO: Ankle–foot orthosis

AHA: American Heart Association

ALIMCO: Artificial Limbs Manufacturing Corporation of India

ALS: Amyotrophic lateral sclerosis

APTA: American Physical Therapy Association

ASA: American Stroke Association

ASIA: American Spinal Injury Association

ATNR: Asymmetrical tonic neck reflex

BBS: Berg Balance Scale

BP: Blood pressure

BPPV: Benign paroxysmal positional vertigo

CDC: Centers for Disease Control and Prevention (USA)

CFCS: Communication Function Classification System

CIDP: Chronic inflammatory demyelinating polyneuropathy

CIMT: Constraint-induced movement therapy

CK: Creatine kinase

CMT: Charcot–Marie–Tooth disease

CNS: Central nervous system

COMT: Catechol-O-methyltransferase

CP: Cerebral palsy

CPP: Cerebral perfusion pressure

CRPS: Complex regional pain syndrome

CSF: Cerebrospinal fluid

CT: Computed tomography

CVA: Cerebrovascular accident (stroke)

DASH: Disabilities of the Arm, Shoulder and Hand questionnaire

DGI: Dynamic Gait Index

DM: Diabetes mellitus

DMD: Duchenne muscular dystrophy

DNA: Deoxyribonucleic acid

DVT: Deep vein thrombosis

EAN: European Academy of Neurology

EAU: European Association of Urology

ECG: Electrocardiogram

EDSS: Expanded Disability Status Scale

EEG: Electroencephalography

EFNS: European Federation of Neurological Societies

EMG: Electromyography

EQ-5D: EuroQol five-dimension questionnaire

ESR: Erythrocyte sedimentation rate

FES: Functional electrical stimulation

FIM: Functional Independence Measure

FVC: Forced vital capacity

GBS: Guillain–Barré syndrome

GCS: Glasgow Coma Scale

GMCH: Government Medical College and Hospital

GMFCS: Gross Motor Function Classification System

HbA1c: Glycated haemoglobin (3-month blood sugar)

HINTS: Head impulse, nystagmus, test of skew (bedside stroke check in acute vertigo)

HIV: Human immunodeficiency virus

HKAFO: Hip–knee–ankle–foot orthosis

HSMN: Hereditary sensory and motor neuropathy

ICF: International Classification of Functioning, Disability and Health

ICMR: Indian Council of Medical Research

ICMR-INDIAB: ICMR–India Diabetes study

ICP: Intracranial pressure

ICU: Intensive care unit

IMT: Inspiratory muscle training

IP: Interphalangeal

IQ: Intelligence quotient

ISNCSCI: International Standards for Neurological Classification of Spinal Cord Injury

IVIG: Intravenous immunoglobulin

IWGDF: International Working Group on the Diabetic Foot

JE: Japanese encephalitis

KAFO: Knee–ankle–foot orthosis

LAQ: Long-answer question

LMN: Lower motor neuron

LSVT: Lee Silverman Voice Treatment

MAO-B: Monoamine oxidase B

MAS: Modified Ashworth Scale

MCA: Middle cerebral artery

MCP: Metacarpophalangeal

MDS: International Parkinson and Movement Disorder Society

MDS-UPDRS: Movement Disorder Society Unified Parkinson's Disease Rating Scale

MEP: Maximal expiratory pressure

MGFA: Myasthenia Gravis Foundation of America

MI: Myocardial infarction

MMSE: Mini-Mental State Examination

MND: Motor neuron disease

MoCA: Montreal Cognitive Assessment

MRC: Medical Research Council

MRI: Magnetic resonance imaging

MRP: Motor Relearning Programme
MS: Multiple sclerosis
NCV: Nerve conduction velocity
NDT: Neurodevelopmental treatment (Bobath)
NGO: Non-governmental organisation
NICE: National Institute for Health and Care Excellence (UK)
NICU: Neonatal intensive care unit
NIHSS: National Institutes of Health Stroke Scale
NIV: Non-invasive ventilation
NMES: Neuromuscular electrical stimulation
OPD: Outpatient department
PAD: Peripheral arterial disease
PCA: Posterior cerebral artery
PD: Parkinson's disease
PDQ-39: Parkinson's Disease Questionnaire (39 items)
PMA: Progressive muscular atrophy
PNF: Proprioceptive neuromuscular facilitation
PNS: Peripheral Nerve Society
PPRP: Post-polio residual paralysis
PT: Physiotherapy
ROM: Range of motion
RPE: Rating of perceived exertion (Borg scale)
RR: Respiratory rate
RTA: Road traffic accident
SA: Sinoatrial
SCI: Spinal cord injury
SF-36: Short Form 36 health survey
SMA: Spinal muscular atrophy
SMART: Specific, measurable, achievable, relevant, time-bound
SOL: Space-occupying lesion
STNR: Symmetrical tonic neck reflex
STREAM: Stroke Rehabilitation Assessment of Movement
TBI: Traumatic brain injury
TENS: Transcutaneous electrical nerve stimulation
TIA: Transient ischaemic attack
TUG: Timed Up and Go test
UK: United Kingdom
UMN: Upper motor neuron
UPDRS: Unified Parkinson's Disease Rating Scale
VP: Ventriculoperitoneal (shunt)
WHO: World Health Organization